

# Universal Field-Size Caps, Two-Sided Sandwiches, and Certificate Grammar

## for Mutual Correlated Agreement on Smooth Reed–Solomon Domains

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### Abstract

We study the mutual correlated agreement (MCA), correlated agreement (CA), and list challenges of Arnon–Boneh–Fenzi [ABF26] for Reed–Solomon codes  $C = \text{RS}[\mathbb{F}, D, k]$  on smooth evaluation domains. The goal is to locate the threshold  $\delta_C^*(\varepsilon)$  at cryptographic targets such as  $2^{-128}$ .

On the unsafe side we prove a field-size-universal cap. For every finite field with  $n < |\mathbb{F}| < 2^{256}$  and every smooth multiplicative row of rate  $\rho \in \{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}\}$  with  $k \leq 2^{40}$ ,

$$\varepsilon_{\text{mca}}(C, \delta) > \frac{1}{2k} \left(1 - \frac{n}{|\mathbb{F}|}\right) \geq 2^{-86} \gg 2^{-128}$$

throughout the printed near-capacity band, with the same bound for  $\varepsilon_{\text{ca}}$  on the integer-admissible subinterval. The proof is elementary: quotient-locator pigeonhole floors are composed with a self-contained deep-point conversion, with no value-set counting, exponential-sum estimate, or equidistribution input. The same mechanism extends to map-smooth domains, Chebyshev line rounds, bivariate circle codes, and rational scale maps relevant to genus-one rows.

On the safe side we prove two self-contained reductions. First, for every linear code, MCA has error at most  $(\lfloor \delta n \rfloor + 1)/q$  below one third of the minimum distance. Second, below one half of the minimum distance, MCA is bounded by ordinary CA up to an explicit tangent term. Combining this with the classical unique-decoding proximity gap of Ben-Sasson–Carmon–Ishai–Kopparty–Saraf [BCIKS20] gives a conditional half-distance safe frontier; a self-contained half-Johnson CA bound gives an import-free alternative.

For the deployed KoalaBear-sextic and Mersenne-31 circle rows, these results give explicit two-sided threshold intervals at the natural target of each row ( $2^{-128}$  and  $2^{-100}$  respectively). The remaining gap is an explicit aperiodic band problem, represented by finite rank-pivot and compressed-eliminant certificate language. The paper also records explicit witness words, explicit certifying pairs in residue-line normal form, an optimized failure profile, and exact integer certificates for the printed deployed-row verdicts; the sole optional non-elementary input is the imported half-distance CA theorem, isolated explicitly.

**Keywords:** Reed–Solomon codes, proximity gaps, correlated agreement, list decoding, FRI, STARKs.

**MSC 2020:** 94B35 (primary); 11T71, 68Q25 (secondary).

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# 1 Introduction

Proximity testing for Reed–Solomon codes underlies the soundness of modern SNARKs, and the sharpest available soundness accounting is phrased through *mutual correlated agreement* (MCA): a random linear combination  $f_1 + \gamma f_2$  being  $\delta$ -close to the code should force  $f_1, f_2$  to be simultaneously explained by codewords on a *common* agreement set, except with small probability  $\varepsilon_{\text{mca}}(C, \delta)$  over  $\gamma$ . The survey of Arnon, Boneh and Fenzi [ABF26] crystallizes the state of the art into a *grand MCA challenge*: for  $C = \text{RS}[\mathbb{F}, L, k]$  over a smooth domain  $L$ , with rate  $\rho \in \{1/2, 1/4, 1/8, 1/16\}$ , degree  $k \leq 2^{40}$ , and field size  $|\mathbb{F}| < 2^{256}$ , determine the largest proximity parameter  $\delta^*$  at which  $\varepsilon_{\text{mca}}(C, \delta) \leq 2^{-128}$  can hold.

**Role of this manuscript.** This is the main submission document for the present RS–MCA package. The companion papers and notes have distinct roles: the no-slack and slack papers identify the obstruction mechanisms and reserve ledgers, the protocol paper explains how those ledgers enter SNARK soundness, and the short *towards-prize* note extracts the current execution plan. The final mathematical claims intended for submission are consolidated here: the field-size-universal unsafe cap, the safe-side pincer, the deployed-row two-sided intervals, the model-transfer statements, and the finite certificate grammar. In particular, whenever a public row, scanner output, or companion note cites the cap or deployed threshold package, this paper is the authority for the hypotheses, endpoint conventions, denominators, and proof status.

Positive results reach the Johnson bound  $1 - \sqrt{\rho}$ . On the negative side, [Cho26b] showed  $\delta_C^*(2^{-128}) < 1 - \rho - 1/64$  for all prime fields of size up to roughly  $2^{150}$  (per-rate reaches between  $2^{150.6}$  and  $2^{161.8}$ ), by exhibiting quotient-locator witnesses whose bad-slope sets are value sets of restricted symmetric sums over small subgroups; it further proved a *conditional* universal cap over every field of size at most  $2^{256}$  — error  $2^{-42}$  at gap  $2^{-11}$ , assuming  $|\mathbb{F}| \geq 2n$  — by composing quotient-core lists with the same conversion isolated and proved directly here (its universal-cap theorem). Above the value-set reach, what remained open was the *error-one* question, identified there with the *per-fiber collision problem*: controlling, at a *fixed* large prime, the collisions of characteristic-zero locator values under reduction. Extension fields, the setting actually deployed ([ABF26, §6.3] works over a sextic extension of KoalaBear), posed an additional obstruction: by the subfield-confinement theorem of [Cho26b], refined in Section 6, every explicit witness of [Cho26a, Cho26b] is  $B$ -rational and contributes only  $p^{-5} < 2^{-154}$  over the deployed extension.

This paper sharpens the universal cap — gap  $2^{-9}$  (resp.  $2^{-10}$ ) in place of  $2^{-11}$ , error  $2^{-86}$  with no condition relating  $n$  and  $|\mathbb{F}|$ , and extension fields handled directly — for every field below  $2^{256}$  at once, by changing what is counted.

**Theorem 1.1** (universal unsafe cap, informal; see Theorem 4.1 and Corollary 5.1). *Let  $\mathbb{F}$  be any finite field with  $|\mathbb{F}| < 2^{256}$ , let  $D \subseteq \mathbb{F}^\times$  be a smooth multiplicative evaluation domain of order*

$n$ , and let  $C = \text{RS}[\mathbb{F}, D, k]$  have  $k = \rho n \leq 2^{40}$  with  $\rho \in \{1/2, 1/4, 1/8, 1/16\}$ . Assume  $2^{10} \mid n$  for  $\rho \in \{1/2, 1/4, 1/8\}$ , and  $2^{11} \mid n$  for  $\rho = 1/16$ . Then

$$\varepsilon_{\text{mca}}(C, \delta) > 2^{-86} \gg 2^{-128}$$

for every  $\delta \in [1 - \rho - 2^{-9}, 1 - \rho]$  at rates  $1/2, 1/4, 1/8$ , and for every  $\delta \in [1 - \rho - 2^{-10}, 1 - \rho]$  at rate  $1/16$ . The same lower bound holds for  $\varepsilon_{\text{ca}}(C, \delta)$  on the integer-admissible subinterval ending at  $1 - \rho - 1/n$ . If  $|\mathbb{F}| \geq 2n$ , the lower bound is  $\geq 2^{-42}$ . Hence  $\delta_C^*(2^{-128}) \leq 1 - \rho - 2^{-9}$  for  $\rho \in \{1/2, 1/4, 1/8\}$  and  $\delta_C^*(2^{-128}) \leq 1 - \rho - 2^{-10}$  for  $\rho = 1/16$ .

Combining the unsafe caps, the safe-side theorems, and the certificate refinements gives the following two-sided summary.

**Theorem 1.2** (two-sided deployed intervals, informal; see Theorem 10.12, Table 5, Corollary 11.5). *For every deployed row — the KoalaBear-sextic multiplicative row and the Mersenne-31 circle rows, at their stated challenge targets — the grand MCA threshold is pinned two-sidedly: at rate  $\rho = \frac{1}{2}$ ,*

$$\begin{aligned} \delta_C^* &\in \left[ \frac{1-\rho}{3}, \frac{15331}{32768} \right] && \text{unconditionally, and} \\ \delta_C^* &\in \left[ \frac{1-\rho}{2}, \frac{15331}{32768} \right] && \text{modulo one classical import [BCIKS20],} \end{aligned}$$

so the threshold is known within a factor smaller than two (the upper edge is  $\frac{15331}{32768} \approx 0.46787$ ). The same two-sided sandwich holds for genus-one (ECFFT) rows and for bivariate circle rows over every challenge field, whether or not it contains  $i$ .

The proof composes two ingredients. The first (Lemma 3.1) is a pigeonhole lower bound on list sizes at a single received word: the quotient locators of [Cho26a] attach to each  $\ell$ -subset  $A$  of the order- $N$  quotient coset  $Q = D^a$  a codeword  $c_A$  at distance  $\leq 1 - \rho - t/N$  from the word  $u_{z_A} = x^{k+ta} + z_A x^{k+(t-1)a}$ , where  $z_A = -e_1(A)$ ; pigeonholing the  $\binom{N}{\ell}$  subsets over their  $\leq |B|$  slope values produces one word  $u_z$  carrying a list of size  $\binom{N}{\ell}/|B|$ . Two refinements of [Cho26a, Main Thm. (d)] matter here: we run the construction at *slack two* ( $t = 2$ ), so that the listed codewords have degree  $\leq k$  and live in  $\text{RS}[\mathbb{F}, D, k+1]$  without any divisibility demand on  $k+1$  (which is odd in all challenge instantiations, while  $a$  is a 2-power); and we pigeonhole over the *subfield*  $B$  containing  $D$  rather than over  $\mathbb{F}$ , which is what makes the extension-field case go through. The second ingredient is the direct deep-point conversion of Theorem 2.6: in the integer-error range  $f \leq n - k - 1$ , if  $\varepsilon_{\text{ca}}(C, \delta)$  is below  $\eta(\frac{1}{k} - \frac{n}{k|\mathbb{F}|})$ , then *every* list of  $\text{RS}[\mathbb{F}, D, k+1]$  at radius  $\delta$  has size at most  $\lceil |\mathbb{F}| \varepsilon_{\text{ca}} / (1 - \eta) \rceil$ . Taking  $\eta = \frac{1}{2}$ , the heavy fiber of Lemma 3.1 violates this whenever  $\binom{N}{\ell} \geq |B|(|\mathbb{F}|/k + 1)$  — an inequality with hundreds of bits of slack throughout the challenge envelope — and the contrapositive yields  $\varepsilon_{\text{ca}}(C, \delta) > \frac{1}{2k}(1 - n/|\mathbb{F}|)$  throughout the integer-admissible range. The MCA cap then persists up to capacity by support-wise MCA monotonicity.

No value-set counting, exponential-sum estimate, or equidistribution hypothesis enters: pigeonhole fibers exist over every field unconditionally, and Theorem 2.6 converts list mass into correlated-agreement error directly. The cost of this route, relative to the error-one results of [Cho26a, Cho26b] below  $2^{150}$ , is an error of order  $1/k$  rather than  $1 - o(1)$ , and a gap  $2^{-9}$  at the first three prize rates (uniformly  $2^{-10}$  across all four), rather than the fixed gap  $1/64$ ; Section 12 quantifies the trade-off and what remains open.

Beyond the universal cap (Corollary 5.1, with a two-tier summary combining it with [Cho26b]), we prove: a deployed-parameter corollary at the KoalaBear-sextic instantiation of

[ABF26, §6.3], with  $\varepsilon_{\text{mca}} > 2^{-22}$  at gap  $2^{-7}$  and the same  $\varepsilon_{\text{ca}}$  lower bound on the integer-admissible subinterval (Corollary 5.4), extended to every interleaved row of [ABF26, Tables 2–3] via an interleaving-transfer lemma (Lemma 3.84 and Corollary 5.5); an independent slacked variant through [BCHKS25, Thm. 1.9] for robustness (Proposition 5.6); and a subfield-confinement lemma (Lemma 6.1) showing that for lines with  $B$ -valued words over an extension  $\mathbb{F}/B$ , every CA- or MCA-bad slope lies in  $B$ . The latter has a structural consequence (Corollary 6.2): at deployed parameters the lines certifying Corollary 5.4 are necessarily *not*  $B$ -rational, so the failure proved here is fiber-borne and currently non-constructive; exhibiting explicit certifying lines is posed as Open Problem 12.1.

A further axis is that the counting side of the composition is not tied to the multiplicative structure of  $D$ . Section 7 proves the fiber lemma for arbitrary polynomial folding maps with complete uniform fibers (Definition 7.1 and Lemma 7.2), removes the divisibility hypothesis  $a \mid k$ , and instantiates the resulting universal cap (Theorem 7.4) on the deployed isogeny-smooth family: twin-coset  $x$ -coordinate domains of the circle group over  $p \equiv 3 \pmod{4}$  carry exact Chebyshev fibers at every dyadic scale, and over challenge fields containing  $i$  the bivariate circle code is diagonally equivalent to a Reed–Solomon code on a torus twin coset. The Mersenne-31/QM31 circle family of [HLP24] thereby acquires the same negative bands as the multiplicative rows — error  $> 2^{-23}$  at gap  $2^{-7}$  at the deployed sizes, gaps  $2^{-9}$ – $2^{-11}$  universally (Corollaries 7.12 and 7.13) — answering the circle case of the isogeny-smooth problem with the elliptic (ECFFT) case reduced to a precise lacunarity obstruction (Remark 7.16, Open Problem 12.4).

The paper has three main components. First, it gives a field-size-universal unsafe construction: quotient-fiber and graded-prefix floors produce large lists at one received word, and a direct deep-point conversion turns that list mass into CA and MCA failure near capacity. Second, it supplies the safe side needed to interpret these failures as a threshold statement: a self-contained deep-regime theorem proves MCA soundness below one third of the distance, while a pincer theorem reduces MCA to ordinary CA up to half the distance, with the half-distance endpoint obtained from a single imported proximity-gap theorem. Third, it packages the deployed consequences as finite certificates: the scanner outputs the safe edge, unsafe edge, and remaining open band for each row, and every printed inequality is checkable by exact integer arithmetic.

After these components, the remaining mathematical question is no longer a list of book-keeping tasks. It is the aperiodic band problem: between the half-distance safe frontier and the entropy–subfield unsafe frontier, prove a correlated-agreement theorem crossing toward capacity, or find a new floor construction beating the graded-prefix envelope (Open Problem 9.11).

**Roadmap.** Section 2 fixes definitions, aligned with [ABF26, Defs. 4.1, 4.3], proves the direct deep-point conversion used for the main cap, and records the separate BCHKS/ABF slacked fallback. Section 3 proves the fiber and interleaving lemmas and the conservative quotient-support upper ledger, Section 4 the main theorem, Section 5 the challenge-level and deployed corollaries, and Section 6 subfield confinement. Section 7 extends the cap beyond multiplicative smoothness: it proves the map-smooth, divisibility-free fiber lemma and universal cap, the exact Chebyshev-fiber and torus twin-coset geometry of circle domains, and the resulting circle-code caps at the deployed Mersenne-31/QM31 parameters. Section 8 formulates the integer staircase and the certificate interface used to compare safe and unsafe radii. Section 9 develops the structural ledgers, extension-line lift, residual-band language, scanner, and finite certificate core. Section 10 proves the safe-side pincer below half the distance and the scale-optimized unsafe floors, then assembles the narrowed two-sided sandwich. Section 11 treats rational scales,

circle-code transport over every challenge field, explicit witnesses, optimized failure profiles, and the deployed-row certificate table. Section 12 discusses scope, limitations, and the remaining open problems. Throughout, companion results of [Cho26a, Cho26b] are cited by their printed theorem names rather than numbers, since the numbering of those manuscripts is not yet frozen.

## 2 Preliminaries

**Codes and distances.** Throughout,  $B \subseteq \mathbb{F}$  are finite fields,  $q := |\mathbb{F}|$ , and  $D = \alpha H \subseteq B^\times$  is a multiplicative coset of a cyclic subgroup  $H$  of order  $n$ ; we write  $D$  for the evaluation domain (denoted  $L$  in [ABF26]). The subgroup case is  $\alpha = 1$ , and taking  $B = \mathbb{F}$  is always allowed. For  $k \leq n$ ,

$$\text{RS}[\mathbb{F}, D, k] := \{(p(x))_{x \in D} : p \in \mathbb{F}[X], \deg p < k\} \subseteq \mathbb{F}^n, \quad \rho := k/n.$$

For  $u, v \in \mathbb{F}^n$ ,  $\Delta(u, v)$  is the relative Hamming distance,  $\Delta(u, C) := \min_{c \in C} \Delta(u, c)$ , and the relative minimum distance of  $C = \text{RS}[\mathbb{F}, D, k]$  is  $\delta_{\min}(C) = 1 - \rho + 1/n$ . For  $S \subseteq D$ ,  $\Delta_S(u, v)$  and  $\Delta_S(u, C)$  denote the corresponding restricted relative distances on  $S$ . Lists are denoted

$$\Lambda(C, \delta, u) := \{c \in C : \Delta(u, c) \leq \delta\}, \quad \Lambda(C, \delta) := \max_{u \in \mathbb{F}^n} |\Lambda(C, \delta, u)|.$$

For  $s \geq 1$  the  $s$ -interleaved code is  $C^{\equiv s} := \{(c^{(1)}, \dots, c^{(s)}) : c^{(i)} \in C\} \subseteq (\mathbb{F}^n)^s$  with the column distance  $\Delta_s(F, G) := \frac{1}{n} |\{j : \exists i, f_j^{(i)} \neq g_j^{(i)}\}|$ , and  $\Delta_s(F, C^{\equiv s}) := \min_{\mathbf{c} \in C^{\equiv s}} \Delta_s(F, \mathbf{c})$ .

**Definition 2.1** (correlated agreement error; [ABF26, Def. 4.1] up to notation). For  $0 \leq \delta_{\text{fld}} \leq \delta_{\text{int}} < 1$ ,

$$\varepsilon_{\text{ca}}(C, \delta_{\text{fld}}, \delta_{\text{int}}) := \max_{f_1, f_2 \in \mathbb{F}^n} \Pr_{\gamma \leftarrow \mathbb{F}} \left[ \Delta(f_1 + \gamma f_2, C) \leq \delta_{\text{fld}} \wedge \Delta_2((f_1, f_2), C^{\equiv 2}) > \delta_{\text{int}} \right],$$

and the no-proximity-loss variant is  $\varepsilon_{\text{ca}}(C, \delta) := \varepsilon_{\text{ca}}(C, \delta, \delta)$ . A slope  $\gamma$  realizing the event for a fixed pair  $(f_1, f_2)$  is called *CA-bad* for that pair.

**Definition 2.2** (mutual correlated agreement error, support-wise; [ABF26, Def. 4.3] up to notation). For  $\delta \in (0, 1)$ ,

$$\varepsilon_{\text{mca}}(C, \delta) := \max_{f_1, f_2 \in \mathbb{F}^n} \Pr_{\gamma \leftarrow \mathbb{F}} \left[ \begin{array}{l} \exists S \subseteq D, |S| \geq (1 - \delta)n, \Delta_S(f_1 + \gamma f_2, C) = 0, \\ \text{and } \Delta_S((f_1, f_2), C^{\equiv 2}) > 0 \end{array} \right].$$

Equivalently,  $\gamma$  is MCA-bad for  $(f_1, f_2)$  if some large agreement set  $S$  explains the point  $f_1 + \gamma f_2$ , but the same set  $S$  does not simultaneously explain  $f_1$  and  $f_2$  by two codewords of  $C$ .

**Definition 2.3** (challenge threshold). Following the grand MCA challenge of [ABF26, §1], for  $\varepsilon^* \in (0, 1)$  set  $\delta_C^*(\varepsilon^*) := \sup\{\delta \in (0, 1 - \rho) : \varepsilon_{\text{mca}}(C, \delta) \leq \varepsilon^*\}$ , with  $\sup \emptyset := 0$ .

All displayed threshold intervals use the integer-ball convention: distances are integral, so a radius  $\delta$  corresponds to the agreement requirement  $\lceil (1 - \delta)n \rceil$  and to  $\lfloor \delta n \rfloor$  permitted errors. When a theorem certifies that every  $\delta \in [\delta_0, 1 - \rho)$  is unsafe at target  $\varepsilon^*$ , it follows that  $\delta_C^*(\varepsilon^*) \leq \delta_0$  under the supremum convention; when it certifies safety at a single radius  $\delta_1 < 1 - \rho$ , it follows that  $\delta_C^*(\varepsilon^*) \geq \delta_1$ .

**Fact 2.4** (cf. [ABF26, Fact 4.5]). *For every  $\delta \in (0, 1)$ :  $\varepsilon_{\text{ca}}(C, \delta) \leq \varepsilon_{\text{mca}}(C, \delta)$ .*

*Proof.* Fix  $(f_1, f_2)$  and let  $\gamma$  be CA-bad:  $\Delta(f_1 + \gamma f_2, C) \leq \delta$  and  $\Delta_2((f_1, f_2), C^{\equiv 2}) > \delta$ . Choose  $S \subseteq D$  of size at least  $(1 - \delta)n$  on which  $f_1 + \gamma f_2$  agrees with a codeword of  $C$ . Since  $(f_1, f_2)$  is not  $\delta$ -close to  $C^{\equiv 2}$ , there is no set of size at least  $(1 - \delta)n$  on which  $f_1$  and  $f_2$  are simultaneously explained by codewords of  $C$ ; in particular this fails on the chosen  $S$ . Thus  $\gamma$  is MCA-bad for the same pair. Taking maxima over pairs gives the claim.  $\square$

**Lemma 2.5** (support-wise MCA monotonicity). *For every linear code  $C \subseteq \mathbb{F}^D$ , the function*

$$\delta \mapsto \varepsilon_{\text{mca}}(C, \delta)$$

*is nondecreasing.*

*Proof.* Fix a pair  $(f_1, f_2)$  and suppose that  $\gamma$  is MCA-bad at radius  $\delta$ . Then there is a set  $S \subseteq D$  with

$$|S| \geq (1 - \delta)n$$

such that  $f_1 + \gamma f_2$  is explained by  $C$  on  $S$ , but  $(f_1, f_2)$  is not simultaneously explained by two codewords of  $C$  on the same  $S$ . If  $\delta' \geq \delta$ , then

$$|S| \geq (1 - \delta)n \geq (1 - \delta')n,$$

so the same set  $S$  witnesses that  $\gamma$  is MCA-bad at radius  $\delta'$ . Taking the maximum over pairs gives the claim.  $\square$

**Conversion theorem.** The main composition below uses the deep-point list-to-CA conversion proved here directly. This is the same consequence previously imported from Crites–Stewart, with the integer-radius admissibility condition made explicit; see Remark 2.17. We keep the slacked BCHKS/ABF fallback as a separate imported route.

**Theorem 2.6** (deep-point list-to-CA conversion). *Let  $C = \text{RS}[\mathbb{F}, D, k]$ , let  $C^+ := \text{RS}[\mathbb{F}, D, k + 1]$ , let  $q := |\mathbb{F}|$  and  $n := |D|$ , and assume  $q > n$ . Let  $\delta \in (0, 1)$  and set  $f_\delta := \lfloor \delta n \rfloor$ . Assume*

$$f_\delta \leq n - k - 1.$$

*For any  $\eta \in [0, 1)$ , if*

$$\varepsilon_{\text{ca}}(C, \delta) \leq \eta \left( \frac{1}{k} - \frac{n}{kq} \right),$$

*then*

$$\Lambda(C^+, \delta) \leq \left\lceil \frac{q \varepsilon_{\text{ca}}(C, \delta)}{1 - \eta} \right\rceil.$$

*Proof.* Put  $\epsilon := \varepsilon_{\text{ca}}(C, \delta)$  and  $f = f_\delta$ . Let  $U : D \rightarrow \mathbb{F}$  be any received word and suppose that  $P_1, \dots, P_L \in \mathbb{F}[X]_{< k+1}$  are distinct polynomials whose restrictions to  $D$  all lie in the Hamming ball of relative radius  $\delta$  around  $U$ . Since distances are integral, each  $P_i$  agrees with  $U$  on at least

$$a := n - f$$

points of  $D$ . The hypothesis  $f \leq n - k - 1$  gives  $a \geq k + 1$ , in particular  $a > k$ .

Let  $\Omega := \mathbb{F} \setminus D$ , so  $|\Omega| = q - n$ . For each  $\alpha \in \Omega$  define the simple-pole line

$$f_\alpha(x) = \frac{U(x)}{x - \alpha}, \quad g_\alpha(x) = -\frac{1}{x - \alpha} \quad (x \in D).$$



If  $P_i$  agrees with  $U$  on a set  $S_i$  of size at least  $a$ , then for  $z_i(\alpha) := P_i(\alpha)$  and every  $x \in S_i$ ,

$$f_\alpha(x) + z_i(\alpha)g_\alpha(x) = \frac{P_i(x) - P_i(\alpha)}{x - \alpha}.$$

The right-hand side is the restriction of a polynomial of degree  $< k$ . Thus each distinct value among the  $P_i(\alpha)$  gives a slope for which the line point is  $\delta$ -close to  $C$ .

We next check the correlated-agreement far condition. If a polynomial  $G \in \mathbb{F}[X]_{<k}$  agreed with  $g_\alpha$  on more than  $k$  points of  $D$ , then

$$(X - \alpha)G(X) + 1$$

would be a polynomial of degree at most  $k$  with more than  $k$  roots, but its value at  $X = \alpha$  is 1, a contradiction. Hence  $(f_\alpha, g_\alpha)$  is not  $\delta$ -close to  $C^{\equiv 2}$ , because any common explanation on a support of size at least  $a > k$  would in particular explain  $g_\alpha$  there. Therefore the distinct values of  $P_i(\alpha)$  are CA-bad slopes.

It remains to lower-bound the number of distinct values for some  $\alpha$ . For  $i \neq j$ , the polynomial  $P_i - P_j$  is nonzero of degree at most  $k$ , so it has at most  $k$  roots in  $\Omega$ . Let  $M(\alpha)$  be the number of distinct values among  $P_1(\alpha), \dots, P_L(\alpha)$ . Over all  $\alpha \in \Omega$ , the total number of colliding unordered pairs  $(i, j)$  with  $P_i(\alpha) = P_j(\alpha)$  is at most  $k \binom{L}{2}$ . Equivalently, if the fiber sizes of the map  $i \mapsto P_i(\alpha)$  are  $m_v(\alpha)$ , then

$$\sum_{\alpha \in \Omega} \sum_v \binom{m_v(\alpha)}{2} \leq k \binom{L}{2}.$$

Choose  $\alpha \in \Omega$  for which the inner collision count is at most  $k \binom{L}{2} / (q - n)$ . If the fiber sizes of the map  $i \mapsto P_i(\alpha)$  are  $m_1, \dots, m_{M(\alpha)}$ , then

$$\sum_r m_r = L, \quad \sum_r m_r^2 = L + 2 \sum_r \binom{m_r}{2} \leq L + \frac{kL(L-1)}{q-n}.$$

Cauchy–Schwarz therefore gives

$$L^2 \leq M(\alpha) \sum_r m_r^2, \quad M(\alpha) \geq \frac{L(q-n)}{q-n+k(L-1)} \geq \frac{L(q-n)}{q-n+kL}.$$

Consequently

$$\epsilon \geq \frac{1}{q} \cdot \frac{L(q-n)}{q-n+kL}.$$

Solving this inequality for  $L$  gives

$$L \leq \frac{\epsilon q(q-n)}{q-n-k\epsilon q},$$

provided  $\epsilon < (q-n)/(kq)$ . Under the stated hypothesis,  $\epsilon \leq \eta(q-n)/(kq)$  with  $\eta < 1$ , so the denominator is at least  $(1-\eta)(q-n)$  and hence

$$L \leq \frac{q\epsilon}{1-\eta}.$$

Since  $U$  and the list were arbitrary, the displayed ceiling bounds the maximum list size of  $C^+$  at radius  $\delta$ .  $\square$



*Remark 2.7* (relation to Crites–Stewart). The displayed conversion is the only consequence of Crites–Stewart needed by the universal-cap argument. The proof above is included to make the main cap self-contained. The separate slacked fallback through BCHKS/ABF remains an independent imported route; it is not used in the final deployed certificate grammar and is retained as a robustness check.

**Proposition 2.8** (quantitative deep-list floor). *Let  $C = \text{RS}[\mathbb{F}, D, k]$ , let  $C^+ := \text{RS}[\mathbb{F}, D, k+1]$ , put  $q = |\mathbb{F}|$  and  $n = |D|$ , and assume  $q > n$ . Let  $\delta_0 \in [0, 1 - k/n)$ , and suppose that for some received word  $U : D \rightarrow \mathbb{F}$  there are  $L \geq 1$  distinct codewords of  $C^+$  in the closed ball of radius  $\delta_0$  around  $U$ . Then, for every  $\delta \in [\delta_0, 1 - k/n)$ ,*

$$\varepsilon_{\text{ca}}(C, \delta) \geq \frac{L(q-n)}{q(q-n+kL)}.$$

For every such  $\delta > 0$ ,

$$\varepsilon_{\text{mca}}(C, \delta) \geq \frac{L(q-n)}{q(q-n+kL)}$$

as well. More precisely, some received line has at least

$$\left\lceil \frac{L(q-n)}{q-n+kL} \right\rceil$$

CA-bad finite slopes at every such radius; for  $\delta > 0$ , the same slopes are MCA-bad.

*Proof.* Fix  $\delta \in [\delta_0, 1 - k/n)$  and choose distinct  $P_1, \dots, P_L \in \mathbb{F}[X]_{<k+1}$  in the  $\delta_0$ -ball around  $U$ . They are also in the  $\delta$ -ball. Since  $\delta < 1 - k/n$ , if  $f = \lfloor \delta n \rfloor$  then  $f \leq n - k - 1$ , so every codeword in this  $\delta$ -ball agrees with  $U$  on at least  $a = n - f > k$  points.

Repeat the simple-pole construction in the proof of Theorem 2.6. For  $\alpha \in \Omega := \mathbb{F} \setminus D$  set

$$f_\alpha(x) = \frac{U(x)}{x - \alpha}, \quad g_\alpha(x) = -\frac{1}{x - \alpha} \quad (x \in D).$$

For every distinct value of  $P_i(\alpha)$ , the word  $f_\alpha + P_i(\alpha)g_\alpha$  is  $\delta$ -close to  $C$ . The pair  $(f_\alpha, g_\alpha)$  is not  $\delta$ -close to  $C^{\equiv 2}$ , because a common explanation on more than  $k$  points would in particular explain  $g_\alpha$  there, and then  $(X - \alpha)G(X) + 1$  would be a polynomial of degree at most  $k$  with more than  $k$  roots and value 1 at  $\alpha$ . Thus the distinct values of the map  $i \mapsto P_i(\alpha)$  are CA-bad slopes.

For  $i \neq j$ , the polynomial  $P_i - P_j$  has degree at most  $k$  and is nonzero, hence it has at most  $k$  roots in  $\Omega$ . Averaging over  $|\Omega| = q - n$  gives an  $\alpha$  for which the number of colliding unordered pairs among the  $L$  values is at most  $k \binom{L}{2} / (q - n)$ . If  $m_1, \dots, m_M$  are the value multiplicities for this  $\alpha$ , then

$$\sum_r m_r = L, \quad \sum_r m_r^2 \leq L + \frac{kL(L-1)}{q-n}.$$

By Cauchy–Schwarz,

$$M \geq \frac{L^2}{L + kL(L-1)/(q-n)} = \frac{L(q-n)}{q-n+k(L-1)} \geq \frac{L(q-n)}{q-n+kL}.$$

Since  $M$  is an integer,  $M \geq \lceil L(q-n)/(q-n+kL) \rceil$ . Dividing by the  $q$  possible finite slopes gives the CA lower bound. For  $\delta > 0$ , the MCA bound follows from Fact 2.4.  $\square$

**Corollary 2.9** (deep-list trigger and ceiling). *In the setting of Proposition 2.8, define*

$$\mathcal{E}_{q,k}(L) := \frac{L(q-n)}{q(q-n+kL)}, \quad \mathcal{N}_{q,k}(L) := \left\lceil \frac{L(q-n)}{q-n+kL} \right\rceil.$$

*Then  $\mathcal{E}_{q,k}(L)$  is increasing in  $L$ , and*

$$\mathcal{E}_{q,k}(L) > \frac{1}{2k} \left(1 - \frac{n}{q}\right) \iff L > \frac{q-n}{k}.$$

*Moreover*

$$\mathcal{E}_{q,k}(L) < \frac{1}{k} \left(1 - \frac{n}{q}\right)$$

*for every finite  $L$ , and the right-hand side is the limiting value as  $L \rightarrow \infty$ .*

*Proof.* The derivative of  $L/(q-n+kL)$  is positive. The trigger inequality is

$$\frac{L}{q-n+kL} > \frac{1}{2k},$$

which is equivalent to  $kL > q-n$ . The upper bound follows from  $q-n+kL > kL$ .  $\square$

*Remark 2.10* (why this is stronger than the contrapositive). Theorem 2.6 is optimized for contradiction: once  $L > (q-n)/k$ , taking  $\eta = 1/2$  already forces the printed  $(2k)^{-1}(1-n/q)$  lower bound. Proposition 2.8 keeps the whole interpolation count. It is the form a staircase scanner should use when a quotient fiber is too small to cross the  $1/(2k)$  trigger but still contributes a nonzero explicit bad-slope numerator, and also when a very large fiber improves the constant toward  $(1/k)(1-n/q)$ .

**Proposition 2.11** (punctured simple-pole deep-list floor). *Let  $C = \text{RS}[\mathbb{F}, D, k]$ , let  $C^+ := \text{RS}[\mathbb{F}, D, k+1]$ , put  $q = |\mathbb{F}|$  and  $n = |D|$ . Let*

$$\Omega_0 \subseteq \mathbb{F} \setminus D, \quad m := |\Omega_0| > 0.$$

*Let  $\delta_0 \in [0, 1-k/n)$ , and suppose that for some received word  $U : D \rightarrow \mathbb{F}$  there are  $L \geq 1$  distinct codewords of  $C^+$  in the closed ball of radius  $\delta_0$  around  $U$ . Then, for every  $\delta \in [\delta_0, 1-k/n)$ ,*

$$\varepsilon_{\text{ca}}(C, \delta) \geq \frac{Lm}{q(m+kL)}.$$

*For every such  $\delta > 0$ ,*

$$\varepsilon_{\text{mca}}(C, \delta) \geq \frac{Lm}{q(m+kL)}.$$

*More precisely, there is some  $\alpha \in \Omega_0$  such that the simple-pole line*

$$f_\alpha(x) = \frac{U(x)}{x-\alpha}, \quad g_\alpha(x) = -\frac{1}{x-\alpha} \quad (x \in D)$$

*has at least*

$$\left\lceil \frac{Lm}{m+kL} \right\rceil$$

*CA-bad finite slopes at every such radius; for  $\delta > 0$ , the same slopes are MCA-bad.*

*Proof.* Repeat the proof of Proposition 2.8, but average only over  $\Omega_0$  instead of over all of  $\mathbb{F} \setminus D$ . The simple-pole argument works for every  $\alpha \notin D$ , so every distinct value among  $P_1(\alpha), \dots, P_L(\alpha)$  gives a CA-bad slope. For  $i \neq j$ , the nonzero polynomial  $P_i - P_j$  has degree at most  $k$ , hence has at most  $k$  roots inside the smaller set  $\Omega_0$ . Therefore the total number of colliding unordered pairs over all  $\alpha \in \Omega_0$  is at most  $k \binom{L}{2}$ . Choose  $\alpha \in \Omega_0$  for which the collision count is at most  $k \binom{L}{2} / m$ . If  $M$  is the number of distinct values at this  $\alpha$ , the same Cauchy–Schwarz calculation gives

$$M \geq \frac{Lm}{m + k(L-1)} \geq \frac{Lm}{m + kL}.$$

Since  $M$  is integral, this gives the displayed bad-slope numerator. Dividing by the  $q$  finite slopes gives the CA lower bound, and the MCA statement for  $\delta > 0$  follows from Fact 2.4.  $\square$

**Corollary 2.12** (extension-pole deep-list floor). *Assume in addition that  $B \subsetneq \mathbb{F}$ ,  $D \subseteq B$ , and  $|D| \geq 2$ . Put  $b := |B|$ . In the setting of Proposition 2.11, take*

$$\Omega_0 := \mathbb{F} \setminus B, \quad m = q - b.$$

*Then for every  $\delta \in [\delta_0, 1 - k/n)$ ,*

$$\varepsilon_{\text{ca}}(C, \delta) \geq \mathcal{E}_{q,b,k}^{\text{ext}}(L) := \frac{L(q-b)}{q(q-b+kL)}.$$

*For  $\delta > 0$  the same bound holds for  $\varepsilon_{\text{mca}}(C, \delta)$ . Moreover the witnessing line may be chosen with pole  $\alpha \in \mathbb{F} \setminus B$ , and its pole vector  $((x-\alpha)^{-1})_{x \in D}$  is not a scalar multiple over  $\mathbb{F}$  of any  $B$ -valued vector. In particular the witness line is genuinely extension-valued, not merely a  $B$ -rational line viewed over  $\mathbb{F}$ .*

*Proof.* The lower bound is Proposition 2.11 with  $\Omega_0 = \mathbb{F} \setminus B$ . It remains to prove genuine extension-valuedness. Suppose that, for some  $\lambda \in \mathbb{F}^\times$ , all numbers  $\lambda/(x-\alpha)$  with  $x \in D$  lay in  $B$ . Choose two distinct points  $x, y \in D$ . Then

$$\frac{y-\alpha}{x-\alpha} = \frac{\lambda/(x-\alpha)}{\lambda/(y-\alpha)} \in B.$$

Writing this ratio as  $r \in B$ , we cannot have  $r = 1$  because  $x \neq y$ . Hence

$$\alpha = \frac{rx - y}{r - 1} \in B,$$

contradicting  $\alpha \notin B$ . Thus the pole vector is not projectively  $B$ -valued. In particular the displayed affine parametrization has no  $B$ -valued direction vector after rescaling.  $\square$

**Corollary 2.13** (extension-pole quotient-remainder floor). *Assume  $B \subsetneq \mathbb{F}$ ,  $D \subseteq B^\times$  is a multiplicative coset of order  $n \geq 2$ , put  $b := |B|$ , and let  $C = \text{RS}[\mathbb{F}, D, k]$ . Fix a quotient-remainder profile as in Lemma 3.7 with  $K = k + 1$  and agreement  $A_0 > k$ . Let*

$$L := H_{c,m,s}^{k+1}$$

*be the certified heaviest prefix-fiber size, or replace  $L$  by either fallback*

$$\left\lceil \frac{M_{c,m,s}}{I_{c,m,s}^{k+1}} \right\rceil, \quad \left\lceil \frac{M_{c,m,s}}{|B|^{w_c(s, A_0 - k - 1)}} \right\rceil.$$

Then, for every

$$\delta \in [1 - A_0/n, 1 - k/n),$$

there is a genuinely extension-valued simple-pole line with at least

$$\mathcal{N}_{q,b,k}^{\text{ext}}(L) := \left\lceil \frac{L(q-b)}{q-b+kL} \right\rceil$$

CA-bad finite slopes. For  $\delta > 0$ , these slopes are MCA-bad. Equivalently,

$$\varepsilon_{\text{ca}}(C, \delta) \geq \frac{L(q-b)}{q(q-b+kL)}, \quad \varepsilon_{\text{mca}}(C, \delta) \geq \frac{L(q-b)}{q(q-b+kL)} \quad (\delta > 0).$$

If  $L > (q-b)/k$ , this gives the clean trigger

$$\varepsilon_{\text{ca}}(C, \delta) > \frac{1}{2k} \left(1 - \frac{b}{q}\right), \quad \varepsilon_{\text{mca}}(C, \delta) > \frac{1}{2k} \left(1 - \frac{b}{q}\right) \quad (\delta > 0).$$

*Proof.* Lemma 3.7 gives a  $B$ -valued received word with at least  $L$  codewords of  $C^+$  in the closed ball of radius  $1 - A_0/n$ . Apply Corollary 2.12. The trigger inequality is

$$\frac{L(q-b)}{q(q-b+kL)} > \frac{q-b}{2kq},$$

which is equivalent to  $kL > q-b$ . □

**Definition 2.14** (extension-pole certificate). An extension-pole certificate for a list  $P_1, \dots, P_L \in \mathbb{F}[X]_{<k+1}$  and a received word  $U$  consists of:

- (i) a declared proper subfield  $B \subsetneq \mathbb{F}$  containing  $D$  and a pole  $\alpha \in \mathbb{F} \setminus B$ ;
- (ii) the agreement supports showing that each  $P_i$  lies in the declared ball around  $U$ ;
- (iii) the bucket table for the values  $P_i(\alpha)$ , with duplicate values coalesced;
- (iv) the simple-pole line  $(f_\alpha, g_\alpha)$  and the proof that  $g_\alpha$  is not projectively  $B$ -valued;
- (v) the final bad-slope numerator, namely the number of distinct buckets in (iii).

The verifier checks  $\alpha \notin B$ , the support agreements, the value buckets, and the simple-pole far condition. For quotient-remainder floors, the list in (ii) may itself be supplied by a prefix-fiber certificate from Definition 3.6.

*Remark 2.15* (what the extension-pole construction changes). The subfield-confinement theorem says that  $B$ -rational lines are inert over a proper extension. Corollary 2.12 gives the matching existence theorem on the failure side: any large list can be converted using poles restricted to  $\mathbb{F} \setminus B$ , so the resulting bad line is genuinely extension-valued. Thus the extension-line bucket in a prize scanner is no longer only an abstract warning; it has an explicit simple-pole source and a stable numerator with  $q-b$  replacing  $q-n$ . What remains open is not existence of extension-valued witnesses, but their sharp classification inside the aperiodic/residue-line ledgers and, for a fully explicit counterexample, printing a concrete pole  $\alpha$  with its value-bucket table.

**Theorem 2.16** ([BCHKS25, Thm. 1.9], as stated in [ABF26, Thm. 5.2]). *Let  $C = \text{RS}[\mathbb{F}, D, k]$ , let  $\rho = k/n$ , and let  $\delta \in (0, 1 - \rho)$  satisfy  $\delta + 2/n \leq 1 - \rho - 1/n$ . If*

$$\varepsilon_{\text{ca}}\left(C, \delta + \frac{2}{n}, 1 - \rho - \frac{1}{n}\right) < \frac{1}{2n}, \quad \text{then} \quad \Lambda(C, \delta) < |\mathbb{F}|.$$

*Remark 2.17* (on imports and radius conventions). Theorem 2.6 is the only Crites–Stewart-type consequence used by the main cap, and it is now proved above by the simple-pole deep-point argument. The important admissibility condition is the integer-radius condition

$$\lfloor \delta n \rfloor \leq n - k - 1.$$

Consequently, the no-loss CA lower bound proved below should be read only in this admissible range. The Proximity-Prize MCA cap is unaffected: we prove the lower bound at the endpoint

$$\delta_N := 1 - \rho - \frac{2}{N}$$

and then use support-wise MCA monotonicity to extend the MCA failure to all larger sub-capacity radii.

Theorem 2.16 remains a separate slacked fallback imported through the BCHKS/ABF route. It is not needed for the main field-size-universal MCA cap.

### 3 Locator fibers and interleaving transfer

Fix  $N \mid n$ , set  $a := n/N$ , and let  $Q := D^a := \{x^a : x \in D\} \subseteq B^\times$ , a multiplicative coset of order  $N$ . The  $a$ -th power map  $D \rightarrow Q$  is surjective and its fibers  $S_b := \{x \in D : x^a = b\}$  have exactly  $a$  elements each. Since  $a \mid n \mid |B| - 1$ , the characteristic of  $\mathbb{F}$  does not divide  $a$ , so  $X^a - b$  is separable for  $b \neq 0$  and its full root set inside  $D$  is  $S_b$  whenever  $b \in Q$ . For a set  $A$ ,  $e_j(A)$  denotes the  $j$ -th elementary symmetric function of its elements.

**Lemma 3.1** (locator fibers are lists). *Let  $B \subseteq \mathbb{F}$ , let  $D \subseteq B^\times$  be a multiplicative coset of order  $N$ , and let  $k$  satisfy  $a \mid k$  and  $\rho = k/n$ . Write  $\ell_1 := \rho N + 1$  and  $\ell_2 := \rho N + 2$ .*

(i) *If  $\ell_1 \leq N$ , there exists  $z \in B$  such that, with  $u_z := (x^{k+a} + z x^k)_{x \in D} \in \mathbb{F}^n$ ,*

$$|\Lambda(\text{RS}[\mathbb{F}, D, k], 1 - \rho - \frac{1}{N}, u_z)| \geq \binom{N}{\ell_1} / |B|.$$

(ii) *If  $\ell_2 \leq N$ , there exists  $z \in B$  such that, with  $u_z := (x^{k+2a} + z x^{k+a})_{x \in D} \in \mathbb{F}^n$ ,*

$$|\Lambda(\text{RS}[\mathbb{F}, D, k+1], 1 - \rho - \frac{2}{N}, u_z)| \geq \binom{N}{\ell_2} / |B|.$$

*In both cases the words  $u_z$  are  $B$ -valued and all listed codewords lie in  $\text{RS}[B, D, k]$  (resp.  $\text{RS}[B, D, k+1]$ ).*

*Proof.* We prove (ii); part (i) is the identical computation one degree rung lower and is [Cho26a, Main Thm. (d)] sharpened by pigeonholing over  $B$  in place of  $\mathbb{F}$ .

Let  $A \subseteq Q$  with  $|A| = \ell_2$ , and put

$$L_A(X) := \prod_{b \in A} (X^a - b) = \sum_{j=0}^{\ell_2} (-1)^j e_j(A) X^{a(\ell_2-j)} = X^{k+2a} - e_1(A) X^{k+a} + R_A(X),$$

where  $a\ell_2 = k + 2a$ ,  $a(\ell_2 - 1) = k + a$ , and  $R_A$  collects the terms with  $j \geq 2$ , so  $\deg R_A \leq a(\ell_2 - 2) = k$ . The polynomial  $L_A$  is squarefree with root set  $S_A := \bigcup_{b \in A} S_b \subseteq D$  of size  $a\ell_2 = k + 2a$ .

Set  $z_A := -e_1(A) \in B$  and  $c_A := (-R_A(x))_{x \in D}$ . Since  $\deg R_A \leq k$ , we have  $c_A \in \text{RS}[B, D, k+1] \subseteq \text{RS}[\mathbb{F}, D, k+1]$ . For every  $x \in S_A$ ,

$$0 = L_A(x) = x^{k+2a} + z_A x^{k+a} + R_A(x), \quad \text{i.e.} \quad u_{z_A}(x) = c_A(x),$$

so  $u_{z_A}$  and  $c_A$  agree on at least  $k + 2a$  points of  $D$  and  $\Delta(u_{z_A}, c_A) \leq 1 - \rho - 2/N$ .

The map  $A \mapsto c_A$  is injective on the  $\ell_2$ -subsets of  $Q$  sharing a common slope value: if  $z_A = z_{A'}$  and  $c_A = c_{A'}$ , then  $R_A$  and  $R_{A'}$  are polynomials of degree  $\leq k < n$  agreeing on all  $n$  points of  $D$ , hence equal as polynomials; then  $L_A = L_{A'}$ , so  $S_A = S_{A'}$ , and  $A = \{x^a : x \in S_A\} = A'$ .

Finally,  $z_A = -e_1(A)$  takes at most  $|B|$  values as  $A$  ranges over the  $\binom{N}{\ell_2}$  subsets; some  $z \in B$  is attained by at least  $\binom{N}{\ell_2}/|B|$  of them, and the corresponding codewords  $c_A$  are pairwise distinct elements of  $\Lambda(\text{RS}[\mathbb{F}, D, k+1], 1 - \rho - 2/N, u_z)$ .  $\square$

*Remark 3.2* (why slack two). Part (i) applied to  $C^+ = \text{RS}[\mathbb{F}, D, k+1]$  directly would require  $a \mid k+1$ ; in every challenge instantiation  $a$  is a power of two and  $k+1$  is odd, so this fails. At slack two the listed codewords have degree  $\leq k$ , i.e. lie in  $C^+$ , while the divisibility demand stays on  $k$ . This is exactly the interface needed by Theorem 2.6, whose list conclusion concerns  $C^+$  while its hypothesis concerns  $C$ .

*Remark 3.3* (the subfield pigeonhole elsewhere). The same one-line strengthening — pigeonholing locator data over the field of definition  $B$  rather than over  $\mathbb{F}$  — also upgrades the coefficient-pigeonhole bound of [Cho26b] to the generated field: the prefix map  $\Phi_\sigma$  there has image in  $B^\sigma$  whenever  $D \subseteq B$ , since the elementary symmetric prefixes of subsets of  $D$  are  $B$ -valued. This is the form consumed by the “necessary” half of the list-profile conjecture of [Cho26b], which states the entropy floor at the generated field  $\tau^*(\rho, q_D)$ ; Lemma 3.1 supplies the missing subfield step.

*Remark 3.4* (lists, not lines). The words  $u_z$  of Lemma 3.1 lie on the  $B$ -rational line  $f + zg$  with  $f = (x^{k+ta})_{x \in D}$ ,  $g = (x^{k+(t-1)a})_{x \in D}$ , but the lemma is purely a statement about lists at a received word and no line structure is used in Section 4. This matters: by Corollary 6.2 below, over a proper extension  $\mathbb{F} \supsetneq B$  the lines that end up certifying the CA failure cannot be these (or any)  $B$ -rational lines.

### 3.1 Quotient-profile floors with remainder supports

The fiber lemma above uses supports that are unions of complete quotient fibers. For threshold work one also needs agreements that are not multiples of the fiber size. The following locator-prefix pigeonhole gives a finite lower ledger for such remainder supports. It is a lower-floor theorem only; the matching upper ledger over all divisor scales remains open; it is the upper-ledger side of the residual band problem.

Let  $c \mid n$  and put  $Q_c := D^c$ , so the map

$$\pi_c : D \rightarrow Q_c, \quad x \mapsto x^c$$

has fibers of size  $c$  and  $|Q_c| = N = n/c$ . For  $0 \leq s < c$  and an integer  $\sigma \geq 0$ , define the prefix weight

$$w_c(s, \sigma) := \#\{h : 1 \leq h \leq \sigma, h \bmod c \in \{0, 1, \dots, s\}\},$$

where residues are taken in  $\{0, 1, \dots, c-1\}$ . Thus  $w_c(0, \sigma) = \lfloor \sigma/c \rfloor$  for full-fiber supports.

**Lemma 3.5** (quotient-remainder locator-prefix lower floor). *Let  $B \subseteq \mathbb{F}$ , let  $D \subseteq B^\times$  be a multiplicative coset of order  $n$ , and let  $K < n$ . Fix  $c \mid n$ , write  $N = n/c$ , and let*

$$A_0 = mc + s, \quad 0 \leq s < c, \quad 0 \leq m \leq N, \quad A_0 \geq K.$$

*If  $s > 0$ , assume  $mc + s \leq n$ . Put  $\sigma = A_0 - K$  and*

$$M_{c,m,s} := \binom{N}{m} \binom{n-mc}{s}.$$

*Then there is a  $B$ -valued received word  $U : D \rightarrow B$  such that*

$$|\Lambda(\text{RS}[\mathbb{F}, D, K], 1 - A_0/n, U)| \geq \left\lceil \frac{M_{c,m,s}}{|B|^{w_c(s, \sigma)}} \right\rceil.$$

*The listed codewords may be chosen in  $\text{RS}[B, D, K]$ .*

*Proof.* For  $M \subseteq Q_c$  with  $|M| = m$ , write

$$P_M(X) = \prod_{b \in M} (X^c - b).$$

For  $R \subseteq D \setminus \pi_c^{-1}(M)$  with  $|R| = s$ , put

$$E_R(X) = \prod_{r \in R} (X - r), \quad L_{M,R}(X) = P_M(X)E_R(X).$$

Then  $L_{M,R}$  is a squarefree locator of degree  $A_0$  whose roots are exactly  $\pi_c^{-1}(M) \cup R$ . Write

$$L_{M,R}(X) = X^{A_0} + \sum_{h=1}^{A_0} \lambda_h(M, R) X^{A_0-h}.$$

The coefficients lie in  $B$ . Moreover, because  $P_M$  has monomials only in degrees congruent to 0 modulo  $c$  and  $E_R$  has degree  $s$ , the coefficient  $\lambda_h(M, R)$  is zero unless  $h \bmod c \in \{0, 1, \dots, s\}$ . Consequently the high-degree prefix

$$(\lambda_h(M, R) : 1 \leq h \leq \sigma, h \bmod c \in \{0, 1, \dots, s\})$$

takes at most  $|B|^{w_c(s, \sigma)}$  values.

Pigeonhole over these prefix values. There is a fixed prefix vector  $\xi$  and a subfamily  $\mathcal{T}_\xi$  of at least  $M_{c,m,s}/|B|^{w_c(s, \sigma)}$  pairs  $(M, R)$  with this prefix. Let

$$U_\xi(X) = X^{A_0} + \sum_{1 \leq h \leq \sigma} \lambda_h(\xi) X^{A_0-h}$$



be the corresponding high-degree part, restricted to  $D$ . For every  $(M, R) \in \mathcal{T}_\xi$ , the difference

$$R_{M,R}(X) := L_{M,R}(X) - U_\xi(X)$$

has degree  $< K$ . Hence the codeword  $-R_{M,R}$  agrees with  $U_\xi$  on all roots of  $L_{M,R}$ , a set of size  $A_0$ .

It remains only to check that the listed codewords are distinct. If two pairs  $(M, R)$  and  $(M', R')$  in the same prefix class gave the same codeword on  $D$ , then the two remainders of degree  $< K < n$  would be equal as polynomials. Since the high prefixes are also equal,  $L_{M,R} = L_{M',R'}$ . Their root sets in  $D$  are therefore equal. A full fiber has size  $c$  while  $|R| = s < c$ , so the full fibers contained in this root set are exactly  $M$ , and then the residual set is forced as well. Thus  $(M, R) = (M', R')$ .  $\square$

**Definition 3.6** (prefix fibers and prefix-image certificates). In the notation of Lemma 3.5, let  $\Phi_{c,m,s}^K$  be the map from pairs  $(M, R)$  to the high-degree prefix

$$(\lambda_h(M, R) : 1 \leq h \leq A_0 - K, h \bmod c \in \{0, 1, \dots, s\}) \in B^{w_c(s, A_0 - K)}.$$

If  $A_0 = K$  this is the unique empty prefix. For a prefix value  $\xi$ , put

$$\mu_{c,m,s}^K(\xi) := |(\Phi_{c,m,s}^K)^{-1}(\xi)|.$$

Define the image size and the heaviest prefix fiber by

$$I_{c,m,s}^K := |\text{im } \Phi_{c,m,s}^K|, \quad H_{c,m,s}^K := \max_{\xi} \mu_{c,m,s}^K(\xi).$$

Thus

$$H_{c,m,s}^K \geq \left\lceil \frac{M_{c,m,s}}{I_{c,m,s}^K} \right\rceil \geq \left\lceil \frac{M_{c,m,s}}{|B|^{w_c(s, A_0 - K)}} \right\rceil.$$

A prefix-fiber certificate is an orbit decomposition, hash table, or other verifiable bucket table for the finite map  $\Phi_{c,m,s}^K$ , including the image size and the maximum fiber size. If only  $I_{c,m,s}^K$  is certified, the middle lower bound is still valid; if no image certificate is supplied, the ambient-box lower bound is valid.

**Lemma 3.7** (heaviest-prefix locator floor). *Under the hypotheses of Lemma 3.5, there is a  $B$ -valued received word  $U : D \rightarrow B$  such that*

$$|\Lambda(\text{RS}[\mathbb{F}, D, K], 1 - A_0/n, U)| \geq H_{c,m,s}^K.$$

*Consequently*

$$|\Lambda(\text{RS}[\mathbb{F}, D, K], 1 - A_0/n, U)| \geq \left\lceil \frac{M_{c,m,s}}{I_{c,m,s}^K} \right\rceil \geq \left\lceil \frac{M_{c,m,s}}{|B|^{w_c(s, A_0 - K)}} \right\rceil$$

*when only the coarser certificates are available. The listed codewords may be chosen in  $\text{RS}[B, D, K]$ .*

*Proof.* Choose a prefix value  $\xi$  with  $\mu_{c,m,s}^K(\xi) = H_{c,m,s}^K$  and form the high-degree word

$$U_\xi(X) = X^{A_0} + \sum_{1 \leq h \leq A_0 - K} \lambda_h(\xi) X^{A_0 - h},$$

where  $\lambda_h(\xi) = 0$  for the high coefficients outside the allowed residue classes. For every pair  $(M, R)$  in this heaviest prefix fiber, the difference  $L_{M,R}(X) - U_\xi(X)$  has degree  $< K$ , so the negative of this remainder is a codeword of  $\text{RS}[B, D, K]$  agreeing with  $U_\xi$  on the  $A_0$  roots of  $L_{M,R}$ . If two pairs in the same prefix fiber gave the same codeword on  $D$ , then their degree- $< K < n$  remainders would be equal as polynomials; together with the common prefix this gives  $L_{M,R} = L_{M',R'}$ . Their root sets in  $D$  are equal. A full quotient fiber has size  $c$  while the residual set has size  $s < c$ , so the full fibers contained in the root set determine  $M$ , and then the residual set determines  $R$ . Thus the codewords are pairwise distinct. The two coarser displayed bounds follow from Definition 3.6.  $\square$

For  $A \in \{k+1, \dots, n\}$  and  $c \mid n$ , write

$$A = m_c(A)c + s_c(A), \quad 0 \leq s_c(A) < c,$$

and set

$$\begin{aligned} M_c(A) &:= \binom{n/c}{m_c(A)} \binom{n - cm_c(A)}{s_c(A)}, \\ L_{c,\text{heavy}}^+(A) &:= H_{c,m_c(A),s_c(A)}^{k+1}, \\ L_{c,\text{image}}^+(A) &:= \left\lceil \frac{M_c(A)}{I_{c,m_c(A),s_c(A)}^{k+1}} \right\rceil, \\ L_{c,\text{box}}^+(A) &:= \left\lceil \frac{M_c(A)}{|B|^{w_c(s_c(A), A-k-1)}} \right\rceil. \end{aligned}$$

The three levels satisfy

$$L_{c,\text{heavy}}^+(A) \geq L_{c,\text{image}}^+(A) \geq L_{c,\text{box}}^+(A).$$

Here “heavy” uses a full prefix-fiber certificate, “image” uses only the prefix-image size, and “box” is the certificate-free bound already implicit in Lemma 3.5.

**Theorem 3.8** (quotient-remainder deep-point lower ledger). *Let  $B \subseteq \mathbb{F}$ , let  $q = |\mathbb{F}| > n$ , let  $D \subseteq B^\times$  be a multiplicative coset of order  $n$ , and let  $C = \text{RS}[\mathbb{F}, D, k]$ . Fix an agreement value  $A \in \{k+1, \dots, n\}$  and a divisor  $c \mid n$ . For each*

$$\star \in \{\text{heavy}, \text{image}, \text{box}\}$$

*and every*

$$\delta \in [1 - A/n, 1 - k/n)$$

*one has*

$$\varepsilon_{\text{ca}}(C, \delta) \geq \mathcal{E}_{q,k}(L_{c,\star}^+(A)).$$

*If  $\delta > 0$ , then also*

$$\varepsilon_{\text{mca}}(C, \delta) \geq \mathcal{E}_{q,k}(L_{c,\star}^+(A)).$$

*Here  $\mathcal{E}_{q,k}$  is as in Corollary 2.9. More precisely, some received line has at least*

$$\mathcal{N}_{q,k}(L_{c,\star}^+(A))$$

CA-bad finite slopes at that radius, and for  $\delta > 0$  these slopes are MCA-bad.

Consequently, for a closed agreement threshold  $a \geq k + 1$ , define

$$B_{Q,\star}^-(a) := \max_{\substack{c|n \\ A \in \{a, a+1, \dots, n\}}} \mathcal{N}_{q,k}(L_{c,\star}^+(A)).$$

Then  $B_{Q,\star}^-(a)$  is a lower numerator for  $q\varepsilon_{\text{ca}}(C, \delta)$  for every  $\delta \in [1 - a/n, 1 - k/n)$ , and for  $q\varepsilon_{\text{mca}}(C, \delta)$  on the same interval with  $\delta > 0$ . The strongest certified ledger uses  $\star = \text{heavy}$ ; the certificate-free ledger uses  $\star = \text{box}$ .

*Proof.* Apply Lemma 3.7 with  $K = k + 1$  and  $A_0 = A$ . This produces a  $C^+ = \text{RS}[\mathbb{F}, D, k + 1]$  list of size at least  $L_{c,\text{heavy}}^+(A)$  at radius  $1 - A/n$ , with the image and ambient-box variants giving the corresponding smaller certified list sizes. Then apply Proposition 2.8. If  $A \geq a$  and  $\delta \geq 1 - a/n$ , then  $\delta \geq 1 - A/n$ , so the same converted floor applies. The maximum over  $(c, A)$  is the correct lower-ledger operation: the definitions of  $\varepsilon_{\text{ca}}$  and  $\varepsilon_{\text{mca}}$  maximize over received lines, so lower witnesses from different divisor scales or exact agreement values need not share one line and must not be summed.  $\square$

**Corollary 3.9** (quotient-remainder trigger). *In the setting of Theorem 3.8, if  $L_{c,\star}^+(A) > (q - n)/k$  for any  $\star \in \{\text{heavy}, \text{image}, \text{box}\}$ , then, for every  $\delta \in [1 - A/n, 1 - k/n)$ ,*

$$\varepsilon_{\text{ca}}(C, \delta) > \frac{1}{2k} \left(1 - \frac{n}{q}\right),$$

and for every such  $\delta > 0$ ,

$$\varepsilon_{\text{mca}}(C, \delta) > \frac{1}{2k} \left(1 - \frac{n}{q}\right).$$

*Proof.* This is Corollary 2.9 applied inside Theorem 3.8.  $\square$

**Corollary 3.10** (quantitative first-grid floor). *Let  $C = \text{RS}[\mathbb{F}, D, k]$ , let  $q = |\mathbb{F}| > n$ , and assume only that  $D \subseteq \mathbb{F}$  has  $n$  distinct points. Then for every  $\delta \in [1 - (k + 1)/n, 1 - k/n)$ ,*

$$\varepsilon_{\text{ca}}(C, \delta) \geq \mathcal{E}_{q,k} \left( \binom{n}{k+1} \right),$$

and for every such  $\delta > 0$ ,

$$\varepsilon_{\text{mca}}(C, \delta) \geq \mathcal{E}_{q,k} \left( \binom{n}{k+1} \right).$$

In particular, if  $\binom{n}{k+1} > (q - n)/k$ , then the printed  $(2k)^{-1}(1 - n/q)$  lower bound holds on this entire first grid interval without any multiplicative smoothness assumption.

*Proof.* Use the ordinary locator identity for all  $(k + 1)$ -subsets of  $D$ ; this is the  $c = 1$  argument behind Lemma 3.7, but it does not use multiplicative smoothness. The prefix is empty, so the list size is at least  $\binom{n}{k+1}$ . Apply Proposition 2.8 and then Corollary 2.9.  $\square$

**Definition 3.11** (quotient lower-ledger certificate). A quotient-remainder lower-ledger certificate for a row and a threshold  $a$  contains, for each divisor scale  $c$  and each exact agreement  $A \geq a$  that is used in the maximum:

- (i) the decomposition  $A = m_c(A)c + s_c(A)$ ;

- (ii) the support count  $M_c(A)$  and prefix weight  $w_c(s_c(A), A - k - 1)$ ;
- (iii) either a full prefix-fiber bucket table certifying  $H_{c, m_c(A), s_c(A)}^{k+1}$ , or an image-size certificate for  $I_{c, m_c(A), s_c(A)}^{k+1}$ , or the declaration that the ambient-box denominator is being used;
- (iv) the resulting list floor  $L_{c, \star}^+(A)$ ;
- (v) the converted slope numerator  $\mathcal{N}_{q, k}(L_{c, \star}^+(A))$  and error floor  $\mathcal{E}_{q, k}(L_{c, \star}^+(A))$ .

The verifier checks the locator-prefix formula, distinctness of codewords inside a prefix fiber, the integer arithmetic, and the final maximum over divisor scales and exact agreement values.

*Remark 3.12* (quotient lower ledger versus quotient upper ledger). The quotient-profile lower side is now scanner-ready at every agreement value  $A = mc + s$ : the generated-field contribution is the actual heaviest prefix fiber  $H_{c, m, s}^{k+1}$  when certified, the exact prefix-image denominator  $I_{c, m, s}^{k+1}$  when only image data are certified, and the ambient box  $|B|^{w_c(s, A - k - 1)}$  when no certificate is supplied. The challenge-field size  $q$  enters only through the independent conversion functions  $\mathcal{E}_{q, k}$  and  $\mathcal{N}_{q, k}$ . Across divisor scales the lower-side operation is a maximum, not a union count, because the witness line may change. The remaining upper-ledger problem is sharper and unchanged in logical status: replace the safe support ledger and the exact declared-branch image ledger by an exhaustive quotient-periodic classification, with duplicate slopes coalesced across all divisor scales.

**Corollary 3.13** (augmented slack-one quotient floor). *Let  $B \subseteq \mathbb{F}$  and let  $D \subseteq B^\times$  be a multiplicative coset of order  $n$ . Put  $C = \text{RS}[\mathbb{F}, D, k]$  and  $C^+ = \text{RS}[\mathbb{F}, D, k + 1]$ . If  $c \mid \gcd(n, k)$ , then there is a  $B$ -valued received word  $U_c$  such that*

$$|\Lambda(C^+, 1 - (k + c)/n, U_c)| \geq \binom{n/c}{k/c + 1}.$$

*One may take  $U_c(x) = x^{k+c}$ . In particular, for  $c = 1$  this gives the universal first-grid floor*

$$|\Lambda(C^+, 1 - (k + 1)/n, x^{k+1})| \geq \binom{n}{k + 1},$$

*with no quotient denominator. The  $c = 1$  statement in fact holds for any set  $D \subseteq B$  of  $n$  distinct points, not only for multiplicative cosets.*

*Proof.* For  $c > 1$ , apply Lemma 3.5 with  $K = k + 1$ ,  $s = 0$ , and  $m = k/c + 1$ . Then  $A_0 = k + c$  and  $\sigma = A_0 - K = c - 1$ , so  $w_c(0, c - 1) = 0$ . The high-degree prefix is only the monomial  $X^{k+c}$ , giving the displayed word. For  $c = 1$ , the same argument is just the ordinary locator identity for all  $(k + 1)$ -subsets of  $D$ : the high-degree prefix of a monic locator of degree  $k + 1$  is  $X^{k+1}$ , and its remainder has degree at most  $k$ .  $\square$

**Corollary 3.14** (first-grid deep-point cap). *Let  $C = \text{RS}[\mathbb{F}, D, k]$ ,  $q = |\mathbb{F}|$ , and suppose  $q > n$ . Let  $c \mid \gcd(n, k)$ . If  $c > 1$ , assume  $D \subseteq B^\times$  is a multiplicative coset as above; for  $c = 1$ , no smoothness is needed. If*

$$\binom{n/c}{k/c + 1} \geq q/k + 1,$$

then

$$\varepsilon_{\text{ca}}(C, 1 - (k + c)/n) > \frac{1}{2k} \left(1 - \frac{n}{q}\right), \quad \varepsilon_{\text{mca}}(C, 1 - (k + c)/n) > \frac{1}{2k} \left(1 - \frac{n}{q}\right).$$

For  $c = 1$  this is the first closed grid point below capacity.

*Proof.* Corollary 3.13 gives a  $C^+$ -list of size  $L \geq q/k + 1$  at agreement  $k + c$ . The proof of the deep-point conversion Theorem 2.6 applies whenever the agreement is  $> k$ , so it applies here for every  $c \geq 1$ . If the displayed CA lower bound failed, then

$$\varepsilon_{\text{ca}}(C, 1 - (k + c)/n) \leq \frac{q - n}{2kq}.$$

Theorem 2.6 with  $\eta = 1/2$  would give

$$\Lambda(C^+, 1 - (k + c)/n) \leq \left\lceil \frac{q - n}{k} \right\rceil < q/k + 1,$$

contradicting the list lower bound. Every no-loss CA-bad slope is also support-wise MCA-bad on the same explaining support, giving the MCA inequality.  $\square$

**Corollary 3.15** (first-grid cap for large challenge-envelope rows). *Let  $\rho \in \{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}\}$ , let  $C = \text{RS}[\mathbb{F}, D, k]$  with  $|D| = n$ ,  $k = \rho n$ ,  $q = |\mathbb{F}| < 2^{256}$ , and  $q > n$ . Assume*

$$\frac{\rho}{k} \geq \begin{array}{c|cccc} \frac{1}{2} & \frac{1}{4} & \frac{1}{8} & \frac{1}{16} \\ \hline 127 & 78 & 58 & 47 \end{array}$$

*for the corresponding rate. Then the first closed grid point below capacity is already CA- and MCA-unsafe:*

$$\varepsilon_{\text{ca}}(C, 1 - \rho - 1/n) > \frac{1}{2k} \left(1 - \frac{n}{q}\right), \quad \varepsilon_{\text{mca}}(C, 1 - \rho - 1/n) > \frac{1}{2k} \left(1 - \frac{n}{q}\right).$$

*If also  $k \leq 2^{40}$ , then this lower bound is  $> 2^{-86}$  and hence is far above the prize target  $2^{-128}$ .*

*Proof.* By Corollary 3.14 with  $c = 1$ , it is enough to check

$$\binom{n}{k+1} \geq q/k + 1.$$

Since  $q < 2^{256}$ , the stronger sufficient condition is  $k(\binom{n}{k+1} - 1) \geq 2^{256}$ . The four printed lower bounds on  $k$  are exactly small finite checks of this inequality at the corresponding rate; the left side is increasing thereafter. For the error size,  $q \geq n + 1$ ,  $n = k/\rho \leq 16k$ , and  $k \leq 2^{40}$  give

$$\frac{1}{2k} \left(1 - \frac{n}{q}\right) \geq \frac{1}{2k(n+1)} > 2^{-86}.$$

$\square$

### 3.2 A conservative quotient-support upper ledger

The remainder lower floor above supplies the unsafe side of the quotient-profile ledger. For a prize threshold one also needs a safe-side accounting rule: if a later argument declares a bad parameter to be quotient-periodic, the declaration must come with a divisor-lattice numerator that is counted with overlaps under control. The next statements give such a conservative upper ledger. They do not prove that every bad line parameter is quotient-periodic; they only say that once a permitted support family has been specified, every ledger term witnessed inside that family is bounded by the size of the family, with a factor  $d$  for finite-parameter degree- $d$  power curves.

Let  $\mathcal{S}$  be a family of subsets of  $D$ , all of size at least  $a$ . For a received line  $f + zg$ , say that a finite slope is  $\mathcal{S}$ -witnessed support-wise bad if it has a support-wise noncontained explanation on some  $S \in \mathcal{S}$ . Define the analogous  $\mathcal{S}$ -witnessed no-loss CA slopes by requiring that  $f + zg$  be explained on some  $S \in \mathcal{S}$  while  $(f, g)$  is not globally explained on any support of size at least  $a$ .

**Lemma 3.16** (one support pays for at most one line parameter). *Let  $C = \text{RS}[\mathbb{F}, D, k]$  and let  $a \geq k$ . Fix a support  $S \subseteq D$  with  $|S| \geq a$ . For any received line  $f + zg$ , at most one finite slope  $z \in \mathbb{F}$  can be support-wise noncontained on the fixed support  $S$ . Likewise, at most one finite slope can be no-loss CA-bad while being explained on this fixed support  $S$ . Consequently, for every support family  $\mathcal{S}$ ,*

$$\#\{\mathcal{S}\text{-witnessed support-wise MCA-bad finite slopes}\} \leq |\mathcal{S}|,$$

and the same bound holds for no-loss CA slopes witnessed by  $\mathcal{S}$ .

*Proof.* Suppose two distinct finite slopes  $z_1 \neq z_2$  are both explained on the same support  $S$ , by codewords  $p_1, p_2 \in C$ :

$$f + z_i g = p_i \quad \text{on } S, \quad i = 1, 2.$$

Then on  $S$ ,

$$g = \frac{p_1 - p_2}{z_1 - z_2}, \quad f = p_1 - z_1 g.$$

Both right-hand sides are codewords of  $C$ . Thus  $(f, g)$  is simultaneously explained on  $S$ . This contradicts support-wise noncontainment on  $S$ . It also contradicts no-loss CA badness at agreement  $a$ , because  $|S| \geq a$  gives a global simultaneous explanation support of the forbidden size. Hence each fixed support contributes at most one bad finite line parameter, and summing over supports gives the displayed bound.  $\square$

For a finite-parameter power curve

$$W_\gamma = f_0 + \gamma f_1 + \cdots + \gamma^d f_d, \quad \gamma \in \mathbb{F},$$

say that a parameter is support-wise curve-bad on  $S$  if  $W_\gamma$  is explained by a codeword on  $S$ , but the coefficient tuple  $(f_0, \dots, f_d)$  is not simultaneously explained by  $C^{\equiv(d+1)}$  on  $S$ . The no-loss curve-CA variant replaces this support-wise noncontainment by the corresponding global far condition at agreement  $a$ .

**Lemma 3.17** (one support pays for at most  $d$  curve parameters). *Let  $C = \text{RS}[\mathbb{F}, D, k]$ , let  $a \geq k$ , and fix a support  $S \subseteq D$  with  $|S| \geq a$ . For a degree- $d$  power curve  $W_\gamma$ , at most  $d$  finite parameters  $\gamma \in \mathbb{F}$  can be support-wise curve-bad on  $S$ . The same bound holds for no-loss curve-CA parameters explained on  $S$ . Consequently any support family  $\mathcal{S}$  witnesses at most  $d|\mathcal{S}|$  bad finite curve parameters.*

*Proof.* If  $d + 1$  distinct parameters  $\gamma_0, \dots, \gamma_d$  were explained on the same support  $S$ , with codewords  $P_j \in C$ , then Lagrange interpolation in the parameter variable gives codewords  $Q_0, \dots, Q_d \in C$  such that

$$\sum_{i=0}^d \gamma_j^i Q_i = P_j \quad (0 \leq j \leq d).$$

On every point of  $S$ , the same interpolation applied to the values of  $W_{\gamma_j}$  gives  $f_i = Q_i$  for every coefficient  $i$ . Hence  $(f_0, \dots, f_d)$  is simultaneously explained by  $C^{\equiv(d+1)}$  on  $S$ . This contradicts support-wise curve-noncontainment on  $S$ , and also contradicts no-loss curve-CA badness at agreement  $a$ . Thus every support contributes at most  $d$  parameters.  $\square$

**Lemma 3.18** (one support pays for at most one interleaved tuple). *Let  $C = \text{RS}[\mathbb{F}, D, k]$ , let  $a \geq k$ , and let  $\mu \geq 1$ . Fix a  $\mu$ -row received word  $U = (U_1, \dots, U_\mu)$ . For a fixed support  $S \subseteq D$  with  $|S| \geq a$ , there is at most one tuple  $(P_1, \dots, P_\mu) \in C^\mu$  agreeing with  $U$  on every row over  $S$ . Hence the number of interleaved list tuples whose common agreement support belongs to a support family  $\mathcal{S}$  is at most  $|\mathcal{S}|$ .*

*Proof.* For each row  $i$ , two degree- $< k$  polynomials agreeing with  $U_i$  on the fixed support  $S$  agree with each other on at least  $k$  points, so they are equal. Thus each row has at most one codeword compatible with  $S$ , and therefore the whole interleaved tuple is unique if it exists.  $\square$

We now instantiate the support-family ledger on quotient-remainder supports. For  $c \mid n$  and an integer  $A$  with  $0 \leq A \leq n$ , write

$$A = m_c(A)c + s_c(A), \quad 0 \leq s_c(A) < c,$$

and define

$$\begin{aligned} \mathcal{S}_c(A) := \{ & \pi_c^{-1}(M) \cup R : M \subseteq Q_c, |M| = m_c(A), \\ & R \subseteq D \setminus \pi_c^{-1}(M), |R| = s_c(A) \}. \end{aligned}$$

For a divisor set  $\mathcal{C} \subseteq \{c : c \mid n\}$  put

$$\mathcal{S}_c^{\geq a} := \bigcup_{c \in \mathcal{C}} \bigcup_{A=a}^n \mathcal{S}_c(A)$$

and

$$U_c^{\text{supp}}(a) := |\mathcal{S}_c^{\geq a}|.$$

When the exact union is inconvenient, use the safe sum

$$U_c^{\text{sum}}(a) := \sum_{c \in \mathcal{C}} \sum_{A=a}^n \binom{n/c}{\lfloor A/c \rfloor} \binom{n - c\lfloor A/c \rfloor}{A - c\lfloor A/c \rfloor},$$

so that  $U_c^{\text{supp}}(a) \leq U_c^{\text{sum}}(a)$ .

**Proposition 3.19** (divisor-union quotient support ledger). *Let  $C = \text{RS}[\mathbb{F}, D, k]$  and let  $a \geq k$ . Fix a divisor set  $\mathcal{C}$  and let  $\mathcal{S}_c^{\geq a}$  be the quotient-remainder support family above. Then, for every received line, the number of finite support-wise MCA-bad slopes whose witness support lies in  $\mathcal{S}_c^{\geq a}$  is at most*

$$U_c^{\text{supp}}(a) \leq U_c^{\text{sum}}(a).$$

The same bound holds for no-loss CA slopes witnessed by this family. For a finite-parameter degree- $d$  power curve, the corresponding support-wise curve-MCA and no-loss curve-CA numerator is at most

$$dU_{\mathcal{C}}^{\text{supp}}(a) \leq dU_{\mathcal{C}}^{\text{sum}}(a).$$

For every interleaving arity  $\mu \geq 1$ , the number of interleaved list tuples whose common agreement support belongs to  $\mathcal{S}_{\mathcal{C}}^{\geq a}$  is at most  $U_{\mathcal{C}}^{\text{supp}}(a)$ .

*Proof.* Apply Lemmas 3.16, 3.17, and 3.18 to the support family  $\mathcal{S}_{\mathcal{C}}^{\geq a}$ . The inequality  $U_{\mathcal{C}}^{\text{supp}}(a) \leq U_{\mathcal{C}}^{\text{sum}}(a)$  is only the union bound over the divisor and agreement-size indices.  $\square$

*Remark 3.20* (what this does and does not finish). Proposition 3.19 supplies a certified safe-side quotient ledger at the support-family level, and it is already enough for a staircase scanner to avoid double-counting divisor scales: one may compute the exact union  $U_{\mathcal{C}}^{\text{supp}}$  or use the printed safe sum. This is not yet the sharp quotient upper theorem needed for a full prize solution. Near capacity the support count can be exponentially larger than the number of distinct bad slopes, so the next refinement is a slope-collision quotient ledger: classify when many quotient-remainder supports map to the same line or curve parameter, and replace the support count by the corresponding distinct-parameter union count.

**Definition 3.21** (block support-profile polynomial). Let  $D$  be a multiplicative coset of order  $n$ , let  $\mathcal{C}$  be a nonempty finite set of divisors of  $n$ , and put

$$h := \text{lcm } \mathcal{C}, \quad N_h := n/h.$$

For each  $c \in \mathcal{C}$  let  $H_c \leq H_h$  denote the subgroup of the cyclic kernel of order  $h$  with  $|H_c| = c$ . If  $T \subseteq H_h$ , define

$$\phi_c(T) := \#\{\text{cosets } uH_c \subseteq H_h : uH_c \subseteq T\},$$

the number of complete  $c$ -subfibers contained in the block subset  $T$ . The block support-profile polynomial of  $\mathcal{C}$  is

$$G_{\mathcal{C},h}(Y, (Z_c)_{c \in \mathcal{C}}) := \sum_{T \subseteq H_h} Y^{|T|} \prod_{c \in \mathcal{C}} Z_c^{\phi_c(T)}.$$

For a vector  $\nu = (\nu_c)_{c \in \mathcal{C}}$  write  $Z^\nu := \prod_{c \in \mathcal{C}} Z_c^{\nu_c}$ , and define  $N_{\mathcal{C},h}(A; \nu)$  by

$$G_{\mathcal{C},h}(Y, Z)^{N_h} = \sum_{A=0}^n \sum_{\nu} N_{\mathcal{C},h}(A; \nu) Y^A Z^\nu.$$

**Lemma 3.22** (block polynomial counts exact quotient profiles). For a support  $S \subseteq D$  and  $c \mid n$ , let

$$F_c(S) := \#\{c\text{-fibers of } D \rightarrow D^c \text{ contained in } S\}.$$

Then  $N_{\mathcal{C},h}(A; \nu)$  is exactly the number of supports  $S \subseteq D$  such that

$$|S| = A, \quad F_c(S) = \nu_c \quad (c \in \mathcal{C}).$$

*Proof.* The  $h$ -fibers partition  $D$  into  $N_h$  blocks, and every  $c$ -fiber with  $c \in \mathcal{C}$  is contained in a unique  $h$ -fiber because  $c \mid h$ . After choosing an identification of each  $h$ -fiber with the cyclic



kernel  $H_h$ , the number of selected points and the number of complete internal  $c$ -subfibers are recorded by the monomial

$$Y^{|T|} \prod_{c \in \mathcal{C}} Z_c^{\phi_c(T)}$$

for the block subset  $T$ . The choices in distinct  $h$ -fibers are independent, and the global statistics  $|S|$  and  $F_c(S)$  are the sums of the block statistics. Multiplying the block polynomial over the  $N_h$  blocks and extracting the coefficient gives the asserted count.  $\square$

**Lemma 3.23** (membership by full-fiber profile). *Fix  $A$  and  $c \mid n$ , and write*

$$A = m_c(A)c + s_c(A), \quad 0 \leq s_c(A) < c.$$

*For a support  $S \subseteq D$  of size  $A$ , one has*

$$S \in \mathcal{S}_c(A) \iff F_c(S) = m_c(A) = \lfloor A/c \rfloor.$$

*Proof.* If  $S \in \mathcal{S}_c(A)$ , then by definition it is the union of  $m_c(A)$  complete  $c$ -fibers and  $s_c(A) < c$  residual points outside those fibers; there is no room for any further complete  $c$ -fiber. Hence  $F_c(S) = m_c(A)$ . Conversely, if  $F_c(S) = m_c(A)$ , take the complete  $c$ -fibers contained in  $S$  as the set  $M$  in the definition of  $\mathcal{S}_c(A)$ ; the remaining  $A - m_c(A)c = s_c(A)$  selected points form the residual set.  $\square$

**Theorem 3.24** (exact divisor-lattice support-union formula). *Let  $\mathcal{C}$  be a nonempty finite set of divisors of  $n$ , and define  $G_{\mathcal{C},h}$  and  $N_{\mathcal{C},h}(A; \nu)$  as in Definition 3.21. Then the exact support-family union from Proposition 3.19 is*

$$U_{\mathcal{C}}^{\text{supp}}(a) = \sum_{A=a}^n \sum_{\nu} N_{\mathcal{C},h}(A; \nu) \mathbf{1}[\exists c \in \mathcal{C} : \nu_c = \lfloor A/c \rfloor].$$

*Consequently the quotient-support branch numerators in Proposition 3.19 may be computed by this coefficient formula instead of by the safe sum  $U_{\mathcal{C}}^{\text{sum}}(a)$ .*

*Proof.* For a fixed exact size  $A$ , the union  $\bigcup_{c \in \mathcal{C}} \mathcal{S}_c(A)$  consists precisely of those supports  $S$  of size  $A$  for which  $S \in \mathcal{S}_c(A)$  for at least one divisor  $c \in \mathcal{C}$ . By Lemma 3.23, this is exactly the condition

$$F_c(S) = \lfloor A/c \rfloor$$

for at least one  $c \in \mathcal{C}$ . By Lemma 3.22, the number of supports with a fixed profile vector  $\nu = (F_c(S))_{c \in \mathcal{C}}$  is  $N_{\mathcal{C},h}(A; \nu)$ . Summing over all exact sizes  $A \geq a$  and over the profile vectors satisfying the displayed predicate gives the formula.  $\square$

**Corollary 3.25** (single-divisor sanity check). *For  $\mathcal{C} = \{c\}$  the block polynomial is*

$$G_{\{c\},c}(Y, Z) = (1 + Y)^c + (Z - 1)Y^c,$$

*and Theorem 3.24 gives*

$$|\mathcal{S}_c(A)| = \binom{n/c}{\lfloor A/c \rfloor} \binom{n - c\lfloor A/c \rfloor}{A - c\lfloor A/c \rfloor},$$

*the summand used in the safe ledger.*

*Proof.* Inside one  $c$ -fiber, every proper subset contributes only  $Y^{|T|}$ , while the full subset contributes  $Y^c Z$ . Thus the block polynomial is  $(1 + Y)^c - Y^c + Y^c Z$ . Raising to  $n/c$  and selecting exactly  $m = \lfloor A/c \rfloor$  full blocks and  $s = A - mc$  residual points in the remaining blocks gives the displayed count.  $\square$

**Definition 3.26** (divisor-block support-ledger certificate). A divisor-block support-ledger certificate for a divisor set  $\mathcal{C}$  and a threshold  $a$  consists of:

- (i) the lcm block size  $h = \text{lcm } \mathcal{C}$  and the block count  $N_h = n/h$ ;
- (ii) the internal coset table of every subgroup  $H_c \leq H_h$  for  $c \in \mathcal{C}$ ;
- (iii) the block polynomial  $G_{\mathcal{C},h}$ , either as a full coefficient table or as an orbit-compressed table under translations of  $H_h$ ;
- (iv) the coefficient extraction from  $G_{\mathcal{C},h}^{N_h}$  for every exact size  $A \geq a$ ;
- (v) the predicate table  $\mathbf{1}[\exists c \in \mathcal{C} : \nu_c = \lfloor A/c \rfloor]$  and the final integer  $U_{\mathcal{C}}^{\text{supp}}(a)$ .

The verifier checks the block enumeration, the subgroup containment relation  $c \mid h$ , coefficient arithmetic, and the final predicate sum. This is an exact support-union certificate; it does not by itself coalesce distinct line or curve parameters that arise from different supports.

*Remark 3.27* (support ledger versus slope ledger). Theorem 3.24 closes the divisor-lattice overlap problem at the support-family level: intersections between quotient descriptions at incomparable divisor scales are now counted by one coefficient extraction in the common  $h$ -block. This is still weaker than the desired distinct-slope quotient ledger. To finish that stronger ledger one must apply the exact support-image maps of Propositions 3.30 and 3.32 to the support set counted here and then prove or certify the resulting parameter image size after duplicate coalescing.

### 3.3 Distinct-parameter quotient image ledger

The support-union ledger of Section 3.2 is deliberately safe but often very loose: many quotient-remainder supports can certify the same line slope or the same curve parameter. The next refinement is to count the image of the actual parameter map. This subsection gives an exact image ledger for the quotient branch. It is still a branch ledger, not an aperiodic upper bound: it counts only witnesses whose agreement support lies in a declared quotient-remainder support family.

We first record the support-to-parameter map in syndrome form. Let  $C = \text{RS}[\mathbb{F}, D, k]$ ,  $|D| = n$ , and put  $r = n - k$ . For each  $x \in D$  set

$$\lambda_x := \left( \prod_{y \in D, y \neq x} (x - y) \right)^{-1}.$$

For a word  $Y : D \rightarrow \mathbb{F}$ , define its syndrome vector

$$\text{Syn}(Y)_m := \sum_{x \in D} \lambda_x x^m Y(x), \quad 0 \leq m < r.$$

Thus  $\text{Syn}(Y) = 0$  if and only if  $Y \in C$ . If  $T \subseteq D$  has size  $j \leq r$ , write

$$L_T(X) = \prod_{x \in T} (X - x) = \ell_0 + \ell_1 X + \cdots + \ell_j X^j, \quad \ell_T = (\ell_0, \dots, \ell_j)^t,$$

and put  $t = r - j$ . For  $w \in \mathbb{F}^r$  let  $H_{t,j}(w)$  be the  $t \times (j+1)$  Hankel window

$$H_{t,j}(w)_{a,b} = w_{a+b}, \quad 0 \leq a < t, \quad 0 \leq b \leq j.$$

When  $S = D \setminus T$ , the integer  $t$  is exactly  $|S| - k$ .

**Lemma 3.28** (support-locator syndrome recurrence). *Let  $Y : D \rightarrow \mathbb{F}$  and let  $T \subseteq D$  with  $|T| = j \leq r$ . Put  $S = D \setminus T$  and  $t = r - j$ . Then  $Y$  is explained by  $C$  on  $S$  if and only if*

$$H_{t,j}(\text{Syn}(Y))\ell_T = 0.$$

*Proof.* Let  $W_T \subseteq \mathbb{F}^r$  be the span of the parity-check columns indexed by  $T$ ,

$$W_T = \text{span}_{\mathbb{F}}\{\lambda_x(1, x, \dots, x^{r-1}) : x \in T\}.$$

The word  $Y$  is explained by  $C$  on  $S$  if and only if  $Y - c$  is supported on  $T$  for some  $c \in C$ , equivalently  $\text{Syn}(Y) \in W_T$ . If  $w \in W_T$ , write  $w_m = \sum_{x \in T} a_x x^m$ . Then, for  $0 \leq h < t$ ,

$$(H_{t,j}(w)\ell_T)_h = \sum_{b=0}^j \ell_b w_{h+b} = \sum_{x \in T} a_x x^h L_T(x) = 0.$$

Conversely, because  $L_T$  is monic, the displayed recurrence determines all coordinates  $w_j, \dots, w_{r-1}$  from the first  $j$  coordinates, so its solution space has dimension at most  $j$ . The  $j$  parity-check columns indexed by the distinct points of  $T$  have Vandermonde rank  $j$ , hence  $W_T$  has dimension  $j$  and is exactly the recurrence space.  $\square$

**Lemma 3.29** (exact affine and projective support-image map). *Let  $f, g : D \rightarrow \mathbb{F}$ , put  $u = \text{Syn}(f)$  and  $v = \text{Syn}(g)$ , and fix  $T \subseteq D$  with  $|T| = j \leq r$  and  $S = D \setminus T$ . Define*

$$A_T := H_{t,j}(u)\ell_T, \quad B_T := H_{t,j}(v)\ell_T.$$

*A finite slope  $z \in \mathbb{F}$  is explained on the fixed support  $S$  and is support-wise noncontained there if and only if*

$$A_T + zB_T = 0, \quad B_T \neq 0.$$

*Consequently the fixed support contributes either no finite bad slope or the unique value*

$$\theta_T = -A_{T,m}/B_{T,m}$$

*for any coordinate  $m$  with  $B_{T,m} \neq 0$ , provided all nonzero coordinates give the same value.*

*For the projective line  $\alpha f + \beta g$ , the fixed support contributes a support-wise noncontained point  $[\alpha : \beta] \in \mathbb{P}^1(\mathbb{F})$  if and only if  $A_T$  and  $B_T$  are linearly dependent and not both zero. In that case the contributed projective parameter is the unique point satisfying*

$$\alpha A_T + \beta B_T = 0.$$

*Proof.* A word  $Y$  is explained by  $C$  on  $S = D \setminus T$  if and only if  $Y - c$  is supported on  $T$  for some  $c \in C$ , which is equivalent to  $\text{Syn}(Y)$  lying in the span of the parity-check columns indexed by  $T$ . The standard locator recurrence says that this is equivalent to

$$H_{t,j}(\text{Syn}(Y))\ell_T = 0.$$

Applying this to  $Y = f + zg$  gives  $A_T + zB_T = 0$ . Under this incidence, simultaneous explanation of  $f$  and  $g$  on  $S$  is equivalent to  $B_T = 0$ , because  $A_T = (A_T + zB_T) - zB_T$ . This proves the finite-slope statement. The projective statement is the same calculation with  $Y = \alpha f + \beta g$ ; if  $A_T = B_T = 0$  then both coefficient words are already explained on  $S$ , so the point is contained rather than bad.  $\square$

Let  $\mathcal{S}_C^{\geq a}$  be the quotient-remainder support family from Section 3.2, and let

$$\mathcal{T}_C^{\geq a} := \{D \setminus S : S \in \mathcal{S}_C^{\geq a}\}$$

be the corresponding co-support family. For a received line define

$$\Theta_C^{\text{aff}}(f, g; a) := \{\theta_T : T \in \mathcal{T}_C^{\geq a}, \theta_T \text{ is defined by Lemma 3.29}\}$$

and define  $\Theta_C^{\text{proj}}(f, g; a)$  analogously using the projective point of Lemma 3.29.

**Proposition 3.30** (distinct-parameter quotient ledger for lines). *For every received line  $f + zg$ , the finite affine slopes that are support-wise MCA-bad and have a witness support in  $\mathcal{S}_C^{\geq a}$  are exactly the elements of  $\Theta_C^{\text{aff}}(f, g; a)$ . Hence the exact quotient-branch line numerator is*

$$|\Theta_C^{\text{aff}}(f, g; a)| \leq U_C^{\text{supp}}(a).$$

*The corresponding projective-slope numerator is*

$$|\Theta_C^{\text{proj}}(f, g; a)| \leq U_C^{\text{supp}}(a).$$

*The same image ledgers upper-bound no-loss CA parameters whose explaining support lies in the same quotient family.*

*Proof.* This is only Lemma 3.29 applied to every co-support in the family, followed by taking a union of the resulting parameter values. The CA claim follows because no-loss CA-badness is stronger than having an explained slope on the support and forbids simultaneous explanation on any support of size at least  $a$ .  $\square$

The curve analogue is just as explicit. For a degree- $d$  power curve

$$W_\gamma = f_0 + \gamma f_1 + \cdots + \gamma^d f_d, \quad \gamma \in \mathbb{F},$$

put  $u_i = \text{Syn}(f_i)$  and, for  $T \subseteq D$ , define

$$V_{i,T} := H_{t,j}(u_i)\ell_T \in \mathbb{F}^t.$$

Let

$$\Gamma_T^{(d)} := \{\gamma \in \mathbb{F} : V_{0,T} + \gamma V_{1,T} + \cdots + \gamma^d V_{d,T} = 0, (V_{0,T}, \dots, V_{d,T}) \neq 0\}.$$

The nonzero tuple condition is exactly support-wise noncontainment of the coefficient tuple on the fixed support.

**Lemma 3.31** (exact support-image map for finite-parameter power curves). *For a fixed support  $S = D \setminus T$ , the set of support-wise noncontained curve-bad finite parameters is exactly  $\Gamma_T^{(d)}$ . In particular  $|\Gamma_T^{(d)}| \leq d$ .*

*Proof.* The same syndrome-locator recurrence applied to  $W_\gamma$  gives the displayed vector equation. If every  $V_{i,T}$  is zero, then every coefficient word  $f_i$  is already explained on  $S$ , so the parameter is contained. Otherwise at least one scalar coordinate of  $V_{0,T} + \gamma V_{1,T} + \cdots + \gamma^d V_{d,T}$  is a nonzero polynomial in  $\gamma$  of degree at most  $d$ , so it has at most  $d$  roots.  $\square$

Define the exact quotient curve image

$$\Gamma_C^{(d)}(f_0, \dots, f_d; a) := \bigcup_{T \in \mathcal{T}_C^{\geq a}} \Gamma_T^{(d)}.$$

**Proposition 3.32** (distinct-parameter quotient ledger for curves). *For a finite-parameter degree- $d$  power curve, the quotient-branch support-wise curve-MCA numerator is exactly*

$$|\Gamma_C^{(d)}(f_0, \dots, f_d; a)|.$$

*It satisfies the safe bounds*

$$|\Gamma_C^{(d)}(f_0, \dots, f_d; a)| \leq d U_C^{\text{supp}}(a) \leq d U_C^{\text{sum}}(a).$$

*The same image ledger upper-bounds no-loss curve-CA parameters whose explaining support lies in the quotient family.*

*Proof.* Apply Lemma 3.31 to each support and take the union of the resulting parameter sets. The displayed inequalities follow from  $|\Gamma_T^{(d)}| \leq d$  and from the support-union bound of Proposition 3.19.  $\square$

**Definition 3.33** (fixed-support parameter gcd). Let  $C = \text{RS}[\mathbb{F}, D, k]$ , let  $q = |\mathbb{F}|$ , and let

$$W_\Gamma = f_0 + \Gamma f_1 + \cdots + \Gamma^d f_d, \quad \Gamma \in \mathbb{F},$$

be a finite-parameter degree- $d$  power curve of received words. Fix a co-support  $T \subseteq D$ , put  $S = D \setminus T$ , and assume  $|S| > k$ . With the notation of Lemma 3.31, write

$$P_{T,m}(\Gamma) := \sum_{i=0}^d \Gamma^i (V_{i,T})_m, \quad 0 \leq m < t, \quad t = |S| - k.$$

Define the fixed-support contribution polynomial  $g_T^{(d)}(\Gamma) \in \mathbb{F}[\Gamma]$  as follows. If all coordinate polynomials  $P_{T,m}$  are identically zero, set  $g_T^{(d)} = 1$ . Otherwise let  $g_T^{(d)}$  be the monic greatest common divisor of the nonzero polynomials among  $P_{T,0}, \dots, P_{T,t-1}$ .

**Lemma 3.34** (fixed-support gcd equals the noncontained parameter set). *In the setting of Definition 3.33, the finite parameters  $\gamma \in \mathbb{F}$  for which  $W_\gamma$  is explained on  $S = D \setminus T$  and the coefficient tuple  $(f_0, \dots, f_d)$  is not simultaneously explained on  $S$  are exactly the roots in  $\mathbb{F}$  of  $g_T^{(d)}$ . In particular*

$$\deg g_T^{(d)} \leq d,$$

*with equality possible only before taking common factors across different supports.*

*Proof.* By Lemma 3.31, the fixed support explains  $W_\gamma$  exactly when

$$P_{T,m}(\gamma) = 0 \quad (0 \leq m < t).$$

If every coordinate polynomial is identically zero, then every coefficient vector  $V_{i,T}$  is zero, so every coefficient word  $f_i$  is already explained on  $S$ ; this is the contained branch and contributes no noncontained parameter. The convention  $g_T^{(d)} = 1$  gives the empty root set, as required.

Otherwise the common zeros of the coordinate polynomials are exactly the roots of their monic gcd after zero coordinates are discarded. Since at least one coordinate polynomial is nonzero of degree at most  $d$ , the gcd also has degree at most  $d$ . The nonzero tuple condition in Lemma 3.31 is precisely the assertion that the coordinate polynomials are not all zero.  $\square$

**Definition 3.35** (quotient image lcm polynomial). Let  $\mathcal{T}$  be any finite family of co-supports  $T \subseteq D$  with  $|D \setminus T| > k$ . For a degree- $d$  finite-parameter power curve define

$$R_{\mathcal{T}}^{(d)}(\Gamma) := \text{rad lcm}_{T \in \mathcal{T}} g_T^{(d)}(\Gamma),$$

where factors equal to 1 are ignored and the empty lcm is 1. For the quotient branch at agreement threshold  $a > k$  and divisor set  $\mathcal{C}$ , we use

$$\mathcal{T} = \mathcal{T}_{\mathcal{C}}^{\geq a} := \{D \setminus S : S \in \mathcal{S}_{\mathcal{C}}^{\geq a}\}.$$

The finite-field root-count polynomial is

$$R_{\mathcal{T},F}^{(d)}(\Gamma) := \text{gcd}(R_{\mathcal{T}}^{(d)}(\Gamma), \Gamma^q - \Gamma).$$

Because  $R_{\mathcal{T}}^{(d)}$  is squarefree, the degree of  $R_{\mathcal{T},F}^{(d)}$  is the number of  $\mathbb{F}$ -rational roots of  $R_{\mathcal{T}}^{(d)}$ .

**Theorem 3.36** (exact finite-parameter quotient image ledger). *Let  $a > k$ , let  $\mathcal{C}$  be a finite divisor set, and let  $\mathcal{T}_{\mathcal{C}}^{\geq a}$  be the quotient co-support family. For every finite-parameter degree- $d$  power curve, the exact quotient-branch support-wise curve-MCA numerator is*

$$\boxed{\left| \Gamma_{\mathcal{C}}^{(d)}(f_0, \dots, f_d; a) \right| = \deg R_{\mathcal{T}_{\mathcal{C}}^{\geq a}, F}^{(d)}}.$$

Consequently

$$\left| \Gamma_{\mathcal{C}}^{(d)}(f_0, \dots, f_d; a) \right| \leq \deg R_{\mathcal{T}_{\mathcal{C}}^{\geq a}}^{(d)} \leq d U_{\mathcal{C}}^{\text{supp}}(a),$$

and the final support count may be evaluated by the exact divisor-block formula of Theorem 3.24. The same lcm ledger upper-bounds no-loss curve-CA parameters whose explaining support lies in the same quotient family.

*Proof.* By Lemma 3.34, for each fixed co-support  $T$  the noncontained parameters witnessed on  $D \setminus T$  are exactly the  $\mathbb{F}$ -roots of  $g_T^{(d)}$ . Taking the union over  $T \in \mathcal{T}_{\mathcal{C}}^{\geq a}$  is therefore the same as taking the  $\mathbb{F}$ -rational root set of the squarefree lcm  $R_{\mathcal{T}_{\mathcal{C}}^{\geq a}}^{(d)}$ . Over a finite field, the  $\mathbb{F}$ -rational roots of a squarefree polynomial are counted by the degree of its gcd with  $\Gamma^q - \Gamma$ , proving the displayed equality.

The degree bound follows from  $\deg g_T^{(d)} \leq d$  and from the fact that a squarefree lcm has degree at most the sum of the degrees of the factors before coalescing. The no-loss curve-CA claim follows because a no-loss CA parameter with an explaining support in the quotient family is, on that support, noncontained; otherwise the coefficient tuple would have a simultaneous explanation on a support of size at least  $a$ .  $\square$

**Corollary 3.37** (affine lines as the degree-one lcm ledger). *For an affine received line  $f + Zg$ , take  $d = 1$ ,  $f_0 = f$ , and  $f_1 = g$ . Then  $R_{\mathcal{T}_C^{\geq a}}^{(1)}(Z)$  is the squarefree product of the linear factors  $Z - z$  for the distinct finite quotient-witnessed support-wise MCA-bad slopes. Hence*

$$\#\{\text{finite quotient-witnessed MCA-bad slopes}\} = \deg R_{\mathcal{T}_C^{\geq a}}^{(1)}.$$

*The same polynomial upper-bounds no-loss CA finite slopes whose explaining support lies in the declared quotient family.*

*Proof.* For  $d = 1$ , each nonconstant fixed-support gcd has degree at most one. Its root, if present, is exactly the unique finite slope supplied by Lemma 3.29. Taking the squarefree lcm over the declared co-support family therefore multiplies one copy of  $Z - z$  for each distinct finite slope and no copies for supports that contribute no finite noncontained slope.  $\square$

**Definition 3.38** (projective line lcm ledger). For a projective received line  $Z_0f + Z_1g$  and a co-support  $T$  with  $|D \setminus T| > k$ , form the homogeneous linear coordinate forms

$$L_{T,m}(Z_0, Z_1) := Z_0(A_T)_m + Z_1(B_T)_m, \quad 0 \leq m < t.$$

If all  $L_{T,m}$  are zero, set  $h_T^{\text{proj}} = 1$ . Otherwise let  $h_T^{\text{proj}}$  be the gcd of the nonzero linear forms, normalized by a fixed monomial order. Thus  $h_T^{\text{proj}}$  is either 1 or a homogeneous linear form. For a co-support family  $\mathcal{T}$  define

$$R_{\mathcal{T}}^{\text{proj}}(Z_0, Z_1) := \text{rad lcm}_{T \in \mathcal{T}} h_T^{\text{proj}}.$$

**Corollary 3.39** (exact projective quotient image ledger). *For projective slopes, the exact quotient-branch support-wise MCA numerator is*

$$\deg R_{\mathcal{T}_C^{\geq a}}^{\text{proj}}.$$

*The same degree upper-bounds no-loss projective-CA parameters whose explaining support lies in the quotient family.*

*Proof.* A nonzero family of homogeneous linear forms on  $\mathbb{P}^1$  has a common zero if and only if the forms have a common homogeneous linear factor; this factor cuts out exactly one  $\mathbb{F}$ -rational projective point. The all-zero case is the contained branch and contributes no bad point, by convention. The squarefree lcm therefore contains one linear factor for each distinct projective parameter and no duplicates.  $\square$

**Definition 3.40** (lcm quotient-image certificate). An lcm quotient-image certificate for a line or finite-parameter curve consists of:

- (i) the divisor set  $\mathcal{C}$ , the agreement threshold  $a$ , and either the exact support-union certificate of Definition 3.26 or an explicit hash/orbit enumeration of the co-support family  $\mathcal{T}_C^{\geq a}$ ;
- (ii) for every represented co-support  $T$ , the locator vector  $\ell_T$  and the syndrome-window coefficient vectors  $V_{i,T}$ ;
- (iii) the coordinate polynomials  $P_{T,m}(\Gamma)$ , the fixed-support gcd  $g_T^{(d)}$ , and a proof that the all-zero case has been assigned to the contained branch;

- (iv) the squarefree lcm  $R_{\mathcal{T}}^{(d)}$  and, over the finite field, the root-count factor  $R_{\mathcal{T},F}^{(d)} = \gcd(R_{\mathcal{T}}^{(d)}, \Gamma^q - \Gamma)$ ;
- (v) the declared numerator  $\deg R_{\mathcal{T},F}^{(d)}$  for finite curves or  $\deg R_{\mathcal{T}}^{\text{proj}}$  for projective lines.

The verifier checks the locator split condition, the syndrome recurrence, the gcd and lcm arithmetic, squarefreeness, and the final finite-field root count. This certificate replaces a support count by an exact distinct-parameter count inside the declared quotient branch.

*Remark 3.41* (status after the lcm image ledger). The quotient ledgers now separate three tasks that were previously entangled. First, Theorem 3.8 supplies lower floors by heaviest generated-field prefix fibers. Second, Theorem 3.24 counts the declared quotient support union with divisor overlaps handled exactly. Third, Theorem 3.36 and Corollaries 3.37 and 3.39 push that support family through the fixed-locator equations and coalesce duplicate finite, projective, and curve parameters by gcd/lcm arithmetic. The remaining quotient-side obstruction is not duplicate coalescing inside a declared quotient family; it is the structural safe-side theorem that every bad parameter in the intended regime either belongs to this quotient image, to the tangent branch, or to the aperiodic/extension buckets bounded elsewhere.

**Definition 3.42** (quotient image certificate). A quotient image certificate for a line or curve ledger consists of:

- (i) the divisor set  $\mathcal{C}$  and the agreement threshold  $a$ ;
- (ii) a deterministic description, hash, or orbit decomposition of  $\mathcal{T}_{\mathcal{C}}^{\geq a}$ ;
- (iii) for every represented co-support  $T$ , the locator coefficients  $\ell_T$  and the syndrome-window vectors  $A_T, B_T$  for a line, or  $V_{0,T}, \dots, V_{d,T}$  for a curve;
- (iv) the declared parameter value, parameter bucket, or the declaration that no parameter is contributed;
- (v) a bucket table proving the number of distinct parameters after duplicate coalescing.

The verifier checks the locator split condition, the Hankel equations, the noncontainment vector, and the bucket consistency. This certificate is the preferred replacement for a raw quotient-support count whenever the support count is much larger than the actual parameter image.

*Remark 3.43* (status of the refinement). The propositions above are exact for the declared quotient branch and are a real refinement over the support ledger: the numerator is an image size, not a support count. They still do not prove that all bad parameters are quotient-periodic. The remaining safe-side theorem for the full prize is the aperiodic Hankel-packing bound: after tangent and quotient image branches are removed, one must bound the parameters coming from co-supports outside the declared quotient families.

### 3.4 Aperiodic Hankel chart atlas

The quotient image ledger of Section 3.3 counts bad parameters whose witness support lies in a declared quotient-remainder family. The complementary safe-side problem is to control all other supports. This subsection is the contributor-facing formulation of that problem. It does not claim that the needed aperiodic eliminants always exist. It gives a finite chart atlas, with



precise ideals and pivots, so that a verifier can check a supplied eliminant and convert it into a bad-parameter upper bound.

The main point is to avoid one huge saturated incidence ideal. We split the aperiodic branch into:

- (i) a regular overdetermined Hankel bucket, where a single maximal minor in the line parameter already gives an eliminant;
- (ii) finite affine pivot charts, one for each noncontainment coordinate;
- (iii) a separate projective-infinity chart, if projective slopes are being counted;
- (iv) finite-parameter curve pivot charts, one for a nonzero coefficient of one coordinate polynomial;
- (v) a singular bucket, where all regular minors vanish or the projection to the parameter line remains positive-dimensional after the previous splits.

Only the last bucket is genuinely unresolved. A staircase scanner should keep it as a named obstruction rather than hiding it inside the word “aperiodic.”

Fix an exact agreement value  $A \geq k$  and put

$$j = j_A := n - A, \quad t = t_A := A - k, \quad R = n - k.$$

Let

$$P_D(X) := \prod_{x \in D} (X - x).$$

A monic degree- $j$  co-support locator is written

$$L_\ell(X) = X^j + \ell_{j-1}X^{j-1} + \cdots + \ell_0, \quad \ell = (\ell_0, \dots, \ell_{j-1}) \in \mathbb{A}^j.$$

Let  $R_{D,j}(\ell; X)$  be the remainder of  $P_D(X)$  modulo  $L_\ell(X)$ , and let  $I_{\text{split},j}$  be the ideal generated by the  $j$  coefficients of  $R_{D,j}$ . Since  $P_D$  is squarefree, the  $F$ -points of  $V(I_{\text{split},j})$  are exactly the squarefree  $D$ -split monic degree- $j$  locators.

For a word  $y : D \rightarrow F$  write  $\text{Syn}(y) \in F^R$  for the parity-check syndrome used above. Let

$$\tilde{\ell} = (\ell_0, \dots, \ell_{j-1}, 1)^t.$$

For an affine line  $f + zg$ , put  $u = \text{Syn}(f)$  and  $v = \text{Syn}(g)$  and define

$$A(\ell) := H_{t,j}(u)\tilde{\ell}, \quad B(\ell) := H_{t,j}(v)\tilde{\ell} \in F^t.$$

For a fixed split locator, the support-wise finite-slope condition is exactly

$$A(\ell) + ZB(\ell) = 0, \quad B(\ell) \neq 0.$$

The inequality is the noncontainment condition.

### 3.4.1 Regular overdetermined bucket

Assume first that

$$t_A \geq j_A + 1, \quad \text{equivalently} \quad 2A \geq n + k + 1.$$

Set

$$M_A(Z) := H_{t_A, j_A}(u) + ZH_{t_A, j_A}(v).$$

This is a  $t_A \times (j_A + 1)$  matrix with entries affine-linear in  $Z$ .

**Definition 3.44** (regular Hankel minor certificate). A regular Hankel minor certificate at exact agreement  $A$  is a row set  $I_A \subseteq \{0, \dots, t_A - 1\}$  of size  $j_A + 1$  such that

$$\Delta_A(Z) := \det M_A(Z)_{I_A, \{0, \dots, j_A\}}$$

is not the zero polynomial in  $F[Z]$ . If no such row set exists, the line is called *Hankel-singular* at  $A$ .

**Lemma 3.45** (regular exact-agreement eliminant). *Assume  $t_A \geq j_A + 1$ , and suppose  $\Delta_A(Z)$  is a regular Hankel minor certificate. Every finite support-wise MCA-bad slope with a witness support of exact size  $A$  is a root of  $\Delta_A$ . Hence the number of such slopes is at most*

$$\deg \Delta_A \leq j_A + 1 = n - A + 1.$$

*The same root containment holds for no-loss CA-bad slopes with exact agreement  $A$ .*

*Proof.* Let  $z$  be a bad finite slope with witness support  $S$  of size  $A$ , and put  $T = D \setminus S$ . Then  $|T| = j_A$ , and the locator vector  $\tilde{\ell}_T$  is a nonzero kernel vector of  $M_A(z)$ . Therefore the column rank of  $M_A(z)$  is less than  $j_A + 1$ , so every  $(j_A + 1) \times (j_A + 1)$  maximal minor vanishes at  $z$ , including  $\Delta_A(z)$ . The degree bound follows because  $\Delta_A$  is the determinant of a  $(j_A + 1) \times (j_A + 1)$  matrix whose entries are affine-linear in  $Z$ .  $\square$

**Theorem 3.46** (regular closed-range Hankel packing). *Let  $a \geq k$  and suppose that for each exact agreement  $A \in \{a, a + 1, \dots, n\}$  with  $t_A \geq j_A + 1$  a regular Hankel minor certificate  $\Delta_A$  is supplied. Then all bad finite slopes whose exact witness size is in this regular bucket lie in the root set of*

$$\Delta_{\geq a}(Z) := \prod_{A=a}^n \Delta_A(Z),$$

*with factors omitted for exact agreements outside the overdetermined range. Consequently their number is at most*

$$\sum_A \deg \Delta_A.$$

*If exact root sets are supplied, roots already charged to the tangent or quotient-image ledgers may be removed from this count by duplicate coalescing.*

*Proof.* Apply Lemma 3.45 at the exact size of the witness support and take the union of the resulting root sets. The degree sum is a safe union bound.  $\square$

**Canonical regular rank-drop gcds.** The regular-minor certificate below can be made canonical. Instead of choosing one nonzero maximal minor, take the greatest common divisor of all nonzero maximal minors. This gives exactly the finite-parameter values at which the Hankel pencil can drop rank, before the split-locator and noncontainment tests are imposed. Thus it is an upper ledger for the regular bucket, and often a much smaller one than the degree of a single selected minor.

Fix an exact agreement value  $A \geq k$ , and keep the notation

$$j = j_A := n - A, \quad t = t_A := A - k, \quad r = n - k.$$

The regular overdetermined range is

$$t_A > j_A, \quad \text{i.e.} \quad t_A \geq j_A + 1, \quad \text{equivalently} \quad 2A \geq n + k + 1.$$

**Definition 3.47** (canonical affine rank-drop gcd). Fix an affine received line  $f + Zg$ , put  $u = \text{Syn}(f)$  and  $v = \text{Syn}(g)$ , and assume  $t_A \geq j_A + 1$ . Set

$$M_A(Z) := H_{t_A, j_A}(u) + ZH_{t_A, j_A}(v).$$

Let  $\mathfrak{M}_A(Z)$  be the finite set of all  $(j_A + 1) \times (j_A + 1)$  maximal minors of  $M_A(Z)$ . If every element of  $\mathfrak{M}_A(Z)$  is the zero polynomial, the exact-agreement bucket is called *rank-drop singular*. Otherwise define

$$G_A^{\text{aff}}(Z) := \text{rad gcd}\{\Delta(Z) : \Delta \in \mathfrak{M}_A(Z), \Delta \neq 0\},$$

with the gcd normalized to be monic. Over the finite field  $\mathbb{F}$  of size  $q$  set

$$G_{A,F}^{\text{aff}}(Z) := \text{gcd}(G_A^{\text{aff}}(Z), Z^q - Z).$$

Thus  $G_{A,F}^{\text{aff}}$  is the squarefree polynomial whose roots are precisely those  $F$ -rational parameters at which all nonzero maximal minors vanish.

**Theorem 3.48** (canonical exact-agreement rank-drop ledger). *Assume  $t_A \geq j_A + 1$  and the exact-agreement bucket is not rank-drop singular. Every finite support-wise MCA-bad slope with an exact witness support of size  $A$  is a root of  $G_A^{\text{aff}}$ . Consequently the number of such slopes is at most*

$$\deg G_{A,F}^{\text{aff}} \leq \deg G_A^{\text{aff}} \leq j_A + 1 = n - A + 1.$$

*The same bound holds for no-loss CA-bad slopes with exact agreement  $A$ .*

*Proof.* Let  $z \in F$  be support-wise bad with exact witness support  $S$  of size  $A$ , and put  $T = D \setminus S$ . By Lemma 3.28, the locator vector  $\ell_T$  is a nonzero kernel vector of  $M_A(z)$ . Therefore  $\text{rank } M_A(z) < j_A + 1$ , so every maximal minor of  $M_A(Z)$  vanishes at  $Z = z$ . In particular every nonzero maximal minor vanishes at  $z$ , and hence their gcd  $G_A^{\text{aff}}$  vanishes at  $z$ .

The finite-field root count is exactly  $\deg G_{A,F}^{\text{aff}}$  because  $G_{A,F}^{\text{aff}}$  is squarefree and divides  $Z^q - Z$ . Each maximal minor has degree at most  $j_A + 1$ , since it is the determinant of a matrix whose entries are affine-linear in  $Z$ ; the gcd cannot have larger degree. The no-loss CA claim is identical, because no-loss CA-badness with exact agreement  $A$  also supplies an explaining support of size  $A$  and hence the same locator-kernel condition.  $\square$

*Remark 3.49* (why this improves the one-minor certificate). Lemma 3.45 used one certified nonzero maximal minor and therefore counted the roots of a single determinant. The polynomial  $G_A^{\text{aff}}$  uses all maximal minors and counts their common finite-field root set. It is never worse than any one-minor certificate and can be much smaller; it is also canonical, so two scanners that use the same syndrome convention print the same regular-bucket polynomial.

**Definition 3.50** (closed-ball rank-drop lcm). For an agreement threshold  $a > k$ , let

$$\mathcal{A}_{\text{reg}}(a) := \{A \in \{a, a+1, \dots, n\} : t_A \geq j_A + 1 \text{ and the bucket } A \text{ is not rank-drop singular}\}.$$

Define

$$G_{\geq a}^{\text{aff}}(Z) := \text{rad lcm}_{A \in \mathcal{A}_{\text{reg}}(a)} G_A^{\text{aff}}(Z), \quad G_{\geq a, F}^{\text{aff}}(Z) := \text{gcd}(G_{\geq a}^{\text{aff}}(Z), Z^q - Z),$$

with the empty lcm equal to 1. Exact agreement values  $A \in \{a, \dots, n\} \setminus \mathcal{A}_{\text{reg}}(a)$  are recorded as residual buckets: they are either underdetermined ( $t_A < j_A + 1$ ) or rank-drop singular.

**Corollary 3.51** (canonical regular closed-ball Hankel packing). *All finite support-wise MCA-bad slopes whose exact witness size lies in  $\mathcal{A}_{\text{reg}}(a)$  are roots of  $G_{\geq a}^{\text{aff}}$ . Hence their number is at most*

$$\deg G_{\geq a, F}^{\text{aff}} \leq \sum_{A \in \mathcal{A}_{\text{reg}}(a)} (n - A + 1).$$

*The same bound holds for no-loss CA-bad slopes in the same regular buckets.*

*Proof.* Apply Theorem 3.48 at the exact agreement size of the witness and take the union over  $A$ . The squarefree lcm coalesces duplicate roots across exact-agreement buckets; the final gcd with  $Z^q - Z$  counts only finite-field roots. The displayed degree sum is the safe bound obtained before coalescing.  $\square$

**Definition 3.52** (paid-root removal for the regular branch). Suppose a squarefree polynomial  $P_{\text{paid}, F}(Z)$  is supplied whose roots are exactly the finite-field parameters already charged to tangent/common-line and quotient-image ledgers. Define the regular residual polynomial

$$G_{\text{reg} \setminus \text{paid}, F}^{\text{aff}}(Z) := \frac{G_{\geq a, F}^{\text{aff}}(Z)}{\text{gcd}(G_{\geq a, F}^{\text{aff}}(Z), P_{\text{paid}, F}(Z))}.$$

Its degree is the number of regular rank-drop parameters not already paid by the previous ledgers.

**Corollary 3.53** (quotient/tangent-removed regular aperiodic numerator). *If  $P_{\text{paid}, F}$  is exact in the sense of Definition 3.52, then the number of finite slopes in the regular overdetermined branch that remain after removing the paid quotient/tangent parameters is at most*

$$\deg G_{\text{reg} \setminus \text{paid}, F}^{\text{aff}}.$$

*This is a safe aperiodic numerator for the regular buckets. The underdetermined and rank-drop singular exact-agreement buckets must still be handled by pivot charts or by a separately named residual obstruction.*

*Proof.* By Corollary 3.51, every regular-bucket bad slope is a root of  $G_{\geq a, F}^{\text{aff}}$ . Removing the roots of the exact paid polynomial deletes precisely those already charged parameters from this finite root set. The remaining root set is counted by the displayed quotient because all polynomials are squarefree divisors of  $Z^q - Z$ .  $\square$

**Definition 3.54** (finite-parameter curve rank-drop gcd). For a finite-parameter degree- $d$  curve

$$W_\Gamma = f_0 + \Gamma f_1 + \cdots + \Gamma^d f_d,$$

put  $u_i = \text{Syn}(f_i)$  and, at exact agreement  $A$ , define

$$M_A^{(d)}(\Gamma) := \sum_{i=0}^d \Gamma^i H_{t_A, j_A}(u_i).$$

When  $t_A \geq j_A + 1$  and not all maximal minors of  $M_A^{(d)}$  vanish identically, set

$$G_A^{(d)}(\Gamma) := \text{rad gcd}\{\Delta(\Gamma) : \Delta \text{ is a nonzero maximal minor of } M_A^{(d)}\},$$

and

$$G_{A, F}^{(d)}(\Gamma) := \text{gcd}(G_A^{(d)}(\Gamma), \Gamma^q - \Gamma).$$

**Theorem 3.55** (canonical curve rank-drop ledger). *In a non-singular overdetermined exact-agreement bucket, every finite curve-MCA-bad parameter is a root of  $G_A^{(d)}$ . Consequently the number of such parameters is at most*

$$\deg G_{A, F}^{(d)} \leq \deg G_A^{(d)} \leq d(j_A + 1).$$

*The same bound holds for no-loss curve-CA parameters in the same bucket. The closed-ball form is obtained by taking the squarefree lcm of the polynomials  $G_A^{(d)}$  over the regular exact-agreement buckets and then taking the gcd with  $\Gamma^q - \Gamma$ .*

*Proof.* A curve-bad parameter with exact witness support  $S$  gives, for  $T = D \setminus S$ , a nonzero locator vector in the kernel of  $M_A^{(d)}(\gamma)$ . Hence every maximal minor vanishes at  $\gamma$ , and the same gcd argument as in Theorem 3.48 applies. Each matrix entry has degree at most  $d$  in  $\Gamma$ , so each maximal minor has degree at most  $d(j_A + 1)$ .  $\square$

**Definition 3.56** (regular rank-drop certificate). A regular rank-drop certificate for a row, a received affine line or finite curve, and an agreement threshold  $a$  consists of:

- (i) the syndrome vectors of the coefficient words and the exact-agreement range  $A \in \{a, \dots, n\}$ ;
- (ii) for every overdetermined exact agreement  $A$ , either a declaration that all maximal minors vanish identically, or the canonical gcd polynomial  $G_A^{\text{aff}}$ , respectively  $G_A^{(d)}$ , together with ideal-membership or raw minor arithmetic proving that it is the gcd of all nonzero maximal minors;
- (iii) the squarefree lcm over all non-singular regular buckets and the finite-root factor obtained by gcd with  $Z^q - Z$  or  $\Gamma^q - \Gamma$ ;
- (iv) if paid branches are removed, the exact paid-root polynomial and the finite root-set quotient of Definition 3.52;

- (v) a residual-bucket table listing all underdetermined and rank-drop singular exact agreements that are not covered by the regular gcd ledger.

The verifier checks the syndrome convention, the Hankel matrices, the maximal minor gcds, squarefree lcms, finite-field root factors, paid-root removal, and the residual-bucket table.

*Remark 3.57* (status after the rank-drop ledger). The regular overdetermined part of the aperiodic Hankel problem is now canonical: one no longer has to choose a minor or accept a degree sum when the common root set of all maximal minors is smaller. This still does not solve the near-capacity underdetermined buckets, which form the hard aperiodic residual class. Those buckets remain assigned to the affine pivot charts, projective-infinity charts, curve coefficient pivots, or named singular residual obstructions.

### 3.4.2 Finite affine pivot charts

The regular minor test is intentionally coarse and only works directly in the overdetermined range. The next atlas is the one contributors should use to construct actual aperiodic eliminants.

**Definition 3.58** (aperiodic locator chart). An aperiodic locator chart at exact agreement  $A$  is a pair

$$X = (E_X, \Delta_X),$$

where  $E_X$  is a finite list of polynomial equations in the locator coefficients  $\ell$ , and  $\Delta_X(\ell)$  is a product of declared nonzero tests. The chart represents the constructible set

$$V(I_{\text{split},j} + \langle E_X \rangle) \setminus V(\Delta_X).$$

The equations and inequations must include, or reference, the removal of already paid tangent/common-code-line and quotient-image branches. A family of charts  $\mathfrak{X}_A$  is a valid aperiodic chart cover if the union of their constructible sets contains every remaining split locator of exact co-support size  $j_A$  not assigned to those earlier ledgers.

For a coordinate  $h \in \{0, \dots, t-1\}$  define the affine pivot consistency ideal

$$\begin{aligned} J_{X,h}^{\text{aff}} &:= I_{\text{split},j} + \langle E_X \rangle + \langle A_m(\ell)B_h(\ell) - A_h(\ell)B_m(\ell) : 0 \leq m < t \rangle \\ &\quad + \langle ZB_h(\ell) + A_h(\ell) \rangle \subseteq F[\ell_0, \dots, \ell_{j-1}, Z], \end{aligned}$$

and saturate it by the pivot denominator:

$$\widehat{J}_{X,h}^{\text{aff}} := J_{X,h}^{\text{aff}} : (\Delta_X B_h)^\infty.$$

The equations  $A_m B_h - A_h B_m = 0$  force  $A$  and  $B$  to be collinear on the pivot chart, and  $ZB_h + A_h = 0$  records the unique finite slope.

**Theorem 3.59** (pivoted affine-line eliminant certificate). *Let  $X$  be an aperiodic locator chart and let  $h$  be a noncontainment pivot. If there is a nonzero polynomial*

$$Q_{X,h}(Z) \in \widehat{J}_{X,h}^{\text{aff}} \cap F[Z]$$

*of degree  $D_{X,h}$ , then the number of finite support-wise MCA-bad slopes in that chart with  $B_h \neq 0$  is at most  $\min\{|F|, D_{X,h}\}$ . The same bound holds for no-loss CA-bad slopes in that pivot chart.*

*Proof.* A bad slope in the chart has a split locator with  $\Delta_X \neq 0$  and, on this pivot chart,  $B_h \neq 0$ . The Hankel incidence equation gives  $A + ZB = 0$ , hence the point  $(\ell, Z)$  satisfies all generators of  $J_{X,h}^{\text{aff}}$  and survives the saturation. Therefore  $Q_{X,h}(Z) = 0$ . A nonzero degree- $D_{X,h}$  polynomial over  $F$  has at most  $D_{X,h}$  roots.  $\square$

*Remark 3.60* (why the pivot split is useful). The saturation by the whole vector  $B$  in the unsplit incidence ideal is replaced by the explicit open cover  $B_h \neq 0$ . A scanner can try all  $h$ , produce small univariate eliminants on successful pivots, and report the residual locus  $B_0 = \dots = B_{t-1} = 0$  as contained for support-wise MCA. For no-loss CA, that residual locus also contributes no bad slope at the same support, because if  $A = B = 0$  then both  $f$  and  $g$  are explained on the witness support.

### 3.4.3 Projective-slope split

For projective parameters  $[Z_0 : Z_1]$  the incidence equation is

$$Z_0 A(\ell) + Z_1 B(\ell) = 0.$$

The finite patch  $Z_0 \neq 0$  is exactly the affine pivot atlas above with  $Z = Z_1/Z_0$ . The extra point at infinity is  $[0 : 1]$ .

**Definition 3.61** (projective infinity chart). For a chart  $X$ , the projective-infinity chart is the constructible locus

$$I_{\text{split},j} + \langle E_X \rangle + \langle B_0, \dots, B_{t-1} \rangle, \quad \Delta_X \neq 0, \quad A \neq 0.$$

If it is nonempty, it contributes the single projective parameter  $[0 : 1]$ . If it is empty, the infinity point contributes nothing on  $X$ .

**Corollary 3.62** (projective eliminant certificate). *For a chart cover  $\mathfrak{X}_A$ , the projective-slope numerator is bounded by the union of the finite-patch root sets from Theorem 3.59 together with at most one extra point  $[0 : 1]$ , provided the infinity charts are certified either empty or nonempty. Equivalently, a homogeneous nonzero eliminant*

$$Q_X(Z_0, Z_1)$$

*of degree  $D_X$  bounds the number of projective bad parameters in that chart by  $D_X$ .*

*Proof.* The finite patch is the affine theorem. The complement of that patch is the single point  $[0 : 1]$ , which is governed by Definition 3.61. For the homogeneous formulation, every bad projective parameter is a zero of the homogeneous eliminant, and a nonzero homogeneous polynomial of degree  $D_X$  on  $\mathbb{P}^1(F)$  has at most  $D_X$  zeros.  $\square$

### 3.4.4 Finite-parameter curve charts

For a degree- $d$  finite power curve

$$W_\Gamma = f_0 + \Gamma f_1 + \dots + \Gamma^d f_d,$$

put

$$V_i(\ell) := H_{t,j}(\text{Syn}(f_i)) \tilde{\ell} \in F^t, \quad 0 \leq i \leq d.$$

The curve incidence equation is

$$V_0(\ell) + \Gamma V_1(\ell) + \cdots + \Gamma^d V_d(\ell) = 0.$$

Curve-MCA noncontainment means that not all coefficient vectors  $V_i(\ell)$  are zero. Thus the noncontained branch is covered by coefficient pivots  $(i, h)$  with  $(V_i)_h \neq 0$ .

For a chart  $X$  and a coefficient pivot  $(i, h)$ , define

$$\begin{aligned} J_{X,i,h}^{\text{curve,d}} &:= I_{\text{split},j} + \langle E_X \rangle \\ &+ \left\langle \sum_{r=0}^d \Gamma^r (V_r(\ell))_m : 0 \leq m < t \right\rangle \subseteq F[\ell_0, \dots, \ell_{j-1}, \Gamma], \end{aligned}$$

and set

$$\widehat{J}_{X,i,h}^{\text{curve,d}} := J_{X,i,h}^{\text{curve,d}} : (\Delta_X(V_i)_h)^\infty.$$

**Theorem 3.63** (pivoted curve eliminant certificate). *If a chart  $X$  and coefficient pivot  $(i, h)$  have a nonzero eliminant*

$$Q_{X,i,h}(\Gamma) \in \widehat{J}_{X,i,h}^{\text{curve,d}} \cap F[\Gamma]$$

*of degree  $D_{X,i,h}$ , then the number of finite degree- $d$  curve-MCA bad parameters in that chart and pivot is at most  $\min\{|F|, D_{X,i,h}\}$ . The same bound applies to no-loss curve-CA parameters in the same pivot chart.*

*Proof.* Every bad curve parameter on the pivot chart gives a split locator satisfying all curve-incidence equations and the nonzero pivot condition  $(V_i)_h \neq 0$ . It therefore survives the saturation and is a root of the supplied eliminant.  $\square$

**Residual rank-pivot charts.** The gcd ledger above removes the regular overdetermined branch. What remains is not a formless exceptional set: every remaining bad parameter is still a point of a finite collection of determinantal rank charts. The point of the next statements is to make this residual bucket verifier-facing. A certificate may now say exactly which rank chart it uses, which noncontainment coordinate is open, and which univariate or homogeneous eliminant cuts out the parameter projection. If no such eliminant is supplied, the chart is a named residual obstruction rather than an unaccounted term.

Fix an exact agreement value  $A \geq k$  and keep

$$j = j_A := n - A, \quad t = t_A := A - k, \quad r = n - k.$$

For a projective received line  $Z_0 f + Z_1 g$ , put

$$U := H_{t,j}(\text{Syn}(f)), \quad V := H_{t,j}(\text{Syn}(g)), \quad M_A^{\text{proj}}(Z_0, Z_1) := Z_0 U + Z_1 V.$$

For a locator vector  $\tilde{\ell}$  write

$$A(\ell) := U\tilde{\ell}, \quad B(\ell) := V\tilde{\ell}.$$

Thus the projective support equation is

$$Z_0 A(\ell) + Z_1 B(\ell) = 0,$$

and projective noncontainment is  $(A(\ell), B(\ell)) \neq (0, 0)$ .



Let  $X = (E_X, \Delta_X)$  be a locator chart in the sense of Definition 3.58. For row and column sets  $I \subseteq \{0, \dots, t-1\}$  and  $J \subseteq \{0, \dots, j\}$  with  $|I| = |J| = s$ , write

$$\Delta_{I,J}(Z_0, Z_1) := \det M_A^{\text{proj}}(Z_0, Z_1)_{I,J},$$

with the convention that the unique  $0 \times 0$  minor is 1. Let  $\mathcal{M}_{s+1}(M_A^{\text{proj}})$  denote the set of all  $(s+1) \times (s+1)$  minors, empty when  $s = \min\{t, j+1\}$ .

For a noncontainment pivot

$$\pi \in \{A_h : 0 \leq h < t\} \cup \{B_h : 0 \leq h < t\},$$

let  $\Pi_\pi(\ell)$  denote the corresponding scalar coordinate. Define the saturated projective rank-pivot ideal

$$\begin{aligned} \widehat{\mathcal{J}}_{X,I,J,\pi}^{\text{proj}} &:= (I_{\text{split},j} + \langle E_X \rangle + \langle M_A^{\text{proj}}(Z_0, Z_1) \tilde{\ell} \rangle \\ &\quad + \langle \Delta : \Delta \in \mathcal{M}_{s+1}(M_A^{\text{proj}}) \rangle) : (\Delta_X \Delta_{I,J} \Pi_\pi)^\infty \subseteq F[\ell_0, \dots, \ell_{j-1}, Z_0, Z_1]. \end{aligned}$$

The open condition  $\Delta_{I,J} \neq 0$  and the vanishing of all larger minors put  $M_A^{\text{proj}}$  on the exact-rank- $s$  stratum on this chart. The saturation by  $\Pi_\pi$  chooses one of the finitely many open noncontainment coordinates.

**Theorem 3.64** (residual rank-pivot cover). *Fix an exact agreement value  $A$  and a chart cover  $\mathfrak{X}_A$  of the remaining locator branch after tangent/common-line and quotient-image removals. Every projective support-wise MCA-bad parameter with exact witness size  $A$  in that remaining branch lies on one of the rank-pivot charts*

$$\widehat{\mathcal{J}}_{X,I,J,\pi}^{\text{proj}}$$

for some  $X \in \mathfrak{X}_A$ , some rank  $s$ , some nonzero  $s \times s$  minor  $\Delta_{I,J}$  at the parameter, and some nonzero noncontainment coordinate  $\Pi_\pi$ . The same cover contains every no-loss projective CA-bad parameter in that branch.

*Proof.* Let  $[z_0 : z_1]$  be bad with witness support  $S$ , co-support  $T$ , and locator vector  $\tilde{\ell}_T$ . Since the chart family covers the remaining branch,  $\tilde{\ell}_T$  lies in some  $X \in \mathfrak{X}_A$  with  $\Delta_X \neq 0$ . The support equation gives  $M_A^{\text{proj}}(z_0, z_1) \tilde{\ell}_T = 0$ . Let  $s = \text{rank } M_A^{\text{proj}}(z_0, z_1)$ . Choose a nonzero  $s \times s$  minor at this parameter; then every  $(s+1) \times (s+1)$  minor vanishes. Finally, support-wise noncontainment says that  $(A(\ell_T), B(\ell_T)) \neq (0, 0)$ , so some coordinate pivot  $\Pi_\pi$  is nonzero. Thus the point survives the displayed saturation and belongs to the indicated rank-pivot chart. The no-loss CA statement uses the same explaining support and same noncontainment conclusion.  $\square$

**Theorem 3.65** (projective residual eliminant certificate). *If a rank-pivot chart has a nonzero homogeneous eliminant*

$$Q_{X,I,J,\pi}(Z_0, Z_1) \in \widehat{\mathcal{J}}_{X,I,J,\pi}^{\text{proj}} \cap F[Z_0, Z_1]$$

of degree  $D_{X,I,J,\pi}$ , then the number of projective line parameters in that chart is at most  $D_{X,I,J,\pi}$ . More exactly, over a finite field  $F$  of size  $q$ , the number of finite-patch roots is

$$\deg \gcd(Q_{X,I,J,\pi}(1, Z), Z^q - Z),$$

and the point at infinity contributes precisely when  $Q_{X,I,J,\pi}(0, 1) = 0$ . Thus the exact finite-field count is

$$\deg \gcd(Q_{X,I,J,\pi}(1, Z), Z^q - Z) + \mathbf{1}_{Q_{X,I,J,\pi}(0,1)=0}.$$

*Proof.* Every parameter represented by the chart gives a point of the saturated incidence variety, hence is a zero of every element of the elimination ideal, including  $Q_{X,I,J,\pi}$ . A nonzero homogeneous polynomial of degree  $D$  on  $\mathbb{P}^1(F)$  has at most  $D$  projective zeros. On the affine patch  $Z_0 = 1$  the finite parameters are exactly the roots of  $Q(1, Z)$  in  $F$ , which are counted by the gcd with  $Z^q - Z$  after squarefree reduction. The complement of the finite patch is the single point  $[0 : 1]$ , giving the displayed indicator.  $\square$

**Corollary 3.66** (finite and infinity patches). *The finite affine rank-pivot certificate is obtained from Theorem 3.65 by setting  $Z_0 = 1$  and using only noncontainment pivots  $B_h \neq 0$ . The infinity chart is obtained by setting  $Z_0 = 0$ ,  $Z_1 = 1$ , imposing*

$$B(\ell) = 0, \quad A_h(\ell) \neq 0$$

*for some  $h$ , and applying the same saturated rank-pivot test to the constant matrix  $V$ . Consequently the projective sampler has no hidden extra case: every finite or infinite bad point is represented by one of these two patch types.*

*Proof.* On the finite patch  $[1 : Z]$ , the support equation is  $A(\ell) + ZB(\ell) = 0$ . If  $B(\ell) = 0$ , then also  $A(\ell) = 0$ , so the fixed support is contained; hence every finite noncontained point has some  $B_h \neq 0$ . At infinity  $[0 : 1]$ , the support equation is  $B(\ell) = 0$  and noncontainment is exactly  $A(\ell) \neq 0$ . These alternatives exhaust  $\mathbb{P}^1(F)$ .  $\square$

**Curve coefficient-pivot rank charts.** For a finite-parameter degree- $d$  power curve

$$W_\Gamma = f_0 + \Gamma f_1 + \cdots + \Gamma^d f_d,$$

put  $U_i := H_{t,j}(\text{Syn}(f_i))$  and

$$M_A^{(d)}(\Gamma) := \sum_{i=0}^d \Gamma^i U_i, \quad V_i(\ell) := U_i \tilde{\ell}.$$

For row and column sets  $I, J$  of size  $s$ , write  $\Delta_{I,J}^{(d)}(\Gamma)$  for the corresponding  $s \times s$  minor of  $M_A^{(d)}(\Gamma)$ , and let  $\mathcal{M}_{s+1}(M_A^{(d)})$  be the set of all larger minors. For a coefficient noncontainment pivot  $(i, h)$  define

$$\begin{aligned} \hat{\mathcal{J}}_{X,I,J,i,h}^{\text{curve,d}} &:= (I_{\text{split},j} + \langle E_X \rangle + \langle M_A^{(d)}(\Gamma) \tilde{\ell} \rangle \\ &\quad + \langle \Delta : \Delta \in \mathcal{M}_{s+1}(M_A^{(d)}) \rangle) : (\Delta_X \Delta_{I,J}^{(d)}(V_i)_h)^\infty. \end{aligned}$$

**Theorem 3.67** (curve residual rank-pivot eliminant). *The curve-bad parameters in a residual exact-agreement branch are covered by the charts  $\hat{\mathcal{J}}_{X,I,J,i,h}^{\text{curve,d}}$ . If such a chart has a nonzero eliminant*

$$Q_{X,I,J,i,h}(\Gamma) \in \hat{\mathcal{J}}_{X,I,J,i,h}^{\text{curve,d}} \cap F[\Gamma]$$

*of degree  $D_{X,I,J,i,h}$ , then that chart contributes at most*

$$\deg \gcd(Q_{X,I,J,i,h}(\Gamma), \Gamma^q - \Gamma) \leq D_{X,I,J,i,h}$$

*finite parameters over  $F$ . The same bound applies to no-loss curve-CA parameters in the same chart.*

*Proof.* A curve-bad parameter with locator  $\tilde{\ell}_T$  satisfies  $M_A^{(d)}(\gamma)\tilde{\ell}_T = 0$ . Taking  $s$  to be the rank of  $M_A^{(d)}(\gamma)$  gives a nonzero  $s \times s$  minor and vanishing of all larger minors. Noncontainment of the coefficient tuple means that some coordinate  $(V_i(\ell_T))_h$  is nonzero. Thus the point lies on one of the saturated rank-pivot charts and any univariate eliminant for that chart must vanish at  $\gamma$ . The finite-field root count is the gcd with  $\Gamma^q - \Gamma$ .  $\square$

**Extension-line Weil restriction.** The previous charts are intrinsic over the line field. For extension-field samplers one also needs a generated-field ledger: a low-dimensional set over the base field  $B$  can be negligible after division by  $|F|$ , even when it is large as a subset of  $B$ . The following is the certificate form of that accounting.

Let  $F/B$  have degree  $m$ , choose a  $B$ -basis  $\omega_0, \dots, \omega_{m-1}$  of  $F$ , and write a finite affine line parameter as

$$Z = Z_0\omega_0 + \dots + Z_{m-1}\omega_{m-1}, \quad Z_i \in B.$$

Assume  $D \subseteq B$  and write every locator coefficient in the same base field coordinates when necessary. Replacing each  $F$ -valued equation in a rank-pivot chart by its  $m$  coordinate equations over  $B$  gives a Weil-restricted saturated ideal

$$\hat{J}_Y^{\text{WR}} \subseteq B[\ell\text{-coordinates}, Z_0, \dots, Z_{m-1}]$$

for a chart  $Y$ ; noncontainment is split by one nonzero  $B$ -coordinate of the chosen  $F$ -valued pivot and included in the saturation. Let  $\pi_Z$  denote projection to the parameter-coordinate space  $\mathbb{A}_B^m$ .

**Theorem 3.68** (extension-line dimension-degree ledger). *Suppose that for a Weil-restricted residual chart  $Y$  a certificate gives a closed set*

$$Y_Z \supseteq \pi_Z(V(\hat{J}_Y^{\text{WR}})) \subseteq \mathbb{A}_B^m$$

*of degree at most  $\Delta_Y$  and dimension at most  $e_Y$ . Then the number of extension-field affine parameters contributed by this chart is at most*

$$\Delta_Y |B|^{e_Y}.$$

*If  $Y_Z$  is zero-dimensional and radical, this bound may be replaced by the exact number of its  $B$ -rational points. After normalization by the line field, this chart contributes at most*

$$\frac{\Delta_Y |B|^{e_Y}}{|F|} = \Delta_Y |B|^{e_Y - m}$$

*to the MCA or CA error numerator.*

*Proof.* The chosen basis identifies  $F$  with  $B^m$ , so affine line parameters over  $F$  are exactly the  $B$ -rational points of  $\mathbb{A}_B^m$ . Every bad parameter represented by the chart lies in the projection of the Weil-restricted incidence variety and hence in  $Y_Z(B)$ . The standard affine degree point-count bound gives  $|Y_Z(B)| \leq \Delta_Y |B|^{e_Y}$ . The normalized statement divides by  $|F| = |B|^m$ .  $\square$

**Corollary 3.69** (base-rational line inertness as a chart certificate). *If  $f$  and  $g$  are  $B$ -valued, then every finite affine bad slope on every rank-pivot chart lies in  $B$ . Equivalently, in the above coordinates all extension-valued charts with  $(Z_1, \dots, Z_{m-1}) \neq 0$  are empty.*

*Proof.* For a fixed support and locator over  $B$ , the vectors  $A(\ell)$  and  $B(\ell)$  are  $B$ -valued. On a finite noncontained chart some coordinate  $B_h$  is nonzero, and the slope is forced by  $Z = -A_h/B_h \in B$ . Hence no genuinely extension-valued finite parameter can survive the chart equations.  $\square$

**Residual aperiodic certificate theorem.** We can now state the safe-side theorem in the form a scanner should print. Let  $P_{\text{paid},F}$  be the exact squarefree finite-root polynomial already charged to tangent/common-line and quotient-image branches, and let  $G_{\text{reg}\backslash\text{paid},F}^{\text{aff}}$  be the regular residual polynomial of Definition 3.52. For the remaining exact agreement levels, let  $\mathfrak{R}_A$  be a finite list of rank-pivot residual charts, each equipped with either a finite/projective/curve eliminant or a Weil-restricted extension projection certificate.

**Theorem 3.70** (scanner-checkable residual aperiodic ledger). *Assume that the chart families  $\mathfrak{R}_A$ , for  $A \geq a$ , cover every remaining locator after tangent/common-line and quotient-image removals. Suppose each residual chart  $Y$  is certified by one of the following data:*

- (i) *an affine eliminant  $Q_Y(Z)$ , contributing  $\deg \gcd(Q_Y, Z^q - Z)$ ;*
- (ii) *a homogeneous projective eliminant  $Q_Y(Z_0, Z_1)$ , contributing  $\deg \gcd(Q_Y(1, Z), Z^q - Z) + \mathbf{1}_{Q_Y(0,1)=0}$ ;*
- (iii) *a finite curve eliminant  $Q_Y(\Gamma)$ , contributing  $\deg \gcd(Q_Y, \Gamma^q - \Gamma)$ ;*
- (iv) *an extension Weil-restricted projection certificate  $(\Delta_Y, e_Y)$ , contributing  $\Delta_Y |B|^{e_Y}$  parameters.*

*Then the finite affine aperiodic numerator after quotient/tangent removals is bounded by*

$$\deg G_{\text{reg}\backslash\text{paid},F}^{\text{aff}} + \sum_{A \geq a} \sum_{Y \in \mathfrak{R}_A} B_Y,$$

*where  $B_Y$  is the corresponding contribution above. The projective and curve variants are obtained by using the projective or curve contributions for all charts. If exact root sets or exact zero-dimensional projections are printed, the displayed safe sum may be replaced by their union after duplicate coalescing.*

*Proof.* The regular nonsingular overdetermined branch is covered by Corollary 3.53. Every remaining bad parameter is covered by the rank-pivot cover theorem, or by its curve analogue, after applying the declared chart cover. Each chart contribution is an upper bound by Theorems 3.65, 3.67 and 3.68. Summing the chart bounds is a safe union bound; replacing the sum by an exact union is valid when the certificate supplies the corresponding finite root or projection sets.  $\square$

**Definition 3.71** (singular residual certificate). A singular or underdetermined residual aperiodic certificate consists of:

- (i) the paid tangent/common-line and quotient-image root ledgers already removed;
- (ii) the regular rank-drop gcd/lcm ledger and its paid-root removal;
- (iii) for each remaining exact agreement  $A$ , a chart cover of the residual locator set after the paid branches are removed;
- (iv) for every chart, its rank  $s$ , pivot minor  $\Delta_{I,J}$ , noncontainment pivot, and either an affine, projective, or curve eliminant, or a Weil-restricted projection degree/dimension certificate;

- (v) a residual table for every chart on which no eliminant or projection bound is supplied, labelled as quotient, tangent/common-line, base-field exceptional, genuinely extension-valued, or new singular obstruction.

The verifier checks chart coverage, rank-stratum equations, saturations by the pivot and non-containment coordinates, eliminant membership, finite-field root counts, extension projection bounds, and duplicate coalescing when claimed.

*Remark 3.72* (status after the singular-bucket charts). This does not prove that the hard aperiodic residual buckets are small. It does remove one ambiguity: underdetermined and rank-drop singular buckets now have a precise finite certificate language, including finite slopes, projective infinity, finite-parameter curves, and genuinely extension-valued parameters. The next mathematical step is to construct low-degree eliminants, or prove structural classification, for these rank-pivot charts in the near-capacity regimes where  $t_A = A - k$  is small.

**Kernel-compressed rank-pivot eliminants.** The rank-pivot charts above are already finite certificate objects, but the variables are still the full locator coordinates. The next reduction removes the linear kernel equations on a pivot chart by Cramer's rule. It is the algebraic form a scanner should use before attempting Gröbner elimination or resultants: all remaining equations are split-locator equations, chart-removal equations, rank-stratum equations, and noncontainment pivots in the free kernel coordinates and the line parameter.

We write the statement first on the finite affine patch. Fix an exact agreement value  $A$ , put  $j = n - A$  and  $t = A - k$ , and set

$$M_A(Z) := U + ZV, \quad U := H_{t,j}(\text{Syn}(f)), \quad V := H_{t,j}(\text{Syn}(g)).$$

Let  $I \subseteq \{0, \dots, t-1\}$  and  $J \subseteq \{0, \dots, j\}$  have the same cardinality  $s$ , and put

$$\Delta_{I,J}(Z) := \det M_A(Z)_{I,J}.$$

Let  $C_J := \{0, \dots, j\} \setminus J$  and introduce free kernel coordinates  $Y = (Y_c)_{c \in C_J}$ . On the open set  $\Delta_{I,J} \neq 0$ , define the polynomial vector

$$\Lambda_{I,J}^\#(Z, Y) \in F[Z, Y]^{j+1}$$

by

$$(\Lambda_{I,J}^\#)_{C_J} := \Delta_{I,J} Y, \quad (\Lambda_{I,J}^\#)_J := -\text{adj}(M_A(Z)_{I,J}) M_A(Z)_{I,C_J} Y.$$

Thus

$$M_A(Z)_{I,*} \Lambda_{I,J}^\#(Z, Y) = 0.$$

If, in addition, all  $(s+1) \times (s+1)$  minors of  $M_A(Z)$  vanish, then every row of  $M_A(Z)$  lies in the span of the pivot rows, and hence

$$M_A(Z) \Lambda_{I,J}^\#(Z, Y) = 0.$$

To encode splitting without dividing by the leading coefficient, use homogeneous locator coordinates. For  $\lambda = (\lambda_0, \dots, \lambda_j)$  set

$$L_\lambda(X) := \sum_{b=0}^j \lambda_b X^b.$$

Let  $Q(X) = \sum_{r=0}^{n-j} q_r X^r$  be an auxiliary quotient polynomial and let  $I_{\text{hsplit},j}$  be the homogeneous split incidence

$$P_D(X) = L_\lambda(X)Q(X), \quad \lambda_j \neq 0,$$

viewed as the coefficient equations saturated by  $\lambda_j$ . Equivalently, its points with  $\lambda_j \neq 0$  are precisely the nonzero scalar multiples of the monic locators  $\prod_{x \in T} (X - x)$  with  $T \subseteq D$  and  $|T| = j$ . For a monic locator chart  $X = (E_X, \Delta_X)$ , write  $E_X^{\text{hom}}$  and  $\Delta_X^{\text{hom}}$  for the usual homogenizations in the coordinates  $\lambda_0, \dots, \lambda_j$ .

For a finite noncontainment pivot  $B_h \neq 0$ , set

$$\Pi_h^\#(Z, Y) := (V\Lambda_{I,J}^\#(Z, Y))_h.$$

The *kernel-compressed affine rank-pivot ideal* is

$$\begin{aligned} \mathcal{K}_{X,I,J,h}^{\text{aff}} := & \langle \langle \text{coeff}_X(P_D - L_{\Lambda_{I,J}^\#} Q) \rangle + \langle E_X^{\text{hom}}(\Lambda_{I,J}^\#) \rangle \\ & + \langle \Delta : \Delta \in \mathcal{M}_{s+1}(M_A) \rangle \rangle : (\Delta_{I,J} \Delta_X^{\text{hom}}(\Lambda_{I,J}^\#)(\Lambda_{I,J}^\#)_j \Pi_h^\#)^\infty \end{aligned}$$

in the ring  $F[Z, Y, (q_r)]$ . Here  $L_{\Lambda_{I,J}^\#}$  means  $L_\lambda$  after substituting  $\lambda = \Lambda_{I,J}^\#(Z, Y)$ .

**Theorem 3.73** (kernel-compressed affine eliminant certificate). *Every finite affine bad slope represented by the rank-pivot chart  $(X, I, J, h)$  is in the projection of  $V(\mathcal{K}_{X,I,J,h}^{\text{aff}})$  to the  $Z$ -line. Consequently, if*

$$Q_{X,I,J,h}(Z) \in \mathcal{K}_{X,I,J,h}^{\text{aff}} \cap F[Z]$$

*is nonzero, then the chart contributes at most*

$$\deg \gcd(Q_{X,I,J,h}(Z), Z^q - Z)$$

*finite slopes over  $F_q$ .*

*Proof.* Let  $z$  be a bad slope in the chart, and let  $\lambda$  be a nonzero scalar multiple of the corresponding locator vector. Since the pivot minor is nonzero, put  $Y_c = \lambda_c / \Delta_{I,J}(z)$  for  $c \in C_J$ . Cramer's rule and the pivot row equations give

$$\Lambda_{I,J}^\#(z, Y) = \lambda.$$

The locator is split, lies in the chart  $X$ , has nonzero leading coefficient, and has the chosen nonzero noncontainment coordinate. The larger minors vanish on the exact-rank stratum. Thus the point survives the displayed saturation and lies in the compressed incidence. Every element of the elimination ideal must vanish on its  $Z$ -coordinate, and the finite-field count is obtained by intersecting with  $Z^q - Z$ .  $\square$

*Remark 3.74* (why the compression is useful). The original chart has  $j$  locator variables, one parameter variable, and the split equations. The compressed chart has only  $j + 1 - s$  free kernel coordinates before imposing the leading-coordinate and chart opens. In the overdetermined rank-drop case this is often a large saving. In the genuinely underdetermined near-capacity case the saving may be small, but the construction still separates three causes of difficulty: many split locators, a dominant parameter projection, or a high-degree finite eliminant.

The projective and curve variants are identical after replacing the parameter ring. For projective lines, replace  $M_A(Z)$  by  $M_A^{\text{proj}}(Z_0, Z_1)$ , use homogeneous pivot minors, and take homogeneous eliminants in  $F[Z_0, Z_1]$ . On the infinity patch  $[0 : 1]$ , this is just the constant matrix  $V$  with equations  $B(\ell) = 0$  and an open pivot  $A_h \neq 0$ . For a degree- $d$  finite power curve, replace  $M_A(Z)$  by

$$M_A^{(d)}(\Gamma) = \sum_{i=0}^d \Gamma^i H_{t,j}(\text{Syn}(f_i))$$

and use coefficient pivots  $(V_i)_h \neq 0$ .

**Corollary 3.75** (kernel-compressed projective and curve eliminants). *The projective rank-pivot chart obtained from the compressed construction has the same parameter projection as the uncompressed rank-pivot chart on the declared opens. A nonzero homogeneous eliminant  $Q(Z_0, Z_1)$  contributes at most*

$$\deg \gcd(Q(1, Z), Z^q - Z) + \mathbf{1}_{Q(0,1)=0}$$

*projective parameters over  $F_q$ . For a finite degree- $d$  curve chart, a nonzero eliminant  $Q(\Gamma)$  contributes at most*

$$\deg \gcd(Q(\Gamma), \Gamma^q - \Gamma)$$

*parameters.*

*Proof.* The same Cramer parametrization applies to the projective pencil and to the curve pencil. The root-count statements are the finite-field root counts for a homogeneous polynomial on  $\mathbb{P}^1(F_q)$  and a univariate polynomial on  $\mathbb{A}^1(F_q)$ .  $\square$

**Exact residual support-image lcm ledger.** The compressed eliminant is a constructive way to find a small parameter polynomial. There is also an exact, definition-level residual image ledger that works for any finite co-support family, quotient or not. This is the clean object against which scanner-produced eliminants should be audited.

Let  $\mathcal{T}$  be any family of co-supports  $T \subseteq D$  of size  $j = n - A$ . For a line  $f + Zg$ , put

$$A_T := H_{t,j}(\text{Syn}(f))\ell_T, \quad B_T := H_{t,j}(\text{Syn}(g))\ell_T.$$

Define the finite affine fixed-support polynomial

$$g_T^{\text{aff}}(Z) := \begin{cases} \text{monic } \gcd\{(A_T)_h + Z(B_T)_h : 0 \leq h < t\}, & (A_T, B_T) \neq (0, 0), \\ 1, & (A_T, B_T) = (0, 0), \end{cases}$$

with the convention that the gcd of constants with no common root is 1. Thus  $g_T^{\text{aff}}$  has degree either 0 or 1. The contained case  $(A_T, B_T) = (0, 0)$  is assigned the polynomial 1, because it contributes no support-wise bad slope. Set

$$R_{\mathcal{T}}^{\text{aff}}(Z) := \text{rad lcm}_{T \in \mathcal{T}} g_T^{\text{aff}}(Z).$$

For projective lines, define  $g_T^{\text{proj}}(Z_0, Z_1)$  as the monic homogeneous gcd of the linear forms

$$Z_0(A_T)_h + Z_1(B_T)_h, \quad 0 \leq h < t,$$

again setting it equal to 1 in the contained case, and put

$$R_{\mathcal{T}}^{\text{proj}} := \text{rad lcm}_{T \in \mathcal{T}} g_T^{\text{proj}}.$$

For a finite degree- $d$  curve, use the vector polynomials

$$\sum_{i=0}^d \Gamma^i(V_{i,T})_h$$

and define  $R_{\mathcal{T}}^{(d)}(\Gamma)$  by the same radical lcm rule, with the all-zero coefficient tuple treated as contained.

**Theorem 3.76** (exact arbitrary-residual image ledger). *Let  $\mathcal{T}$  be any finite co-support family. The exact finite affine line parameters witnessed by  $\mathcal{T}$  are the roots in  $F$  of  $R_{\mathcal{T}}^{\text{aff}}$ . Hence their number is*

$$\deg \gcd(R_{\mathcal{T}}^{\text{aff}}(Z), Z^q - Z).$$

*The exact projective parameters witnessed by  $\mathcal{T}$  are the projective zeros of  $R_{\mathcal{T}}^{\text{proj}}$ , counted over  $F_q$  by*

$$\deg \gcd(R_{\mathcal{T}}^{\text{proj}}(1, Z), Z^q - Z) + \mathbf{1}_{R_{\mathcal{T}}^{\text{proj}}(0,1)=0}.$$

*For finite degree- $d$  curves, the exact finite parameter count is*

$$\deg \gcd(R_{\mathcal{T}}^{(d)}(\Gamma), \Gamma^q - \Gamma).$$

*The same ledgers upper-bound no-loss CA parameters witnessed by the same co-support family.*

*Proof.* For a fixed co-support  $T$ , the syndrome-locator recurrence says that the line point is explained on  $D \setminus T$  precisely when  $A_T + ZB_T = 0$ . If  $A_T = B_T = 0$ , then both coefficient words are explained on the same support and the fixed support is contained, so it is correctly charged as 1. Otherwise the common roots of the scalar coordinates are exactly the finite bad slopes contributed by  $T$ , and this common-root set is represented by  $g_T^{\text{aff}}$ . Taking the radical lcm over  $T \in \mathcal{T}$  coalesces duplicate slopes. The projective and curve statements are identical, using the homogeneous or degree- $d$  coordinate polynomials. No-loss CA badness is stronger than existence of an explaining support with noncontainment, so the same image set is an upper ledger.  $\square$

**Corollary 3.77** (closed-ball residual exact image after paid-branch removal). *Fix a closed-ball agreement threshold  $a$ . For each exact agreement  $A \in \{a, a+1, \dots, n\}$ , let  $\mathcal{T}_{\text{all},A}$  be the family of co-supports of size  $n - A$ , let  $\mathcal{T}_{\text{paid},A} \subseteq \mathcal{T}_{\text{all},A}$  be a certified paid support family, and put*

$$\mathcal{T}_{\text{res},A} := \mathcal{T}_{\text{all},A} \setminus \mathcal{T}_{\text{paid},A}.$$

*Let  $R_{\text{res},A}^{\text{aff}}$  be the polynomial of Theorem 3.76 for the fixed-size family  $\mathcal{T}_{\text{res},A}$ . Define the closed-ball residual polynomial*

$$R_{\text{res},\geq a}^{\text{aff}}(Z) := \text{rad lcm}_{A=a}^n R_{\text{res},A}^{\text{aff}}(Z), \quad R_{\text{res},\geq a,F}^{\text{aff}}(Z) := \gcd(R_{\text{res},\geq a}^{\text{aff}}(Z), Z^q - Z).$$

*Then the exact finite affine residual numerator after paid-support removal is*

$$\deg R_{\text{res},\geq a,F}^{\text{aff}}.$$



The projective and finite-curve residual numerators are obtained by the same exact-agreement radical-lcm construction, using  $R_{\text{res},A}^{\text{proj}}$  and  $R_{\text{res},A}^{(d)}$  respectively, followed by the finite-field root count of Theorem 3.76. If a scanner also prints an exact paid root polynomial  $P_{\text{paid},F}$  for the paid branch, the total distinct finite affine numerator is

$$\deg \text{rad lcm}(P_{\text{paid},F}, R_{\text{res},\geq a,F}^{\text{aff}}),$$

not the sum of the paid and residual degrees.

*Proof.* Apply Theorem 3.76 separately at each exact agreement size  $A$ , where the co-support size  $j = n - A$  and the Hankel window height  $t = A - k$  are fixed. The radical lcm over  $A$  is exactly union and duplicate coalescing across exact agreement levels. Intersecting with  $Z^q - Z$  counts only  $F$ -rational finite slopes. The final displayed lcm performs the same duplicate coalescing between paid roots and residual roots.  $\square$

**Elimination trichotomy for residual charts.** The exact image lcm exists because split locators are a finite set. A compressed rank-pivot certificate is useful when it finds a much smaller polynomial without enumerating all residual supports. The following trichotomy is the scanner contract.

**Proposition 3.78** (compressed-chart elimination trichotomy). *For a compressed finite affine residual chart over the algebraic closure, exactly one of the following occurs:*

- (i) *the saturated compressed ideal is the unit ideal, and the chart is empty;*
- (ii) *the saturated compressed ideal is proper and the elimination ideal in  $F[Z]$  contains a nonzero polynomial, and its squarefree generator gives a finite parameter envelope containing the exact support-image lcm of that chart;*
- (iii) *the saturated compressed ideal is proper and the elimination ideal in  $F[Z]$  is zero, equivalently the projection of the relaxed chart to the  $Z$ -line is Zariski dense.*

*If the full homogeneous split equations are present, case (iii) can occur only through a positive-dimensional relaxation or through a contained component not removed by the noncontainment saturation; an exact split, noncontained chart has finite parameter image.*

*Proof.* The first three alternatives are the standard elimination theorem for the projection of an affine variety to a line. If the full split equations are present and the leading coordinate is nonzero, the locator coordinate belongs to the finite set of scalar multiples of the finitely many degree- $j$  locators supported on  $D$ . After projectivizing the locator scale, each fixed support contributes at most one finite noncontained slope by Lemma 3.16. Hence the exact noncontained split image is a finite subset of the parameter line. A dense projection can therefore only come from a relaxation of the split condition or from an unremoved contained component on which every parameter is explained but no noncontainment pivot should survive.  $\square$

**Remark 3.79** (compressed-residual status). The singular bucket has now been reduced to two concrete certificate tasks. The first is computational-algebraic: use the kernel-compressed ideals to find small univariate or homogeneous eliminants. The second is enumerative but exact: when compression is not yet sharp enough, print the residual support-image lcm for a finite residual family and coalesce it with the paid quotient/tangent roots. Neither task proves the final aperiodic packing theorem by itself, but both replace the phrase “singular residual bucket” by checkable objects with exact root-count semantics.

### 3.4.5 Fallback and final certificate

The eliminant certificates above are strongest when the projection to the parameter line is finite. A failed eliminant search should not be discarded; it should be split further or recorded as a singular bucket. The following fallback is safe but usually too weak for a final  $2^{-128}$  threshold.

**Proposition 3.80** (dimension-degree fallback). *Let  $Y \subseteq \mathbb{A}_F^N$  be an affine variety of degree at most  $\Delta$  and Krull dimension at most  $e$ . Then*

$$|Y(F)| \leq \Delta |F|^e.$$

*Consequently, if a saturated Hankel incidence chart for line or curve parameters has degree  $\Delta_X$  and dimension  $e_X$ , then the number of bad parameters in that chart is at most  $\Delta_X |F|^{e_X}$ .*

*Proof.* This is the standard affine degree point-count bound, proved by induction on dimension using generic affine hyperplane sections. The parameter image is no larger than the incidence set.  $\square$

For a threshold statement, use finitely many charts and agreement levels. Let  $\mathfrak{X}_A$  be a chart cover of the declared aperiodic branch at exact agreement  $A$ . If each finite affine pivot chart has a univariate eliminant, set

$$B_{\text{ap}}^{\text{aff}, \text{elim}}(a) := \sum_{A=a}^n \sum_{X \in \mathfrak{X}_A} \sum_{h=0}^{t_A-1} \deg Q_{X,h}.$$

For projective slopes add the certified infinity contribution, and for degree- $d$  curves replace the inner sum by the coefficient-pivot sum  $\sum_{i=0}^d \sum_h \deg Q_{X,i,h}$ . Exact root-union tables may replace these safe sums whenever duplicate parameters across charts are certified.

**Definition 3.81** (aperiodic Hankel-packing certificate). An aperiodic Hankel-packing certificate for a row, a line or curve family, and an agreement threshold  $a$  consists of:

- (i) the tangent/common-code-line and quotient-image ledgers that have already been removed;
- (ii) for each exact agreement  $A \in \{a, \dots, n\}$ , either a regular minor certificate  $\Delta_A$  or a declaration that the exact-agreement bucket is Hankel-singular;
- (iii) for every singular or near-capacity bucket, a finite constructible locator chart cover  $\mathfrak{X}_A$  as in Definition 3.58;
- (iv) for affine lines, a list of pivot eliminants  $Q_{X,h}(Z)$ , or a failed-pivot record explaining which residual locus remains unsplit;
- (v) for projective slopes, the finite-patch eliminants plus the infinity-chart empty/nonempty certificate;
- (vi) for curves, coefficient-pivot eliminants  $Q_{X,i,h}(\Gamma)$ ;
- (vii) a root-union table or a safe degree-sum table producing the declared aperiodic numerator  $B_{\text{ap}}(a)$ ;
- (viii) if any chart uses the dimension-degree fallback, its degree, dimension, and an explicit statement that the resulting bound is only a fallback, not a sharp threshold certificate.

The verifier checks locator divisibility by  $P_D$ , chart coverage of the declared aperiodic complement, nonzero pivot tests, ideal membership of eliminants after saturation, and final root-count arithmetic.

*Remark 3.82* (contributor workflow). For each row and exact agreement  $A$ , contributors should try the following in order. First test the overdetermined regular minors. If a nonzero minor is found, record its root set and remove roots already paid by tangent or quotient ledgers. If all maximal minors vanish identically, move the instance to the singular bucket. On the singular or near-capacity bucket, build locator charts, then split by affine pivots  $B_h \neq 0$ ; for projective samplers also test the infinity chart  $B = 0, A \neq 0$ ; for curves split by coefficient pivots  $(V_i)_h \neq 0$ . A chart is complete only when it has either a nonzero eliminant, a certified empty locus, or a named residual obstruction. Residual obstructions should be labelled as quotient-periodic, tangent/common-code-line, subfield/extension exceptional, or candidate new obstruction.

*Remark 3.83* (status). Theorems 3.46, 3.59 and 3.63 and Corollary 3.62 are proved certificate theorems. They do not prove that all rows admit small aperiodic eliminants. They make the remaining MCA safe-side task concrete: after tangent and quotient branches are removed, construct small eliminants for the pivot charts, or identify the singular chart that prevents such an eliminant.

**Lemma 3.84** (interleaving transfer). *For every linear code  $C \subseteq \mathbb{F}^n$ , every  $s \geq 1$  and every  $\delta \in (0, 1)$ :*

$$\varepsilon_{\text{ca}}(C^{\equiv s}, \delta) \geq \varepsilon_{\text{ca}}(C, \delta), \quad \varepsilon_{\text{mca}}(C^{\equiv s}, \delta) \geq \varepsilon_{\text{mca}}(C, \delta).$$

*Proof.* For  $h \in \mathbb{F}^n$  write  $h^\Delta := (h, \dots, h) \in (\mathbb{F}^n)^s$ . For any codeword tuple  $\mathbf{c} = (c^{(1)}, \dots, c^{(s)}) \in C^{\equiv s}$ , the columns where  $h^\Delta$  and  $\mathbf{c}$  agree are  $\bigcap_i \{j : c_j^{(i)} = h_j\} \subseteq \{j : c_j^{(1)} = h_j\}$ , so  $\Delta_s(h^\Delta, C^{\equiv s}) \geq \Delta(h, C)$ ; taking  $c^{(1)} = \dots = c^{(s)}$  gives equality:  $\Delta_s(h^\Delta, C^{\equiv s}) = \Delta(h, C)$ .

Fix  $(f_1, f_2)$  attaining  $\varepsilon_{\text{ca}}(C, \delta)$  (resp.  $\varepsilon_{\text{mca}}$ ) and consider the pair  $(f_1^\Delta, f_2^\Delta)$  for  $C^{\equiv s}$ , with the slope  $\gamma$  acting row-wise, so  $f_1^\Delta + \gamma f_2^\Delta = (f_1 + \gamma f_2)^\Delta$ . By the displayed equality,  $\Delta_s((f_1 + \gamma f_2)^\Delta, C^{\equiv s}) \leq \delta$  iff  $\Delta(f_1 + \gamma f_2, C) \leq \delta$ . For the correlated condition, the pair  $(f_1^\Delta, f_2^\Delta)$  viewed as a  $2s$ -row interleaved word has rows  $f_1$  ( $s$  times) and  $f_2$  ( $s$  times); by the same column argument its distance to  $(C^{\equiv s})^{\equiv 2} = C^{\equiv 2s}$  equals  $\Delta_2((f_1, f_2), C^{\equiv 2})$ . For the MCA condition, a common set  $S$  explains  $f_1^\Delta$  and  $f_2^\Delta$  iff it explains  $f_1$  and  $f_2$ . Hence  $\gamma$  is CA-bad (resp. MCA-bad) for  $(f_1^\Delta, f_2^\Delta)$  iff it is so for  $(f_1, f_2)$ , and the maxima over pairs can only grow.  $\square$

## 4 The main theorem

**Theorem 4.1** (universal cap). *Let  $B \subseteq \mathbb{F}$  be finite fields, let  $q = |B|$ , let  $D \subseteq B^\times$  be a multiplicative coset of order  $n$ , and let  $k$  have  $\rho = k/n \in (0, 1)$ ; set  $C := \text{RS}[\mathbb{F}, D, k]$ . Let  $N \mid n$  satisfy  $a := n/N \mid k$  and  $(1 - \rho)N \geq 3$ . If*

$$\binom{N}{\rho N + 2} \geq |B| \left( \frac{q}{k} + 1 \right), \tag{1}$$

*then, writing*

$$\delta_N := 1 - \rho - \frac{2}{N},$$

*we have*

$$\varepsilon_{\text{ca}}(C, \delta) > \frac{1}{2k} \left( 1 - \frac{n}{q} \right)$$

for every integer-admissible radius

$$\delta_N \leq \delta \leq 1 - \rho - \frac{1}{n}.$$

In particular,

$$\varepsilon_{\text{mca}}(C, \delta) > \frac{1}{2k} \left(1 - \frac{n}{q}\right)$$

for every

$$\delta_N \leq \delta < 1 - \rho.$$

Consequently,

$$\delta_C^*(\varepsilon^*) \leq 1 - \rho - \frac{2}{N}$$

for every

$$\varepsilon^* \leq \frac{1}{2k} \left(1 - \frac{n}{q}\right).$$

*Proof.* Note first that  $\rho N = k/a \in \mathbb{Z}$ ,  $\ell_2 = \rho N + 2 \leq N - 1$  by  $(1 - \rho)N \geq 3$ , and  $\delta_N = 1 - \rho - 2/N > 0$  for the same reason. Also  $k + 1 \leq n$ .

Fix

$$\delta \in [\delta_N, 1 - \rho - \frac{1}{n}].$$

Then

$$\lfloor \delta n \rfloor \leq n - k - 1,$$

so Theorem 2.6 is applicable.

Suppose toward a contradiction that

$$\varepsilon_{\text{ca}}(C, \delta) \leq \frac{1}{2k} \left(1 - \frac{n}{q}\right).$$

Apply Theorem 2.6 with  $\eta = \frac{1}{2}$ . Its hypothesis holds, so

$$\Lambda(C^+, \delta) \leq \left\lceil 2q \cdot \varepsilon_{\text{ca}}(C, \delta) \right\rceil \leq \left\lceil \frac{q - n}{k} \right\rceil < \frac{q - n}{k} + 1 \leq \frac{q}{k} + 1.$$

On the other hand, by Lemma 3.1(ii) and monotonicity of lists in the radius, since  $\delta \geq \delta_N$ ,

$$\Lambda(C^+, \delta) \geq |\Lambda(C^+, \delta_N, u_z)| \geq \binom{N}{\ell_2} / |B| \geq \frac{q}{k} + 1$$

by (1). This is a contradiction. Hence

$$\varepsilon_{\text{ca}}(C, \delta) > \frac{1}{2k} \left(1 - \frac{n}{q}\right)$$

for every  $\delta \in [\delta_N, 1 - \rho - 1/n]$ .

In particular, the above holds at  $\delta = \delta_N$ . By Fact 2.4,

$$\varepsilon_{\text{mca}}(C, \delta_N) \geq \varepsilon_{\text{ca}}(C, \delta_N) > \frac{1}{2k} \left(1 - \frac{n}{q}\right).$$

By support-wise MCA monotonicity, Lemma 2.5, the same lower bound for  $\varepsilon_{\text{mca}}(C, \delta)$  holds for every

$$\delta \in [\delta_N, 1 - \rho).$$

Thus no sub-capacity radius

$$\delta \geq 1 - \rho - \frac{2}{N}$$

is  $\varepsilon^*$ -admissible whenever

$$\varepsilon^* \leq \frac{1}{2k} \left(1 - \frac{n}{q}\right),$$

which proves the claimed bound on  $\delta_C^*(\varepsilon^*)$ .  $\square$

*Remark 4.2* (constants). Choosing  $\eta = 1 - \theta$  in Theorem 2.6 improves the conclusion to  $\varepsilon_{\text{ca}} > \frac{1-\theta}{k}(1 - \frac{n}{q})$  at the price of strengthening (1) to  $\binom{N}{\rho N+2} \geq |B|(\frac{q}{\theta k} + 1)$ ; given the hundreds of bits of slack in (1) throughout the challenge envelope (Table 1), the error constant is effectively  $(1 - o(1))/k$ . We fix  $\eta = \frac{1}{2}$  for cleanliness.

*Remark 4.3* (scope). No smoothness is assumed beyond the existence of a single divisor  $N \mid n$  with  $a = n/N \mid k$  and (1). The theorem applies verbatim to 2-adic, mixed-radix, or any other FFT-friendly multiplicative domain, and — via the subfield pigeonhole — to any extension field  $\mathbb{F} \supseteq B$  with  $D \subseteq B$ .

The deep-point conversion is an integer-radius theorem with the condition  $\lfloor \delta n \rfloor \leq n - k - 1$ . Therefore the no-loss CA lower bound is stated for

$$\delta \leq 1 - \rho - \frac{1}{n}.$$

The MCA threshold cap is stronger in the range needed for the Proximity Prize: once the support-wise MCA lower bound is proved at

$$\delta_N = 1 - \rho - \frac{2}{N},$$

monotonicity of support-wise MCA extends it to every larger  $\delta < 1 - \rho$ .

## 5 Consequences for the grand MCA challenge

**Corollary 5.1** (universal field-size cap for the challenge envelope). *Let  $\rho \in \{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}\}$  and set*

$$N_\rho := 1024 \quad (\rho \in \{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}\}), \quad N_{1/16} := 2048.$$

*Let  $\mathbb{F}$  be any finite field with  $q < 2^{256}$ , let  $B \subseteq \mathbb{F}$  be any subfield, let  $D \subseteq B^\times$  be a multiplicative coset of order  $n$  with  $N_\rho \mid n$ , and let  $k = \rho n \leq 2^{40}$ . Then  $C = \text{RS}[\mathbb{F}, D, k]$  satisfies*

$$\varepsilon_{\text{mca}}(C, \delta) > \frac{1}{2k} \left(1 - \frac{n}{q}\right) \geq 2^{-86} \gg 2^{-128}$$

*for every*

$$\delta \in \left[1 - \rho - \frac{2}{N_\rho}, 1 - \rho\right).$$

Moreover, the same lower bound holds for  $\varepsilon_{\text{ca}}(C, \delta)$  throughout the integer-admissible subinterval

$$\delta \in \left[1 - \rho - \frac{2}{N_\rho}, 1 - \rho - \frac{1}{n}\right].$$

If  $q \geq 2n$ , the lower bound improves to  $\geq 2^{-42}$ . In particular,

$$\delta_C^*(2^{-128}) \leq 1 - \rho - 2^{-9} \quad (\rho \in \{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}\}), \quad \delta_C^*(2^{-128}) \leq 1 - \rho - 2^{-10} \quad (\rho = \frac{1}{16}).$$

*Proof.* We verify the hypotheses of Theorem 4.1 with  $N = N_\rho$ . Divisibility:  $a = n/N_\rho$  divides  $k$  because  $k/a = \rho N_\rho \in \mathbb{Z}$  (equal to 512, 256, 128, 128 respectively);  $(1 - \rho)N_\rho \geq 128 \geq 3$ . For (1), the entropy bound  $\binom{N}{\ell} \geq 2^{NH_2(\ell/N)}/(N+1)$  together with the conservative evaluations of  $H_2$  recorded in Table 1 gives  $\log_2 \binom{N_\rho}{\rho N_\rho + 2} \geq 552$  in all four cases, while

$$|B|\left(\frac{q}{k} + 1\right) \leq q\left(\frac{q}{k} + 1\right) \leq q^2 + q < 2^{512} + 2^{256} < 2^{513} \leq 2^{552},$$

so (1) holds with at least 39 bits to spare (and over 500 bits at rate  $\frac{1}{2}$ ). Since  $D \subseteq \mathbb{F}^\times$ , we have  $n \leq q - 1$ ; for fixed  $n$ , the function  $1 - n/q$  is minimized at  $q = n + 1$ , hence  $1 - n/q \geq 1/(n + 1)$ . Also  $n = k/\rho \leq 16k \leq 2^{44}$ . Therefore

$$\frac{1}{2k}\left(1 - \frac{n}{q}\right) \geq \frac{1}{2k(n + 1)} > 2^{-86}.$$

If  $q \geq 2n$  then  $1 - n/q \geq 1/2$ , giving the sharper  $1/(4k) \geq 2^{-42}$ . The gap is  $2/N_\rho = 2^{-9}$  for  $\rho \in \{1/2, 1/4, 1/8\}$  and  $2/N_\rho = 2^{-10}$  for  $\rho = 1/16$ . Theorem 4.1 gives the MCA lower bound throughout the stated sub-capacity interval, and gives the CA lower bound on the integer-admissible subinterval ending at  $1 - \rho - 1/n$ .  $\square$

| $\rho$ | $N_\rho$ | $\ell_2 = \rho N_\rho + 2$ | $H_2(\ell_2/N_\rho) \geq$ | $\log_2 \binom{N_\rho}{\ell_2} \geq$ | needed | gap $2/N_\rho$ |
|--------|----------|----------------------------|---------------------------|--------------------------------------|--------|----------------|
| 1/2    | 1024     | 514                        | 0.99998                   | 1013                                 | 513    | $2^{-9}$       |
| 1/4    | 1024     | 258                        | 0.8143                    | 823                                  | 513    | $2^{-9}$       |
| 1/8    | 1024     | 130                        | 0.5490                    | 552                                  | 513    | $2^{-9}$       |
| 1/16   | 2048     | 130                        | 0.34105                   | 687                                  | 513    | $2^{-10}$      |

Table 1: Numerics for Corollary 5.1:  $\log_2 \binom{N}{\ell} \geq NH_2(\ell/N) - \log_2(N + 1)$ , rounded down. The “needed” column is the universal upper bound  $\log_2(|B|(q/k + 1)) < 513$  valid throughout the envelope  $q < 2^{256}$ ,  $k \geq 1$ .

*Remark 5.2* (hypothesis  $N_\rho \mid n$ ). For the 2-adic domains of the challenge this holds whenever  $n \geq 2^{10}$  (resp.  $2^{11}$ ), i.e. for all but very small instances. For mixed-radix smooth  $n$ , replace  $N_\rho$  by any divisor  $N \mid n$  of comparable size with  $\rho N \in \mathbb{Z}$ ; the entropy count in Table 1 is insensitive to the exact value of  $N$  at fixed  $\log_2 N$ . The divisibility-free fiber lemma removes the integrality requirement  $\rho N \in \mathbb{Z}$  altogether: Lemma 7.2, Corollary 7.6 caps every mixed-radix row possessing *any* divisor in the printed dyadic windows.

*Remark 5.3* (two-tier answer to the grand challenge). Combining Corollary 5.1 with the grand-challenge cap of [Cho26b, Prop. “Grand-challenge cap”] (gap  $\frac{1}{64}$ , prime fields up to  $\approx 2^{150}$ ,

via value-set coverage) yields the current state of the negative side for smooth multiplicative domains:

$$\delta_C^*(2^{-128}) \leq \begin{cases} 1 - \rho - \frac{1}{64}, & \mathbb{F} \text{ prime, } q \lesssim 2^{150}, \\ 1 - \rho - 2^{-9}, & \rho \in \{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}\}, \text{ every } \mathbb{F}, q < 2^{256}, \\ 1 - \rho - 2^{-10}, & \rho = \frac{1}{16}, \text{ every } \mathbb{F}, q < 2^{256}. \end{cases}$$

Together with the positive results collected in [ABF26, Table 1], this pins the grand MCA threshold between the Johnson region and the explicit capacity-edge caps above, throughout the stated challenge envelope. The second and third tiers supersede the conditional cap of [Cho26b, Thm. “Universal cap”] ( $\delta^* < 1 - \rho - 2^{-11}$  under  $|\mathbb{F}| \geq 2n$ ), which was the previous state in this band. Circle rows — line rounds and bivariate circle codes — now carry an analogous negative band; see Corollary 7.12 and Remark 7.15.

**Corollary 5.4** (deployed parameters: KoalaBear sextic). *Let  $B = \mathbb{F}_p$  with  $p = 2^{31} - 2^{24} + 1$ , let  $\mathbb{F} = \mathbb{F}_{p^6}$  (so  $q = p^6 \approx 2^{185.93}$ ), let  $D \subseteq B^\times$  be the subgroup of order  $n = 2^{21}$ , and let  $k = 2^{20}$ ,  $\rho = \frac{1}{2}$ , as in [ABF26, §6.3]. Then*

$$\varepsilon_{\text{mca}}(\text{RS}[\mathbb{F}_{p^6}, D, 2^{20}], \delta) > (1 - 2^{-164}) 2^{-21} > 2^{-22}$$

for every

$$\delta \in \left[ \frac{1}{2} - 2^{-7}, \frac{1}{2} \right).$$

Moreover, the same lower bound holds for  $\varepsilon_{\text{ca}}$  throughout the integer-admissible subinterval

$$\delta \in \left[ \frac{1}{2} - 2^{-7}, \frac{1}{2} - 2^{-21} \right).$$

*Proof.* Apply Theorem 4.1 with  $N = 256$ ,  $a = 2^{13} \mid k$ ,  $(1 - \rho)N = 128 \geq 3$ ,  $\text{gap } 2/N = 2^{-7}$ . Hypothesis (1):  $\log_2 \binom{256}{130} \geq 256 \cdot H_2(130/256) - \log_2 257 \geq 256 \cdot 0.99982 - 8.01 > 247$ , while  $|B|(q/k + 1) < 2^{31}(2^{166} + 1) < 2^{198}$ . The error bound is  $\frac{1}{2k}(1 - n/q)$  with  $n/q = 2^{21}/p^6 < 2^{-164}$ . Theorem 4.1 gives the stated MCA bound on  $[\frac{1}{2} - 2^{-7}, \frac{1}{2})$  and the stated CA bound on  $[\frac{1}{2} - 2^{-7}, \frac{1}{2} - 2^{-21})$ .  $\square$

**Corollary 5.5** (all interleaved rows of [ABF26, Tables 2–3]). *In the setting of Corollary 5.4, for each  $s = 2^j$  with  $0 \leq j \leq 12$  let  $C_s := \text{RS}[\mathbb{F}_{p^6}, D_s, k_s]$  with  $|D_s| = n_s = 2^{21}/s$  and  $k_s = 2^{20}/s$  (rate  $\frac{1}{2}$ ,  $s n_s = 2^{21}$  as deployed). Then for every  $\delta \in [\frac{1}{2} - 2^{-7}, \frac{1}{2})$ ,*

$$\varepsilon_{\text{mca}}(C_s^{\equiv s}, \delta) > \frac{s}{2^{21}}(1 - 2^{-164}) \geq 2^{-21}(1 - 2^{-164}).$$

Moreover, the same lower bound holds for  $\varepsilon_{\text{ca}}(C_s^{\equiv s}, \delta)$  on the integer-admissible subinterval

$$\delta \in \left[ \frac{1}{2} - 2^{-7}, \frac{1}{2} - \frac{1}{n_s} \right).$$

Every row of the deployed parameter family therefore fails the  $2^{-128}$  target at gap  $2^{-7}$  below capacity.

*Proof.* For each  $s \leq 2^{12}$ :  $256 \mid n_s$  (as  $n_s \geq 2^9$ ),  $a_s = n_s/256 = 2^{13}/s$  divides  $k_s = 2^{20}/s$ , and  $(1 - \rho) \cdot 256 = 128 \geq 3$ . Hypothesis (1) for  $C_s$ :  $|B|(q/k_s + 1) < 2^{31}(2^{166}s + 1) < 2^{198}s \leq 2^{210} \leq 2^{247} \leq \binom{256}{130}$ . Theorem 4.1 gives the MCA lower bound for  $C_s$  throughout  $[\frac{1}{2} - 2^{-7}, \frac{1}{2})$ , and the CA lower bound throughout  $[\frac{1}{2} - 2^{-7}, \frac{1}{2} - 1/n_s]$ . Lemma 3.84 transfers the corresponding lower bounds to the  $s$ -interleaved code.  $\square$

**Proposition 5.6** (independent slacked variant via Theorem 2.16). *In the setting of Theorem 4.1, assume instead  $(1 - \rho)N \geq 2$ ,  $n \geq 3N$ , and*

$$\binom{N}{\rho N + 1} \geq |B| \cdot q.$$

*Then*

$$\varepsilon_{\text{ca}}\left(C, 1 - \rho - \frac{1}{N} + \frac{2}{n}, 1 - \rho - \frac{1}{n}\right) \geq \frac{1}{2n}.$$

*In particular, with  $N = N_\rho$  as in Corollary 5.1 and  $n \geq 3N_\rho$ , every challenge-envelope instantiation satisfies this slacked CA lower bound, and  $\frac{1}{2n} \geq 2^{-45} \gg 2^{-128}$ .*

*Proof.* Set  $\delta_0 := 1 - \rho - \frac{1}{N}$ . By  $(1 - \rho)N \geq 2$ ,  $\delta_0 \in (0, 1 - \rho)$ . By  $n \geq 3N$ , we also have  $\delta_0 + 2/n \leq 1 - \rho - 1/n$ , so the parameters in Theorem 2.16 are admissible. By Lemma 3.1(i), some  $u_z$  carries at least  $\binom{N}{\rho N + 1}/|B| \geq q$  codewords of  $C$  within radius  $\delta_0$ , so  $\Lambda(C, \delta_0) \geq q$ . The contrapositive of Theorem 2.16 gives

$$\varepsilon_{\text{ca}}\left(C, \delta_0 + \frac{2}{n}, 1 - \rho - \frac{1}{n}\right) \geq \frac{1}{2n}.$$

For the numerical claim, the worst case of  $\log_2 \binom{N_\rho}{\rho N_\rho + 1}$  over the four rates is at  $\rho = 1/8$ , where the entropy bound used in Table 1 certifies  $\log_2 \binom{1024}{129} \geq 1024 H_2(129/1024) - \log_2 1025 \geq 549$  (the exact value is  $\approx 554.7$ ); hence  $\binom{N_\rho}{\rho N_\rho + 1} \geq 2^{549} > 2^{512} > |B|q$  throughout  $q < 2^{256}$ . Finally  $n = k/\rho \leq 16k \leq 2^{44}$ , so  $1/(2n) \geq 2^{-45}$ .  $\square$

*Remark 5.7.* Proposition 5.6 is logically independent of Theorem 2.6 and yields a weaker error ( $\frac{1}{2n}$  versus  $\frac{1}{2k}$ ) in the *maximally slacked* form of correlated agreement — the internal radius sits at capacity minus  $\frac{1}{n}$  — which is the weakest failure mode one can certify. That even this fails by 83 orders of magnitude (in bits) above the  $2^{-128}$  target gives a separate fallback route through the imported BCHKS/ABF theorem.

## 6 Subfield confinement and the shape of certifying lines

The deployed instantiations run the protocol over an extension  $\mathbb{F}/B$  while keeping the domain in the base:  $D \subseteq B$ . The subfield-confinement theorem of [Cho26b, Thm. “Subfield confinement”] pins down where bad slopes can live for lines whose words are base-rational: all support-wise MCA-bad slopes lie in  $B$ . The following lemma refines this by tracking two radii, so that it also covers the proximity-loss form of correlated agreement (Definition 2.1); part (b) is the statement of [Cho26b], reproved here for completeness in the present notation.

**Lemma 6.1** (subfield confinement; refines [Cho26b]). *Let  $B \subseteq \mathbb{F}$  be finite fields,  $D \subseteq B^\times$ ,  $C = \text{RS}[\mathbb{F}, D, k]$ , and let  $f, g \in B^n \subseteq \mathbb{F}^n$  be  $B$ -valued words. Then:*

- (a) *for any  $0 \leq \delta_{\text{fld}} \leq \delta_{\text{int}} < 1$ , every CA-bad slope for  $(f, g)$  at radii  $(\delta_{\text{fld}}, \delta_{\text{int}})$  lies in  $B$ ;*
- (b) *for any  $\delta \in (0, 1)$ , every MCA-bad slope for  $(f, g)$  at radius  $\delta$  lies in  $B$ .*

*Consequently the bad-slope set of any  $B$ -valued pair has density at most  $|B|/q$  in  $\mathbb{F}$ . The same holds for pairs  $(\lambda f, \lambda g)$  with  $\lambda \in \mathbb{F}^\times$  and  $f, g$   $B$ -valued, by  $\mathbb{F}$ -linearity of  $C$ .*



*Proof.* Fix a  $B$ -basis  $1 = \beta_0, \beta_1, \dots, \beta_{m-1}$  of  $\mathbb{F}$ . Every word  $w \in \mathbb{F}^n$  decomposes uniquely as  $w = \sum_i \beta_i w_i$  with  $w_i \in B^n$ , and every codeword  $c \in \text{RS}[\mathbb{F}, D, k]$  decomposes as  $c = \sum_i \beta_i c_i$  with  $c_i \in \text{RS}[B, D, k]$ : write the coefficient vector of the underlying polynomial in the basis and evaluate at  $D \subseteq B$ .

Let  $z \in \mathbb{F} \setminus B$ , say  $z = \sum_i z_i \beta_i$  with  $z_{i^*} \neq 0$  for some  $i^* \geq 1$ . We first record the support-wise consequence. Suppose that  $c \in C$  agrees with  $f + zg$  on some set  $S \subseteq D$ . Since  $f_j, g_j \in B$ , the basis components of  $(f + zg)_j$  are  $f_j + z_0 g_j$  (component 0) and  $z_i g_j$  (components  $i \geq 1$ ). Agreement on  $S$  holds componentwise, so on  $S$ :

$$f + z_0 g = c_0, \quad z_{i^*} g = c_{i^*}.$$

Hence  $g$  agrees on  $S$  with  $z_{i^*}^{-1} c_{i^*} \in C$ , and then  $f = (f + z_0 g) - z_0 g$  agrees on  $S$  with  $c_0 - z_0 z_{i^*}^{-1} c_{i^*} \in C$ . Thus every support on which the line point  $f + zg$  is explained by  $C$  also explains the pair  $(f, g)$  on that same support.

For CA, if  $\Delta(f + zg, C) \leq \delta_{\text{fld}}$ , choose such an  $S$  with  $|S| \geq (1 - \delta_{\text{fld}})n$ ; the preceding paragraph gives  $\Delta_2((f, g), C^{\equiv 2}) \leq \delta_{\text{fld}} \leq \delta_{\text{int}}$ , so  $z$  is not CA-bad. For MCA, the preceding paragraph rules out every possible bad support  $S$  in the support-wise definition. Hence  $z$  is not MCA-bad either.  $\square$

**Corollary 6.2** (certifying lines over extensions are genuinely  $\mathbb{F}$ -valued). *At the parameters of Corollary 5.4,  $|B|/q = p^{-5} < 2^{-154}$ . Hence any pair  $(f, g)$  whose CA- or MCA-bad-slope density at some radius*

$$\delta \in \left[ \frac{1}{2} - 2^{-7}, \frac{1}{2} \right)$$

*exceeds  $2^{-154}$  contains a word that is not  $B$ -valued, even after removing a common scalar factor. In particular, any pair attaining the MCA lower bound  $> 2^{-22}$  of Corollary 5.4 is genuinely  $\mathbb{F}$ -valued. On the integer-admissible subinterval*

$$\delta \in \left[ \frac{1}{2} - 2^{-7}, \frac{1}{2} - 2^{-21} \right),$$

*the same statement applies to pairs attaining the CA lower bound. All explicit failure witnesses currently known on these domains (the locator lines of [Cho26a, Cho26b] and the fiber words of Lemma 3.1) are  $B$ -rational and are therefore inert here: the failure certified by Corollary 5.4 is fiber-borne and, at present, non-constructive.*

*Proof.* Immediate from Lemma 6.1 and Corollary 5.4: a  $B$ -valued pair, up to common scalar, has CA- and MCA-bad-slope density at most  $|B|/q = p^{-5} < 2^{-154} < 2^{-22}$ . Thus any pair witnessing the deployed lower bound cannot be of this form.  $\square$

**Remark 6.3.** Lemma 6.1 and Corollary 6.2 sharpen the division of labor between the two grand challenges of [ABF26] over deployed extensions: the *list* side does not deflate  $(\text{RS}[B, D, k] \subseteq \text{RS}[\mathbb{F}, D, k])$ , so all base-field list lower bounds, including Lemma 3.1, persist verbatim), while every sub- $2^{-128}$  MCA attack by explicit base-rational witnesses is killed by confinement — and yet MCA still fails at  $2^{-22}$ . The failure can also be made constructive once a large list is supplied: Corollaries 2.12 and 2.13 allow one to choose a simple pole in  $\mathbb{F} \setminus B$  and print its value-bucket table. The deflation and decoupling corollaries of [Cho26b] record the same division at the no-loss radius, using the same corrected rounding  $p^{-5} = 2^{-154.94\dots} < 2^{-154}$  as in Corollary 6.2. Exhibiting deployed-size bucket tables for such lines is Open Problem 12.1.

## 7 Beyond multiplicative smoothness: map-smooth domains, Chebyshev fibers, and the circle-code cap

The fiber lemma Lemma 3.1 uses exactly two features of the power map  $x \mapsto x^a$  on a multiplicative coset: its fibers inside the domain are complete and of uniform size  $a$ , and the locators  $\prod_{b \in A} (X^a - b)$  are *lacunary* — expanded in powers of  $X^a$ , their two top terms occupy degrees  $al$  and  $a(\ell - 1)$ , leaving everything from degree  $a(\ell - 2)$  downward to the listed codeword. This section isolates exactly these two features. Lemma 7.2 proves the fiber lemma for an arbitrary polynomial folding map  $\varphi$  whose fibers inside the domain are complete and uniform, and simultaneously removes the divisibility hypothesis  $a \mid k$  by rounding the locator length; Theorem 7.4 is the corresponding universal cap, containing Theorem 4.1 as the special case  $\varphi = X^a$  with  $a \mid k$ . We then verify both features for the deployed isogeny-smooth family, the circle-group domains of Circle STARKs over  $p = 2^{31} - 1$  [HLP24]: twin-coset  $x$ -coordinate domains carry exact Chebyshev fibers at every dyadic scale (Lemma 7.8), and over any challenge field containing  $i$  the bivariate circle code is diagonally equivalent to a Reed–Solomon code on a torus twin coset, an equivalence preserving lists and both correlated-agreement errors exactly (Lemmas 7.10 and 7.11). The resulting universal caps (Corollary 7.12) and deployed instantiations (Corollary 7.13) settle the circle-group case of the isogeny-smooth problem posed below; what remains of that problem is isolated, with a precise obstruction, in Remark 7.16 and restated as Open Problem 12.4.

### 7.1 Map-smooth domains and a divisibility-free fiber lemma

**Definition 7.1** (map-smooth domain). Let  $B \subseteq \mathbb{F}$  be finite fields, let  $\varphi \in B[X]$  have degree  $a \geq 1$ , and let  $D \subseteq B$  be a finite set of size  $n$ . The set  $D$  is  $(\varphi, a)$ -smooth if every nonempty fiber  $\varphi^{-1}(w) \cap D$ ,  $w \in \varphi(D)$ , has exactly  $a$  elements. In that case  $Q := \varphi(D) \subseteq B$  has size  $N := n/a$ , and  $D$  is the disjoint union of the  $N$  fibers  $S_b := \{x \in D : \varphi(x) = b\}$ ,  $b \in Q$ .

Two remarks on scope. First, a multiplicative coset  $D \subseteq B^\times$  of order  $n$  is  $(X^a, a)$ -smooth for every  $a \mid n$ : the fibers are the cosets of the order- $a$  subgroup contained in  $D$ 's underlying group. Second, nothing is assumed about the roots of  $\varphi - w$  outside  $D$ : the definition constrains fibers inside  $D$  only, so separability or squarefreeness of  $\varphi - w$  plays no role below.

**Lemma 7.2** (map-smooth fiber lemma, divisibility-free). *Let  $B \subseteq \mathbb{F}$  and let  $D \subseteq B$  be a  $(\varphi, a)$ -smooth domain of size  $n$ , with  $Q = \varphi(D)$  and  $N = n/a$  as above. Let  $1 \leq k < n$  and  $\rho := k/n$ . Put*

$$\ell_1 := \left\lfloor \frac{k-1}{a} \right\rfloor + 2, \quad \ell_2 := \left\lfloor \frac{k}{a} \right\rfloor + 2, \quad A_1 := a\ell_1, \quad A_2 := a\ell_2,$$

*so that  $k + a \leq A_1 \leq k + 2a - 1$  and  $k + a + 1 \leq A_2 \leq k + 2a$ , with  $A_1 = k + a$  and  $A_2 = k + 2a$  when  $a \mid k$ .*

- (i) *If  $\ell_1 \leq N - 1$ , there exists  $z \in B$  such that, with  $u_z := (\varphi(x)^{\ell_1} + z \varphi(x)^{\ell_1-1})_{x \in D} \in B^n$ ,*

$$|\Lambda(\text{RS}[\mathbb{F}, D, k], 1 - \frac{A_1}{n}, u_z)| \geq \binom{N}{\ell_1} / |B|.$$

- (ii) *If  $\ell_2 \leq N - 1$ , there exists  $z \in B$  such that, with  $u_z := (\varphi(x)^{\ell_2} + z \varphi(x)^{\ell_2-1})_{x \in D} \in B^n$ ,*

$$|\Lambda(\text{RS}[\mathbb{F}, D, k+1], 1 - \frac{A_2}{n}, u_z)| \geq \binom{N}{\ell_2} / |B|.$$

In both cases  $u_z$  is  $B$ -valued, and the listed codewords lie in  $\text{RS}[B, D, a(\ell_1 - 2) + 1] \subseteq \text{RS}[B, D, k]$  (resp.  $\text{RS}[B, D, a(\ell_2 - 2) + 1] \subseteq \text{RS}[B, D, k + 1]$ ).

*Proof.* First the bracketing of  $A_1, A_2$ : writing  $k = a\lfloor k/a \rfloor + s$  with  $0 \leq s \leq a - 1$  gives  $A_2 = k - s + 2a \in [k + a + 1, k + 2a]$ ; writing  $k - 1 = a\lfloor (k - 1)/a \rfloor + s_1$  with  $0 \leq s_1 \leq a - 1$  gives  $A_1 = k - 1 - s_1 + 2a \in [k + a, k + 2a - 1]$ ; and  $a \mid k$  forces  $s = 0$  and  $s_1 = a - 1$ .

We prove (ii); part (i) is the identical computation one degree rung lower, with  $k + 1$  replaced by  $k$  throughout. Let  $A \subseteq Q$  with  $|A| = \ell_2$ , and put

$$P_A(Y) := \prod_{b \in A} (Y - b) = Y^{\ell_2} - e_1(A) Y^{\ell_2 - 1} + R_A(Y), \quad R_A(Y) := \sum_{j=2}^{\ell_2} (-1)^j e_j(A) Y^{\ell_2 - j},$$

so  $\deg_Y R_A \leq \ell_2 - 2$ , and set  $L_A(X) := P_A(\varphi(X)) \in B[X]$ . The polynomial  $L_A$  vanishes on

$$S_A := \varphi^{-1}(A) \cap D = \bigsqcup_{b \in A} S_b, \quad |S_A| = a\ell_2 = A_2,$$

since the fibers  $S_b$  are complete, pairwise disjoint, and of size exactly  $a$ .

Set  $z_A := -e_1(A) \in B$  and  $c_A := (-R_A(\varphi(x)))_{x \in D}$ . Since  $\deg_X R_A(\varphi(X)) \leq a(\ell_2 - 2) = A_2 - 2a \leq k$ , we have  $c_A \in \text{RS}[B, D, a(\ell_2 - 2) + 1] \subseteq \text{RS}[B, D, k + 1] \subseteq \text{RS}[\mathbb{F}, D, k + 1]$ . For every  $x \in S_A$ ,

$$0 = L_A(x) = \varphi(x)^{\ell_2} + z_A \varphi(x)^{\ell_2 - 1} + R_A(\varphi(x)), \quad \text{i.e.} \quad u_{z_A}(x) = c_A(x),$$

so  $u_{z_A}$  and  $c_A$  agree on at least  $A_2$  points of  $D$  and  $\Delta(u_{z_A}, c_A) \leq 1 - A_2/n$ .

The assignment  $A \mapsto c_A$  is injective among subsets sharing a common value of  $z_A$ . Indeed, suppose  $z_A = z_{A'}$  and  $c_A = c_{A'}$ . Then  $R_A(\varphi(X))$  and  $R_{A'}(\varphi(X))$  are polynomials of degree  $\leq k < n$  agreeing at all  $n$  points of  $D$ , hence equal as polynomials. Composition with the nonconstant  $\varphi$  is injective on  $\mathbb{F}[Y]$ : if  $R := R_A - R_{A'} \neq 0$  has degree  $d \geq 1$ , then  $R(\varphi(X))$  has degree  $ad \geq 1$  with leading coefficient  $\text{lc}(R) \text{lc}(\varphi)^d \neq 0$ , and if  $d = 0$  then  $R(\varphi(X)) = R$  is a nonzero constant; either way  $R(\varphi(X)) \neq 0$ . Hence  $R_A = R_{A'}$ , so  $P_A = Y^{\ell_2} + z_A Y^{\ell_2 - 1} + R_A(Y) = P_{A'}$ , and the root multisets give  $A = A'$ .

Finally,  $z_A = -e_1(A)$  lies in  $B$  because  $A \subseteq Q \subseteq B$ , so it takes at most  $|B|$  values as  $A$  ranges over the  $\binom{N}{\ell_2}$  subsets of  $Q$  of size  $\ell_2$  (here  $\ell_2 \leq N - 1 \leq N$  is used to have this many subsets). Some value  $z \in B$  is attained by at least  $\binom{N}{\ell_2}/|B|$  subsets, and the corresponding codewords  $c_A$  are pairwise distinct elements of  $\Lambda(\text{RS}[\mathbb{F}, D, k + 1], 1 - A_2/n, u_z)$ .  $\square$

*Remark 7.3* (the divisibility hypothesis is removed). With  $\varphi = X^a$ ,  $D$  a multiplicative coset, and  $a \mid k$ , Lemma 7.2 is Lemma 3.1 verbatim:  $A_1 = k + a$ ,  $A_2 = k + 2a$ , and  $u_z$  has the printed monomial shape. The rounding  $\ell_2 = \lfloor k/a \rfloor + 2$  removes the hypothesis  $a \mid k$  at the cost of a certified agreement  $A_2 \in [k + a + 1, k + 2a]$  rather than exactly  $k + 2a$ ; the certified radius therefore satisfies

$$1 - \rho - \frac{2}{N} \leq 1 - \frac{A_2}{n} \leq 1 - \rho - \frac{1}{N} - \frac{1}{n},$$

so the listed ball always sits at least one quotient step  $\approx 1/N$  below capacity. This matters precisely where  $a \mid k$  genuinely fails: odd-dimension circle codes (Corollary 7.12) and mixed-radix multiplicative rows without an integral  $\rho N$  (Corollary 7.6). Remark 3.2 already used slack two to move the divisibility demand from  $k + 1$  onto  $k$ ; the present rounding removes it altogether.

**Theorem 7.4** (universal cap for map-smooth domains). *Let  $B \subseteq \mathbb{F}$  be finite fields,  $q := |\mathbb{F}|$ , and let  $D \subseteq B$  be a  $(\varphi, a)$ -smooth domain of size  $n < q$ . Let  $1 \leq k < n$ ,  $\rho := k/n$ ,  $C := \text{RS}[\mathbb{F}, D, k]$ , and let  $N = n/a$ ,  $\ell_2 = \lfloor k/a \rfloor + 2$ ,  $A_2 = a\ell_2$  be as in Lemma 7.2. Assume  $\ell_2 \leq N - 1$  and*

$$\binom{N}{\ell_2} \geq |B| \left( \frac{q}{k} + 1 \right). \quad (2)$$

Then

$$\varepsilon_{\text{ca}}(C, \delta) > \frac{1}{2k} \left( 1 - \frac{n}{q} \right) \quad \text{for every } \delta \in \left[ 1 - \frac{A_2}{n}, 1 - \rho - \frac{1}{n} \right],$$

and

$$\varepsilon_{\text{mca}}(C, \delta) > \frac{1}{2k} \left( 1 - \frac{n}{q} \right) \quad \text{for every } \delta \in \left[ 1 - \frac{A_2}{n}, 1 - \rho \right).$$

Consequently

$$\delta_C^*(\varepsilon^*) \leq 1 - \frac{A_2}{n} \leq 1 - \rho - \frac{a+1}{n} \quad \text{for every } \varepsilon^* \leq \frac{1}{2k} \left( 1 - \frac{n}{q} \right).$$

*Proof.* Since  $\ell_2 \leq N - 1$  we have  $A_2 \leq n - a < n$ , and since  $A_2 \geq k + a + 1 \geq k + 2$  both stated intervals are nonempty, with  $1 - A_2/n \leq 1 - \rho - (a+1)/n \leq 1 - \rho - 2/n$ .

Fix  $\delta \in [1 - A_2/n, 1 - \rho - 1/n]$ ; then  $\lfloor \delta n \rfloor \leq n - k - 1$ , so Theorem 2.6 is applicable to  $C$  at radius  $\delta$ . Suppose toward a contradiction that  $\varepsilon_{\text{ca}}(C, \delta) \leq \frac{1}{2k} (1 - \frac{n}{q})$ . Applying Theorem 2.6 with  $\eta = \frac{1}{2}$  gives, for  $C^+ := \text{RS}[\mathbb{F}, D, k+1]$ ,

$$\Lambda(C^+, \delta) \leq \lceil 2q \varepsilon_{\text{ca}}(C, \delta) \rceil \leq \left\lceil \frac{q-n}{k} \right\rceil < \frac{q-n}{k} + 1 \leq \frac{q}{k} + 1.$$

On the other hand, by Lemma 7.2(ii) and monotonicity of lists in the radius (using  $\delta \geq 1 - A_2/n$ ),

$$\Lambda(C^+, \delta) \geq |\Lambda(C^+, 1 - \frac{A_2}{n}, u_z)| \geq \binom{N}{\ell_2} / |B| \geq \frac{q}{k} + 1$$

by (2), a contradiction. Hence the  $\varepsilon_{\text{ca}}$  lower bound holds throughout the stated integer-admissible interval. In particular it holds at  $\delta_0 := 1 - A_2/n$ , where Fact 2.4 gives the same lower bound for  $\varepsilon_{\text{mca}}(C, \delta_0)$ , and support-wise MCA monotonicity (Lemma 2.5) extends it to every  $\delta \in [\delta_0, 1 - \rho]$ . The bound on  $\delta_C^*$  follows exactly as in Theorem 4.1, using  $A_2 \geq k + a + 1$ .  $\square$

*Remark 7.5* (relation to Theorem 4.1). With  $\varphi = X^a$ ,  $D$  a multiplicative coset, and  $a \mid k$ , Theorem 7.4 is Theorem 4.1 verbatim:  $A_2 = k + 2a$ , the left endpoint is  $\delta_N = 1 - \rho - 2/N$ , hypothesis (2) is (1), and  $\ell_2 \leq N - 1$  is the printed condition  $(1 - \rho)N \geq 3$ . The generalization is strict in two independent directions — the folding map and the removal of  $a \mid k$  — at no cost in constants. As in Remark 4.2, the constant  $\frac{1}{2k}$  can be pushed to  $\frac{1-o(1)}{k}$  given slack in (2).

**Corollary 7.6** (mixed-radix multiplicative rows, divisibility-free). *Let  $\rho \in \{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}\}$ , let  $\mathbb{F}$  be any finite field with  $q := |\mathbb{F}| < 2^{256}$ , let  $B \subseteq \mathbb{F}$  be any subfield, let  $D \subseteq B^\times$  be a multiplicative coset of order  $n$ , and let  $k = \rho n \leq 2^{40}$ ,  $C = \text{RS}[\mathbb{F}, D, k]$ . Suppose  $n$  has some divisor  $N$  with*

$$2^{10} \leq N \leq 2^{11} \quad (\rho \in \{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}\}), \quad \text{resp.} \quad 2^{11} \leq N \leq 2^{12} \quad (\rho = \frac{1}{16}),$$

*with no divisibility condition relating  $N$  and  $k$  (in particular  $\rho N$  need not be an integer). Then*

$$\varepsilon_{\text{mca}}(C, \delta) > \frac{1}{2k} \left( 1 - \frac{n}{q} \right) \geq 2^{-86} \gg 2^{-128} \quad \text{for every } \delta \in \left[ 1 - \rho - \frac{1}{N}, 1 - \rho \right),$$

and the same lower bound holds for  $\varepsilon_{\text{ca}}(C, \delta)$  on the integer-admissible subinterval  $[1 - \frac{A_2}{n}, 1 - \rho - \frac{1}{n}]$ , with  $A_2$  as in Lemma 7.2 for  $a = n/N$ . In particular

$$\delta_C^*(2^{-128}) \leq 1 - \rho - 2^{-11} \quad (\rho \in \{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}\}), \quad \delta_C^*(2^{-128}) \leq 1 - \rho - 2^{-12} \quad (\rho = \frac{1}{16}).$$

*Proof.* Apply Theorem 7.4 with  $\varphi = X^{n/N}$  and  $a = n/N$ ; the coset  $D$  is  $(X^a, a)$ -smooth, and  $n < q$  as in Corollary 5.1. Since  $\lfloor k/a \rfloor \in (k/a - 1, k/a]$ , we have  $\ell_2/N \in (\rho + \frac{1}{N}, \rho + \frac{2}{N}]$ , so  $\ell_2 \leq \rho N + 2 \leq N/2 + 2 \leq N - 1$ . For (2): the interval  $(\rho + \frac{1}{N}, \rho + \frac{2}{N}]$  is contained in  $[\frac{1}{8}, 0.502]$  for the first three rates and in  $[\frac{1}{16}, 0.0645]$  for  $\rho = \frac{1}{16}$ ; on each interval the concave function  $H_2$  attains its minimum at an endpoint, and  $H_2(0.502) \geq 0.999 \geq H_2(\frac{1}{8})$  while  $H_2(0.0645) \geq H_2(\frac{1}{16})$ , so  $H_2(\ell_2/N) \geq H_2(\frac{1}{8}) \geq 0.5435$ , resp.  $H_2(\ell_2/N) \geq H_2(\frac{1}{16}) \geq 0.3372$ . Hence, using the worst  $N$  in each window,

$$\log_2 \binom{N}{\ell_2} \geq N H_2(\ell_2/N) - \log_2(N+1) \geq \begin{cases} 2^{10} \cdot 0.5435 - \log_2(2^{11} + 1) \geq 545, \\ 2^{11} \cdot 0.3372 - \log_2(2^{12} + 1) \geq 678, \end{cases}$$

while  $|B|(q/k + 1) \leq q(q/k + 1) \leq q^2 + q < 2^{513}$ , exactly as in Corollary 5.1. Theorem 7.4 gives the lower bounds on  $[1 - \frac{A_2}{n}, 1 - \rho - \frac{1}{n}]$  and  $[1 - \frac{A_2}{n}, 1 - \rho)$ ; since  $A_2 \geq k + a + 1$ , the MCA interval contains  $[1 - \rho - \frac{1}{N}, 1 - \rho)$ , and  $\frac{1}{N} \geq 2^{-11}$ , resp.  $2^{-12}$ . The error numerics  $\frac{1}{2k}(1 - \frac{n}{q}) \geq \frac{1}{2k(n+1)} > 2^{-86}$  (using  $n \leq 16k \leq 2^{44}$ ) are those of Corollary 5.1.  $\square$

For FFT-friendly mixed-radix orders  $n = 2^\alpha 3^\beta 5^\gamma \dots$  the divisor set is dense on the logarithmic scale, so the window hypothesis is mild; and taking the smallest available  $N$  in the window recovers the smallest gap.

## 7.2 Twin cosets on the circle, Chebyshev fibers, and torus uniformization

Throughout this subsection  $p$  is a prime with  $p \equiv 3 \pmod{4}$ , so that  $X^2 + 1$  is irreducible over  $\mathbb{F}_p$  and  $\mathbb{F}_{p^2} = \mathbb{F}_p(i)$  with  $i^2 = -1$ . Let

$$\mathbb{U} := \{u \in \mathbb{F}_{p^2}^\times : u^{p+1} = 1\}$$

be the norm-one torus, a cyclic group of order  $p+1$ ; for the deployed Mersenne prime  $p = 2^{31} - 1$  one has  $p+1 = 2^{31}$ . The circle  $\mathcal{C}(\mathbb{F}_p) := \{(x, y) \in \mathbb{F}_p^2 : x^2 + y^2 = 1\}$  is in bijection with  $\mathbb{U}$  via

$$\iota(x, y) := x + iy, \quad \iota^{-1}(u) = \left( \frac{u + u^{-1}}{2}, \frac{u - u^{-1}}{2i} \right),$$

using  $u^p = u^{-1}$  for  $u \in \mathbb{U}$  and  $i^p = -i$ . Write  $\chi(u) := \frac{u+u^{-1}}{2} \in \mathbb{F}_p$  for the  $x$ -coordinate map;  $\chi(u) = \chi(v)$  holds if and only if  $v \in \{u, u^{-1}\}$ . The Chebyshev polynomials of the first kind, normalized by  $T_0 = 1$ ,  $T_1 = X$ ,  $T_{a+1} = 2XT_a - T_{a-1}$ , have integer coefficients and degree  $a$ , reduce modulo  $p$  to polynomials in  $\mathbb{F}_p[X]$ , and satisfy the semiconjugacy

$$T_a(\chi(u)) = \chi(u^a) \quad (u \in \mathbb{U}, a \geq 0), \quad T_{ab} = T_a \circ T_b.$$

Following the twin-coset geometry of circle FFTs [HLP24], for a subgroup  $H \leq \mathbb{U}$  of order  $M$  and  $g \in \mathbb{U}$  with  $g^2 \notin H$  we call

$$\mathcal{D} := gH \cup g^{-1}H \subseteq \mathbb{U}$$

a *twin coset*. It has size  $2M$ , is closed under inversion, and contains no self-inverse element: if  $u \in gH$  satisfied  $u = u^{-1}$ , then  $u \in gH \cap (gH)^{-1} = gH \cap g^{-1}H$ , forcing  $g^2 \in H$ ; likewise for  $u \in g^{-1}H$ . Consequently  $\chi$  is exactly two-to-one on  $\mathcal{D}$ , and the  $x$ -coordinate domain  $D := \chi(\mathcal{D}) \subseteq \mathbb{F}_p$  has size exactly  $M$ . For  $a \mid M$  write  $H^{(a)} := \{h^a : h \in H\}$  for the image subgroup, of order  $M/a$ , and  $\mathcal{D}^{(a)} := \{u^a : u \in \mathcal{D}\}$ .

**Lemma 7.7** (power fibers on torus twin cosets). *Let  $\mathcal{D} = gH \cup g^{-1}H$  be a twin coset of size  $2M$  and let  $a \mid M$ .*

- (a) *If  $g^{2a} \notin H^{(a)}$ , then  $\mathcal{D}$  is  $(X^a, a)$ -smooth (over any field containing  $\mathbb{F}_{p^2}$ ), and  $\mathcal{D}^{(a)} = g^a H^{(a)} \sqcup g^{-a} H^{(a)}$  is again a twin coset, of size  $2M/a$ .*
- (b) *If  $g^{2a} \in H^{(a)}$ , then  $\mathcal{D}$  is  $(X^a, 2a)$ -smooth, with  $\mathcal{D}^{(a)} = g^a H^{(a)}$  a single coset of size  $M/a$ .*

Moreover,  $g^{2a} \notin H^{(a)}$  holds for every  $a \mid M$  simultaneously if and only if  $g^{2M} \neq 1$ , i.e.  $\text{ord}(g) \nmid 2M$ .

*Proof.* Since  $a \mid M \mid p+1$  and  $\mathbb{U}$  is cyclic of order  $p+1$ , the kernel  $K_a$  of  $u \mapsto u^a$  on  $\mathbb{U}$  has order  $a$  and, being the unique subgroup of  $\mathbb{U}$  of that order, is contained in  $H$ . Hence the  $a$ -th power map sends the coset  $gH$  onto  $g^a H^{(a)}$  with every fiber a coset of  $K_a$ , i.e. exactly  $a$ -to-one, and likewise sends  $g^{-1}H$  onto  $g^{-a} H^{(a)}$ . The two image cosets of the subgroup  $H^{(a)}$  are disjoint or equal according to whether  $g^{2a} \notin H^{(a)}$  or  $g^{2a} \in H^{(a)}$ . In the disjoint case, every  $v \in \mathcal{D}^{(a)}$  lies in exactly one of the two image cosets and has exactly  $a$  preimages in  $\mathcal{D}$ ; the twin-coset condition for  $\mathcal{D}^{(a)}$ , namely  $(g^a)^2 \notin H^{(a)}$ , is the hypothesis itself. In the coincident case, every  $v \in g^a H^{(a)}$  has exactly  $a$  preimages in  $gH$  and exactly  $a$  in  $g^{-1}H$ , hence exactly  $2a$  in  $\mathcal{D}$ . For the last claim: at  $a = M$  we have  $H^{(M)} = \{1\}$ , so the scale- $M$  condition reads  $g^{2M} \neq 1$ ; conversely, if  $g^{2a} = h^a$  for some  $a \mid M$  and  $h \in H$ , then  $g^{2M} = (g^{2a})^{M/a} = h^M = 1$ .  $\square$

**Lemma 7.8** (exact Chebyshev fibers on  $x$ -coordinate twin-coset domains). *Let  $\mathcal{D} = gH \cup g^{-1}H$  be a twin coset of size  $2M$ , let  $D := \chi(\mathcal{D}) \subseteq \mathbb{F}_p$ ,  $|D| = M$ , and let  $a \mid M$  satisfy  $g^{2a} \notin H^{(a)}$ . Then  $D$  is  $(T_a, a)$ -smooth over  $\mathbb{F}_p$ , and*

$$T_a(D) = \chi(\mathcal{D}^{(a)})$$

*is the  $x$ -coordinate domain of the twin coset  $\mathcal{D}^{(a)}$ , of size  $M/a$ . In particular, if  $\text{ord}(g) \nmid 2M$ , this holds at every scale  $a \mid M$  simultaneously, and for  $M = 2^m$  the tower*

$$D \xrightarrow{T_2} T_2(D) \xrightarrow{T_2} T_4(D) \xrightarrow{T_2} \dots$$

*consists of  $x$ -coordinate twin-coset domains of sizes  $M, M/2, M/4, \dots$  — the line-round tower of circle FRI [HLP24].*

*Proof.* The semiconjugacy gives  $T_a(D) = T_a(\chi(\mathcal{D})) = \chi(\mathcal{D}^{(a)})$ ; by Lemma 7.7(a),  $\mathcal{D}^{(a)}$  is a twin coset, so  $\chi$  is exactly two-to-one on it and  $|T_a(D)| = |\mathcal{D}^{(a)}|/2 = M/a$ .

Fix  $w \in T_a(D)$  and write  $w = \chi(v)$  with  $v \in \mathcal{D}^{(a)} = g^a H^{(a)} \sqcup g^{-a} H^{(a)}$ . Since  $(g^a H^{(a)})^{-1} = g^{-a} H^{(a)}$ , the set  $\mathcal{D}^{(a)}$  is closed under inversion, and after replacing  $v$  by  $v^{-1}$  if necessary (which does not change  $w$ ) we may assume  $v \in g^a H^{(a)}$ . Then  $v \neq v^{-1}$ : otherwise  $v$  would lie in both of the disjoint cosets  $g^a H^{(a)}$  and  $g^{-a} H^{(a)}$ . For  $x = \chi(u) \in D$  with  $u \in \mathcal{D}$ ,

$$T_a(x) = w \iff \chi(u^a) = \chi(v) \iff u^a \in \{v, v^{-1}\}.$$



By the fiber structure in the proof of Lemma 7.7, the equation  $u^a = v$  has exactly  $a$  solutions  $u \in gH$  and none in  $g^{-1}H$  (whose  $a$ -th powers fill the disjoint coset  $g^{-a}H^{(a)}$ ), and the equation  $u^a = v^{-1}$  has exactly  $a$  solutions in  $g^{-1}H$  — the inverses of the former — and none in  $gH$ . Hence

$$E_w := \{u \in \mathcal{D} : u^a \in \{v, v^{-1}\}\}$$

has exactly  $2a$  elements, is closed under inversion, and, being a subset of  $\mathcal{D}$ , contains no self-inverse element. Since  $\chi(u) = \chi(u')$  forces  $u' \in \{u, u^{-1}\}$ , the image  $\chi(E_w)$  — which is precisely the fiber  $T_a^{-1}(w) \cap D$  — has exactly  $|E_w|/2 = a$  elements. As  $w \in T_a(D)$  was arbitrary,  $D$  is  $(T_a, a)$ -smooth with  $|T_a(D)| = M/a = |D|/a$ , as required. The tower statement follows from the all-scales equivalence in Lemma 7.7 together with  $T_{2j+1} = T_2 \circ T_{2j}$ .  $\square$

*Remark 7.9* (standard-position cosets are twin cosets with  $\text{ord}(g) \nmid 2M$ ). A standard-position coset of circle FFT [HLP24] is a coset  $gH_0$ ,  $H_0 := \langle g^2 \rangle$ , of size  $2M$  with  $\text{ord}(g) = 4M$  (so  $|H_0| = 2M$ ). Let  $K \leq H_0$  be the index-two subgroup, of order  $M$ . Then  $g^{-1}K = g g^{-2}K$ , and  $g^{-2}$  generates  $H_0$ , hence  $g^{-2} \notin K$ , so

$$gH_0 = gK \sqcup g^{-1}K$$

is exactly the twin coset attached to  $(K, g)$ : the condition  $g^2 \notin K$  holds because  $g^2$  generates  $H_0 \supsetneq K$ , and  $\text{ord}(g) = 4M \nmid 2M$ . Thus every standard-position domain of [HLP24], at every folding level, satisfies the hypotheses of Lemmas 7.7 and 7.8 at every dyadic scale simultaneously.

**Lemma 7.10** (diagonal invariance of lists and correlated agreement). *Let  $C \subseteq \mathbb{F}^n$  be a linear code and  $t \in (\mathbb{F}^\times)^n$ ; write  $t \circ v$  for the coordinatewise product and  $t \circ C := \{t \circ c : c \in C\}$ . Then for every received word  $U$ , every radius  $\delta$ , and every pair of radii,*

$$\Lambda(t \circ C, \delta, t \circ U) = t \circ \Lambda(C, \delta, U), \quad \varepsilon_{\text{ca}}(t \circ C, \cdot, \cdot) = \varepsilon_{\text{ca}}(C, \cdot, \cdot), \quad \varepsilon_{\text{mca}}(t \circ C, \cdot) = \varepsilon_{\text{mca}}(C, \cdot).$$

*Proof.* The map  $v \mapsto t \circ v$  is an  $\mathbb{F}$ -linear bijection of  $\mathbb{F}^n$  that preserves coordinatewise agreement:  $(t \circ v)_j = (t \circ w)_j \iff v_j = w_j$ . Hence it preserves  $\Delta$  and every restricted distance  $\Delta_S$ , maps  $C$  onto  $t \circ C$ , acts row-wise on pairs, and commutes with the line structure:  $t \circ (f_1 + \gamma f_2) = (t \circ f_1) + \gamma (t \circ f_2)$ . It therefore carries every witness of the defining events of  $\Lambda$ ,  $\varepsilon_{\text{ca}}$ , and  $\varepsilon_{\text{mca}}$  for  $C$  bijectively onto a witness for  $t \circ C$  with the same slope  $\gamma$  and the same agreement supports, and conversely via  $t^{-1}$ .  $\square$

**Lemma 7.11** (torus uniformization of circle codes). *Let  $p \equiv 3 \pmod{4}$ , let  $\mathbb{F} \supseteq \mathbb{F}_{p^2}$  be a finite field (so  $i \in \mathbb{F}$ ), let  $E \subseteq \mathcal{C}(\mathbb{F}_p)$  be a set of circle points, and let  $E' := \iota(E) \subseteq \mathbb{U} \subseteq \mathbb{F}_{p^2}^\times$ . For  $w \geq 0$  define the  $\mathbb{F}$ -coefficient circle code*

$$\mathcal{C}_w(\mathbb{F}, E) := \{(f(P))_{P \in E} : f \in \mathbb{F}[x, y]/(x^2 + y^2 - 1), \deg f \leq w\}.$$

*Then, under the coordinate bijection  $\iota : E \rightarrow E'$  and with the diagonal vector  $t := (u^{-w})_{u \in E'}$ ,*

$$\mathcal{C}_w(\mathbb{F}, E) = t \circ \text{RS}[\mathbb{F}, E', 2w + 1].$$

*Consequently, by Lemma 7.10, the circle code and the Reed–Solomon code  $\text{RS}[\mathbb{F}, E', 2w + 1]$  of odd dimension  $2w + 1$  on the torus domain  $E'$  have identical list sizes at every radius and identical  $\varepsilon_{\text{ca}}$  and  $\varepsilon_{\text{mca}}$  at every (pair of) radii.*

*Proof.* Since  $i \in \mathbb{F}$  and  $x + iy$  is invertible in the quotient ring (with inverse  $x - iy$ , as  $(x + iy)(x - iy) = x^2 + y^2 = 1$ ), the substitution  $u := x + iy$  defines a ring isomorphism

$$\mathbb{F}[x, y]/(x^2 + y^2 - 1) \xrightarrow{\sim} \mathbb{F}[u, u^{-1}], \quad x \mapsto \frac{u + u^{-1}}{2}, \quad y \mapsto \frac{u - u^{-1}}{2i},$$

with inverse  $u \mapsto x + iy$ ; it matches evaluation at  $P \in E$  on the left with evaluation at  $\iota(P) \in E'$  on the right. Every class of degree  $\leq w$  has the canonical form  $f_0(x) + y f_1(x)$  with  $\deg f_0 \leq w$  and  $\deg f_1 \leq w - 1$  (reduce  $y^2 = 1 - x^2$ ), and  $\{h_0(x) + y h_1(x)\}$  is a free-module canonical form of the quotient ring over  $\mathbb{F}[x]$ , so this space has dimension exactly  $2w + 1$ . Under the isomorphism,

$$x^j = 2^{-j} u^{-j} (u^2 + 1)^j \in u^{-j} \mathbb{F}[u]_{\leq 2j}, \quad y x^j \in u^{-(j+1)} \mathbb{F}[u]_{\leq 2j+2},$$

so the degree- $\leq w$  subspace maps into  $u^{-w} \mathbb{F}[u]_{\leq 2w}$ ; both spaces have dimension  $2w + 1$  and the map is injective, hence the image is exactly  $u^{-w} \mathbb{F}[u]_{\leq 2w}$ . Evaluating on  $E'$  gives

$$\mathcal{C}_w(\mathbb{F}, E) = \{(u^{-w} h(u))_{u \in E'} : h \in \mathbb{F}[u], \deg h \leq 2w\} = t \circ \text{RS}[\mathbb{F}, E', 2w + 1]. \quad \square$$

Note the parity: the uniformized dimension  $k_c = 2w + 1$  is always odd, while the useful folding scales  $a$  on a dyadic torus twin coset are even, so  $a \nmid k_c$  *always*. The divisibility-free form of Lemma 7.2 is therefore not a luxury here: the original Lemma 3.1 can never be applied to a uniformized circle code directly.

### 7.3 Universal caps for circle codes and circle-FRI line rounds

**Corollary 7.12** (universal circle-row cap). *Let  $p \equiv 3 \pmod{4}$  be prime, let  $\mathcal{D} = gH \cup g^{-1}H \subseteq \mathbb{U}$  be a twin coset of size  $2M$  with  $\text{ord}(g) \nmid 2M$  (e.g. any standard-position coset of [HLP24], by Remark 7.9), and let  $\mathbb{F}$  be any finite field with  $q := |\mathbb{F}| < 2^{256}$ .*

- (a) (Line rounds.) *Assume  $\mathbb{F} \supseteq \mathbb{F}_p$ . Let  $D := \chi(\mathcal{D})$ ,  $n := M$ , and  $C := \text{RS}[\mathbb{F}, D, k]$  with  $k \leq 2^{40}$  and  $\rho := k/n \in [\frac{1}{16}, \frac{1}{2}]$ . If  $\rho \geq \frac{1}{8}$ , assume  $2^{10} \mid M$  and set  $N := 2^{10}$ ; if  $\rho < \frac{1}{8}$ , assume  $2^{11} \mid M$  and set  $N := 2^{11}$ . Then*

$$\varepsilon_{\text{mca}}(C, \delta) > \frac{1}{2k} \left(1 - \frac{n}{q}\right) \geq 2^{-86} \gg 2^{-128} \quad \text{for every } \delta \in \left[1 - \rho - \frac{1}{N}, 1 - \rho\right),$$

*and the same lower bound holds for  $\varepsilon_{\text{ca}}(C, \delta)$  on the integer-admissible subinterval  $[1 - \frac{A_2}{n}, 1 - \rho - \frac{1}{n}]$ , with  $A_2$  as in Lemma 7.2 for  $a = n/N$ . If moreover  $\frac{n}{N} \mid k$  — as in every 2-power instantiation at the four challenge rates — the MCA band extends to  $[1 - \rho - \frac{2}{N}, 1 - \rho)$ . In particular*

$$\delta_C^*(2^{-128}) \leq 1 - \rho - 2^{-10} \quad (\rho \geq \frac{1}{8}), \quad \delta_C^*(2^{-128}) \leq 1 - \rho - 2^{-11} \quad (\rho < \frac{1}{8}),$$

*improving to  $1 - \rho - 2^{-9}$  (resp.  $1 - \rho - 2^{-10}$ ) under  $\frac{n}{N} \mid k$ .*

- (b) (Circle codes.) *Assume  $\mathbb{F} \supseteq \mathbb{F}_{p^2}$ . Let  $C := \mathcal{C}_w(\mathbb{F}, \iota^{-1}(\mathcal{D}))$  with  $k_c := 2w + 1 \leq 2^{41}$ ,  $n_c := 2M$ , and  $\rho_c := k_c/n_c \in [\frac{1}{16}, \frac{1}{2}]$ . If  $\rho_c \geq \frac{1}{8}$ , assume  $2^9 \mid M$  and set  $N := 2^{10}$ ; if  $\rho_c < \frac{1}{8}$ , assume  $2^{10} \mid M$  and set  $N := 2^{11}$ . Then*

$$\varepsilon_{\text{mca}}(C, \delta) > \frac{1}{2k_c} \left(1 - \frac{n_c}{q}\right) \geq 2^{-88} \gg 2^{-128} \quad \text{for every } \delta \in \left[1 - \rho_c - \frac{1}{N}, 1 - \rho_c\right),$$

*with the corresponding  $\varepsilon_{\text{ca}}$  bound on  $[1 - \frac{A_2}{n_c}, 1 - \rho_c - \frac{1}{n_c}]$  (here  $a := n_c/N \nmid k_c$  always, since  $k_c$  is odd). In particular  $\delta_C^*(2^{-128}) \leq 1 - \rho_c - 2^{-10}$  for  $\rho_c \geq \frac{1}{8}$ , and  $\delta_C^*(2^{-128}) \leq 1 - \rho_c - 2^{-11}$  for  $\rho_c < \frac{1}{8}$ .*



*Proof.* (a) Put  $a := n/N = M/N$ , so  $a \mid M$ , and  $g^{2a} \notin H^{(a)}$  by the all-scales equivalence of Lemma 7.7; Lemma 7.8 then makes  $D$  a  $(T_a, a)$ -smooth domain over  $B := \mathbb{F}_p \subseteq \mathbb{F}$ . We check the hypotheses of Theorem 7.4. First,  $n < q$ : since  $M \mid p+1$  and  $\text{ord}(g) \mid p+1$ , the case  $M = p+1$  would give  $\text{ord}(g) \mid p+1 \mid 2M$ , contradicting  $\text{ord}(g) \nmid 2M$ ; hence  $M \leq \frac{p+1}{2}$  and  $n \leq \frac{p+1}{2} < p \leq q$ . Second,  $\ell_2 \leq \rho N + 2 \leq N/2 + 2 \leq N - 1$ . Third, for (2):  $\ell_2/N \in (\rho + \frac{1}{N}, \rho + \frac{2}{N}]$ , which is contained in  $[\frac{1}{8}, 0.502]$  when  $\rho \geq \frac{1}{8}$  and in  $[\frac{1}{16}, 0.126]$  when  $\rho < \frac{1}{8}$ ; on each interval the concave  $H_2$  is minimized at an endpoint, and both right endpoints have larger  $H_2$ -value than the left, so  $H_2(\ell_2/N) \geq H_2(\frac{1}{8}) \geq 0.5435$ , resp.  $H_2(\ell_2/N) \geq H_2(\frac{1}{16}) \geq 0.3372$ . Hence

$$\log_2 \binom{N}{\ell_2} \geq N H_2(\ell_2/N) - \log_2(N+1) \geq 546, \text{ resp. } 679,$$

while  $|B|(q/k+1) \leq q(q/k+1) \leq q^2 + q < 2^{513}$ . The error numerics are those of Corollary 5.1:  $\rho \geq \frac{1}{16}$  gives  $n \leq 16k \leq 2^{44}$ , so  $\frac{1}{2k}(1 - \frac{n}{q}) \geq \frac{1}{2k(n+1)} > 2^{-86}$ . Theorem 7.4 now yields the  $\varepsilon_{\text{ca}}$  and  $\varepsilon_{\text{mca}}$  lower bounds on  $[1 - \frac{A_2}{n}, 1 - \rho - \frac{1}{n}]$  and  $[1 - \frac{A_2}{n}, 1 - \rho]$  respectively; since  $A_2 \geq k + a + 1$ , the MCA interval contains  $[1 - \rho - \frac{1}{N}, 1 - \rho]$ , and when  $\frac{n}{N} \mid k$  one has  $A_2 = k + 2a$ , so it equals  $[1 - \rho - \frac{2}{N}, 1 - \rho]$ .

(b) By Lemma 7.11, all list sizes,  $\varepsilon_{\text{ca}}$ , and  $\varepsilon_{\text{mca}}$  of  $C$  coincide with those of  $C' := \text{RS}[\mathbb{F}, \mathcal{D}, k_c]$  on the torus twin coset  $\mathcal{D} \subseteq \mathbb{F}_{p^2}$ . Put  $a := n_c/N = 2M/N$ ; then  $a \mid M$  (indeed  $M/a = N/2 \in \mathbb{Z}$ ), and  $g^{2a} \notin H^{(a)}$  as before, so  $\mathcal{D}$  is  $(X^a, a)$ -smooth over  $B := \mathbb{F}_{p^2} \subseteq \mathbb{F}$  by Lemma 7.7(a). Also  $n_c = 2M < p+1 \leq p^2 \leq q$ : the case  $2M = p+1$  would again give  $\text{ord}(g) \mid p+1 = 2M$ . The verification of (2) is identical to (a), with the same two entropy cases and  $|B| = p^2 \leq q$ ; the error numerics use  $n_c \leq 16k_c \leq 2^{45}$ , so  $\frac{1}{2k_c}(1 - \frac{n_c}{q}) \geq \frac{1}{2k_c(n_c+1)} > 2^{-88}$ . Theorem 7.4 applied to  $C'$ , transported back through Lemmas 7.10 and 7.11, gives the stated bands;  $A_2 \geq k_c + a + 1$  again yields the containment of  $[1 - \rho_c - \frac{1}{N}, 1 - \rho_c]$ .  $\square$

**Corollary 7.13** (deployed circle parameters: Mersenne-31 with QM31). *Let  $p = 2^{31} - 1$ , let  $\mathbb{F} = \mathbb{F}_{p^4}$  be the deployed QM31 extension (so  $q = p^4 > 2^{123.9}$  and  $\mathbb{F} \supseteq \mathbb{F}_{p^2}$ ), and let  $\mathcal{D}$  be a standard-position twin coset with  $M = 2^{21}$  (so  $\text{ord}(g) = 2^{23} \nmid 2^{22} = 2M$ ).*

(a) (Line round, rate  $\frac{1}{2}$ .) *With  $D = \chi(\mathcal{D})$ ,  $|D| = 2^{21}$ , and  $C = \text{RS}[\mathbb{F}_{p^4}, D, 2^{20}]$ ,*

$$\varepsilon_{\text{mca}}(C, \delta) > (1 - 2^{-102}) 2^{-21} > 2^{-22} \quad \text{for every } \delta \in \left[\frac{1}{2} - 2^{-7}, \frac{1}{2}\right),$$

*and the same lower bound holds for  $\varepsilon_{\text{ca}}(C, \delta)$  for every  $\delta \in [\frac{1}{2} - 2^{-7}, \frac{1}{2} - 2^{-21}]$ .*

(b) (Circle code.) *With  $C = \mathcal{C}_w(\mathbb{F}_{p^4}, \iota^{-1}(\mathcal{D}))$ ,  $w = 2^{20}$ ,  $k_c = 2^{21} + 1$ ,  $n_c = 2^{22}$ ,  $\rho_c = \frac{1}{2} + 2^{-22}$ ,*

$$\varepsilon_{\text{mca}}(C, \delta) > 2^{-23} \quad \text{for every } \delta \in \left[\frac{1}{2} - 2^{-7}, 1 - \rho_c\right),$$

*and the same lower bound holds for  $\varepsilon_{\text{ca}}(C, \delta)$  for every  $\delta \in [\frac{1}{2} - 2^{-7}, \frac{1}{2} - 2^{-21}]$ .*

*Proof.* Both parts use  $N = 256$  and, as in Corollary 5.4,  $\log_2 \binom{256}{130} \geq 247$ .

(a) Here  $a = n/N = 2^{13}$  divides  $k = 2^{20}$ , so  $\ell_2 = \rho N + 2 = 130$ ,  $A_2 = k + 2a = 2^{20} + 2^{14}$ , and  $1 - A_2/n = \frac{1}{2} - 2^{-7}$ . The standard-position hypothesis gives exact Chebyshev fibers at scale  $a$  (Remark 7.9 and Lemma 7.8), with  $B = \mathbb{F}_p$ . Hypothesis (2):  $|B|(q/k+1) = p(p^4/2^{20}+1) < 2^{31} \cdot 2^{105} = 2^{136} \leq 2^{247} \leq \binom{256}{130}$ . The error bound is  $\frac{1}{2k}(1 - n/q)$  with  $n/q = 2^{21}/p^4 < 2^{-102}$ .

Theorem 7.4 gives the MCA band  $[\frac{1}{2} - 2^{-7}, \frac{1}{2})$  and the  $\varepsilon_{\text{ca}}$  band  $[\frac{1}{2} - 2^{-7}, \frac{1}{2} - 2^{-21}]$ , since  $1 - \rho - \frac{1}{n} = \frac{1}{2} - 2^{-21}$ .

(b) Here  $a = n_c/N = 2^{14} \mid M = 2^{21}$ ,  $\ell_2 = \lfloor (2^{21} + 1)/2^{14} \rfloor + 2 = 2^7 + 2 = 130$ ,  $A_2 = 2^{14} \cdot 130 = 2^{21} + 2^{15}$ , and  $1 - A_2/n_c = \frac{1}{2} - 2^{-7}$ . Lemma 7.11 transports everything to  $C' = \text{RS}[\mathbb{F}_{p^4}, \mathcal{D}, 2^{21} + 1]$  on the torus twin coset, which is  $(X^a, a)$ -smooth over  $B = \mathbb{F}_{p^2}$  by Lemma 7.7 and Remark 7.9. Hypothesis (2):  $|B|(q/k_c + 1) = p^2(p^4/(2^{21} + 1) + 1) < 2^{62} \cdot 2^{104} = 2^{166} \leq 2^{247}$ . The error bound:  $n_c/q = 2^{22}/p^4 < 2^{-101}$ , so

$$\frac{1}{2k_c} \left(1 - \frac{n_c}{q}\right) > \frac{1 - 2^{-101}}{2^{22} + 2} > 2^{-22}(1 - 2^{-21})(1 - 2^{-101}) > 2^{-23}.$$

Theorem 7.4 gives the MCA band  $[\frac{1}{2} - 2^{-7}, 1 - \rho_c)$  and the  $\varepsilon_{\text{ca}}$  band  $[\frac{1}{2} - 2^{-7}, 1 - \rho_c - \frac{1}{n_c}] = [\frac{1}{2} - 2^{-7}, \frac{1}{2} - 2^{-21}]$ .  $\square$

**Corollary 7.14** (circle certifying lines are genuinely extension-valued). *The subfield-confinement lemma Lemma 6.1 holds verbatim for an arbitrary evaluation set  $D \subseteq B$ : its proof uses only that the evaluation points lie in  $B$ , not that they are invertible. Consequently, at the parameters of Corollary 7.13:*

- (a) *any pair of words on  $D = \chi(\mathcal{D})$  that is  $\mathbb{F}_p$ -valued up to a common scalar factor has MCA-bad-slope density at most  $p/q = p^{-3} < 2^{-92}$  at every radius; in particular, every pair attaining the bound  $> 2^{-22}$  of Corollary 7.13(a) is genuinely  $\mathbb{F}_{p^4}$ -valued;*
- (b) *the diagonal vector  $t = (u^{-w})_u$  of Lemma 7.11 is  $\mathbb{F}_{p^2}$ -valued, so any pair of circle words whose  $\iota$ -transports are  $\mathbb{F}_{p^2}$ -valued up to a common scalar factor has MCA-bad-slope density at most  $p^2/q = p^{-2} < 2^{-61}$  at every radius; in particular, every circle pair attaining the bound  $> 2^{-23}$  of Corollary 7.13(b) contains a word whose  $\iota$ -transport is not  $\mathbb{F}_{p^2}$ -valued, even after removing a common scalar.*

As in Remark 2.15, the extension-pole floors Corollaries 2.12 and 2.13 supply the matching constructive mechanism: the fiber received words of Lemma 7.2 on circle domains are  $B$ -valued with  $B$ -valued lists, and any such list converts through a simple pole  $\alpha \in \mathbb{F} \setminus B$  into genuinely extension-valued bad lines, with denominator  $q - b$  ( $b = |B| \in \{p, p^2\}$ ) in place of  $q - n$ .

*Proof.* Inspecting the proof of Lemma 6.1: it decomposes words and codewords over a  $B$ -basis of  $\mathbb{F}$  and evaluates polynomials at the points of  $D \subseteq B$ ; no step uses  $D \subseteq B^\times$ , so the two-radius statement holds for arbitrary  $D \subseteq B$ , in particular for the Chebyshev domain  $\chi(\mathcal{D}) \subseteq \mathbb{F}_p$  and the torus domain  $\mathcal{D} \subseteq \mathbb{F}_{p^2}$ . (a) is then Lemma 6.1 with  $B = \mathbb{F}_p$  inside  $q = p^4$ : the bad-slope density of a  $B$ -confined pair is at most  $|B|/q = p^{-3} < 2^{-92} < 2^{-22}$ . (b) A circle pair whose  $\iota$ -transports are  $\mathbb{F}_{p^2}$ -valued up to a common scalar corresponds, after multiplying by the  $\mathbb{F}_{p^2}$ -valued diagonal  $t$ , to an  $\mathbb{F}_{p^2}$ -confined pair for  $\text{RS}[\mathbb{F}_{p^4}, \mathcal{D}, k_c]$ ; Lemma 7.10 preserves the bad-slope set, and Lemma 6.1 with  $B = \mathbb{F}_{p^2}$  bounds its density by  $p^2/q = p^{-2} < 2^{-61} < 2^{-23}$ . The final claim is Corollary 2.12 applied with  $B \in \{\mathbb{F}_p, \mathbb{F}_{p^2}\}$  to the  $B$ -valued received words and lists of Lemma 7.2.  $\square$

*Remark 7.15* (interleaving, slacked fallback, and the two-tier picture for circle rows). The interleaving-transfer lemma Lemma 3.84 is stated for arbitrary linear codes, so every bound of Corollaries 7.12 and 7.13 transfers verbatim to  $s$ -interleaved circle rows, exactly as in Corollary 5.5. The slacked BCHKS/ABF fallback also carries over: Lemma 7.2(i) supplies a list of size

$\geq \binom{N}{\ell_1}/|B|$  in  $\text{RS}[\mathbb{F}, D, k]$  at radius  $1 - \frac{A_1}{n}$ , and whenever  $A_1 \geq k + 3$  (automatic in the 2-power line rounds with  $a \geq 4$ , where  $A_1 = k + a$ ) and  $\binom{N}{\ell_1} \geq |B|q$ , the contrapositive of Theorem 2.16 gives  $\varepsilon_{\text{ca}}(C, 1 - \frac{A_1}{n} + \frac{2}{n}, 1 - \rho - \frac{1}{n}) \geq \frac{1}{2n}$ , as in Proposition 5.6. Finally, the two-tier summary of Remark 5.3 acquires a circle column: the deployed Mersenne-31 circle family carries the same kind of near-capacity negative band as the multiplicative rows — gap  $2^{-7}$  at the deployed sizes,  $2^{-9}$ – $2^{-11}$  universally — over every challenge field  $q < 2^{256}$  containing the relevant base field.

## 7.4 The lacunarity obstruction for genus-one (ECFFT) domains

*Remark 7.16* (what blocks the elliptic case). The proof of Lemma 7.2 works because  $\varphi$  is a *polynomial*: in the expansion  $\prod_{b \in A}(\varphi - b) = \sum_j (-1)^j e_j(A) \varphi^{\ell-j}$ , consecutive terms drop by a full  $a = \deg \varphi$  in degree, so the received word occupies the top two degree slots and everything else is a codeword sitting  $2a$  degrees down — gap  $\approx 2/N$ . Consider instead a rational folding map  $\psi = f/g$  of degree  $a$ , with  $e := \deg g < \deg f = a$  and  $g$  nonvanishing on  $D$ , whose fibers inside  $D$  are complete and of uniform size  $a$ . The natural locators are  $\Lambda_A := \prod_{b \in A}(f - bg)$ , of degree  $a\ell$ , vanishing on the  $\psi$ -fibers over  $A$ ; their expansion  $\sum_j (-1)^j e_j(A) f^{\ell-j} g^j$  has  $j$ -th term of degree  $a\ell - j(a - e)$ , so the slack-two window has width  $2(a - e)$  in degree and the certified gap is  $2(a - e)/n$ . Polynomial maps have  $e = 0$  and recover gap  $2/N$ . The FFTree domains of ECFFT [BCKL21] fold by  $x$ -coordinate maps of 2-isogenies of elliptic curves, which are rational of degree type  $(a, e) = (2, 1)$ ; every composition of such maps is again the  $x$ -map of an isogeny, of type  $(a, a - 1)$ . For these,  $a - e = 1$  at every scale: the window collapses to width 2, and the construction certifies only the first staircase step — a large list at agreement  $k + 2$ , hence an  $\varepsilon_{\text{ca}}$  floor at gap  $2/n$  — a nonvacuous but microscopic band, far from the  $2^{-9}$ – $2^{-11}$  bands above. The circle escapes this because its folding tower becomes polynomial after the  $x$ -projection:  $T_2$  fixes  $\infty$  and is totally ramified there. The elliptic  $x$ -maps also fix  $\infty$  but are *not* totally ramified there ( $\psi^{-1}(\infty) = \{\infty, x(T)\}$  for a 2-isogeny with kernel generator  $T$ ), and the locator expansion sees exactly this defect as  $e = a - 1$ . Obtaining a macroscopic universal cap for ECFFT rows therefore requires either a lacunary basis of isogeny locators, a different received-word family, or a matching safe-side theorem showing the collapse is essential. The second exit is realized by the rational graded floor of Proposition 11.1 (Corollary 11.2); the residual band form is recorded as Open Problem 12.4.

## 8 Threshold formulation and certificate interface

The Proximity Prize asks for a threshold, not merely for a near-capacity obstruction. We therefore use the integer staircase below to state both sides of the result: lower certificates mark agreements where MCA or list decoding is unsafe, while upper certificates mark agreements where the corresponding error is below the target. The paper does not determine the staircase to one step in the remaining band, but it reduces the unknown part to a single aperiodic correlated-agreement problem and gives exact finite certificates for all printed edges.

### 8.1 Established inputs

For the grand MCA challenge at target error  $\varepsilon^* = 2^{-128}$ , the unconditional output of Corollary 5.1 is the following negative certificate: for every challenge-envelope row satisfying the

printed divisor hypothesis,

$$\delta \geq 1 - \rho - 2^{-9} \quad (\rho \in \{1/2, 1/4, 1/8\})$$

and

$$\delta \geq 1 - \rho - 2^{-10} \quad (\rho = 1/16)$$

are impossible MCA radii. Equivalently, any full determination of  $\delta_C^*(2^{-128})$  for these smooth multiplicative rows must search below these gaps. This conclusion uses only the self-contained deep-point conversion, the quotient-fiber pigeonhole, and support-wise MCA monotonicity; it no longer rests on importing the Crites–Stewart conversion.

The theorem also distinguishes two different grades of information. First, the *threshold grade* is already enough for the prize inequality: the certified error is  $\gg 2^{-128}$ . Second, the *failure-profile grade* is still weak: the present proof gives error of order  $1/k$ , not error  $1 - o(1)$ . Closing the error-one profile at fixed gaps such as  $1/64$  is a different, stronger problem. It would improve our understanding of the failure landscape, but is not needed to obtain the stated upper bound on  $\delta_C^*(2^{-128})$ .

The negative certificate also covers a wider row family. The same conclusion — an unsafe near-capacity band at error  $\gg 2^{-128}$  — now holds for mixed-radix multiplicative rows without the integrality condition  $\rho N \in \mathbb{Z}$  (Corollary 7.6, gaps  $2^{-11}/2^{-12}$ ), for the  $x$ -coordinate line rounds of circle FRI over any  $p \equiv 3 \pmod{4}$ , and for the bivariate circle codes over any challenge field containing  $i$  (Corollary 7.12, gaps  $2^{-9}$ – $2^{-11}$ ), including the deployed Mersenne-31/QM31 family at gap  $2^{-7}$  (Corollary 7.13). These additions use the same three ingredients; the only new inputs are the exact Chebyshev-fiber geometry of twin cosets and the torus uniformization, both elementary.

The treatment in Section 9 adds the safe side. Theorem 9.6 certifies  $\varepsilon_{\text{mca}}(C, \delta) \leq (\lfloor \delta n \rfloor + 1)/q$  whenever  $3\lfloor \delta n \rfloor \leq n - k$ , for every row at once, and Theorem 9.14 certifies the interleaved-list challenge up to the Johnson radius uniformly in the arity; Table 4 records the per-obligation outcome, and Theorem 9.26 assembles the resulting two-sided sandwich.

## 8.2 Integer-staircase formulation

Let

$$a(\delta) := \lceil (1 - \delta)n \rceil, \quad r(\delta) := n - a(\delta)$$

be the agreement and integer-radius forms of the same radius. For a fixed finite-slope line field  $F_{\text{line}}$ , write  $q_{\text{line}} = |F_{\text{line}}|$  and let

$$B_{\text{mca}}(a)$$

be the exact maximum number of MCA-bad line parameters at agreement at least  $a$ , for the line family used in the challenge. Then

$$\varepsilon_{\text{mca}}(C, 1 - a/n) = B_{\text{mca}}(a)/q_{\text{line}}$$

under the closed finite-slope convention. For projective-slope or curve samplers, the same staircase formulation applies after replacing  $q_{\text{line}}$  and  $B_{\text{mca}}$  by the corresponding sampler denominator and numerator. Thus the MCA prize problem is an integer staircase problem: find the first agreement  $a$  for which

$$B_{\text{mca}}(a) \leq 2^{-128} q_{\text{line}}.$$

The present paper supplies lower bounds on  $B_{\text{mca}}(a)$  near capacity by routing quotient-fiber list mass through deep points.

**Proposition 8.1** (one-step threshold certificate). *Fix a row  $C$  and a denominator  $q_{\text{line}}$ . Suppose that for some agreement  $a_0$  one has a lower certificate*

$$B_{\text{mca}}(a_0) > 2^{-128} q_{\text{line}}$$

*and an upper theorem*

$$B_{\text{mca}}(a_0 + 1) \leq 2^{-128} q_{\text{line}}.$$

*Then, under the closed integer-radius convention, radius  $1 - a_0/n$  is unsafe and radius  $1 - (a_0 + 1)/n$  is safe. As a real-radius supremum, the threshold is  $1 - a_0/n$  and is not attained if the closed ball at that endpoint is counted.*

*Proof.* The function  $B_{\text{mca}}(a)$  is nonincreasing in  $a$ , since raising the required agreement only removes witnesses. At the closed radius  $1 - a/n$ , the MCA error is  $B_{\text{mca}}(a)/q_{\text{line}}$ . The two displayed inequalities are therefore exactly the unsafe/safe transition at consecutive integer agreement levels.  $\square$

### 8.3 Conditional MCA one-step criterion

The following theorem is a deliberately explicit proof blueprint. Each hypothesis is a concrete mathematical task, not a black-box “large field” assumption.

**Theorem 8.2** (conditional MCA threshold theorem). *Fix a smooth multiplicative row  $C = \text{RS}[F, D, k]$ , a line field  $F_{\text{line}}$ , and a target  $\varepsilon^* = 2^{-128}$ . Suppose the following four inputs are proved for this row, uniformly in the received line:*

- (i) **Quotient-profile lower and upper ledger.** *For every divisor  $c \mid n$  and every agreement  $a = mc + s$  with  $0 \leq s < c$ , the quotient-periodic bad slopes, including remainder supports consisting of  $m$  full fibers plus  $s$  residual points, are counted with overlaps across divisors accounted for. Write the resulting divisor-lattice upper numerator as  $B_Q^{\text{all}}(a)$ ; individual divisor contributions  $B_Q(a; c)$  may also be printed, but the upper ledger must count their union or a certified safe sum rather than only their maximum.*
- (ii) **Aperiodic residue-line packing.** *After removing tangent and quotient-periodic locators, the Hankel/residue-line incidence variety has at most  $B_{\text{ap}}(a)$  bad line parameters.*
- (iii) **Extension-line accounting.** *If  $F_{\text{line}}$  is a proper extension of the field generated by  $D$ , either all genuinely extension-valued line witnesses are included in  $B_{\text{ap}}(a)$ , or a separate extension term  $B_{\text{ext}}(a)$  is proved.*
- (iv) **No missing line family.** *The line family in the theorem is the same finite-slope or projective-slope family used by the challenge or protocol under consideration.*

*Assume moreover that the resulting upper numerator*

$$B^+(a) := B_{\text{tan}}(a) + B_Q^{\text{all}}(a) + B_{\text{ap}}(a) + B_{\text{ext}}(a)$$

*and a corresponding lower numerator  $B^-(a)$  satisfy*

$$B^-(a_0) > \varepsilon^* q_{\text{line}}, \quad B^+(a_0 + 1) \leq \varepsilon^* q_{\text{line}}$$

*for some  $a_0$ . Then  $\delta_C^*(\varepsilon^*)$  is determined at the one-step accuracy of Proposition 8.1.*

*Proof.* The four hypotheses classify every possible MCA-bad line parameter into one of four buckets: tangent, quotient-periodic, aperiodic, or genuinely extension-line. The quotient bucket is counted by the divisor-lattice union ledger  $B_Q^{\text{all}}$ , so overlaps or simultaneous quotient descriptions cannot invalidate the upper bound. The displayed  $B^+(a)$  is therefore an upper bound for  $B_{\text{mca}}(a)$ , while  $B^-(a)$  is a certified lower bound. The conclusion is exactly Proposition 8.1.  $\square$

The present paper supplies part of the lower numerator  $B^-(a)$  in the near-capacity band: quotient-fiber list mass forces MCA mass above the printed universal caps. It does not prove the aperiodic upper input (ii), and therefore does not by itself determine the exact staircase.

#### 8.4 Conditional interleaved-list one-step criterion

The grand list-decoding challenge asks for the largest  $\delta$  such that an interleaved list, divided by the field used for the relevant challenge or list ledger, is below  $2^{-128}$ . In the simplest same-field form this is

$$|\Lambda(C^{\equiv m}, \delta)| \leq 2^{-128}|F|.$$

The quotient-fiber construction in this paper already gives lower floors for ordinary lists, and ordinary lists embed into interleaved lists by placing the same received word in one row and zero words in the others. On the MCA side, Section 3.4 supplies a certified eliminant route for aperiodic Hankel packing, but it does not prove that the needed eliminants always exist. The missing positive list direction is an arbitrary-word locator-fiber theorem.

**Theorem 8.3** (conditional interleaved-list threshold theorem). *Fix an interleaving arity  $m$ . Suppose that for every received word  $U$  and agreement  $a = k + \sigma$  one has a base-code list bound of the form*

$$|\Lambda(C, 1 - a/n, U)| \leq L_Q(a, U) + L_{\text{ap}}(a),$$

*where  $L_Q$  is an explicit quotient-profile list contribution and  $L_{\text{ap}}$  is a uniform aperiodic locator-fiber bound. Suppose also that the same bound is available at the higher agreement levels that occur in the codegree recursion, starting with  $2a - k$ . Then the interleaved-list threshold is obtained by applying the codegree recursion row by row and comparing the resulting numerator to  $2^{-128}|F|$ .*

*Proof.* For two rows, split the second-row list according to whether its agreement set has size below or above  $2a - k$ . Below  $2a - k$ , two first-row codewords would agree on more than  $k$  positions and hence are equal; above  $2a - k$ , use the base-code list bound at agreement  $2a - k$ . Iterating gives the stated  $m$ -row recursion. This is the same codegree mechanism used in the experimental ledger; the only missing input is the arbitrary-word locator-fiber bound.  $\square$

#### 8.5 Residual ingredients for one-step threshold certificates

The one-step criteria above consume the following residual ingredients, which we record with their current status. The lower-floor half of the periodic-support ledger is supplied in scanner-ready form by Lemma 3.7 and Theorem 3.8: every quotient-plus-remainder agreement  $a = mc + s$  has a generated-field prefix-fiber floor, an image-size fallback, an ambient-box fallback, and a converted CA/MCA lower numerator. The support-family upper half is exact at the divisor-lattice level by Theorem 3.24: overlaps between quotient descriptions at different divisor scales are counted by a common-block coefficient extraction rather than by a raw safe sum. The



distinct-parameter branch image is exact inside every declared quotient family by the lcm ledger of Theorem 3.36 and Corollaries 3.37 and 3.39; duplicate finite slopes, projective slopes, and finite curve parameters are coalesced before the numerator is printed. The residual-band analysis is split into four certified pieces. The regular overdetermined nonsingular branch is canonical by Theorems 3.48 and 3.55 and Corollary 3.51. The underdetermined and rank-drop singular branches are covered by the rank-pivot certificate theorem Theorems 3.64 and 3.70. The new compressed-residual layer adds the kernel-compressed eliminant construction of Theorem 3.73 and Corollary 3.75 and the exact arbitrary-residual support-image lcm ledger of Theorem 3.76 and Corollary 3.77. Thus a residual chart must now be discharged by a small compressed eliminant, by an exact residual image polynomial, by a Weil-restricted extension projection bound, or by a named structural obstruction. Extension-line accounting is handled on the safe side by Theorem 3.68: after choosing a  $B$ -basis of  $F$ , each extension chart prints a parameter-projection degree and dimension over  $B$ , or an exact zero-dimensional point count. The curve-sampler ledger is upgraded for finite-parameter curves and projective lines in parallel: quotient branches have exact gcd/lcm image ledgers, regular curve rank-drop branches have canonical maximal-minor gcd ledgers, and residual curve branches have both coefficient-pivot rank charts and kernel-compressed curve eliminants. What remains open is no longer a bookkeeping ambiguity: prove that the compressed residual eliminants are small enough in the near-capacity rows, or classify any large residual image as quotient, tangent/common-line, base-field exceptional, genuinely extension-valued, or a new obstruction.

- T1. Remainder quotient-profile theorem.** Generalize the fiber lemma from full quotient-fiber unions to all agreements  $a = mc + s$ . The target is an exact lower and upper quotient ledger over the divisor lattice of  $D$ . The present quotient-ledger package supplies the scanner-ready lower half via heaviest prefix fibers and the support-union upper half via the exact divisor-block coefficient formula; inside a declared quotient family, the gcd/lcm image ledger also coalesces duplicate finite, projective, and curve parameters. What remains is the structural exhaustion theorem saying that the declared quotient family captures all quotient-periodic bad parameters in the intended regime, with the subfield denominator  $|B|$  printed separately from the line denominator  $q$ .
- T2. Aperiodic Hankel-packing theorem.** Use the syndrome/Hankel normal form of [Cho26b] to prove that, after quotient-periodic and tangent locators are removed, the number of remaining bad line parameters is at most  $B_{\text{ap}}(a)$ . The regular overdetermined nonsingular branch is now canonical by Theorem 3.48 and Corollary 3.51: a scanner takes the gcd of all nonzero maximal Hankel minors and then an lcm over exact agreement levels. The underdetermined and rank-drop singular buckets now have the rank-pivot cover of Theorem 3.64, projective-infinity and curve coefficient pivots, and the kernel-compressed eliminant ideals of Theorem 3.73 and Corollary 3.75. The remaining proof problem is constructive and structural: produce small eliminants for these compressed residual charts, use the exact residual image lcm of Theorem 3.76 when finite enumeration is feasible, or classify every large residual component as quotient, tangent/common-line, base-field exceptional, genuinely extension-valued, or a new obstruction.
- T3. Extension-line lift or counterexample.** For  $B \subset F$  and  $D \subset B$ , either prove that genuinely  $F$ -valued lines are covered by the aperiodic ledger with a stable numerator, or exhibit explicit extension-valued counterexamples. Corollary 6.2 shows that deployed failures certified by this paper must be genuinely extension-valued; Corollaries 2.12 and 2.13

provide an explicit lower-side mechanism from any large list by taking the pole in  $F \setminus B$ . On the safe side, Theorem 3.68 gives the certificate format: Weil-restrict each residual extension chart to  $B$ , print a parameter-projection degree/dimension pair or an exact zero-dimensional point count, and divide by the line field size. The remaining structural problem is to prove that all genuinely extension-valued residual charts have small projected dimension/degree or to exhibit the chart that violates this expectation.

- T4. Arbitrary-word locator-fiber theorem.** Prove the base-code list bound needed by Theorem 8.3, first for monomial-prefix received words and quotient-free dimensions, then for arbitrary received words with quotient cores budgeted.
- T5. Curve and projective-line ledgers.** Extend the affine-line Hankel classification to projective slopes and to finite-parameter power curves  $f_0 + \gamma f_1 + \cdots + \gamma^d f_d$ . The projective and curve ledger package gives exact gcd/lcm ledgers for projective and finite-curve quotient branches, canonical rank-drop gcfs for regular overdetermined finite-curve buckets, projective-infinity residual charts, coefficient-pivot curve rank charts, and kernel-compressed projective/curve eliminant certificates. The remaining work is to prove smallness or structural classification for the residual projective and curve charts, including exceptional vanishing-vector strata and genuinely extension-valued components.
- T6. Certificate scanner.** Implement a finite staircase scanner that, for each challenge row, prints tangent, quotient, generated-field entropy, aperiodic, extension, curve, and interleaved-list numerators at every agreement  $a$ . The desired output is an interval

$$a_{\text{unsafe}} < a_C^* \leq a_{\text{safe}}$$

that shrinks to one step when Theorem 8.2 or Theorem 8.3 applies.

- T7. Formal finite core.** Formalize the self-contained deep-point conversion, the quotient-fiber pigeonhole, support-wise MCA monotonicity, and the integer staircase lemma. These are small enough for proof-assistant checking and directly strengthen a formal certificate package.
- T8. Circle-row completion and genus-one floors.** Extend the quotient-support and quotient-image upper ledgers (item 1) and the aperiodic Hankel classification (item 2) from power locators on multiplicative divisor lattices to Chebyshev locator towers on  $x$ -coordinate twin-coset domains, where the divisor lattice of  $D$  is replaced by the lattice of dyadic Chebyshev scales; and settle the genus-one lower-floor question of Open Problem 12.4, for which the lacunarity collapse of Remark 7.16 blocks the present route beyond the first staircase step.

## 8.6 Scope of claims

We state precisely what this paper establishes and what it does not. Established: a universal unsafe band at capacity for the grand MCA challenge, over smooth multiplicative rows in the stated field-size envelope (Corollaries 5.1 and 7.6) and, by the later sections, over Chebyshev, circle, and genus-one rows as well (Corollaries 7.12, 11.2 and 11.5); an unconditional safe region below one third of the distance (Theorem 9.6) with a list-side safe region up to the Johnson radius, uniform in the interleaving arity (Theorem 9.14); a safe region up to one half of the



distance modulo a single classical import (Theorem 10.1 and Corollary 10.7); and the resulting sandwich, which determines the grand threshold of every deployed row within a factor smaller than two modulo that import, and within a factor of three unconditionally (Theorems 9.26 and 10.12, Table 5). Not established, and not claimed: the exact threshold, a positive theorem inside the remaining open band, protocol soundness, or a curve-MCA theorem. What separates the sandwich from a one-step determination is precisely Open Problem 9.11.

## 9 Structural ledgers, residual bands, and certificate machinery

This section collects the technical infrastructure used by the two-sided threshold statements, organized by mathematical function. Stabilizer and quotient-descent lemmas isolate the periodic supports and make their numerators field-size independent. The deep-regime theorem controls MCA directly below one third of the distance and removes the singular Hankel branches there. The restriction-of-scalars lift shows that extension-valued lines are not a separate class of witnesses. The Johnson and recursion statements control the list side outside the band. The curve theorem covers projective and finite-parameter samplers in the deep regime. Finally, the scanner and finite-certificate grammar package all deployed verdicts as exact integer comparisons. The only mathematical component not closed by these ledgers is the aperiodic band problem, Open Problem 9.11 below.

### 9.1 Stabilizer exhaustion and exact quotient descent

Throughout this subsection, *multiplicative case* means:  $D = \alpha H$  is a multiplicative coset of a cyclic group  $H$  of order  $n$ , acting freely on  $D$  by multiplication; for  $c \mid n$ ,  $H_c \leq H$  is the unique subgroup of order  $c$ , equal to the full group  $\mu_c$  of  $c$ -th roots of unity in  $B$ , and the fibers of  $\pi_c : x \mapsto x^c$  on  $D$  are exactly the  $H_c$ -orbits. *Chebyshev case* means:  $D = \chi(\mathcal{D})$  for a twin coset  $\mathcal{D} = gH \cup g^{-1}H$  of size  $2M$  with  $\text{ord}(g) \nmid 2M$  as in Section 7.2; for  $c \mid M$ ,  $K_c \leq \mathbb{U}$  is the order- $c$  kernel of the  $c$ -th power map, and for  $S \subseteq D$  we write  $\tilde{S} := \chi^{-1}(S) \cap \mathcal{D}$  for the (automatically inversion-closed) lift. By Lemma 7.8, the  $T_c$ -fibers in  $D$  are exactly the  $\chi$ -images of the  $\langle K_c, u \mapsto u^{-1} \rangle$ -orbits on  $\mathcal{D}$ . In both cases the *scale lattice* is the divisor lattice of  $n$  (resp. of  $M$ ).

**Definition 9.1** (periodicity scale of a support). For  $S \subseteq D$ , the periodicity scale  $c(S)$  is  $|\{h \in H : hS = S\}|$  in the multiplicative case and  $|\{\zeta \in K_M : \zeta\tilde{S} = \tilde{S}\}|$  in the Chebyshev case. A support is *aperiodic* if  $c(S) = 1$ .

**Theorem 9.2** (stabilizer partition and structural exhaustion). *In either case:*

- (i) *The stabilizer in the definition is a subgroup of a cyclic group, so  $c(S)$  divides  $n$  (resp.  $M$ ), and for every scale  $c$ ,*

$$S \text{ is a union of complete scale-}c \text{ fibers} \iff c \mid c(S).$$

*In particular  $c(S)$  is the unique maximal such scale, and the supports of each size are partitioned by the value of  $c(S)$ , with  $c(S) = 1$  the aperiodic class.*

- (ii) *Fix a received line and an agreement threshold  $a \geq k$ , and call a bad parameter periodically witnessed if at least one of its witness supports  $S$  (in the sense of Definitions 2.1 and 2.2) has  $c(S) \geq 2$ . Then every bad parameter is either periodically witnessed — and hence*

counted, with multiplicity one per support, by the family of supports of size  $\geq a$  that are unions of complete fibers at some scale  $\geq 2$  — or all of its witness supports are aperiodic, in which case it belongs to the aperiodic bucket by definition. The two classes are exhaustive and disjoint.

*Proof.* (i) The set  $\{h \in H : hS = S\}$  is a subgroup of the cyclic group  $H$  (resp.  $\{\zeta \in K_M : \zeta\tilde{S} = \tilde{S}\} \leq K_M$ ), hence equals  $H_{c_0}$  (resp.  $K_{c_0}$ ) for a unique divisor  $c_0$ , which is  $c(S)$ . In the multiplicative case, the  $\pi_c$ -fibers in  $D$  are the  $H_c$ -orbits, and a set is a union of complete  $H_c$ -orbits if and only if it is  $H_c$ -invariant, i.e.  $H_c \leq H_{c_0}$ , i.e.  $c \mid c_0$ . In the Chebyshev case,  $S$  is a union of complete  $T_c$ -fibers if and only if  $\tilde{S}$  is a union of complete  $\langle K_c, \text{inv} \rangle$ -orbits; since  $\tilde{S}$  is inversion-closed automatically, this holds if and only if  $\tilde{S}$  is  $K_c$ -invariant, i.e.  $c \mid c_0$ . The partition statement is immediate.

(ii) Exhaustiveness and disjointness are definitional given (i). The counting clause is Lemma 3.16 applied to the support family  $\{S : |S| \geq a, c(S) \geq 2\}$ : each fixed support pays for at most one finite line parameter, so the number of periodically witnessed bad parameters is at most the size of that family.  $\square$

**Corollary 9.3** (exact size of the periodic support family;  $q$ -free numerator). *Let  $P$  be the set of primes dividing  $n$  (resp.  $M$ ). The number of supports of exact size  $A$  with  $c(S) \geq 2$  is*

$$\Pi(A) := \sum_{\emptyset \neq T \subseteq P} (-1)^{|T|+1} \mathbf{1}[\text{lcm}(T) \mid A] \binom{n/\text{lcm}(T)}{A/\text{lcm}(T)},$$

and consequently, for every received line and every agreement threshold  $a \geq k$ ,

$$\#\{\text{periodically witnessed bad finite parameters}\} \leq \sum_{A=a}^n \Pi(A),$$

with an extra factor  $d$  for degree- $d$  power-curve samplers (Lemma 3.17). The numerator is independent of  $q$  and of  $|B|$ ; the subfield order enters only the lower floors, as the periodic-support ledger demands.

*Proof.*  $c(S) \geq 2$  if and only if  $H_p S = S$  for some  $p \in P$ , i.e.  $S$  lies in the union over  $p \in P$  of the families of complete- $p$ -fiber unions. For  $T \subseteq P$ , invariance under every  $H_p$ ,  $p \in T$ , is invariance under the subgroup they generate, which is  $H_{\text{lcm}(T)}$ ; the number of unions of complete  $\text{lcm}(T)$ -fibers of total size  $A$  is  $\binom{n/\text{lcm}(T)}{A/\text{lcm}(T)}$  when  $\text{lcm}(T) \mid A$  and 0 otherwise. Inclusion–exclusion gives  $\Pi(A)$ ; the parameter bound is Theorem 9.2(ii) (resp. Lemma 3.17). The Chebyshev case is identical with  $K_p \leq K_M$ .  $\square$

This complements Theorem 3.24: that formula counts a *declared* remainder-profile family exactly; Corollary 9.3 counts the *intrinsic* periodic family exactly, and Theorem 9.2 proves that nothing quotient-periodic escapes it. What the periodic-support ledger still lacked was the passage from support-level to image-level counting. The next theorem supplies it exactly on the class of received data for which the quotient family was designed.

**Theorem 9.4** (exact quotient descent for fiber-constant data). *Work in the multiplicative or Chebyshev case, write  $\varphi_c$  for the scale- $c$  map ( $X^c$ , resp.  $T_c$ ) and  $Q_c := \varphi_c(D)$ ,  $N_c := n/c$ . Let  $c$*

be a scale, let  $p \in \mathbb{F}[X]_{<k}$ , let  $S \subseteq D$  be a union of complete  $c$ -fibers with  $|S| \geq k$ , and suppose  $p$  agrees on  $S$  with a fiber-constant word  $W \circ \varphi_c$ . Then there is a unique  $g \in \mathbb{F}[Y]$  with

$$p = g \circ \varphi_c, \quad \deg g < \lceil k/c \rceil,$$

and  $g$  agrees with  $W$  on  $\varphi_c(S)$ , a set of  $|S|/c$  points of  $Q_c$ . Consequently, for a received line  $f + zg_0$  with  $f$  and  $g_0$  both constant on  $c$ -fibers, and any agreement threshold  $a \geq k$ : a finite slope  $z$  is MCA-bad with some witness support that is a union of complete  $c$ -fibers if and only if  $z$  is MCA-bad for the quotient line  $(f_c, g_{0,c})$  on  $\text{RS}[\mathbb{F}, Q_c, \lceil k/c \rceil]$  at the corresponding agreement  $\geq \lceil a/c \rceil$ , with witness  $\varphi_c(S)$ ; and the same holds verbatim for lists:  $p \mapsto g$  is a bijection between scale- $c$ -witnessed list elements and quotient list elements.

*Proof. Multiplicative case.* For  $h \in H_c = \mu_c$  and  $x \in S$  we have  $hx \in S$  and  $\varphi_c(hx) = \varphi_c(x)$ , so  $p(hx) = W(\varphi_c(hx)) = W(\varphi_c(x)) = p(x)$ . Thus  $p(hX) - p(X)$ , of degree  $< k$ , has at least  $|S| \geq k$  roots, hence vanishes identically:  $p(hX) = p(X)$  for every  $h \in \mu_c$ . Comparing coefficients,  $a_i h^i = a_i$  for all  $h \in \mu_c$ , so  $a_i = 0$  whenever  $c \nmid i$  (take  $h$  a generator of  $\mu_c$ ), i.e.  $p = g(X^c)$  with  $c \deg g = \deg p < k$ , so  $\deg g < \lceil k/c \rceil$ ; uniqueness is clear. For  $x \in S$ ,  $g(\varphi_c(x)) = p(x) = W(\varphi_c(x))$ , and  $\varphi_c(S)$  has  $|S|/c$  elements since  $S$  is a union of complete  $c$ -fibers.

*Chebyshev case.* Work over the compositum  $\mathbb{F}' := \mathbb{F} \cdot \mathbb{F}_{p^2}$  and put  $P(u) := p(\chi(u)) \in \mathbb{F}'[u, u^{-1}]$ , a Laurent polynomial with exponents in  $[-(k-1), k-1]$ . For  $\zeta \in K_c = \mu_c$  and  $u \in \tilde{S}$  we have  $\zeta u \in \tilde{S}$  and  $T_c(\chi(\zeta u)) = \chi(\zeta^c u^c) = T_c(\chi(u))$ , so  $P(\zeta u) = P(u)$  on  $\tilde{S}$ ; since  $|\tilde{S}| = 2|S| \geq 2k$  exceeds the degree  $2(k-1)$  of  $u^{k-1}(P(\zeta u) - P(u))$ , the identity  $P(\zeta u) = P(u)$  holds in  $\mathbb{F}'[u, u^{-1}]$ . Hence only exponents divisible by  $c$  occur, and  $P$  is also inversion-symmetric because  $\chi$  is, so  $P$  lies in the symmetric Laurent ring  $\mathbb{F}'[u^c + u^{-c}]$ :  $P(u) = G(u^c + u^{-c})$  for a unique  $G$ . Since  $u^c + u^{-c} = 2T_c(\chi(u))$ , setting  $g(Y) := G(2Y)$  gives  $p(\chi(u)) = g(T_c(\chi(u)))$  identically, and injectivity of the ring map  $r(x) \mapsto r((u+u^{-1})/2)$  yields  $p = g \circ T_c$  in  $\mathbb{F}'[X]$ ; as  $g$  is the unique solution of an  $\mathbb{F}$ -linear system with data in  $\mathbb{F}$ , in fact  $g \in \mathbb{F}[Y]$ , and  $c \deg g = \deg p < k$  as before. Agreement on  $\varphi_c(S) = T_c(S)$  follows as above, with  $|T_c(S)| = |S|/c$  by Lemma 7.8.

*Lines and lists.* If  $z$  has an MCA witness  $S$  that is a union of complete  $c$ -fibers, the explaining codeword for the point  $f + zg_0$  (a fiber-constant word) descends by the above; conversely a quotient explanation composes with  $\varphi_c$ . For noncontainment, if the pair  $(f, g_0)$  were explained on  $S$  by  $(p_1, p_2)$ , both  $p_i$  descend (each agrees on  $S$  with a fiber-constant word), so the quotient pair would be explained on  $\varphi_c(S)$ ; and conversely quotient explanations compose. Hence badness with a scale- $c$  witness corresponds exactly. For lists,  $p \mapsto g$  is well defined and injective (uniqueness of  $g$ ), and  $g \mapsto g \circ \varphi_c$  is its inverse from the quotient list, since the full agreement set of  $g \circ \varphi_c$  with  $W \circ \varphi_c$  is the full  $\varphi_c$ -preimage of the agreement set of  $g$  with  $W$ .  $\square$

**Corollary 9.5** (structural exhaustion is exact). *With the intrinsic Definition 9.1, the structural exhaustion is a theorem: every bad parameter is quotient-periodic (periodically witnessed, hence counted exactly at the support level by Corollary 9.3, and exactly at the image level by Theorem 9.4 whenever the received line is fiber-constant) or belongs to the aperiodic bucket. The subfield order  $|B|$  appears only in lower floors; all upper numerators above are  $q$ -free. The entire residual content of the periodic-support ledger is thereby the aperiodic bucket, treated next.*

## 9.2 The deep regime, singular pencils, and the residual band

The following theorem is the safe-side counterpart of the paper's unsafe caps. It is self-contained, uses nothing beyond interpolation and double counting, and holds for arbitrary linear codes —

in particular for every multiplicative row, every Chebyshev line-round row, every bivariate circle code over any field (including fields not containing  $i$ ), and every ECFFT row.

**Theorem 9.6** (deep-regime mutual correlated agreement, unconditional). *Let  $C \subseteq \mathbb{F}^n$  be any  $\mathbb{F}$ -linear code of minimum Hamming weight  $w$ , let  $q = |\mathbb{F}|$ , let  $\delta \in (0, 1)$ , and put  $r := \lfloor \delta n \rfloor$ . Assume*

$$3r \leq w - 1.$$

*Then for every pair  $(f_1, f_2) \in \mathbb{F}^n \times \mathbb{F}^n$ , at least one of the following alternatives holds:*

- (a) (Tangent pair.) *There are codewords  $c_1, c_2 \in C$  and a set  $T' \subseteq \{1, \dots, n\}$  with  $|T'| \leq r$  such that  $f_i = c_i$  off  $T'$  for  $i = 1, 2$ ; in this case  $(f_1, f_2)$  has no CA-bad slope at radii  $(\delta, \delta_{\text{int}})$  for any  $\delta_{\text{int}} \geq \delta$ , and at most  $|T'| \leq r$  MCA-bad slopes at radius  $\delta$ , namely a subset of  $\{-e_{1,j}/e_{2,j} : j \in T', e_{2,j} \neq 0\}$  where  $e_i := f_i - c_i$ .*
- (b) (Generic pair.)  $\#\{\gamma \in \mathbb{F} : \Delta(f_1 + \gamma f_2, C) \leq \delta\} \leq r + 1$ .

*Consequently*

$$\varepsilon_{\text{ca}}(C, \delta) \leq \frac{r+1}{q}, \quad \varepsilon_{\text{mca}}(C, \delta) \leq \frac{r+1}{q}.$$

*For  $C = \text{RS}[\mathbb{F}, D, k]$  on any evaluation set  $D$  of size  $n$ , the hypothesis reads  $3\lfloor \delta n \rfloor \leq n - k$ .*

*Proof.* Write  $S_{\text{cl}} := \{\gamma : \Delta(f_1 + \gamma f_2, C) \leq \delta\}$ ; note  $\Delta \leq \delta$  means disagreement on at most  $r$  coordinates, by integrality. If  $|S_{\text{cl}}| \leq 1$ , alternative (b) holds ( $r + 1 \geq 1$ ). Otherwise fix  $\gamma_1 \neq \gamma_2$  in  $S_{\text{cl}}$  with codewords  $p_1, p_2$  and error sets  $E_1, E_2$  ( $|E_i| \leq r$ ), and set

$$c_2 := \frac{p_1 - p_2}{\gamma_1 - \gamma_2}, \quad c_1 := p_1 - \gamma_1 c_2, \quad e_i := f_i - c_i.$$

Off  $T := E_1 \cup E_2$  ( $|T| \leq 2r$ ) we can solve the two linear equations  $f_1 + \gamma_i f_2 = p_i$  coordinatewise, giving  $f_2 = c_2$  and  $f_1 = c_1$  there; hence  $e_1, e_2$  are supported in  $T$ . Let  $T' := \{j : (e_{1,j}, e_{2,j}) \neq (0, 0)\} \subseteq T$ .

*Step 1: every close slope rides the recovered line.* Let  $\gamma \in S_{\text{cl}}$  with codeword  $p$  and error set  $E$ ,  $|E| \leq r$ . Off  $T \cup E$  we have  $p = f_1 + \gamma f_2 = c_1 + \gamma c_2$ , so the codeword  $p - (c_1 + \gamma c_2)$  has weight at most  $|T| + |E| \leq 3r \leq w - 1 < w$  and is therefore zero. Hence

$$f_1 + \gamma f_2 - (c_1 + \gamma c_2) = e_1 + \gamma e_2 \quad \text{has weight at most } r \quad \text{for every } \gamma \in S_{\text{cl}}. \quad (*)$$

*Step 2: dichotomy.* Suppose  $|T'| \geq r + 1$ . For  $j \in T'$ , the affine function  $\gamma \mapsto e_{1,j} + \gamma e_{2,j}$  is not identically zero, so it vanishes for at most one  $\gamma$ . By (\*), each  $\gamma \in S_{\text{cl}}$  needs at least  $|T'| - r \geq 1$  vanishing coordinates inside  $T'$ . Double counting vanishings,

$$|S_{\text{cl}}| (|T'| - r) \leq |T'|, \quad \text{hence} \quad |S_{\text{cl}}| \leq \frac{|T'|}{|T'| - r} \leq r + 1,$$

which is alternative (b). Otherwise  $|T'| \leq r$  and  $(f_1, f_2)$  is a tangent pair as in (a) with this  $T'$ .

*Step 3: bad slopes of tangent pairs.* Let  $(f_1, f_2)$  be tangent with data  $(c_1, c_2, T')$ ,  $|T'| \leq r$ . First,  $\Delta_2((f_1, f_2), C^{\equiv 2}) \leq |T'|/n \leq r/n \leq \delta \leq \delta_{\text{int}}$ , so no slope is CA-bad at radii  $(\delta, \delta_{\text{int}})$ . For MCA, note  $2r \leq w - 1$  (from  $3r \leq w - 1$  and  $r \geq 0$ ), so two codewords agreeing on at least  $n - 2r$  coordinates are equal. Let  $\gamma$  be MCA-bad with witness  $S$ ,  $|S| \geq n - r$ . On  $S \setminus T'$ , which has at least  $n - 2r$  elements,  $f_1 + \gamma f_2 = c_1 + \gamma c_2$ ; hence the codeword explaining the point

on  $S$  equals  $c_1 + \gamma c_2$ , and thus  $e_1 + \gamma e_2$  vanishes on  $S \cap T'$ . Moreover, any pair explanation  $(p'_1, p'_2)$  on  $S$  satisfies  $p'_i = c_i$  on the  $\geq n - 2r$  points of  $S \setminus T'$ , forcing  $p'_i = c_i$ ; so the pair fails to be explained on  $S$  exactly when  $S \cap T' \neq \emptyset$  (every  $j \in T'$  has  $(e_{1,j}, e_{2,j}) \neq 0$ ). Therefore  $\gamma$  is MCA-bad if and only if  $e_{1,j} + \gamma e_{2,j} = 0$  for some  $j \in T'$ : necessity was just shown; for sufficiency take  $S := (\{1, \dots, n\} \setminus T') \cup \{j\}$ , of size  $\geq n - r$ , on which  $c_1 + \gamma c_2$  explains the point while the pair is not explained. Each  $j \in T'$  contributes at most one slope, giving at most  $|T'| \leq r$  MCA-bad slopes.

Finally, generic pairs have at most  $r + 1$  close slopes, hence at most  $r + 1$  CA- or MCA-bad slopes (an MCA witness of size  $\geq n - r$  implies  $\Delta \leq r/n \leq \delta$ ); tangent pairs have 0 CA-bad and at most  $r$  MCA-bad slopes. Taking the maximum over pairs gives the two displayed error bounds. For  $\text{RS}[\mathbb{F}, D, k]$ ,  $w = n - k + 1$ , so  $3r \leq w - 1$  is  $3r \leq n - k$ .  $\square$

*Remark 9.7* (two radii; scope). The CA statement holds in the two-radius form: for any  $\delta_{\text{int}} \geq \delta_{\text{fld}} = \delta$  with  $3\lfloor \delta n \rfloor \leq w - 1$ ,  $\varepsilon_{\text{ca}}(C, \delta, \delta_{\text{int}}) \leq (r + 1)/q$ , since the tangent alternative kills the far condition at every  $\delta_{\text{int}} \geq \delta$ . The constant  $\frac{1}{3}$  can be improved toward  $\frac{1}{2}$  of the distance by the methods of [BCHKS25]; we deliberately keep the three-line-per-step, formalization-ready proof — it feeds the formal finite core — and record the import separately. Note also that Theorem 9.6 applies to the bivariate circle code over  $\mathbb{F}_p$  itself (minimum weight  $n_c - k_c + 1$ , by extending scalars to  $\mathbb{F}_{p^2}$  and using Lemma 7.11), so the deep regime of Open Problem 12.4(b) is closed as well.

We next resolve a named obstruction of the residual-band analysis — the singular Hankel buckets — in the regime where Theorem 9.6 applies. First, the structural normal form of the identically singular branch.

**Lemma 9.8** (identically singular pencils are locator families). *In the setting of Section 3.4, fix an exact agreement  $A$  with  $t_A \geq j_A + 1$  and set  $M(Z) := H_{t_A, j_A}(u) + ZH_{t_A, j_A}(v)$ . The following are equivalent:*

- (i) *every  $(j_A + 1) \times (j_A + 1)$  minor of  $M(Z)$  vanishes identically in  $Z$ ;*
- (ii) *there is a nonzero polynomial vector  $\tilde{v}(Z) = \sum_{i=0}^d Z^i v_i \in \mathbb{F}[Z]^{j_A+1}$  with  $M(Z)\tilde{v}(Z) \equiv 0$ , equivalently a chain*

$$H_{t_A, j_A}(u) v_0 = 0, \quad H_{t_A, j_A}(u) v_i = -H_{t_A, j_A}(v) v_{i-1} \quad (1 \leq i \leq d), \quad H_{t_A, j_A}(v) v_d = 0.$$

*In that case, writing  $\lambda(Z)$  for the last coordinate of  $\tilde{v}(Z)$ , every  $Z = \gamma$  with  $\lambda(\gamma) \neq 0$  admits a monic degree- $j_A$  syndrome annihilator for  $u + \gamma v$ , i.e. every such slope passes the rank test at exact agreement  $A$ ; the at most  $\deg \lambda$  remaining slopes pass it after column truncation. The rank test is necessary for badness at agreement  $A$  but not sufficient (the annihilator must in addition be  $D$ -split), so membership in the singular bucket never certifies badness by itself.*

*Proof.* (i) says  $\text{rank}_{\mathbb{F}(Z)} M(Z) \leq j_A$ , which holds if and only if the kernel of  $M(Z)$  over the rational function field is nonzero; clearing denominators and dividing by common factors produces a nonzero polynomial kernel vector, and expanding  $M(Z)\tilde{v}(Z) \equiv 0$  in powers of  $Z$  gives the chain. The converse direction is immediate. Specializing  $Z = \gamma$  with  $\lambda(\gamma) \neq 0$  and normalizing the last coordinate to 1 produces the vector  $\ell$  of the chart atlas with  $A(\ell) + \gamma B(\ell) = 0$ , i.e. a monic annihilator;  $\lambda \neq 0$  has at most  $\deg \lambda$  roots. The final sentence restates the convention of Section 3.4.  $\square$

**Corollary 9.9** (the singular buckets are resolved in the deep regime). *Let  $C = \text{RS}[\mathbb{F}, D, k]$  and let  $A$  be an exact agreement with  $3(n - A) \leq n - k$ . Then for every received line — in particular for every line in the underdetermined or identically singular buckets of Section 3.4 — the number of finite slopes that are bad at agreement threshold  $A$  is at most  $n - A + 1$  if the pair is not  $\frac{n-A}{n}$ -tangent, and at most  $n - A$  MCA-bad (zero CA-bad) slopes if it is. In particular, in this regime the singular bucket contributes at most  $n - A + 1$  to  $B_{\text{ap}}(A)$  per line, and no line contributes  $\Theta(q)$ : the “totally bad line” scenario that the singular bucket was guarding against does not occur below one third of the distance.*

*Proof.* Immediate from Theorem 9.6 with  $r = n - A$  and  $\delta = r/n$ : badness at agreement threshold  $A$  implies distance at most  $r/n$ .  $\square$

**Remark 9.10** (calibration: near-capacity unsafety is quotient-borne). The unsafe mass produced by this paper does not press on the aperiodic bucket. The fiber witnesses of Lemmas 3.1 and 7.2 have agreement supports that are unions of complete fibers, hence periodicity scale  $\geq a \geq 2$ ; by Theorem 9.2 they are charged to the quotient side of the ledger, and by Theorem 9.4 even at the image level. So a polynomially bounded aperiodic numerator in the band remains consistent with every theorem in this paper. This motivates stating the entire residual band as a single conjecture.

**Open Problem 9.11** (aperiodic band conjecture — the residual kernel). Fix a multiplicative or Chebyshev row  $C = \text{RS}[\mathbb{F}, D, k]$ . Prove or refute: there is a polynomial  $P$ , independent of  $q$ , such that for every agreement  $a$  in the open band

$$k + \frac{2n}{N} < a < n - \left\lfloor \frac{n - k}{3} \right\rfloor,$$

the number of MCA-bad finite slopes of any received line, all of whose witness supports are aperiodic in the sense of Definition 9.1, is at most  $P(n)$ . By Corollary 9.5 and Theorem 9.12 (structural exhaustion and the extension-line lift) and Theorem 9.6 (deep regime discharged), this conjecture is exactly the missing input (ii) of Theorem 8.2 throughout the band, for every challenge row.

### 9.3 The exact extension-line lift

**Theorem 9.12** (restriction of scalars for lines). *Let  $B \subseteq \mathbb{F}$  with  $e := [\mathbb{F} : B]$  and  $D \subseteq B$ , and fix a  $B$ -basis  $1 = \beta_0, \beta_1, \dots, \beta_{e-1}$  of  $\mathbb{F}$ . Decompose words as  $w = \sum_i \beta_i w_i$  with  $w_i \in B^n$ , and let  $\mu_{ij}(\gamma) \in B$  be the multiplication tensor,  $\gamma \beta_i = \sum_j \mu_{ij}(\gamma) \beta_j$ , which is  $B$ -linear and injective in  $\gamma$ . Then:*

- (i) *A word  $c$  lies in  $\text{RS}[\mathbb{F}, D, k]$  if and only if every component  $c_j$  lies in  $\text{RS}[B, D, k]$ , and for all words  $w$ ,*

$$\Delta(w, \text{RS}[\mathbb{F}, D, k]) = \Delta_e((w_0, \dots, w_{e-1}), \text{RS}[B, D, k]^{\equiv e}).$$

*More generally, a set  $S \subseteq D$  explains  $w$  over  $\mathbb{F}$  if and only if  $S$  column-explains the component tuple over  $B$ .*

- (ii) *For a pair  $(f_1, f_2)$  over  $\mathbb{F}$  and  $\gamma \in \mathbb{F}$ , the components of  $f_1 + \gamma f_2$  are*

$$(f_1 + \gamma f_2)_j = f_{1,j} + \sum_i \mu_{ij}(\gamma) f_{2,i},$$

*an  $e$ -dimensional  $B$ -linear (in  $\gamma$ ) family of  $e$ -interleaved words over  $B$ .*



- (iii) For every radius (and pair of radii),  $\gamma$  is CA-bad (resp. MCA-bad) for  $(f_1, f_2)$  over  $\mathbb{F}$  if and only if the parameter  $\gamma$  of the sampler in (ii) is bad in the corresponding interleaved sense over  $B$ , with pair noncontainment read on the  $2e$ -tuple  $(f_{1,0}, \dots, f_{1,e-1}, f_{2,0}, \dots, f_{2,e-1})$  against  $\text{RS}[B, D, k]^{\equiv 2e}$ , on the same supports. For  $\gamma \in B$  the sampler restricts to the diagonal  $e$ -interleaved line  $(f_{1,j} + \gamma f_{2,j})_j$ .

Consequently the extension term  $B_{\text{ext}}(a)$  of Theorem 8.2 is not an independent quantity: it equals, definitionally, a base-field interleaved-sampler numerator. In particular no “extension-valued counterexample” can exist outside the base ledgers: every genuinely  $\mathbb{F}$ -valued line is charged, exactly, to a base bucket.

*Proof.* (i) is the componentwise decomposition already used in the proof of Lemma 6.1 (valid for any  $D \subseteq B$ , cf. Corollary 7.14): the coefficients of a polynomial over  $\mathbb{F}$  decompose in the basis, evaluation at  $D \subseteq B$  commutes with the decomposition, and pointwise agreement over  $\mathbb{F}$  is simultaneous componentwise agreement, i.e. column agreement. (ii) is the definition of  $\mu$ . (iii) The point condition, the pair condition, and every support-restricted condition in Definitions 2.1 and 2.2 are, by (i), equalities of components on sets of coordinates, and these translate verbatim; injectivity of  $\gamma \mapsto \mu(\gamma)$  makes the parameterizations match one-to-one. The diagonal claim is  $\mu_{ij}(\gamma) = \gamma \delta_{ij}$  for  $\gamma \in B$ .  $\square$

**Corollary 9.13** (status of the extension-line lift). *The extension-line question is discharged on the “lift” side of its dichotomy: extension lines are covered — exactly, not merely up to a stable numerator — by base-field ledgers, namely the interleaved-sampler ledgers of the list and curve sides via Theorem 9.12. In the deep regime nothing at all remains: Theorem 9.6 applies over  $\mathbb{F}$  directly, uniformly in the field of definition of the pair. On the lower side, Corollaries 2.12 and 2.13 already produce genuinely extension-valued bad lines from any supplied list, and Lemma 6.1 (with Corollaries 6.2 and 7.14) shows the certified deployed failures must be of this kind. What the extension-line lift leaves behind is exactly the band input of the list and curve ledgers, i.e. Open Problem 9.11.*

## 9.4 Uniform interleaved-list bounds, exact recursion, and calibration

**Theorem 9.14** (elementary Johnson list bound, uniform in the interleaving arity). *Let  $C = \text{RS}[\mathbb{F}, D, k]$  on any evaluation set  $D$  of size  $n$ , let  $m \geq 1$ , and let  $a$  be an agreement threshold with  $a^2 > (k-1)n$ . Then for every received  $m$ -tuple  $U$ ,*

$$|\Lambda(C^{\equiv m}, 1 - \frac{a}{n}, U)| \leq \frac{n(a-k+1)}{a^2 - (k-1)n},$$

where the list is taken in the column distance  $\Delta_m$ . The same bound holds for any code family in which two distinct codewords column-agree on at most  $k-1$  coordinates; in particular it holds for interleaved bivariate circle codes with  $k$  replaced by  $k_c$ .

*Proof.* Let  $\mathbf{c}^{(1)}, \dots, \mathbf{c}^{(L)}$  be distinct listed codewords with column-agreement sets  $S_1, \dots, S_L \subseteq D$ ,  $|S_i| \geq a$ . Two distinct interleaved codewords column-agree exactly where all their rows agree; some pair of rows consists of distinct polynomials of degree  $< k$ , so  $|S_i \cap S_j| \leq k-1$  for  $i \neq j$ . Let  $\deg(x) := \#\{i : x \in S_i\}$ , so  $\sum_x \deg(x) \geq La$  and

$$\sum_{x \in D} \binom{\deg(x)}{2} = \sum_{i < j} |S_i \cap S_j| \leq \binom{L}{2} (k-1).$$

By convexity of  $t \mapsto t(t-1)$ ,  $\sum_x \deg(x)(\deg(x)-1) \geq n \cdot \frac{La}{n} (\frac{La}{n} - 1)$  (the right side is nonpositive when  $La \leq n$ , and the average-value bound applies otherwise). Hence  $La(La - n) \leq L(L-1)(k-1)n$ , i.e.  $a(La - n) \leq (L-1)(k-1)n$ , i.e.

$$L(a^2 - (k-1)n) \leq an - (k-1)n = n(a - k + 1),$$

which is the claim when  $a^2 > (k-1)n$ .  $\square$

**Theorem 9.15** (exact divisor-lattice list recursion for fiber-constant words). *Multiplicative case; let  $c_0 \mid n$ , let  $U = W \circ \pi_{c_0}$  be constant on  $c_0$ -fibers, and let  $a \geq k$ . For  $p \in \Lambda(C, 1 - a/n, U)$  let  $S_p := \{x \in D : p(x) = U(x)\}$  be the full agreement set and put  $c_p := \gcd(c(S_p), c_0)$ . For  $c \mid c_0$  write  $W_c(y) := W(y^{c_0/c})$  on  $Q_c$ , so  $U = W_c \circ \pi_c$ . Then, for each  $c \mid c_0$ , the descent map of Theorem 9.4 is a bijection from  $\{p \in \Lambda(C, 1 - a/n, U) : c_p = c\}$  onto*

$$\left\{ g \in \Lambda(\text{RS}[\mathbb{F}, Q_c, \lceil k/c \rceil], 1 - \lceil a/c \rceil / N_c, W_c) : \gcd(c(\bar{S}_g), c_0/c) = 1 \right\},$$

where  $\bar{S}_g$  is the full agreement set of  $g$  with  $W_c$  and its scale is computed in the quotient lattice. In particular

$$|\Lambda(C, 1 - \frac{a}{n}, U)| = \sum_{c \mid c_0} \#\{\cdots\},$$

and if moreover  $a^2 > kn$ , then every bucket obeys the level- $c$  Johnson bound and

$$|\Lambda(C, 1 - \frac{a}{n}, U)| \leq \sum_{c \mid c_0} \frac{n(a - k + 2c - 1)}{a^2 - kn} \leq \tau(c_0) \frac{n(a - k + 2c_0 - 1)}{a^2 - kn},$$

$\tau$  the number-of-divisors function.

*Proof.* Fix  $p$  with  $c_p = c$ . Then  $c \mid c(S_p)$ , so  $S_p$  is a union of complete  $c$ -fibers by Theorem 9.2(i), and  $U = W_c \circ \pi_c$  is constant on  $c$ -fibers; Theorem 9.4 (with  $S = S_p$ ,  $|S_p| \geq a \geq k$ ) gives  $p = g \circ \pi_c$  with  $g$  in the displayed quotient list, and  $S_p = \pi_c^{-1}(\bar{S}_g)$  because  $p(x) = U(x) \iff g(x^c) = W_c(x^c)$ . The power map is equivariant,  $\pi_c(hx) = h^c \pi_c(x)$ , and  $Q_c$  is a coset of  $H^{(c)} := \{h^c : h \in H\}$  with kernel  $H_c$ ; hence  $hS_p = S_p \iff h^c \bar{S}_g = \bar{S}_g$  and  $c(S_p) = c \cdot c(\bar{S}_g)$ . Since  $c \mid c_0$ ,  $\gcd(c \cdot c(\bar{S}_g), c_0) = c \cdot \gcd(c(\bar{S}_g), c_0/c)$ , so  $c_p = c$  is exactly the displayed coprimality. Conversely, for  $g$  in the displayed set,  $p := g \circ \pi_c$  has full agreement set  $\pi_c^{-1}(\bar{S}_g)$  of size  $c|\bar{S}_g| \geq c\lceil a/c \rceil \geq a$ , and the same computation returns  $c_p = c$ ; the two maps are mutually inverse. Summing the buckets gives the exact formula. For the Johnson clause: at level  $c$  the code has dimension  $\lceil k/c \rceil$  and length  $N_c = n/c$ , and

$$\left\lceil \frac{a}{c} \right\rceil^2 - \left( \left\lceil \frac{k}{c} \right\rceil - 1 \right) \frac{n}{c} \geq \frac{a^2}{c^2} - \frac{kn}{c^2} = \frac{a^2 - kn}{c^2} > 0,$$

using  $\lceil k/c \rceil - 1 < k/c$ , so Theorem 9.14 applies at every level; with  $\lceil a/c \rceil - \lceil k/c \rceil + 1 \leq (a - k + 2c - 1)/c$  the level- $c$  bound is at most  $n(a - k + 2c - 1)/(a^2 - kn)$ .  $\square$

**Proposition 9.16** (calibration: the quotient list term is necessarily word-dependent and exponential). *Let the hypotheses of Lemma 7.2(ii) hold and let  $U = u_z$  be the pigeonholed received word at agreement  $A_2$ . Every listed codeword  $c_A = -R_A(\varphi)$  lies in the scale- $a$  quotient image  $\{g \circ \varphi : \deg g \leq \ell_2 - 2\}$ . Hence in any bound of the shape demanded by the interleaved-list ledger,*

$$|\Lambda(C^+, 1 - \frac{A_2}{n}, U)| \leq L_Q(A_2, U) + L_{\text{ap}}(A_2),$$



in which  $L_Q$  upper-bounds the quotient-image sublist and  $L_{\text{ap}}$  the rest, one necessarily has

$$L_Q(A_2, u_z) \geq \binom{N}{\ell_2} / |B|.$$

No uniform subexponential  $L_Q$  exists in the band; only the aperiodic term  $L_{\text{ap}}$  can be uniformly polynomial, which is consistent with Open Problem 9.11 and with Remark 9.10.

*Proof.* Immediate from Lemma 7.2: the  $\geq \binom{N}{\ell_2} / |B|$  listed codewords are of the displayed composed form, hence are counted by  $L_Q$ .  $\square$

**Corollary 9.17** (the grand interleaved-list staircase outside the band). *Fix any row satisfying the hypotheses of Corollary 5.1, Corollary 7.6, or Corollary 7.12, and any interleaving arity  $m \geq 1$ .*

(i) (Unsafe near capacity.) *For every agreement  $a \leq k + n/N$  (radius  $\geq 1 - \rho - 1/N$ ),*

$$|\Lambda(C^{\equiv m}, 1 - \frac{a}{n})| \geq \binom{N}{\ell_1} / |B| > 2^{-128} q,$$

*by Lemma 7.2(i) and the entropy counts of the corresponding corollary, together with the embedding  $c \mapsto (c, 0, \dots, 0)$  of base lists into interleaved lists.*

(ii) (Safe in the Johnson regime.) *For every agreement  $a$  with  $a^2 > (k - 1)n$  and  $n(a - k + 1)/(a^2 - (k - 1)n) \leq 2^{-128} q$ ,*

$$|\Lambda(C^{\equiv m}, 1 - \frac{a}{n})| \leq 2^{-128} q,$$

*by Theorem 9.14, uniformly in  $m$ .*

Thus the interleaved-list staircase is determined outside the same Johnson-to-capacity band as the MCA staircase; inside the band, Proposition 9.16 shows that any resolution of the interleaved-list ledger must carry an exponential, word-dependent quotient term, and Theorem 9.15 shows that for fiber-constant words this term is exactly a divisor-lattice recursion.

*Proof.* (i) At agreement  $A_1 \leq k + n/N$  the base-field list floor is  $\binom{N}{\ell_1} / |B|$  by Lemma 7.2(i); list sizes are nondecreasing as the agreement threshold decreases, and the zero-padding embedding preserves column agreement, so the interleaved list is at least as large. The entropy counts of the cited corollaries give  $\binom{N}{\ell_1} \geq 2^{513} \geq |B| \cdot 2^{-128} q \cdot 2^0$  throughout the challenge envelope, since  $|B| q 2^{-128} \leq q^2 2^{-128} < 2^{384}$ . (ii) is Theorem 9.14.  $\square$

## 9.5 Curves and projective slopes in the deep regime

**Theorem 9.18** (deep-regime theorem for degree- $d$  power-curve samplers). *Let  $C \subseteq \mathbb{F}^n$  be  $\mathbb{F}$ -linear with minimum weight  $w$ , let  $d \geq 1$ , let  $W_\gamma = f_0 + \gamma f_1 + \dots + \gamma^d f_d$  be a power-curve sampler, let  $\delta \in (0, 1)$ ,  $r := \lfloor \delta n \rfloor$ , and assume*

$$(d + 2)r \leq w - 1.$$

*Then either there are codewords  $c_0, \dots, c_d \in C$  and a set  $T'$  with  $|T'| \leq r$  such that  $f_i = c_i$  off  $T'$  for every  $i$  (curve-tangent tuple; at most  $d|T'| \leq dr$  MCA-bad and no CA-bad parameters), or*

$$\#\{\gamma \in \mathbb{F} : \Delta(W_\gamma, C) \leq \delta\} \leq d(r + 1).$$

In particular the curve analogues of  $\varepsilon_{\text{ca}}$  and  $\varepsilon_{\text{mca}}$  (with noncontainment read on the coefficient tuple against  $C^{\equiv(d+1)}$ , as in Lemma 3.17) are at most  $d(r+1)/q$  at radius  $\delta$ . Adjoining the projective parameter at infinity (the leading word  $f_d$ ) adds at most one to every count. For  $C = \text{RS}[\mathbb{F}, D, k]$  the hypothesis reads  $(d+2)\lfloor \delta n \rfloor \leq n - k$ .

*Proof.* Let  $S_{\text{cl}}$  be the displayed set. If  $|S_{\text{cl}}| \leq d$  the second alternative holds. Otherwise pick distinct  $\gamma_0, \dots, \gamma_d \in S_{\text{cl}}$  with codewords  $p_j$  and error sets  $E_j$  ( $|E_j| \leq r$ ), and let  $c_i := \sum_j \lambda_{ij} p_j \in C$ , where  $(\lambda_{ij})$  is the inverse of the Vandermonde matrix  $(\gamma_j^i)$ ; off  $T := \bigcup_j E_j$  ( $|T| \leq (d+1)r$ ) the same inversion applied to values gives  $f_i = c_i$ . Put  $e_i := f_i - c_i$ , supported in  $T$ , and  $T' := \{x : (e_{0,x}, \dots, e_{d,x}) \neq 0\}$ . For any  $\gamma \in S_{\text{cl}}$  with codeword  $p$  and error set  $E$ : off  $T \cup E$ ,  $p = W_\gamma = \sum_i \gamma^i c_i$ , so the codeword  $p - \sum_i \gamma^i c_i$  has weight at most  $(d+2)r < w$ , hence is zero, and

$$W_\gamma - \sum_i \gamma^i c_i = \sum_i \gamma^i e_i \quad \text{has weight at most } r.$$

For  $x \in T'$  the nonzero polynomial  $\gamma \mapsto \sum_i e_{i,x} \gamma^i$  has at most  $d$  roots. If  $|T'| \geq r+1$ , each  $\gamma \in S_{\text{cl}}$  needs at least  $|T'| - r$  vanishing coordinates in  $T'$ , so  $|S_{\text{cl}}|(|T'| - r) \leq d|T'|$  and  $|S_{\text{cl}}| \leq d|T'|/(|T'| - r) \leq d(r+1)$ . Otherwise  $|T'| \leq r$  and the tuple is curve-tangent. In the tangent case,  $2r \leq w-1$  makes two codewords agreeing on  $\geq n-2r$  coordinates equal; arguing exactly as in Step 3 of Theorem 9.6 (with  $n-(d+2)r \geq n-w+1$  forcing both the point codeword and any tuple explanation to coincide with the  $c_i$  off  $T'$ ), a parameter is MCA-bad if and only if  $\sum_i e_{i,x} \gamma^i = 0$  for some  $x \in T'$ , giving at most  $d|T'|$  parameters, and the coefficient tuple is  $\delta$ -close to  $C^{\equiv(d+1)}$  so no parameter is CA-bad. MCA-bad parameters are close, so the counts cover both notions; the parameter at infinity is a single additional value. For Reed–Solomon,  $w = n - k + 1$ .  $\square$

Together with Theorem 9.12, Theorem 9.18 also covers the multi-parameter samplers into which extension lines decompose, since those are handled over  $\mathbb{F}$  directly by Theorem 9.6; what the curve-sampler ledger still lacks is only the band-regime aperiodic pivot atlas, i.e. the curve instance of Open Problem 9.11, together with the exceptional vanishing-vector strata already flagged in Section 3.4.

## 9.6 The certificate scanner, its soundness, and deployed output

**Proposition 9.19** (small-field degeneracy). *Let  $C \subsetneq \mathbb{F}^n$  be any linear code and  $q = |\mathbb{F}|$ . For every  $\delta \in (0, 1)$ ,  $\varepsilon_{\text{mca}}(C, \delta) \geq 1/q$ . Consequently  $\delta_C^*(\varepsilon^*) = 0$  whenever  $\varepsilon^* < 1/q$ ; in particular every row over a field of size  $q < 2^{128}$  — including the deployed QM31 circle rows,  $q = p^4 < 2^{124}$  — has a degenerate grand-challenge threshold  $\delta_C^*(2^{-128}) = 0$ , and the meaningful staircase question there is posed at a larger target such as  $\varepsilon^* = 2^{-100}$ . At equality  $1/q = \varepsilon^*$  the construction below gives only a floor meeting the target exactly, not strict unsafety; all degeneracy statements therefore require  $1/q > \varepsilon^*$ .*

*Proof.* Pick  $f_2 \notin C$ , any  $c_0 \in C$  and  $\gamma_0 \in \mathbb{F}$ , and set  $f_1 := c_0 - \gamma_0 f_2$ . With  $S = D$ : the point  $f_1 + \gamma_0 f_2 = c_0$  is explained on  $S$ , while the pair is explained on  $S = D$  only if both words lie in  $C$ , which fails. Hence  $\gamma_0$  is MCA-bad at every radius, and  $\varepsilon_{\text{mca}} \geq 1/q$ .  $\square$

**Definition 9.20** (staircase scanner). The scanner takes a row header ( $D$ -type,  $n, k, |B|, q, q_{\text{line}} = q$ ), a target  $\varepsilon^*$ , and the applicable divisor data, and for each agreement  $a \in \{k+1, \dots, n\}$  evaluates the following verdicts, each tagged by its certifying statement:

- (V1) **Cap-unsafe.** If  $a \leq A_2$  for an instantiated cap (Theorems 4.1 and 7.4 via Corollaries 5.1, 7.6, 7.12 and 7.13) and  $\frac{1}{2k}(1 - \frac{n}{q}) > \varepsilon^*$ : mark  $a$  MCA-unsafe.
- (V2) **Floor-unsafe.** If a remainder floor applies at  $a$  (Theorem 3.8) with converted numerator exceeding  $\varepsilon^*q$ : mark  $a$  MCA-unsafe; if a fiber list floor at  $a$  exceeds  $\varepsilon^*q$  (Lemmas 3.1 and 7.2 with the entropy count): mark  $a$  list-unsafe.
- (V3) **Deep-safe.** If  $3(n - a) \leq n - k$  and  $n - a + 1 \leq \varepsilon^*q$ : mark  $a$  MCA-safe (Theorem 9.6); with  $(d + 2)(n - a) \leq n - k$  and  $d(n - a + 1) \leq \varepsilon^*q$ : curve- $d$ -safe (Theorem 9.18).
- (V4) **Johnson-list-safe.** If  $a^2 > (k - 1)n$  and  $n(a - k + 1)/(a^2 - (k - 1)n) \leq \varepsilon^*q$ : mark  $a$  list-safe for every interleaving arity (Theorem 9.14).

The output per challenge type is the pair  $a_{\text{unsafe}} := \max\{a \text{ marked unsafe}\}$  and  $a_{\text{safe}} := \min\{a \text{ marked safe}\}$ , together with the open interval between them. If  $\varepsilon^* < 1/q$  the scanner instead reports the degeneracy of Proposition 9.19.

**Theorem 9.21** (scanner soundness). *Every verdict printed by the scanner of Definition 9.20 is correct: agreements marked unsafe satisfy  $\varepsilon_{\text{mca}}(C, 1 - a/n) > \varepsilon^*$  (resp. list size  $> \varepsilon^*q$ ), agreements marked safe satisfy  $\varepsilon_{\text{mca}}(C, 1 - a/n) \leq \varepsilon^*$  (resp. list size  $\leq \varepsilon^*q$ , for every  $m$ ), and consequently  $a_{\text{unsafe}} < a_C^* \leq a_{\text{safe}}$  in the staircase formulation of Proposition 8.1.*

*Proof.* Each verdict cites a theorem of this paper whose hypotheses are verified numerically by the scanner; monotonicity across agreements is Lemma 2.5 for MCA and radius-monotonicity of lists. The bracketing of  $a_C^*$  is Proposition 8.1.  $\square$

| Row / challenge                                                                                                                                                                                                                         | Radius range $\delta$                       | Verdict | Certified numerator                    | Source            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|---------|----------------------------------------|-------------------|
| <i>KoalaBear sextic: <math>n = 2^{21}</math>, <math>k = 2^{20}</math>, <math>q = p^6 &gt; 2^{185.9}</math>, <math>\varepsilon^* = 2^{-128}</math>, <math>\varepsilon^*q \approx 2^{57.9}</math></i>                                     |                                             |         |                                        |                   |
| MCA                                                                                                                                                                                                                                     | $[\frac{1}{2} - 2^{-7}, \frac{1}{2})$       | unsafe  | $> 2^{-22}q \approx 2^{164}$           | Corollary 5.4     |
| MCA                                                                                                                                                                                                                                     | $\delta \leq 349525/2^{21} \approx 0.16666$ | safe    | $\leq 349526 < 2^{-167}q$              | Theorem 9.6       |
| MCA                                                                                                                                                                                                                                     | $(0.16667, 0.49218)$                        | open    | —                                      | Open Problem 9.11 |
| List (any $m$ )                                                                                                                                                                                                                         | $\delta \geq \frac{1}{2} - 2^{-8}$          | unsafe  | $\geq \binom{256}{129}/p \geq 2^{216}$ | Lemma 3.1(i)      |
| List (any $m$ )                                                                                                                                                                                                                         | $\delta \leq 0.29289$ ( $a \geq 1482910$ )  | safe    | $\leq 2^{20}$                          | Theorem 9.14      |
| List (any $m$ )                                                                                                                                                                                                                         | $(0.29290, 0.49609)$                        | open    | —                                      | Open Problem 9.11 |
| <i>M31 circle line round: <math>n = 2^{21}</math>, <math>k = 2^{20}</math>, <math>q = p^4 &lt; 2^{124}</math>; at <math>2^{-128}</math>: <math>\delta^* = 0</math> (Proposition 9.19); below: <math>\varepsilon^* = 2^{-100}</math></i> |                                             |         |                                        |                   |
| MCA                                                                                                                                                                                                                                     | $[\frac{1}{2} - 2^{-7}, \frac{1}{2})$       | unsafe  | $> 2^{-22}q \approx 2^{102}$           | Corollary 7.13    |
| MCA                                                                                                                                                                                                                                     | $\delta \leq 0.16666$                       | safe    | $\leq 349526 < 2^{-105}q$              | Theorem 9.6       |
| List (any $m$ )                                                                                                                                                                                                                         | $\delta \geq \frac{1}{2} - 2^{-8}$          | unsafe  | $\geq 2^{216}$                         | Lemma 7.2(i)      |
| List (any $m$ )                                                                                                                                                                                                                         | $\delta \leq 0.29289$                       | safe    | $\leq 2^{20} \leq 2^{-100}q$           | Theorem 9.14      |

Table 2: Baseline scanner output (Definition 9.20) for the two deployed rows, before the pin-cer and the optimized graded-prefix floors; it is retained to display the self-contained deep-safe and Johnson-list-safe certificates, and the final deployed threshold bands are the narrowed bands of Table 5. The open MCA band for the circle row at  $\varepsilon^* = 2^{-100}$  is the same interval  $(0.16667, 0.49218)$  as above.

The table realizes the interval output demanded by the staircase interface: for the KoalaBear MCA staircase,  $a_{\text{unsafe}} = 1,064,960$  ( $\delta = \frac{1}{2} - 2^{-7}$ ) and  $a_{\text{safe}} = 1,747,627$  ( $\delta \approx 0.16666$ ); the interval shrinks to one step exactly when Open Problem 9.11 is settled, via Theorem 8.2 whose other inputs are now discharged (Remark 9.27).

## 9.7 The formal finite core

**Proposition 9.22** (finite certificates and decidability). *Fix a row  $(C, q, q_{\text{line}})$  and an agreement  $a$ . The numerator  $B_{\text{mca}}(a)$  of Section 8 is a well-defined integer, computable by finite enumeration over pairs, slopes, supports, and codewords; hence the one-step certificate of Proposition 8.1 is a finite object. A verifiable prize certificate consists of: the row header; the integer  $a_0$ ; a lower witness — a pair  $(f_1, f_2)$ , a set of more than  $\varepsilon^* q_{\text{line}}$  slopes, and for each slope a codeword and a support, checkable by evaluation — or a theorem tag from this paper with its numerically verified hypotheses; and an upper witness — a theorem tag with numerically verified hypotheses certifying  $B_{\text{mca}}(a_0 + 1) \leq \varepsilon^* q_{\text{line}}$ . Verification of a certificate is polynomial in its length.*

*Proof.* All quantifiers in Definitions 2.1 and 2.2 range over the finite sets  $\mathbb{F}^n$ ,  $\mathbb{F}$ ,  $2^D$ , and  $C$ ; the maximum defining  $B_{\text{mca}}(a)$  is therefore attained and computable. Checking a lower witness is evaluation and counting; checking a theorem tag is checking finitely many integer inequalities.  $\square$

| Core statement                        | Proof ingredients                                             |
|---------------------------------------|---------------------------------------------------------------|
| Theorem 2.6 (deep-point conversion)   | polynomial division, pigeonhole over $\mathbb{F} \setminus D$ |
| Lemma 3.1, Lemma 7.2 (fiber floors)   | elementary symmetric functions, subfield pigeonhole           |
| Fact 2.4, Lemma 2.5                   | set manipulation                                              |
| Proposition 8.1 (staircase)           | monotone integer staircase                                    |
| Theorem 9.6 (deep MCA)                | linear interpolation, double counting of vanishings           |
| Theorem 9.14 (Johnson lists)          | convexity double counting of incidences                       |
| Theorem 9.2, Theorem 9.4              | cyclic group actions, root counting                           |
| Theorem 9.12 (restriction of scalars) | basis decomposition of $\mathbb{F}/B$                         |
| Definition 9.20, Theorem 9.21         | finite case analysis over the statements above                |

Table 3: The formal finite core: each statement quantifies over explicitly bounded finite sets and uses only the listed elementary ingredients; the dependency order is top to bottom. Formalizing this table in a proof assistant is the outstanding transcription task.

The list deliberately includes the new safe-side theorems: a certificate package that contains both an unsafe certificate (Corollary 5.1 and its relatives) and a machine-checked safe region (Theorems 9.6 and 9.14) documents the staircase from both sides in the sense of Proposition 8.1.

## 9.8 Tower transfer and the genus-one first step

The tower-transfer ledger half is already discharged: Theorems 9.2 and 9.4 and Corollary 9.3 were stated and proved for the Chebyshev case alongside the multiplicative case, with the divisor lattice of  $D$  replaced by the dyadic scale lattice of the twin coset, exactly as the tower transfer requires; and Open Problem 9.11 is likewise stated for both cases. It remains to treat the genus-one floor. We first extend Definition 7.1 to rational maps.

**Definition 9.23** (rationally smooth domain). Let  $B \subseteq \mathbb{F}$ , let  $\psi = f/g$  with  $f, g \in B[X]$  coprime,  $a := \deg f > e := \deg g \geq 0$ , and let  $D \subseteq B$  with  $g$  nonvanishing on  $D$ . The set  $D$  is  $(\psi, a)$ -smooth if every nonempty fiber  $\psi^{-1}(w) \cap D$ ,  $w \in \psi(D)$ , has exactly  $a$  elements; then  $N := |D|/a = |\psi(D)|$ . The FFTree domains of ECFFT [BCKL21] are  $(\psi, 2)$ -smooth for the  $x$ -coordinate map  $\psi$  of each 2-isogeny in the chain, of degree type  $(a, e) = (2, 1)$ .

**Proposition 9.24** (rational-map fiber floor; unified  $(a, e)$  form). *Let  $D \subseteq B$  be  $(\psi, a)$ -smooth of size  $n$ , with  $\psi = f/g$  of degree type  $(a, e)$ , and let  $2 \leq \ell \leq N - 1$ . Put*

$$k' := a\ell - 2(a - e), \quad u_z := (f(x)^\ell + z f(x)^{\ell-1} g(x))_{x \in D} \in B^n.$$

*Then there exists  $z \in B$  with*

$$|\Lambda(\text{RS}[\mathbb{F}, D, k' + 1], 1 - \frac{a\ell}{n}, u_z)| \geq \binom{N}{\ell} / |B|,$$

*i.e. a list at agreement  $a\ell = k' + 2(a - e)$ . For  $e = 0$  this is Lemma 7.2(ii) (window  $2a$ , gap  $\approx 2/N$ ); for  $e = a - 1$  the window is 2 and the certified agreement is  $k' + 2$  — the first staircase step, in accordance with the collapse of Remark 7.16.*

*Proof.* For  $A \subseteq \psi(D)$  with  $|A| = \ell$  put  $\Lambda_A := \prod_{b \in A} (f - bg)$ , of degree  $a\ell$  with leading coefficient  $\text{lc}(f)^\ell \neq 0$ , vanishing at every  $x \in D$  with  $\psi(x) \in A$  — exactly  $a\ell$  points by smoothness (and  $g(x) \neq 0$ ). Expanding,

$$\Lambda_A = \sum_{j=0}^{\ell} (-1)^j e_j(A) f^{\ell-j} g^j,$$

with  $j$ -th term of degree  $a\ell - j(a - e)$ . Set  $z_A := -e_1(A) \in B$  and

$$c_A := \left( - \sum_{j \geq 2} (-1)^j e_j(A) f^{\ell-j}(x) g^j(x) \right)_{x \in D},$$

a polynomial of degree at most  $a\ell - 2(a - e) = k'$ , so  $c_A \in \text{RS}[B, D, k' + 1]$ ; on the  $a\ell$  fiber points,  $u_{z_A} = c_A$ . For injectivity at fixed  $z$ : if  $c_A = c_{A'}$  as words then as polynomials ( $\deg \leq k' \leq a(N - 1) < n$ ), so  $\sum_{j \geq 2} (-1)^j (e_j(A) - e_j(A')) f^{\ell-j} g^j = 0$ ; dividing by  $g^\ell$  in  $\mathbb{F}(X)$  gives a polynomial identity in  $\psi$ , and  $\psi$  is nonconstant, hence transcendental over  $\mathbb{F}$  inside  $\mathbb{F}(X)$ , so all coefficients vanish:  $e_j(A) = e_j(A')$  for  $j \geq 2$ , and  $e_1$  agrees via  $z$ ; then  $A = A'$  from the root sets of  $\prod_b (Y - b)$ . Pigeonholing  $z_A$  over  $B$  finishes as in Lemma 7.2.  $\square$

**Corollary 9.25** (every ECFFT row has an unsafe first staircase step). *Let  $D$  be an ECFFT domain that is  $(\psi, 2)$ -smooth (Definition 9.23) over  $B$ , let  $C = \text{RS}[\mathbb{F}, D, k]$  with  $2 \mid k$ ,  $\rho = k/n \in [\frac{1}{16}, \frac{1}{2}]$ ,  $n < q = |\mathbb{F}| < 2^{256}$ , and  $n \geq 2^{12}$ . Then, with  $\ell = (k + 2)/2 \leq N - 1$ ,*

$$\binom{n/2}{(k+2)/2} \geq |B| \left( \frac{q}{k} + 1 \right),$$

*and consequently*

$$\varepsilon_{\text{ca}} \left( C, 1 - \rho - \frac{2}{n} \right) > \frac{1}{2k} \left( 1 - \frac{n}{q} \right), \quad \varepsilon_{\text{mca}}(C, \delta) > \frac{1}{2k} \left( 1 - \frac{n}{q} \right) \text{ for every } \delta \in \left[ 1 - \rho - \frac{2}{n}, 1 - \rho \right),$$

*so  $\delta_C^*(\varepsilon^*) \leq 1 - \rho - 2/n$  for every  $\varepsilon^* \leq \frac{1}{2k} (1 - \frac{n}{q})$ ; and, provided  $k < 2^{128}$ , the list challenge is likewise unsafe at that step, since  $\binom{n/2}{(k+2)/2} / |B| \geq q/k > 2^{-128} q$ . For odd  $k$  the same argument with  $\ell = \lfloor k/2 \rfloor + 1$  gives the step at  $1 - \rho - 1/n$ .*

*Proof.* Apply Proposition 9.24 with  $(a, e) = (2, 1)$ :  $k' = 2\ell - 2 = k$ , so the list lives in  $\text{RS}[\mathbb{F}, D, k+1] = C^+$  at radius  $\delta_0 := 1 - (k+2)/n$ , and  $\lfloor \delta_0 n \rfloor = n - k - 2 \leq n - k - 1$  is admissible for Theorem 2.6. The entropy count:  $\ell/(n/2) = (k+2)/n \in (\rho, \rho + 2^{-11}] \subseteq [\frac{1}{16}, 0.502]$ , so  $H_2(\ell/(n/2)) \geq H_2(\frac{1}{16}) \geq 0.3372$  by the endpoint argument of Corollary 7.6, and

$$\log_2 \binom{n/2}{\ell} \geq \frac{n}{2} H_2\left(\frac{\ell}{n/2}\right) - \log_2 \left(\frac{n}{2} + 1\right) \geq 0.3372 \cdot \frac{n}{2} - \log_2 \left(\frac{n}{2} + 1\right) \geq 690 - 12 \geq 513,$$

since the middle expression is increasing in  $n$  and equals  $690.6 \dots - 11.0 \dots$  at  $n = 2^{12}$ , while  $|B|(q/k+1) \leq q^2 + q < 2^{513}$ . The contradiction with Theorem 2.6 at  $\eta = \frac{1}{2}$ , then Fact 2.4 and Lemma 2.5, run exactly as in Theorem 7.4. The list-challenge clause follows because  $\binom{n/2}{\ell}/|B| \geq q/k+1 > 2^{-128}q$  whenever  $k < 2^{128}$ , which covers every challenge-envelope row.  $\square$

## 9.9 The two-sided picture and what remains

**Theorem 9.26** (two-sided sandwich for the grand MCA challenge). *Let  $C = \text{RS}[\mathbb{F}, D, k]$  be a row for which one of the capacity caps of this paper applies at target  $\varepsilon^* = 2^{-128}$  — Corollary 5.1, Corollary 7.6, Corollary 7.12 or Corollary 7.13, or Corollary 9.25 — with printed gap  $G$  (respectively  $2^{-9}/2^{-10}$ ,  $2^{-11}/2^{-12}$ ,  $2^{-9} \cdot 2^{-11}$  or  $2^{-7}$ , and  $2/n$ ). Put*

$$r^* := \min\left(\left\lfloor \frac{n-k}{3} \right\rfloor, \lfloor 2^{-128}q \rfloor - 1\right).$$

*If  $q > 2^{128}$  then*

$$\frac{r^*}{n} \leq \delta_C^*(2^{-128}) \leq 1 - \rho - G,$$

*so, whenever  $\lfloor 2^{-128}q \rfloor - 1 \geq \lfloor (n-k)/3 \rfloor$  — e.g. whenever  $q \geq 2^{128}n$ , as at every deployed multiplicative row — the grand MCA threshold of such a row is determined up to a multiplicative factor of three, up to the microscopic gap  $G$ ; for  $q$  nearer  $2^{128}$  the lower edge degrades to  $r^*/n$  and no factor-of-three claim is made. If  $q < 2^{128}$  then  $\delta_C^*(2^{-128}) = 0$  (Proposition 9.19); at the boundary  $q = 2^{128}$  the pair of Proposition 9.19 gives  $\varepsilon_{\text{mca}} \geq 2^{-128}$  at every radius, which meets the threshold condition with equality and does not force degeneracy.*

*Proof.* Lower bound:  $3r^* \leq n - k$  and  $r^* + 1 \leq \lfloor 2^{-128}q \rfloor \leq 2^{-128}q$ , so Theorem 9.6 gives  $\varepsilon_{\text{mca}}(C, r^*/n) \leq (r^* + 1)/q \leq 2^{-128}$ , and  $r^*/n < (n-k)/n = 1 - \rho$ , so  $\delta^* \geq r^*/n$  by Definition 2.3 (the case  $r^* = 0$  being trivial). Upper bound: the applicable cap certifies  $\varepsilon_{\text{mca}}(C, \delta) > 2^{-128}$  for every  $\delta \in [1 - \rho - G, 1 - \rho)$ , so no such  $\delta$  belongs to the supremum set, and by Lemma 2.5 neither does any larger sub-capacity radius; hence  $\delta^* \leq 1 - \rho - G$ . The degenerate clause is Proposition 9.19 with  $\varepsilon^* = 2^{-128} < 1/q$ ; the boundary remark is the same construction, whose error floor  $1/q$  equals the target exactly at  $q = 2^{128}$ .  $\square$

For  $\rho = \frac{1}{2}$  and  $q \geq 2^{128}n$  — e.g. the KoalaBear-sextic row,  $q > 2^{185}$  — the sandwich reads  $\delta_C^*(2^{-128}) \in [0.1666 \dots, \frac{1}{2} - G]$ : the first two-sided determination of a grand-challenge MCA threshold, exact to within a factor of three.

*Remark 9.27* (dependency status of the one-step criteria). The one-step MCA criterion of Theorem 8.2 has the following dependency status. The periodic-support input is supplied by Corollary 9.3 at the support level and by Theorem 9.4 at the image level for fiber-constant lines. The deep-regime input is supplied by Theorem 9.6 and Corollary 9.9; outside that range it is exactly



| Component                 | Outcome                                                                                                                                                                                                        | Where                       |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| Periodic-support ledger   | Stabilizer exhaustion is exact at the support level; for fiber-constant lines it descends to an exact quotient-image ledger; upper numerators are independent of $q$ and $ B $ .                               | Section 9.1                 |
| Deep MCA core             | MCA is safe below one third of the distance; singular and rank-drop Hankel branches are removed in that range; the remaining band is Open Problem 9.11, with rank-pivot and compressed-eliminant certificates. | Section 9.2 and Theorem 9.6 |
| Extension-line lift       | Extension-valued lines are handled exactly by restriction of scalars; there is no independent extension bucket.                                                                                                | Section 9.3                 |
| Interleaved-list side     | Johnson safety is uniform in the interleaving arity; fiber-constant recursion is exact; the quotient term is necessarily exponential in the band.                                                              | Section 9.4                 |
| Curve/projective samplers | The deep regime is controlled for degree- $d$ curves and projective parameters; residual curve charts are part of the same aperiodic band problem.                                                             | Section 9.5                 |
| Scanner                   | The finite staircase scanner is specified, proved sound, and run on the deployed rows.                                                                                                                         | Section 9.6                 |
| Certificate core          | The finite certificate format, decidability statement, and dependency chart are supplied; machine formalization is separate implementation work.                                                               | Section 9.7                 |
| Tower transfer            | The ledger half is transferred to the Chebyshev tower; the genus-one first-step cap is proved, later strengthened to a macroscopic cap by the rational graded floors.                                          | Sections 9.8 and 11.1       |

Table 4: Structural and certificate ingredients used in the two-sided threshold statements.

Open Problem 9.11, represented by the residual rank-pivot and kernel-compressed certificate language of Theorems 3.64, 3.73 and 3.76. The extension-line input is supplied by Theorem 9.12. The sampler input is definitional once the protocol’s line sampler is fixed. For the one-step list criterion of Theorem 8.3, the required base-code bound is unconditional throughout the Johnson regime and uniform in  $m$  (Theorem 9.14, making the codegree recursion unnecessary there), is exactly recursive for fiber-constant words (Theorem 9.15), and is provably exponential in the band (Proposition 9.16), so any resolution must take that shape.

After these ingredients are assembled, the remaining mathematical gap has a compact form. The central residue is Open Problem 9.11, the aperiodic band conjecture, now represented by the finite rank-pivot, compressed-eliminant, and exact-image certificate language of Theorems 3.64, 3.73 and 3.76; this is the analytic kernel separating the two-sided interval of Theorem 9.26 from a one-step threshold determination. The other residues are narrower: the failure-profile upgrade from error  $1/k$  to error  $1 - o(1)$  (Open Problem 12.2) and explicit deployed-size certifying pairs (Open Problem 12.1); the genus-one band, formerly a separate residue, is settled at macroscopic scale in Section 11.1. The next section attacks the main band directly, from both sides at once.

## 10 Safe-side pincer and scale-optimized unsafe floors

The residue identified at the end of Section 9 is a single open band per row. This section attacks the band from both sides and narrows it substantially, with three new results. On the safe side, we prove that *mutual* correlated agreement coincides with plain correlated agreement, up to an explicit tangent term, all the way up to half the minimum distance (Theorem 10.1); this eliminates the support-mismatch layer of the problem on that range and, combined with the imported unique-decoding proximity gap of Ben-Sasson–Carmon–Ishai–Kopparty–Saraf [BCIKS20], moves the certified safe frontier of every deployed row from one third to one half of the distance. We also prove a self-contained correlated-agreement bound at half the Johnson radius (Theorem 10.4), which improves the *unconditional* safe frontier for every row of rate below  $\frac{1}{4}$ . On the unsafe side, we extend the graded locator-prefix pigeonhole of Lemma 3.5 to every map-smooth scale — in particular to the Chebyshev scales of the circle tower — and optimize it over the scale (Proposition 10.8); at deployed sizes this widens the certified unsafe band by a factor of four with a hand-checkable certificate, and by a factor of about sixteen with an exactly verified one, saturating an intrinsic entropy–subfield envelope that we identify in closed form (Remark 10.11). Theorem 10.12 and Table 5 assemble the narrowed sandwich.

### 10.1 Mutual equals correlated agreement up to half the distance

The deep-regime theorem (Theorem 9.6) controls  $\varepsilon_{\text{mca}}$  directly, but only below one third of the distance: its alternatives identify every close point-codeword with a value of a single interpolated line, and that identification costs a factor of three in the radius. The next theorem decouples the two layers of the problem. It shows that below *half* the distance, the support-mismatch phenomenon that separates mutual from plain correlated agreement is carried entirely by the tangent slopes of Theorem 9.6(a), of which there are at most  $\lfloor \delta n \rfloor$ ; everything else is already visible to  $\varepsilon_{\text{ca}}$ .

**Theorem 10.1** (mutual from correlated agreement, up to half the distance). *Let  $C \subseteq \mathbb{F}^n$  be any  $\mathbb{F}$ -linear code of minimum Hamming weight  $w$ , let  $q = |\mathbb{F}|$ , let  $\delta \in (0, 1)$ , and put  $r := \lfloor \delta n \rfloor$ . Assume*

$$2r \leq w - 1.$$

*Then for every pair  $(f_1, f_2) \in \mathbb{F}^n \times \mathbb{F}^n$ :*

- (a) *if  $\Delta_2((f_1, f_2), C^{\equiv 2}) \leq \delta$ , the pair has no CA-bad slope at radius  $\delta$ , and at most  $r$  MCA-bad slopes at radius  $\delta$ : writing  $(p_1, p_2)$  for a pair of codewords column-agreeing with  $(f_1, f_2)$  outside a set  $E$  with  $|E| \leq r$  and  $e_i := f_i - p_i$ , the MCA-bad slopes form a subset of  $\{-e_{1,j}/e_{2,j} : j \in E, e_{2,j} \neq 0\}$ ;*
- (b) *if  $\Delta_2((f_1, f_2), C^{\equiv 2}) > \delta$ , every MCA-bad slope at radius  $\delta$  is CA-bad at radii  $(\delta, \delta)$ .*

*Consequently*

$$\varepsilon_{\text{mca}}(C, \delta) \leq \max\left(\varepsilon_{\text{ca}}(C, \delta), \frac{r}{q}\right).$$

*For  $C = \text{RS}[\mathbb{F}, D, k]$  on any evaluation set  $D$  of size  $n$ , the hypothesis reads  $2\lfloor \delta n \rfloor \leq n - k$ : mutual and plain correlated agreement coincide, up to the explicit tangent term  $r/q$ , on the entire unique-decoding range of the interleaved code  $C^{\equiv 2}$ .*



*Proof.* Part (b) is immediate from the definitions: an MCA-bad slope  $\gamma$  has a witness set  $S$  with  $|S| \geq \lceil (1-\delta)n \rceil = n-r$  on which  $f_1 + \gamma f_2$  agrees with a codeword, so  $\Delta(f_1 + \gamma f_2, C) \leq r/n \leq \delta$ ; if additionally the pair is  $\delta$ -far in column distance,  $\gamma$  is CA-bad at radii  $(\delta, \delta)$  by Definition 2.1.

For part (a), fix codewords  $(p_1, p_2)$  column-agreeing with  $(f_1, f_2)$  outside the column error set  $E := \{j : (e_{1,j}, e_{2,j}) \neq (0, 0)\}$ ,  $e_i := f_i - p_i$ ,  $|E| \leq r$ . The pair is  $\delta$ -close, so it has no CA-bad slope. Let  $\gamma$  be MCA-bad with witness  $(S, c)$ :  $|S| \geq n-r$ ,  $c \in C$ ,  $c = f_1 + \gamma f_2$  on  $S$ , and no pair of codewords of  $C$  explains  $(f_1, f_2)$  on  $S$ .

First,  $c = p_1 + \gamma p_2$ . Indeed,  $z := c - (p_1 + \gamma p_2) \in C$  vanishes on  $S \setminus E$ : there  $f_i = p_i$ , so  $f_1 + \gamma f_2 = p_1 + \gamma p_2$ , while  $c = f_1 + \gamma f_2$  on all of  $S$ . Since  $|S \setminus E| \geq n-2r$ , the codeword  $z$  has weight at most  $2r \leq w-1$ , hence  $z = 0$ .

Next,  $S \not\subseteq D \setminus E$ : otherwise  $(p_1, p_2)$  would explain the pair on  $S$ . So there is  $j \in S \cap E$ . On  $S$  we have  $0 = (f_1 + \gamma f_2) - c = (e_1 + \gamma e_2)$ , so at this  $j$ ,

$$e_{1,j} + \gamma e_{2,j} = 0, \quad (e_{1,j}, e_{2,j}) \neq (0, 0).$$

If  $e_{2,j} = 0$  the left side is the nonzero constant  $e_{1,j}$ , which cannot vanish; hence  $e_{2,j} \neq 0$  and  $\gamma = -e_{1,j}/e_{2,j}$ . The MCA-bad slopes of the pair therefore form a subset of the at most  $|E| \leq r$  listed values. Taking the maximum over pairs gives the displayed bound.  $\square$

*Remark 10.2* (scope and sharpness). Three comments. (i) The hypothesis  $2r \leq w-1$  is exactly the unique-decoding radius of the interleaved code  $C^{\equiv 2}$  in column weight, and the proof uses it twice, both times through the statement “a codeword of weight  $\leq 2r$  vanishes”; beyond half the distance a  $\delta$ -close pair can admit several column explanations with incomparable agreement sets, and witness codewords need not ride any explanation line, which is precisely where the band begins. (ii) The theorem is an unconditional *implication*, with no list bound and no field-size hypothesis; instantiated with the self-contained  $\varepsilon_{\text{ca}}$  bound of Theorem 9.6 it returns the deep theorem (with the marginally better error  $\max((r+1)/q, r/q)$  replaced by  $(r+1)/q$ ), and its value is that it accepts *any* stronger correlated-agreement input on the range  $(w-1)/3 < 2r \leq w-1$ , where no self-contained input is available. (iii) The same proof gives the two-radius statement: for  $\delta_{\text{int}} \geq \delta$ , every MCA-bad slope of a  $\delta_{\text{int}}$ -far pair is CA-bad at radii  $(\delta, \delta_{\text{int}})$ , while a  $\delta_{\text{int}}$ -close pair that is also  $\delta$ -close contributes at most  $r$  slopes; only pairs in the annulus between the two radii require the no-loss form.

**Corollary 10.3** (the band conjecture loses its support-mismatch layer below half the distance). *Let  $C = \text{RS}[\mathbb{F}, D, k]$  and let  $\delta$  satisfy  $2\lfloor \delta n \rfloor \leq n-k$ . Then any correlated-agreement bound  $\varepsilon_{\text{ca}}(C, \delta) \leq \varepsilon$  implies  $\varepsilon_{\text{mca}}(C, \delta) \leq \max(\varepsilon, \lfloor \delta n \rfloor / q)$ . In particular, on the agreement subrange  $a \geq \lceil (n+k)/2 \rceil$  of Open Problem 9.11, the aperiodic band conjecture reduces to its correlated-agreement form: column-close pairs contribute at most  $\lfloor \delta n \rfloor$  explicit tangent slopes, and what must be bounded polynomially is only, for each column-far pair, the number of slopes  $\gamma$  for which  $f_1 + \gamma f_2$  has an aperiodically witnessed agreement set of size at least  $\lceil (1-\delta)n \rceil$ .*

*Proof.* Immediate from Theorem 10.1: case (a) pairs contribute at most  $r = \lfloor \delta n \rfloor$  slopes, each explicitly a tangent ratio; case (b) identifies every remaining MCA-bad slope, together with its witness support, as a CA-bad slope with the same witness.  $\square$

## 10.2 A self-contained correlated-agreement bound at half the Johnson radius

Theorem 10.1 converts the safe side of the band problem into a demand for correlated-agreement inputs. The strongest known inputs are the proximity-gap theorems of [BCIKS20] (unique-decoding range with error  $n/q$ ; Johnson range with error  $\text{poly}(n)/q$ ), whose proofs run through

a Berlekamp–Welch decoder over the rational function field  $\mathbb{F}(Z)$  and an induction over curves, and the recent strengthening [BCHKS25]. Those we import. But the certificate interface of Section 8 also prices *self-contained* inputs, since the formal core (Table 3) is built from elementary counting; the following theorem is the elementary member of the family. Its only ingredients are the pair-interleaved Johnson bound of Theorem 9.14 and a rider double count, and it reaches half the Johnson radius with an explicit list-size error.

**Theorem 10.4** (elementary correlated agreement at half the Johnson radius). *Let  $C = \text{RS}[\mathbb{F}, D, k]$  on any evaluation set  $D$  of size  $n$ , let  $\delta \in (0, 1)$ , put  $r := \lfloor \delta n \rfloor$ , and assume*

$$n - 2r > 0 \quad \text{and} \quad (n - 2r)^2 > (k - 1)n.$$

*Then, with*

$$L_2 := \frac{n(n - 2r - k + 1)}{(n - 2r)^2 - (k - 1)n}$$

*the pair-interleaved Johnson bound of Theorem 9.14 at agreement  $n - 2r$ ,*

$$\varepsilon_{\text{ca}}(C, \delta) \leq \frac{1 + (r + 1)L_2}{q},$$

*and the same bound holds for  $\varepsilon_{\text{ca}}(C, \delta, \delta_{\text{int}})$  for every  $\delta_{\text{int}} \geq \delta$ .*

*Proof.* Fix a pair  $(f_1, f_2)$  with  $\Delta_2((f_1, f_2), C^{\equiv 2}) > \delta_{\text{int}} \geq \delta$ ; only such pairs have CA-bad slopes. Let  $B \subseteq \mathbb{F}$  be the set of CA-bad slopes; for each  $\gamma \in B$  fix a codeword  $c_\gamma \in C$  and an agreement set  $S_\gamma$  with  $|S_\gamma| \geq n - r$  and  $c_\gamma = f_1 + \gamma f_2$  on  $S_\gamma$ . If  $|B| \leq 1$  we are done, so fix  $\gamma_1 \in B$  and consider any  $\gamma \in B \setminus \{\gamma_1\}$ . Define

$$P_2 := \frac{c_{\gamma_1} - c_\gamma}{\gamma_1 - \gamma} \in C, \quad P_1 := c_{\gamma_1} - \gamma_1 P_2 \in C,$$

so that, identically as codewords,

$$P_1 + \gamma_1 P_2 = c_{\gamma_1}, \quad P_1 + \gamma P_2 = c_\gamma. \tag{3}$$

On  $S_{\gamma_1} \cap S_\gamma$ , which has size at least  $n - 2r$ , the two coordinatewise linear equations  $f_1 + \gamma_1 f_2 = c_{\gamma_1}$  and  $f_1 + \gamma f_2 = c_\gamma$  have the unique solution  $f_2 = P_2$ ,  $f_1 = P_1$ ; hence the codeword pair  $(P_1, P_2)$  column-agrees with  $(f_1, f_2)$  on at least  $n - 2r$  coordinates, i.e.

$$(P_1, P_2) \in \mathcal{L} := \Lambda\left(C^{\equiv 2}, 1 - \frac{n - 2r}{n}, (f_1, f_2)\right), \quad |\mathcal{L}| \leq L_2$$

by Theorem 9.14 with  $m = 2$  and  $a = n - 2r$ . Say that  $\gamma$  *rides* a member  $(P_1, P_2) \in \mathcal{L}$  if  $c_\gamma = P_1 + \gamma P_2$  as codewords; by (3), every  $\gamma \in B \setminus \{\gamma_1\}$  rides some member of  $\mathcal{L}$ .

Fix  $(P_1, P_2) \in \mathcal{L}$  and let  $E := \{j : (f_{1,j}, f_{2,j}) \neq (P_{1,j}, P_{2,j})\}$  be its column error set. Since the pair is  $\delta$ -far,  $|E| \geq r + 1$ . If  $\gamma$  rides  $(P_1, P_2)$ , then  $(f_1 - P_1) + \gamma(f_2 - P_2) = (f_1 + \gamma f_2) - c_\gamma$  vanishes on  $S_\gamma$ , hence on at least  $|E| - r \geq 1$  coordinates of  $E$ ; and at each  $j \in E$  the map  $\gamma \mapsto (f_{1,j} - P_{1,j}) + \gamma(f_{2,j} - P_{2,j})$  is a nonzero affine function, vanishing for at most one slope. Double counting vanishing incidences between riders of  $(P_1, P_2)$  and coordinates of  $E$ ,

$$\#\{\text{riders of } (P_1, P_2)\} \cdot (|E| - r) \leq |E|, \quad \text{so} \quad \#\{\text{riders}\} \leq \frac{|E|}{|E| - r} \leq r + 1.$$

Summing over  $\mathcal{L}$  and adding back  $\gamma_1$ :  $|B| \leq 1 + (r + 1)L_2$ . Taking the maximum over pairs completes the proof.  $\square$

*Remark 10.5* (provenance and comparison). The argument is elementary and in the folklore spirit “a list bound implies a proximity gap with a list-size loss”; we do not claim novelty of the mechanism, only of the exact constants in this normalization and of its place in the formal core: its only inputs are Theorem 9.14 and counting, so it enters the dependency chart of Table 3 without enlarging the trusted base. The theorems of [BCIKS20, BCHKS25] are strictly stronger where they apply — they reach the full Johnson radius  $1 - \sqrt{\rho}$  with error  $\text{poly}(n)/q$  — at the price of a substantially deeper proof. Note also the built-in factor of two: the recovered pair  $(P_1, P_2)$  lives at column radius  $2r$ , so the list bound is invoked at agreement  $n - 2r$ , which is why the theorem stops at half the Johnson radius rather than at the Johnson radius.

**Corollary 10.6** (self-contained safe frontier at half the Johnson radius). *Let  $C = \text{RS}[\mathbb{F}, D, k]$  with  $n \geq k + 2$ , and let  $\delta, r, L_2$  be as in Theorem 10.4, so  $n - 2r > 0$  and  $(n - 2r)^2 > (k - 1)n$ . Then*

$$\varepsilon_{\text{mca}}(C, \delta) \leq \frac{1 + (r + 1)L_2}{q}.$$

*For the challenge rows this improves the self-contained safe frontier exactly when  $\rho < \frac{1}{4}$ : the frontier moves from  $(1 - \rho)/3$  to  $\approx (1 - \sqrt{\rho})/2$ , i.e. from 0.2917 to 0.3232 at  $\rho = \frac{1}{8}$  and from 0.3125 to 0.3750 at  $\rho = \frac{1}{16}$ , while at  $\rho \in \{\frac{1}{2}, \frac{1}{4}\}$  the deep frontier  $(1 - \rho)/3$  remains the better self-contained certificate. At the deployed KoalaBear row the half-Johnson certificate reads:  $r = 307121$ ,  $\delta = r/n \approx 0.14645$ ,  $L_2 \leq 1001282$ , error at most  $(1 + 307122 \cdot 1001282)/q < 2^{-147}$  — dominated by Theorem 9.6 at this rate, but recorded as an independent certificate of the same verdict.*

*Proof.* The hypothesis gives  $n - 2r > 0$  and  $(n - 2r)^2 > (k - 1)n$ , hence  $n - 2r > \sqrt{(k - 1)n}$ ; since  $n \geq k + 2$  we also have  $\sqrt{(k - 1)n} \geq k$ , so  $n - 2r > k$ , i.e.  $2r \leq n - k - 1$ : the radius lies below half the distance. Theorem 10.1 therefore applies, and with the input of Theorem 10.4,

$$\varepsilon_{\text{mca}}(C, \delta) \leq \max\left(\frac{1 + (r + 1)L_2}{q}, \frac{r}{q}\right) = \frac{1 + (r + 1)L_2}{q}.$$

The frontier comparison is the elementary inequality  $(1 - \sqrt{\rho})/2 > (1 - \rho)/3 \iff \sqrt{\rho} < \frac{1}{2}$ ; the KoalaBear numbers are  $n = 2^{21}$ ,  $k = 2^{20}$ ,  $(n - 2r)^2 - (k - 1)n = 909700$  at  $r = 307121$ , and  $L_2 = \lfloor n(n - 2r - k + 1)/909700 \rfloor \leq 1001282$ , with  $(1 + 307122 \cdot 1001282) < 2^{38.2}$  and  $q > 2^{185.9}$ .  $\square$

**Corollary 10.7** (conditional safe frontier at half the distance). *Let  $C = \text{RS}[\mathbb{F}, D, k]$ ,  $\rho = k/n$ . Import the unique-decoding proximity gap of [BCIKS20, Thm. 4.1]: in the normalization of Definition 2.1,  $\varepsilon_{\text{ca}}(C, \delta) \leq n/q$  for every  $\delta \leq (1 - \rho)/2$ . (Only  $\varepsilon_{\text{ca}}(C, \delta) \leq \text{poly}(n)/q$  is used below, so the conclusion is robust to the exact constant.) Then for every  $\delta$  with  $2\lfloor \delta n \rfloor \leq n - k$ ,*

$$\varepsilon_{\text{mca}}(C, \delta) \leq \max\left(\frac{n}{q}, \frac{\lfloor \delta n \rfloor}{q}\right) = \frac{n}{q}.$$

*Deployed instantiations: for the KoalaBear sextic row,  $\varepsilon_{\text{mca}}(C, \delta) \leq n/q < 2^{-164} < 2^{-128}$  for every  $\delta \leq \frac{1}{4}$  (the endpoint  $r = 2^{19}$ ,  $2r = n - k$ , is included); for the circle line-round row over  $\mathbb{F}_{p^4}$ ,  $p = 2^{31} - 1$ ,  $\varepsilon_{\text{mca}} \leq n/q < 2^{-102} < 2^{-100}$  for every  $\delta \leq \frac{1}{4}$ . In both cases the certified safe frontier moves from one third to one half of the distance.*

*Proof.* Combine the import with Theorem 10.1;  $\lfloor \delta n \rfloor \leq n$  makes the maximum  $n/q$ . For the numbers:  $n/q = 2^{21}/p^6 < 2^{-164.9}$  with  $p = 2^{31} - 2^{24} + 1$ , and  $n/q = 2^{21}/(2^{31} - 1)^4 < 2^{-102.9}$ .

The circle case uses the diagonal equivalence of Lemma 7.11, which preserves both correlated-agreement errors exactly, so the import applies verbatim to the equivalent Reed–Solomon code on the torus twin coset.  $\square$

### 10.3 Widening the unsafe band: graded prefix floors on every scale

The deployed caps of Corollaries 5.4 and 7.13 certify the unsafe band  $[\frac{1}{2} - 2^{-7}, \frac{1}{2})$  using only the top *two* slots of the locator expansion at the single top scale ( $N = 256$ ). The quotient-remainder ledger already prices deeper prefixes at the multiplicative level (Lemma 3.5); two observations remain to be made. First, the graded pigeonhole runs verbatim on *every* map-smooth scale — the locator expansion  $\prod_{b \in M} (\varphi - b) = \sum_j (-1)^j e_j(M) \varphi^{m-j}$  is graded by  $\deg \varphi$  for any polynomial  $\varphi$ , in particular for the Chebyshev scales of the circle tower — which extends the tower ledger from the first staircase step to every depth. Second, the floor can be *optimized over the scale*: small scales trade prefix cost  $|B|$  per slot against fiber entropy  $\binom{n/c}{m}$ , and the optimum widens the certified unsafe band by an order of magnitude at deployed sizes. The route has an intrinsic ceiling, which we identify exactly in Remark 10.11.

**Proposition 10.8** (graded prefix floor for map-smooth scales). *Let  $B \subseteq \mathbb{F}$  be a subfield, let  $D \subseteq B$  with  $|D| = n$ , and let  $\varphi \in B[X]$  of degree  $c \geq 2$  whose fibers within  $D$  are complete and of uniform size  $c$  (the map-smooth hypothesis of Lemma 7.2); put  $N := n/c$  and  $Q := \varphi(D) \subseteq B$ , so  $|Q| = N$ . Let  $1 \leq K \leq n$  and let  $m$  satisfy  $\lceil K/c \rceil \leq m \leq N$ ; put*

$$w := m - \lceil K/c \rceil \geq 0.$$

*Then there is a  $B$ -valued received word  $U : D \rightarrow B$  with*

$$|\Lambda(\text{RS}[\mathbb{F}, D, K], 1 - \frac{mc}{n}, U)| \geq \left\lceil \frac{\binom{N}{m}}{|B|^w} \right\rceil.$$

*For  $\varphi = X^c$  on a multiplicative coset this is Lemma 3.5 with  $s = 0$  (note  $w = \lfloor (mc - K)/c \rfloor = m - \lceil K/c \rceil$ ); the content here is that the same graded pigeonhole runs on every polynomial scale, in particular on the Chebyshev scales  $T_c$  of the circle tower (Lemma 7.8), whose locators are not lacunary in the monomial basis but are perfectly graded in powers of  $T_c$ .*

*Proof.* For  $M \subseteq Q$  with  $|M| = m$ , the locator  $\Lambda_M := \prod_{b \in M} (\varphi - b) \in B[X]$  vanishes exactly on the union of the  $\varphi$ -fibers over  $M$  inside  $D$ , a set of size  $mc$ , and expands as  $\Lambda_M = \sum_{j=0}^m (-1)^j e_j(M) \varphi^{m-j}$  with  $e_j(M) \in B$  the elementary symmetric functions of  $M \subseteq B$ . For  $z = (z_1, \dots, z_w) \in B^w$  define the  $B$ -valued word

$$U_z := \left( \varphi^m + \sum_{j=1}^w z_j \varphi^{m-j} \right) \Big|_D.$$

Assign to each  $M$  the vector  $z(M) := ((-1)^j e_j(M))_{1 \leq j \leq w} \in B^w$ ; by pigeonhole some  $z^*$  is assigned by at least  $\binom{N}{m} / |B|^w$  of the  $\binom{N}{m}$  sets. For each such  $M$  put

$$c_M := U_{z^*} - \Lambda_M \Big|_D = - \sum_{j=w+1}^m (-1)^j e_j(M) \varphi^{m-j} \Big|_D,$$

a polynomial of degree at most  $c(m - w - 1) = c(\lceil K/c \rceil - 1) \leq K - 1$ , hence a codeword of  $\text{RS}[\mathbb{F}, D, K]$ ; it agrees with  $U_{z^*}$  on the  $mc$  zeros of  $\Lambda_M$  in  $D$ , so it lies in the displayed list.

The map  $M \mapsto c_M$  is injective: if  $c_M = c_{M'}$  then  $\sum_{j>w} (-1)^j (e_j(M) - e_j(M')) \varphi^{m-j} = 0$  as a polynomial of degree  $< n$  vanishing on  $D$ , hence identically; the powers  $\varphi^{m-j}$  have pairwise distinct degrees  $c(m-j)$  and are therefore linearly independent over  $\mathbb{F}$ , so  $e_j(M) = e_j(M')$  for  $j > w$ , while for  $j \leq w$  equality holds by the choice of  $z^*$ ; thus  $\prod_{b \in M} (Y - b) = \prod_{b \in M'} (Y - b)$  and  $M = M'$ .  $\square$

**Corollary 10.9** (widened unsafe band at deployed sizes). (i) (KoalaBear sextic, MCA.)  
*With  $B, \mathbb{F}, D, n = 2^{21}, k = 2^{20}$  as in Corollary 5.4,*

$$\varepsilon_{\text{mca}}(\text{RS}[\mathbb{F}_{p^6}, D, 2^{20}], \delta) > 2^{-22} \quad \text{for every} \quad \delta \in \left[\frac{15}{32}, \frac{1}{2}\right),$$

*a certified gap of  $2^{-5}$ , four times the gap  $2^{-7}$  of Corollary 5.4, by a hand-checkable certificate at scale  $c = 16$ .*

- (ii) (KoalaBear sextic, grand list challenge.) *For every  $\delta \in [\frac{15}{32}, 1 - \rho)$  there is a  $B$ -valued word  $U$  with  $|\Lambda(C, \delta, U)| > 2^{-128}q$ .*
- (iii) (Circle line-round row.) *With  $p' = 2^{31} - 1, \mathbb{F} = \mathbb{F}_{p'^4}$ , and the twin-coset  $x$ -domain of size  $n = 2^{21}, k = 2^{20}$  as in Corollary 7.13, the same two statements hold verbatim on  $[\frac{15}{32}, \frac{1}{2})$  resp.  $[\frac{15}{32}, 1 - \rho)$ , at the Chebyshev scale  $\varphi = T_{16}$ , against the target  $2^{-100}$ .*

*Proof.* All four claims instantiate Proposition 10.8 at scale  $c = 16, N = n/c = 2^{17}, m = 69632$ , so  $mc = k + 2^{16}$  and  $m/N = \frac{17}{32}$ , and convert the floor as in Corollary 5.4. The common entropy estimate is

$$\log_2 \binom{N}{m} \geq N H_2\left(\frac{17}{32}\right) - \log_2(N + 1) \geq 131072 \cdot 0.99718 - 17 > 130,685.$$

(i) Take  $K = k + 1$ , so  $w = m - \lceil (k + 1)/16 \rceil = 69632 - 65537 = 4095$ . The threshold for the deep-point conversion (Theorem 2.6 with  $\eta = \frac{1}{2}$ , admissible since  $n - mc \leq n - k - 1$ ) is a floor exceeding  $q/k + 1$ :

$$w \log_2 p + \log_2(q/k + 1) < 4095 \cdot 30.989 + 166 < 127,116 < 130,685,$$

with over 3500 bits of room. So  $\varepsilon_{\text{ca}}(C, 1 - mc/n) > \frac{1}{2k}(1 - n/q) > 2^{-22}$ , and Fact 2.4 with Lemma 2.5 extend the same lower bound to  $\varepsilon_{\text{mca}}(C, \delta)$  for every  $\delta \in [1 - mc/n, \frac{1}{2}) = [\frac{15}{32}, \frac{1}{2})$ . (ii) Take  $K = k$ , so  $w = 4096$ ; the list threshold is  $\binom{N}{m}/|B|^w > 2^{-128}q$ , i.e.  $w \log_2 p + \log_2 q - 128 < 4096 \cdot 30.989 + 58 < 127,000 < 130,685$ ; lists only grow with the radius, giving the full range. (iii)  $T_{16}$  is a polynomial of degree 16 over  $B = \mathbb{F}_{p'}$  with complete uniform fibers on the twin-coset  $x$ -domain at every dyadic scale (Lemma 7.8), so Proposition 10.8 applies with the same  $N, m, w$ ; the thresholds are weaker ( $\log_2(q/k + 1) < 104$  and  $\log_2 q - 100 < 24$ ), and the conversion runs through the diagonal equivalence of Lemma 7.11 exactly as in Corollary 7.13.  $\square$

*Remark 10.10* (exactly verified frontier). The hand-checkable rows above deliberately stop at the round gap  $2^{-5}$ . Optimizing  $(c, m)$  under exact integer verification of  $\binom{N}{m} > p^w(q/k + 1)$  resp.  $\binom{N}{m} \cdot 2^{128} > p^w q$  — a finite certificate in the sense of Definition 9.20 and Proposition 9.22 — widens the band further, to the following edges (all verified by exact arithmetic;  $\Delta := mc - k$ ):

| row / challenge             | scale $c$ | $m$   | $w$  | $\Delta$ | certified unsafe range                                                         |
|-----------------------------|-----------|-------|------|----------|--------------------------------------------------------------------------------|
| KoalaBear, MCA              | 16        | 69748 | 4211 | 67392    | $\delta \in [\frac{15331}{32768}, \frac{1}{2}) = [0.46786 \dots, \frac{1}{2})$ |
| KoalaBear, list             | 32        | 34874 | 2106 | 67392    | $\delta \in [\frac{15331}{32768}, 1 - \rho)$                                   |
| circle $p'=2^{31}-1$ , MCA  | 32        | 34873 | 2104 | 67360    | $\delta \in [\frac{30663}{65536}, \frac{1}{2}) = [0.46787 \dots, \frac{1}{2})$ |
| circle $p'=2^{31}-1$ , list | 32        | 34874 | 2106 | 67392    | $\delta \in [\frac{15331}{32768}, 1 - \rho)$                                   |

The certified gap is thus  $\approx 0.03214 \approx 2^{-4.96}$  throughout, versus  $2^{-7}$  before.

*Remark 10.11* (the entropy–subfield frontier). The optimized route has a closed-form ceiling. At scale  $c$  the floor beats the conversion threshold iff

$$\log_2 \binom{n/c}{m} \gtrsim w \log_2 |B| + \log_2(q/k),$$

and writing  $mc =: (\rho + g)n$  — so  $m/N = \rho + g$  and  $w = gN + O(1)$  — and dividing by  $N$ , this reads

$$H_2(\rho + g) \geq g \log_2 |B| + o(1)$$

for challenge-envelope field sizes ( $\log_2 q \leq 256$ , so the additive terms are  $O(1)$  against  $N \rightarrow \infty$ ). Define

$$g^*(\rho, \beta) := \sup \{ g \in (0, 1 - \rho) : H_2(\rho + g) \geq \beta g \}, \quad \beta := \log_2 |B|.$$

Every gap  $g < g^*$  is certified at large enough deployed sizes, and *no* gap beyond  $g^*$  is certifiable by graded prefix floors at any scale: the frontier is intrinsic to the route, not to our optimization. For  $\beta = 31$  (both deployed base primes):  $g^*(\frac{1}{2}) \approx 0.03216$ ,  $g^*(\frac{1}{4}) \approx 0.02748$ ,  $g^*(\frac{1}{8}) \approx 0.01920$ ,  $g^*(\frac{1}{16}) \approx 0.01238$ . The deployed KoalaBear row achieves 0.03214 at  $n = 2^{21}$ : the envelope is saturated to three digits already at deployed sizes. Two structural consequences: the left edge of the band in Open Problem 9.11 moves from  $k + 2n/N$  to  $k + g^*n - o(n)$ ; and the frontier depends on the base field only through  $\beta$ , so larger base fields *narrow* the certifiable unsafe band — a quantitative form of the subfield-confinement phenomenon of Section 6.

## 10.4 The narrowed sandwich

**Theorem 10.12** (narrowed two-sided sandwich). *Let  $\delta_C^*$  be the challenge threshold of Definition 2.3.*

- (i) (KoalaBear sextic MCA row,  $\varepsilon^* = 2^{-128}$ .) *Self-contained:*

$$\delta_C^*(2^{-128}) \in \left[ \frac{349525}{2^{21}}, \frac{15331}{32768} \right] \approx [0.16666, 0.46787].$$

*With the imported unique-decoding proximity gap (Corollary 10.7):*

$$\delta_C^*(2^{-128}) \in \left[ \frac{1}{4}, \frac{15331}{32768} \right],$$

*an open band of width  $< 0.218$ , down from 0.326 in Theorem 9.26.*

- (ii) (Circle line-round row,  $\varepsilon^* = 2^{-100}$ .) *Self-contained:*  $\delta_C^* \in [\frac{349525}{2^{21}}, \frac{30663}{65536}]$ ; *with the import:*  $\delta_C^* \in [\frac{1}{4}, \frac{30663}{65536}]$ .
- (iii) (General envelope rows.) *For any row with  $q \geq 2^{128}(n+1)$  and a map-smooth scale lattice: the safe edge is  $\max(\lfloor (n-k)/3 \rfloor, r_{\text{hJ}})/n$  self-contained, where  $r_{\text{hJ}}$  is the largest radius passing Corollary 10.6, and  $\lfloor (n-k)/2 \rfloor/n$  with the import; the unsafe edge is  $1 - \rho - g$  for every exactly certified  $g$ , with envelope  $g^*(\rho, \log_2 |B|)$  of Remark 10.11.*



| row                   | challenge        | unsafe (widened)         | safe, self-contained  | safe, with import  | open band          |
|-----------------------|------------------|--------------------------|-----------------------|--------------------|--------------------|
| KoalaBear sextic      | MCA, $2^{-128}$  | $[0.46787, \frac{1}{2})$ | $\delta \leq 0.16666$ | $\delta \leq 0.25$ | (0.25, 0.46787)    |
| KoalaBear sextic      | list, $2^{-128}$ | $\delta \geq 0.46787$    | $\delta \leq 0.29289$ | —                  | (0.29290, 0.46787) |
| circle, $p'=2^{31}-1$ | MCA, $2^{-100}$  | $[0.46788, \frac{1}{2})$ | $\delta \leq 0.16666$ | $\delta \leq 0.25$ | (0.25, 0.46788)    |
| circle, $p'=2^{31}-1$ | list, $2^{-100}$ | $\delta \geq 0.46787$    | $\delta \leq 0.29289$ | —                  | (0.29290, 0.46787) |

Table 5: Scanner output after the pincer, superseding Table 2. “Unsafe (widened)” is certified by Corollary 10.9 and Remark 10.10 (edges rounded outward to five digits: the exact edges are  $\frac{15331}{32768}$  and  $\frac{30663}{65536}$ ). “Safe, self-contained” is Theorem 9.6 resp. Theorem 9.14 as in Table 2; at rates below  $\frac{1}{4}$ , Corollary 10.6 would replace the MCA entries. “Safe, with import” is Corollary 10.7; the list rows need no import, their safe edge being the self-contained Johnson bound. Open bands are stated for the stronger certificate in each column.

*Proof.* Lower edges: Theorem 9.6 at  $r = 349525 = \lfloor (n - k)/3 \rfloor$  (errors  $349526/q < 2^{-167}$  resp.  $< 2^{-105}$ , below the respective targets, as in Table 2); Corollary 10.6 where it exceeds the deep edge; Corollary 10.7 for the imported edge  $\frac{1}{4}$  (errors  $2^{-164}$  resp.  $2^{-102}$ , below the respective targets). Upper edges: Corollary 10.9 and Remark 10.10 certify  $\varepsilon_{\text{mca}} > 2^{-22} \geq \varepsilon^*$  (resp.  $> 2^{-100}$  via the circle thresholds) on  $[\text{edge}, \frac{1}{2})$ , so  $\delta_C^*$  cannot exceed the printed edge. Part (iii) collects the corresponding general statements.  $\square$

The reduced kernel, restated once. After the pincer, the open MCA band of each deployed row is  $(\frac{1}{4}, \text{edge})$  with edge  $\approx 0.4679$ , modulo one import whose own proof is independent of everything here; self-contained, it is  $(0.1667, \text{edge})$ , and at rates below  $\frac{1}{4}$  the left endpoint improves to half the Johnson radius. Below half the distance the mutual and plain correlated-agreement questions are now provably the same question up to an explicit tangent term (Theorem 10.1), so Open Problem 9.11 survives only above half the distance, and there the certified unsafe mass presses down to the entropy-subfield envelope  $g^*$  (Remark 10.11), which the deployed rows already saturate. Closing the band therefore requires one of exactly two kinds of new mathematics: a floor construction whose received words carry more than a graded locator prefix — anything beating  $g^*$  — or a correlated-agreement theorem crossing half the distance toward capacity, to which Theorem 10.1 would immediately add the mutual layer. This is, for the first time in the program, a two-sided quantitative target with both envelopes made explicit.

## 11 Further refinements and finite certificate closure

This section records three groups of consequences that are not part of the main unsafe/safe sandwich but complete the account. First, graded rational floors extend the unsafe construction to genus-one scale maps, so the ECFFT obstruction of Remark 7.16 is a one-slot obstruction rather than a macroscopic barrier (Section 11.1), and a stereographic model treats circle codes over challenge fields that do not contain  $i$ , completing the circle-code transfer (Section 11.2). Second, explicit head words, simple-pole certifying pairs, and the optimized failure profile clarify what is constructive in the failure mechanism and what remains beyond the present method (Sections 11.3 and 11.4). Third, Section 11.5 closes the finite certificate grammar over every verdict used in the narrowed sandwich, prints the deployed certificates as integer tuples, bounds verification complexity, and isolates the single imported input to one optional clause.

### 11.1 Graded floors on rational scales: the genus-one band is macroscopic

The graded prefix floor of Proposition 10.8 pays one subfield factor per locator slot and buys  $c$  degrees of agreement per slot on a polynomial scale of degree  $c$ . On a rational scale of degree type  $(a, e)$  the slot spacing is  $a - e$  rather than  $a$ ; nothing else changes.

**Proposition 11.1** (graded prefix floor for rational scales). *Let  $B \subseteq \mathbb{F}$  be a subfield and let  $D \subseteq B$  with  $|D| = n$  be  $(\psi, a)$ -smooth in the sense of Definition 9.23, with  $\psi = f/g$  of degree type  $(a, e)$ ,  $a > e \geq 0$ ; put  $N := n/a$  and  $Q := \psi(D) \subseteq B$ . Let  $1 \leq K \leq n$  and let  $m$  satisfy*

$$am \geq K, \quad em \leq K, \quad m \leq N, \quad \text{and put} \quad w := \left\lfloor \frac{am - K}{a - e} \right\rfloor \leq m.$$

*Then there is a  $B$ -valued received word  $U : D \rightarrow B$  with*

$$|\Lambda(\text{RS}[\mathbb{F}, D, K], 1 - \frac{am}{n}, U)| \geq \left\lceil \frac{\binom{N}{m}}{|B|^w} \right\rceil.$$

*For  $e = 0$  this is Proposition 10.8 (then  $w = m - \lceil K/a \rceil$ ); for  $w \leq 1$  it recovers the slack-two window of Proposition 9.24.*

*Proof.* For  $M \subseteq Q$  with  $|M| = m$  put  $\Lambda_M := \prod_{b \in M} (f - bg)$ . Each factor has degree exactly  $a$  with leading coefficient  $\text{lc}(f)$ , so  $\deg \Lambda_M = am$  with leading coefficient  $\text{lc}(f)^m \neq 0$ ; and  $\Lambda_M \in B[X]$  vanishes at  $x \in D$  if and only if  $\psi(x) \in M$ , i.e. at exactly  $am$  points of  $D$ , by smoothness and  $g \neq 0$  on  $D$ . Expanding,

$$\Lambda_M = \sum_{j=0}^m (-1)^j e_j(M) f^{m-j} g^j,$$

with  $j$ -th term of degree  $am - j(a - e)$ , strictly decreasing in  $j$ . For  $z = (z_1, \dots, z_w) \in B^w$  define the  $B$ -valued word

$$U_z := \left( f(x)^m + \sum_{j=1}^w z_j f(x)^{m-j} g(x)^j \right)_{x \in D},$$

which makes sense because  $w \leq m$ : indeed  $am - K \leq m(a - e)$  is the hypothesis  $em \leq K$ . By pigeonhole over the prefix map  $M \mapsto z(M) := ((-1)^j e_j(M))_{j \leq w} \in B^w$  there is  $z^*$  attained by at least  $\lceil \binom{N}{m} / |B|^w \rceil$  sets  $M$ . For each such  $M$  set

$$c_M := U_{z^*} - \Lambda_M|_D = \left( - \sum_{j > w} (-1)^j e_j(M) f^{m-j} g^j \right)_{x \in D},$$

of degree at most  $am - (w + 1)(a - e) \leq K - 1$  by the choice of  $w$ , so  $c_M \in \text{RS}[B, D, K] \subseteq \text{RS}[\mathbb{F}, D, K]$ ; and  $U_{z^*}$  agrees with  $c_M$  on the  $am$  fiber points of  $M$ , i.e. within relative radius  $1 - am/n$ . Injectivity of  $M \mapsto c_M$  on the pigeonhole class: equality of words of degree  $\leq K - 1 \leq n - 1$  forces equality of polynomials, hence  $\sum_{j > w} (-1)^j (e_j(M) - e_j(M')) f^{m-j} g^j = 0$ ; dividing by  $g^m$  in  $\mathbb{F}(X)$  gives a polynomial identity in  $\psi$ , and  $\psi$  is nonconstant, hence transcendental over  $\mathbb{F}$  inside  $\mathbb{F}(X)$ , so all coefficients vanish, exactly as in the proof of Proposition 9.24. Thus  $e_j(M) = e_j(M')$  for  $j > w$ ; for  $j \leq w$  they agree through  $z^*$ ; hence  $\prod_{b \in M} (Y - b) = \prod_{b \in M'} (Y - b)$  and  $M = M'$ . The  $c_M$  are therefore distinct listed codewords for the single word  $U_{z^*}$ .  $\square$



**Corollary 11.2** (macroscopic universal cap for genus-one rows). *Let  $D$  be  $(\psi, 2)$ -smooth over  $B$  with  $\psi$  of degree type  $(2, 1)$  (Definition 9.23; in particular every FFTree domain of ECFFT [BCKL21]), let  $C = \text{RS}[\mathbb{F}, D, k]$  with  $\rho := k/n \in [\frac{1}{16}, \frac{1}{2}]$ ,  $n \geq 2^{14}$ ,  $n < q = |\mathbb{F}| < 2^{256}$ , and  $\beta := \log_2 |B| \leq 64$ . Put  $m := \lceil (k + 1 + 2^{-9}n)/2 \rceil$  and  $N := n/2$ . Then*

$$\binom{N}{m} > |B|^{2m-k-1} \left( \frac{q}{k} + 1 \right) \quad \text{and} \quad \binom{N}{m} > |B|^{2m-k} 2^{-128} q,$$

and consequently:

- (i) (MCA/CA cap)  $\varepsilon_{\text{ca}}(C, \delta) > \frac{1}{2k} (1 - \frac{n}{q})$  for every integer-admissible  $\delta \in [1 - \frac{2m}{n}, 1 - \rho - \frac{1}{n}]$ , and  $\varepsilon_{\text{mca}}(C, \delta)$  obeys the same lower bound for every  $\delta \in [1 - \rho - 2^{-9}, 1 - \rho]$ ;
- (ii) (list cap) there is a  $B$ -valued word  $U$  with  $|\Lambda(C, 1 - \frac{2m}{n}, U)| > 2^{-128} q$ , so the grand list challenge is unsafe at target  $2^{-128}$  on the same band.

The gap  $2^{-9}$  coincides with the universal multiplicative gap of Corollary 5.1: genus-one rows are no longer behind the polynomial-scale rows.

*Proof.* Write  $w_1 := 2m - k - 1$  and  $w_0 := 2m - k$  for the two prefix depths ( $K = k + 1$  and  $K = k$ ). From  $k + 1 + 2^{-9}n \leq 2m \leq k + 2^{-9}n + 3$ : first,  $w_0 \leq 2^{-9}n + 3 = N/256 + 3$ , so  $\beta w_0 \leq N/4 + 192$ ; second,  $m/N \in [\rho + 2^{-9}, \rho + 2^{-9} + 2^{-12}]$  using  $3/n \leq 3 \cdot 2^{-14} < 2^{-12}$ . On the interval  $[\frac{1}{16} + 2^{-9}, \frac{1}{2} + 2^{-9} + 2^{-12}]$  the concave  $H_2$  is minimized at an endpoint, and  $H_2(\frac{1}{16} + 2^{-9}) \geq 0.344$ ,  $H_2(\frac{1}{2} + 2^{-9} + 2^{-12}) \geq 0.999$ , so  $H_2(m/N) \geq 0.344$ . Hence

$$\log_2 \binom{N}{m} \geq N H_2(m/N) - \log_2(N + 1) \geq 0.344 N - \log_2(N + 1),$$

while both right-hand sides are at most  $2^{\beta w_0 + 256} \leq 2^{N/4 + 448}$  crudely (using  $q/k + 1 \leq q < 2^{256}$  and  $2^{-128} q < 2^{128}$ ). Since  $0.094 N \geq 0.094 \cdot 2^{13} = 770 > 448 + \log_2(N + 1)$  for all  $N \geq 2^{13}$ , both displayed inequalities hold. The hypotheses of Proposition 11.1 hold for both  $K \in \{k, k + 1\}$ :  $2m \geq k + 1 \geq K$ ;  $em = m \leq k \leq K$  because  $2^{-9}n + 3 \leq k$  for  $\rho \geq \frac{1}{16}$ ,  $n \geq 2^{14}$ ; and  $m \leq N$  because  $2m \leq k + 2^{-9}n + 3 \leq n$ . Part (ii) is the floor at  $K = k$ . For (i), the floor at  $K = k + 1$  exceeds  $q/k + 1 \geq \lceil 2q \varepsilon_{\text{ca}} \rceil$  whenever  $\varepsilon_{\text{ca}}(C, \delta) \leq \frac{1}{2} (\frac{1}{k} - \frac{n}{kq})$ , contradicting Theorem 2.6 at every integer-admissible  $\delta \geq \delta_0 := 1 - 2m/n$  (admissible since  $n - 2m \leq n - k - 1$ ); the assembly through Fact 2.4 at  $\delta_0$  and Lemma 2.5 up to  $1 - \rho$  is verbatim the proof of Theorem 7.4. The band containment is  $1 - 2m/n \leq 1 - \rho - 2^{-9}$ , i.e.  $2m \geq k + 2^{-9}n$ .  $\square$

*Remark 11.3* (the trichotomy of Remark 7.16, resolved). Remark 7.16 offered three exits from the lacunarity collapse: a lacunary basis of isogeny locators, a different received-word family, or a matching safe-side theorem showing the collapse essential. The second exit is the one that exists, and the collapse was not essential: the one-slot analysis priced the window width  $2(a - e)$  as the *reach* of the route, when it only sets the *exchange rate* —  $a - e$  degrees of agreement per subfield factor, against an entropy budget of  $H_2$  bits per fiber. Optimizing the number of slots gives the closed-form frontier

$$H_2(\rho + g^*) = g^* \cdot \beta \cdot \frac{a}{a - e},$$

so the degree type enters only through the ramification-defect ratio  $a/(a - e)$ : polynomial scales ( $e = 0$ ) have ratio 1 and recover Remark 10.11; genus-one  $x$ -maps have  $e = a - 1$ ,

ratio  $a$ , optimized at the first fold  $a = 2$ , i.e. an effective subfield cost of  $2\beta$ . Numerically  $g^*(\frac{1}{2}, 62) \approx 0.0161 \approx 2^{-5.96}$  at  $\beta = 31$  and  $g^*(\frac{1}{2}, 128) \approx 0.0078 \approx 2^{-7}$  at  $\beta = 64$ : macroscopic in every case, against the  $2/n$  of the one-slot route. The safe side needs no new work: Theorems 9.6, 10.1 and 10.4 and Corollary 10.7 are domain-agnostic, so every  $(\psi, 2)$ -smooth row carries the full narrowed sandwich of Theorem 10.12's shape — self-contained safe frontier at  $(1 - \rho)/3$ , safe at  $(1 - \rho)/2$  modulo the single import, unsafe from gap  $2^{-9}$  (and, row by row, from the optimized frontier  $g^*(\rho, 2\beta)$ ). This answers Open Problem 12.4(a) as posed.

## 11.2 The stereographic model: circle codes over every challenge field

The torus uniformization of Lemma 7.11 needs  $i \in \mathbb{F}$ . The circle is a conic with the rational point  $(1, 0)$ , so it also has a uniformization defined over  $\mathbb{F}_p$  itself — the stereographic projection — and this one transports the code theory to *every* challenge field. The price is that the resulting Reed–Solomon domain is no longer a multiplicative coset; but it is rationally smooth of the same  $(2, 1)$  type as the genus-one rows, so Section 11.1 covers it.

**Lemma 11.4** (stereographic uniformization of circle codes; no  $i$  required). *Let  $p \equiv 3 \pmod{4}$ , let  $\mathcal{D} = gH \cup g^{-1}H \subseteq \mathbb{U}$  be a twin coset of size  $2M$  with  $\text{ord}(g) \nmid 2M$  and  $4 \mid M$ , let  $E := \iota^{-1}(\mathcal{D}) \subseteq \mathcal{C}(\mathbb{F}_p)$ , and let  $\mathbb{F} \supseteq \mathbb{F}_p$  be any finite field. Define  $s(x, y) := y/(1 + x)$  and  $T := s(E)$ .*

- (i) *The four torsion points  $(\pm 1, 0), (0, \pm 1)$  lie in  $\iota^{-1}(H)$  and hence not in  $E$ ;  $s$  is injective on  $E$  with inverse  $s \mapsto (\frac{1-s^2}{1+s^2}, \frac{2s}{1+s^2})$ ; and  $T \subseteq \mathbb{F}_p \setminus \{0, \pm 1\}$  with  $|T| = 2M$ . (Over  $\mathbb{F}_p$  one has  $1 + s^2 \neq 0$  throughout, since  $p \equiv 3 \pmod{4}$ .)*
- (ii) *With the diagonal vector  $d := ((1+s(P)^2)^{-w})_{P \in E}$  and the coordinate bijection  $s|_E : E \rightarrow T$ ,*

$$\mathcal{C}_w(\mathbb{F}, E) = d \circ \text{RS}[\mathbb{F}, T, 2w + 1].$$

*Consequently, by Lemma 7.10, the bivariate circle code on  $E$  and  $\text{RS}[\mathbb{F}, T, 2w + 1]$  have identical list sizes at every radius and identical  $\varepsilon_{\text{ca}}$  and  $\varepsilon_{\text{mca}}$  at every (pair of) radii — with no hypothesis on  $i \in \mathbb{F}$ .*

- (iii)  *$T$  is  $(\psi, 2)$ -smooth over  $B := \mathbb{F}_p$  in the sense of Definition 9.23 for the tangent-halving map  $\psi(s) := \frac{s^2-1}{2s}$ , of degree type  $(2, 1)$ : the  $\psi$ -fibers within  $T$  are the pairs  $\{s(P), s(P\tau)\} = \{s, -1/s\}$ ,  $\tau := (-1, 0)$ , complete because  $\tau \in \iota^{-1}(H)$  and of uniform size 2 because  $s^2 \neq -1$  in  $\mathbb{F}_p$ ; and  $N = M$ ,  $Q = \psi(T) \subseteq \mathbb{F}_p$ .*

*Proof.* (i)  $H$  is the unique subgroup of order  $M$  of the cyclic group  $\mathbb{U}$  of order  $p + 1$ ; since  $4 \mid M$  it contains all elements of order dividing 4, whose  $\iota^{-1}$ -images are exactly the four torsion points. From  $\text{ord}(g) \nmid 2M$  we get  $g, g^{-1} \notin H$ , so  $gH \cap H = g^{-1}H \cap H = \emptyset$  and  $\mathcal{D} \cap H = \emptyset$ ; in particular  $(-1, 0) \notin E$ , so  $s$  is defined on  $E$ , and  $(1, 0), (0, \pm 1) \notin E$ , so  $0, \pm 1 \notin T$ . For the inverse formula: from  $x^2 + y^2 = 1$ ,  $1 + s^2 = \frac{(1+x)^2 + y^2}{(1+x)^2} = \frac{2}{1+x}$ , whence  $\frac{1-s^2}{1+s^2} = (1+x) - 1 = x$  and  $\frac{2s}{1+s^2} = s(1+x) = y$ ; and  $1 + s^2 \neq 0$  for  $s \in \mathbb{F}_p$  because  $-1$  is a nonsquare when  $p \equiv 3 \pmod{4}$ .

(ii) Every class of degree  $\leq w$  has the canonical form  $f_0(x) + y f_1(x)$  with  $\deg f_0 \leq w$ ,  $\deg f_1 \leq w - 1$ , a space of dimension  $2w + 1$  (proof of Lemma 7.11). Substituting the inverse of (i) and clearing denominators,

$$(1 + s^2)^w (f_0(x) + y f_1(x)) \Big|_{x=\frac{1-s^2}{1+s^2}, y=\frac{2s}{1+s^2}}$$

is a polynomial in  $s$  of degree at most  $2w$ : the monomial  $x^j$  contributes  $(1-s^2)^j(1+s^2)^{w-j}$  and  $y x^j$  contributes  $2s(1-s^2)^j(1+s^2)^{w-1-j}$ . These  $2w+1$  polynomials form a basis of  $\mathbb{F}[s]_{\leq 2w}$ : the first family is even, the second odd, so it suffices to check each family separately; and in the variable  $v := s^2$  the first family is  $\{(1-v)^j(1+v)^{w-j}\}_{j \leq w}$ , the image of the monomial basis  $\{Y^j Z^{w-j}\}$  under the invertible substitution  $(Y, Z) = (1-v, 1+v)$  ( $p$  odd), and similarly for the odd family. Hence  $F \mapsto (1+s^2)^w F$  is a linear isomorphism from the circle space onto  $\mathbb{F}[s]_{\leq 2w}$ , and evaluating at  $P \in E$  gives  $F(P) = (1+s(P)^2)^{-w} f(s(P))$  with  $f \in \mathbb{F}[s]_{\leq 2w}$  ranging over all of  $\text{RS}[\mathbb{F}, T, 2w+1]$ : the displayed code identity. The vector  $d$  is  $\mathbb{F}_p$ -valued and everywhere nonzero, so Lemma 7.10 applies.

(iii) The group law is  $(x_1, y_1) \cdot (x_2, y_2) = (x_1 x_2 - y_1 y_2, x_1 y_2 + x_2 y_1)$ , so  $P\tau = (-x, -y)$  and

$$s(P) s(P\tau) = \frac{y}{1+x} \cdot \frac{-y}{1-x} = \frac{-y^2}{1-x^2} = -1,$$

using  $x^2 + y^2 = 1$  and  $y \neq 0$  on  $E$ . Since  $\tau \in \iota^{-1}(H)$  ( $2 \mid M$ ) and  $\mathcal{D}H = \mathcal{D}$  coset-wise,  $E\tau = E$ , so the involution  $s \mapsto -1/s$  preserves  $T$ . The fibers of  $\psi$ :  $\psi(s) = b$  if and only if  $s^2 - 2bs - 1 = 0$ , whose two roots multiply to  $-1$ ; hence every fiber is  $\{s, -1/s\}$ , of size exactly 2 since  $s^2 = -1$  is impossible in  $\mathbb{F}_p$ , and complete within  $T$  by the closure just shown. Finally  $f = X^2 - 1$  and  $g = 2X$  are coprime of degrees  $2 > 1$ , and  $g$  is nonvanishing on  $T$  because  $0 \notin T$ .  $\square$

**Corollary 11.5** (circle codes over every challenge field; Open Problem 12.4(b)). *Let  $E$  be a standard-position circle domain as in Lemma 11.4 with  $M = 2^{21}$  (the deployed shape:  $n_c = 2M = 2^{22}$ ,  $k_c = 2^{21} + 1$ ), and let  $\mathbb{F} \supseteq \mathbb{F}_p$ ,  $p = 2^{31} - 1$ , be any challenge field,  $q = |\mathbb{F}|$ .*

- (i) *If  $q < 2^{100}$  — in particular  $\mathbb{F} = \mathbb{F}_p$  and  $\mathbb{F} = \mathbb{F}_{p^3}$ , hence every field without  $i$  among the extensions  $\mathbb{F}_{p^r}$ ,  $r \leq 4$ , that deployed verifiers sample from, since  $i \in \mathbb{F}_{p^r}$  if and only if  $2 \mid r$  — the row is degenerately unsafe at every deployed target by Proposition 9.19, and Open Problem 12.4(b) is out of scope there as posed.*
- (ii) *If  $i \in \mathbb{F}$ , Corollaries 7.12 and 7.13 apply verbatim through Lemma 7.11.*
- (iii) *If  $i \notin \mathbb{F}$  and  $q \geq 2^{100}$  (the first such field is  $\mathbb{F}_{p^5}$ ), the stereographic model of Lemma 11.4 transports the row to  $\text{RS}[\mathbb{F}, T, k_c]$  on the  $(\psi, 2)$ -smooth tangent domain  $T$ , and Proposition 11.1 with*

$$m = 2^{20} + 2^{15} + 1, \quad w = 2m - (k_c + 1) = 2^{16}, \quad \binom{2^{21}}{m} > p^{2^{16}} \cdot 2^{256}$$

(verified exactly in integer arithmetic; the margin exceeds  $2^{63000}$ ) gives, for every  $q \in (n_c, 2^{256})$  at once, an MCA and list cap on

$$\delta \in [1 - \rho_c - 2^{-6}, 1 - \rho_c), \quad \text{error} > \frac{1}{2k_c} \left(1 - \frac{n_c}{q}\right) > 2^{-23},$$

by the assembly of Corollary 11.2 (here  $\beta = \log_2 p$ , effective cost  $2\beta$ , asymptotic frontier  $g^* \approx 2^{-5.96}$ ). The safe side is generic: Theorem 9.6 applies to the circle code directly (as already noted after its proof), and Theorems 10.1 and 10.4 and Corollary 10.7 apply to the diagonal model. Every bivariate circle row over every challenge field therefore carries a two-sided sandwich, and the deep-point conversion Theorem 2.6 does extend to the bivariate setting — through the stereographic model rather than a new proof.

*Proof.* (i) is Proposition 9.19 with  $q = p < 2^{31}$  resp.  $q = p^3 < 2^{93}$ :  $\varepsilon_{\text{mca}} \geq 1/q > 2^{-100}$  at every radius, so  $\delta_C^*(\varepsilon^*) = 0$  for every deployed target  $\varepsilon^* \leq 2^{-100}$ ; the parity criterion is  $\text{ord}(i) = 4 \mid p^r - 1$  if and only if  $r$  is even, as  $p \equiv 3 \pmod{4}$ . (ii) is quoted. (iii): the transport is Lemma 11.4(ii), the smoothness is (iii), and the hypotheses of Proposition 11.1 hold with  $K \in \{k_c, k_c + 1\}$ :  $2m = k_c + 1 + 2^{16} \geq K$ ,  $em = m \leq k_c \leq K$ ,  $m \leq N = 2^{21}$ . The single displayed integer inequality dominates both thresholds for every  $q < 2^{256}$ , since  $q/k_c + 1 \leq q < 2^{256}$  and  $2^{-100}q < 2^{156}$ ; the edge satisfies  $1 - 2m/n_c = 1 - \rho_c - 2^{-6} - 2^{-21} + 2^{-22} \leq 1 - \rho_c - 2^{-6}$ , and  $\lfloor \delta_0 n_c \rfloor = n_c - 2m \leq n_c - k_c - 1$ , so Theorem 2.6, Fact 2.4, and Lemma 2.5 assemble exactly as in Corollary 11.2; the error numerics use  $q \geq p^5$ , so  $\frac{1}{2k_c}(1 - \frac{n_c}{q}) > 2^{-22}(1 - 2^{-21})(1 - 2^{-133}) > 2^{-23}$ .  $\square$

### 11.3 Explicit witness words and explicit certifying pairs

Every floor so far selects its received word by pigeonhole over a prefix. At the first staircase step the pigeonhole can be removed entirely: if the support is assembled from *antipodal pairs*, its first elementary symmetric function vanishes identically, and the received word collapses to a pure power.

**Theorem 11.6** (explicit head-and-pairs floor). *Let  $D$  be  $(\varphi, c)$ -smooth over  $B$  (Definition 7.1) with  $N := n/c$  even, and suppose  $Q := \varphi(D)$  satisfies  $-Q = Q$  and  $0 \notin Q$ , so that  $Q$  is partitioned into  $N/2$  antipodal classes  $\{y, -y\}$ . Let  $K \leq n$  and let  $m$  satisfy  $2 \leq m \leq N - 2$  and  $cm \leq K - 1 + 2c$ .*

(i) *If  $m$  is even, the pure power word  $u := \varphi^m|_D$  satisfies*

$$|\Lambda(\text{RS}[\mathbb{F}, D, K], 1 - \frac{cm}{n}, u)| \geq \binom{N/2}{m/2},$$

*the listed codewords being  $c_M := u - \Lambda_M|_D$  for  $M$  a union of  $m/2$  antipodal classes.*

(ii) *If  $m$  is odd, then for every  $t \in Q$  the one-head word  $u := (\varphi^m - t\varphi^{m-1})|_D$  satisfies*

$$|\Lambda(\text{RS}[\mathbb{F}, D, K], 1 - \frac{cm}{n}, u)| \geq \binom{N/2 - 1}{(m-1)/2},$$

*with  $M = \{t\} \cup ((m-1)/2 \text{ antipodal classes distinct from } \{t, -t\})$ .*

*In both cases the word, the family, and every listed codeword are explicit closed forms, with no pigeonhole and no subfield division. Both deployed towers qualify: on a multiplicative coset,  $Q$  is a coset of the even-order subgroup  $\mu_N$ , so  $-Q = Q$  and  $0 \notin Q$ ; on a Chebyshev domain  $D = \chi(\mathcal{D})$  with  $2c \mid M$ , the antipodal classes of  $Q = T_c(D)$  are exactly the  $T_2$ -fibers inside  $Q$ , complete of size 2 by Lemma 7.8 at scales  $c$  and  $2c$  (two elements with equal  $T_2$ -value satisfy  $y' = -y$ , and  $0 \notin Q$  since a  $T_2$ -fiber  $\{0\}$  would have size 1).*

*Proof.* In case (i),  $M$  is a union of antipodal pairs, so  $e_1(M) = 0$  and  $\Lambda_M = \varphi^m + \sum_{j \geq 2} (-1)^j e_j(M) \varphi^{m-j}$ , whence  $c_M := u - \Lambda_M|_D = -\sum_{j \geq 2} (-1)^j e_j(M) \varphi^{m-j}|_D$  has degree at most  $c(m-2) \leq K-1$  and agrees with  $u$  on the  $cm$  fiber points of  $M$ . In case (ii),  $e_1(M) = t + \sum(\text{pair sums}) = t$ , so the slot-1 coefficient of  $\Lambda_M$  is  $-t\varphi^{m-1}$  for every  $M$  in the family, and  $c_M := u - \Lambda_M|_D$  again consists of the slots  $j \geq 2$  only, of degree at most  $c(m-2) \leq K-1$ ; note that  $e_2(M)$  and beyond are permitted to vary, since they land in the

codeword rather than in the received word. Injectivity:  $c_M = c_{M'}$  as words forces  $\Lambda_M = \Lambda_{M'}$  on all of  $D$ , and both are polynomials of degree  $cm < cN = n$ , hence equal as polynomials, with equal root multisets, hence  $M = M'$  through the fibers. The family counts are the number of choices of antipodal classes.  $\square$

**Corollary 11.7** (explicit witnesses at the deployed rows). *At the KoalaBear row of Corollary 5.4 ( $n = 2^{21}$ ,  $k = 2^{20}$ ,  $B = \mathbb{F}_p$ ,  $q = p^6$ ) and at the Chebyshev line-round row of Corollary 7.13(a) ( $n = 2^{21}$ ,  $k = 2^{20}$ ,  $q = p'^4$ ,  $p' = 2^{31} - 1$ , target  $2^{-100}$ ):*

- (i) (*MCA, gap  $2^{-8}$* ) With  $c = 2^{12}$ ,  $m = k/c + 2 = 258$ ,  $K = k + 1$ : the explicit word  $u = X^{k+2^{13}}|_D$  (resp.  $u = T_c^{258}|_D$ ) has list at agreement  $k + 2^{13}$  at least  $\binom{256}{129} \geq 2^{251}$ , exceeding  $q/k + 1 < 2^{166}$  (resp.  $< 2^{104}$ ); by Theorem 2.6 with  $\eta = \frac{1}{2}$ , exactly as in Theorem 4.1,  $\varepsilon_{\text{ca}}(C, \delta) > \frac{1}{2k}(1 - \frac{n}{q})$  on the integer-admissible part of  $[\frac{1}{2} - 2^{-8}, \frac{1}{2})$  and  $\varepsilon_{\text{mca}}$  obeys the same bound throughout it.
- (ii) (*lists, gap  $2^{-7}$* ) With  $c = 2^{14}$ ,  $m = k/c + 1 = 65$ ,  $K = k$ : for any  $t \in Q$  the explicit word  $u = (X^{k+2^{14}} - tX^k)|_D$  (resp. its  $T_c$  analogue) has list at agreement  $k + 2^{14}$  at least  $\binom{63}{32} = 916312070471295267 > 2^{59}$ , exceeding  $2^{-128}q < 2^{58}$  (resp.  $2^{-100}q < 2^{24}$ ): the first staircase step of the grand list challenge, at its original printed edge  $\frac{1}{2} - 2^{-7}$ , is certified by a fully explicit word.

All four inequalities are verified exactly in integer arithmetic.

*Proof.* The hypotheses of Theorem 11.6 hold with  $N = 512$  resp. 128: parity as displayed,  $cm \leq K - 1 + 2c$  (with equality in (i)), and the Chebyshev scale conditions  $2c \mid M = 2^{21}$ . The threshold comparisons are the displayed integer inequalities; the MCA conversion and monotone assembly are as quoted.  $\square$

The words above certify lists explicitly, and Theorem 2.6 converts large lists into CA failure — but by contradiction, without exhibiting the failing pair. Open Problem 12.1, however, asks for the *pair*. The proof of Theorem 2.6 is more generous than its statement: it converts through the completely explicit simple-pole lines  $(U/(x - \alpha), -1/(x - \alpha))$  — precisely the residue-line normal form with extension denominator that Corollary 6.2 predicts — and its collision count can be read as a lower bound holding for *most* poles.

**Theorem 11.8** (explicit certifying pairs, up to the choice of a pole). *At the KoalaBear row of Corollary 5.4, let  $u := X^{k+2^{13}}|_D$  be the pure power word of Corollary 11.7(i), let  $L_0 := \lceil (q - n)/k \rceil < 2^{166}$ , and fix any explicit subfamily of  $L_0$  of its listed codewords  $P_M \in \mathbb{F}[X]_{<k+1}$  (e.g. the lexicographically least  $L_0$  antipodal-class sets;  $L_0 \leq \binom{256}{129}$ ). For  $\alpha \in \Omega := \mathbb{F} \setminus D$  define the explicit pair*

$$f_\alpha(x) := \frac{u(x)}{x - \alpha}, \quad g_\alpha(x) := -\frac{1}{x - \alpha} \quad (x \in D).$$

*Then for at least half of all  $\alpha \in \Omega$ , the pair  $(f_\alpha, g_\alpha)$  has at least*

$$\left\lceil \frac{q - n}{3k} \right\rceil$$

*distinct CA-bad slopes at radius  $\delta = \frac{1}{2} - 2^{-8}$ , namely among the explicit values  $\{P_M(\alpha)\}$ ; its CA-bad-slope density exceeds  $2^{-21.6} > 2^{-22}$  there and at every larger integer-admissible radius, and its MCA-bad-slope density exceeds the same bound throughout  $[\frac{1}{2} - 2^{-8}, \frac{1}{2})$ . All but at most*

$p$  of these poles have  $\alpha \notin B$ , and for those the pair is genuinely  $\mathbb{F}$ -valued, in accordance with Corollary 6.2. The same construction on the circle line-round row gives density  $> 2^{-21.6} > 2^{-23}$  for at least half of all poles.

*Proof.* Each  $P_M$  agrees with  $u$  on at least  $a = k + 2^{13}$  points, so, as in the proof of Theorem 2.6, for every  $\alpha \in \Omega$  and every  $M$  the slope  $P_M(\alpha)$  carries the point  $f_\alpha + P_M(\alpha)g_\alpha$  to within radius  $1 - a/n = \frac{1}{2} - 2^{-8}$  of  $C$ , while the pair is farther than every  $\delta < 1 - k/n = \frac{1}{2}$  from  $C^{\equiv 2}$  by the  $(X - \alpha)G + 1$  argument there; so every distinct value  $P_M(\alpha)$  is a CA-bad slope at every integer-admissible  $\delta \in [\frac{1}{2} - 2^{-8}, \frac{1}{2})$ , hence (as failing CA is failing MCA, Fact 2.4) an MCA-bad slope throughout that interval. It remains to count distinct values for many  $\alpha$  at once. Distinct sets  $M$  give distinct polynomials  $P_M$  (Theorem 11.6), and  $P_M - P_{M'}$  is nonzero of degree at most  $k$ , so the total number of colliding pairs over all  $\alpha \in \Omega$  is at most  $k \binom{L_0}{2}$ . Let  $x^* := \lceil 2k \binom{L_0}{2} / (q - n) \rceil$ ; by Markov, fewer than  $(q - n)/2$  poles have collision count at least  $x^*$ , so at least half have collision count at most  $x^* - 1$ . For such  $\alpha$ , Cauchy–Schwarz as in Theorem 2.6 gives

$$M(\alpha) \geq \frac{L_0^2}{L_0 + 2(x^* - 1)} \geq \frac{L_0}{1 + 2k(L_0 - 1)/(q - n)} > \left\lceil \frac{q - n}{3k} \right\rceil - 1,$$

so, since  $M(\alpha)$  is an integer,  $M(\alpha) \geq \lceil (q - n)/(3k) \rceil$ . The last comparison and  $\lceil (q - n)/(3k) \rceil / q > 2^{-21.6}$  are verified exactly in integer arithmetic at both rows. The subfield count is  $|\Omega \cap B| \leq |B| = p$ ; for  $\alpha \notin B$  the word  $g_\alpha$  is not  $B$ -valued (its value at any  $x \in D \subseteq B$  lies in  $B$  only if  $x - \alpha \in B$ ), and no common scalar repairs this while keeping  $f_\alpha$  a  $B$ -multiple, so the pair falls outside the inert class of Lemma 6.1, as Corollary 6.2 requires of any pair with this density.  $\square$

*Remark 11.9* (what remains of Open Problem 12.1). Theorem 11.8 answers Open Problem 12.1 at  $\delta = \frac{1}{2} - 2^{-8}$ , on the upper half of the stated band, in the predicted normal form, and for both densities the problem asks about; the non-constructiveness has collapsed from a received word — an object of  $n \log_2 q$  bits with no known certificate — to the choice of one field element in an explicit majority subset of  $\mathbb{F} \setminus D$ . Two residues remain. First, a certified *individual* pair: derandomizing the pole. Second, the left half of the band: at gap exactly  $2^{-7}$  the explicit family at scale  $2^{13}$  has only  $\binom{128}{65} < 2^{125}$  members, short of the  $\approx 2^{166}$  the counting needs, so explicitness there — and throughout the widened band of Corollary 10.9, where the floors are pigeonhole-selected deep prefixes — would require explicit families with prescribed deep prefixes, i.e. explicit subsets of  $\mu_N$  with many prescribed elementary symmetric functions. We record this as the surviving form of Open Problem 12.1.

## 11.4 The failure profile: the deep-point ceiling

Theorem 2.6 was used throughout with  $\eta = \frac{1}{2}$ , giving error floors of  $\frac{1}{2k}(1 - \frac{n}{q})$ . The dichotomy improves when the list is far above threshold, and the improvement is exactly quantifiable.

**Proposition 11.10** (optimized failure profile). *Let  $C = \text{RS}[\mathbb{F}, D, k]$ , let  $\delta$  be integer-admissible ( $\lfloor \delta n \rfloor \leq n - k - 1$ ,  $q > n$ ), and put  $T := \frac{1}{k}(1 - \frac{n}{q})$ . If some received word has  $|\Lambda(C^+, \delta, u)| \geq 1 + \kappa q T$  for a real  $\kappa > 0$ , then*

$$\varepsilon_{\text{ca}}(C, \delta) \geq \frac{\kappa}{\kappa + 1} T.$$

*Proof.* Write  $x := \varepsilon_{\text{ca}}(C, \delta)$ ; if  $x \geq T$  there is nothing to prove, so assume  $x < T$  and set  $\eta := x/T \in [0, 1)$ . Then  $x = \eta T$ , so Theorem 2.6 applies and gives  $L \leq \lceil qx/(1 - \eta) \rceil \leq qx/(1 - \eta) + 1$ , i.e.  $\kappa q T \leq L - 1 \leq qx/(1 - x/T)$ . Rearranging,  $\kappa(T - x) \leq x$ , i.e.  $x \geq \kappa T / (\kappa + 1)$ .  $\square$



**Corollary 11.11** (deployed profile at the first staircase step). *At the KoalaBear row, the fiber floor of Lemma 7.2 at scale  $2^{13}$  gives  $L \geq \lceil \binom{256}{130}/p \rceil > 2^{220}$  at agreement  $k + 2^{14}$ , hence  $\kappa \geq 2^{54}$  and*

$$\varepsilon_{\text{ca}}(C, \delta) \geq (1 - 2^{-54}) \frac{1}{k} \left(1 - \frac{n}{q}\right)$$

*for every integer-admissible  $\delta \in [\frac{1}{2} - 2^{-7}, \frac{1}{2} - 2^{-21}]$ , with the same lower bound for  $\varepsilon_{\text{mca}}(C, \delta)$  throughout  $[\frac{1}{2} - 2^{-7}, \frac{1}{2}]$ : twice the printed constant of Corollary 5.4, saturating the dichotomy's ceiling to relative error  $2^{-54}$ . On the circle line-round row the same floor gives  $\kappa \geq 2^{116}$  and the factor  $1 - 2^{-116}$ .*

*Proof.*  $\kappa = \lfloor (L - 1)k/(q - n) \rfloor \geq 2^{54}$  (resp.  $2^{116}$ ) is verified exactly;  $\kappa/(\kappa + 1) \geq 1 - 2^{-54}$ ; the list persists at every larger radius, and Fact 2.4 with Lemma 2.5 extend as usual.  $\square$

*Remark 11.12* (the ceiling, and what keeps Open Problem 12.2 open). Two ceilings frame Open Problem 12.2. First, the route ceiling: as  $\kappa \rightarrow \infty$  the profile of Proposition 11.10 tends to  $T = \frac{1}{k}(1 - \frac{n}{q})$  and no list, however large, certifies more through Theorem 2.6 — error beyond  $\approx \frac{1}{k}$  requires strengthening the dichotomy itself, which is exactly the unnamed problem following Open Problem 12.2. Second, at the specific radius  $1 - \rho - \frac{1}{64}$  of Open Problem 12.2 over *prime* fields  $2^{150} \leq p \leq 2^{256}$ , the present floors certify nothing at all: the only subfield is  $B = \mathbb{F}_p$ , so every prefix frontier obeys  $g^* \leq 1/\beta = 1/\log_2 p \leq \frac{1}{150} < \frac{1}{64}$ , and the radius lies strictly outside the entropy-subfield envelope. (There is also no confinement barrier there: Lemma 6.1 caps  $B$ -valued pairs at  $|B|/q$ , which is vacuous when  $B = \mathbb{F}$ .) So Open Problem 12.2 is genuinely beyond both mechanisms of this paper: it needs either non-prefix floors at macroscopic gaps over prime fields, or conversions past  $\frac{1}{k}$ . The certified state after this section: error at least  $(1 - 2^{-54})\frac{1}{k}(1 - \frac{n}{q})$  at gap  $2^{-7}$ , error at least  $\frac{1}{2k}(1 - \frac{n}{q})$  at the widened gaps, and error one still open everywhere.

## 11.5 Certificate closure of the two-sided sandwich

Proposition 9.22 supplied the initial certificate format and Table 3 the formal core. The pincer and the refinements above add new verdict types. This subsection closes the certificate grammar over the statements used by the two-sided sandwich, prints the deployed certificates, and isolates the single imported proximity-gap input.

**Definition 11.13** (certificate grammar, second edition). Extend Definition 9.20 by the following verdict types, each a tuple of integers together with a theorem tag:

- (V5) **Prefix-floor (PF) / rational-floor (RF)**. A scale datum (polynomial or Chebyshev  $c$ , or rational type  $(a, e)$ ), integers  $(m, w, K)$ , and the single inequality  $\binom{N}{m} > |B|^w \cdot \Theta$ ,  $\Theta \in \{q/k + 1, \varepsilon^* q\}$ : mark the band  $[1 - am/n, 1 - \rho)$  MCA-unsafe (via Propositions 10.8 and 11.1, Theorem 2.6) resp.  $1 - am/n$  list-unsafe.
- (V6) **Explicit-head (XH)**. Integers  $(c, m, K)$  with  $cm \leq K - 1 + 2c$  and the inequality  $\binom{N/2-\epsilon}{\lfloor m/2 \rfloor} > \Theta$ ,  $\epsilon \in \{0, 1\}$  by parity: as (V5), with the received word printed in closed form (Theorem 11.6).
- (V7) **Mutual-from-correlated (MC)**. An integer  $r$  with  $2r \leq w_{\min} - 1$  and a CA-verdict handle: transfer a CA-safe verdict at radius  $r/n$  to MCA-safe with error  $\max\{\varepsilon_{\text{CA}}, r/q\}$  (Theorem 10.1). The CA handle is either **HJ** — integers  $(r, L_2)$  with  $n - 2r > 0$ ,  $(n - 2r)^2 >$

$(k-1)n, (L_2+1)((n-2r)^2-(k-1)n) > n(n-2r-k+1)$ , and  $(1+(r+1)L_2)/q \leq \varepsilon^*$ , certifying the self-contained half-Johnson CA bound of Theorem 10.4 — or **CH** — the inequality  $2r \leq n - k$  with error  $n/q \leq \varepsilon^*$ , tagged IMPORT: [BCIKS20, Thm. 4.1] (Corollary 10.7).

(V8) **Profile (PR).** A floor handle and an integer  $\kappa$  with  $L \geq 1 + \kappa qT$ : certify error  $\geq \frac{\kappa}{\kappa+1}T$  (Proposition 11.10).

**Theorem 11.14** (certificate closure of the two-sided sandwich). *With the grammar of Proposition 9.22 extended by Definition 11.13:*

- (i) (Soundness) *Every verdict of Definition 11.13 is implied by its cited theorem once its integer inequalities hold; Theorem 9.21 extends verbatim.*
- (ii) (Completeness for the printed results) *Every clause of Theorem 10.12, of Table 5, of Corollary 11.2, of Corollary 11.5, and of Corollaries 11.7 and 11.11 is certified by a certificate of Proposition 9.22 or Definition 11.13; the deployed certificates are printed in Table 7.*
- (iii) (Complexity) *A (V5)/(V6) certificate is checked with  $O(N)$  multiplications of integers of  $O(N + \log q)$  bits (the binomial by its product formula, the power  $|B|^w$  by repeated squaring); every other certificate with  $O(1)$  arithmetic operations on integers of  $O(\log q + \log n)$  bits. A full row is checked in  $O(n)$  big-integer multiplications.*
- (iv) (Import isolation) *Exactly one imported statement occurs anywhere in the grammar: the IMPORT tag of the CH handle. Deleting it degrades only the safe edge, from  $(1 - \rho)/2$  to  $\max\{(1 - \rho)/3, \text{half-Johnson}\}$ , and leaves every other verdict intact. The remaining statements — Table 3 extended by the rows of Table 6 — quantify over explicitly bounded finite sets and use only the listed elementary ingredients; their dependency closure is import-free.*

*Proof.* (i) is quotation: each verdict's hypotheses are the hypotheses of its theorem, specialized to integers. (ii) is inspection of the quoted proofs against Table 7: each proof consists of its printed integer inequalities plus the cited assembly steps, all of which are verdicts or core statements. (iii):  $\binom{N}{m} = \prod_{j=1}^m \frac{N-m+j}{j}$  uses  $2m \leq 2N$  multiplications and divisions of integers of at most  $N + O(\log N)$  bits; thresholds and powers are  $O(\log w)$  squarings of  $O(w \log |B| + \log q)$ -bit integers, and  $w \log |B| = O(N)$  in every certificate above; the remaining verdicts compare  $O(1)$  products. (iv): the only IMPORT tag is CH; Theorem 10.12 lists its conditional clause separately, and the self-contained clauses cite only Theorems 9.6, 10.1 and 10.4 on the safe side. The new core rows are in Table 6; each proof above is elementary in the stated ingredients, as the reader may verify line by line.  $\square$

**Remaining residues.** The refinements above reduce the remaining landscape to a small number of explicit residues. The genus-one and circle-code questions are reduced to the same aperiodic-band shape after the rational-floor and stereographic transports of Corollaries 11.2 and 11.5. The explicit-witness question is resolved at the level of closed-form words and residue-line pairs in Corollary 11.7 and Theorem 11.8, with the remaining scope stated in Remark 11.9. The error-one profile remains open: Corollary 11.11 raises the certified profile to  $(1 - 2^{-54})^{\frac{1}{k}}(1 - \frac{n}{q})$ , while Remark 11.12 explains why reaching  $1 - o(1)$  requires new input. The central open question remains Open Problem 9.11, in the narrowed CA-form shape of



| Added core statement                  | Proof ingredients                       |
|---------------------------------------|-----------------------------------------|
| Theorem 10.1 (mutual from correlated) | minimum weight, two-point interpolation |
| Theorem 10.4 (half-Johnson CA)        | pair-list double counting, convexity    |
| Proposition 10.8, Proposition 11.1    | symmetric functions, graded pigeonhole  |
| Theorem 11.6 (explicit heads)         | antipodal pairing, no pigeonhole        |
| Theorem 11.8 (explicit pairs)         | Theorem 2.6’s collision count, Markov   |
| Proposition 11.10 (optimized profile) | one-line rearrangement of Theorem 2.6   |
| Lemma 11.4 (stereographic model)      | conic parametrization, basis change     |
| Definition 11.13, Theorem 11.14       | finite case analysis over the above     |

Table 6: The formal finite core, continued: statements added by the pincer and the refinements, extending Table 3. Each row quantifies over explicitly bounded finite sets; the dependency order is top to bottom, below the rows of Table 3. The one import (CH) is not a core row.

Corollary 10.3 and Theorem 10.12. The certificate material above is finite and checkable; a proof-assistant transcription would be implementation work rather than a new mathematical ingredient.

## 12 Discussion

**Values versus fibers.** Above the per-rate value-set reach of [Cho26b] (roughly  $2^{150}$ – $2^{162}$ ), the value route — showing that the slope multiset  $e_1(\ell^\wedge Q)$  covers a  $2^{-128}$  fraction of  $\mathbb{F}$  at a fixed large prime — runs into the per-fiber collision problem ([Cho26b, Rem. “The remaining band”]): characteristic-zero locator values can collide arbitrarily badly under reduction at a worst-case prime. [Cho26b] therefore closed the band at the  $\delta^*$  grade only conditionally and with weaker constants — error  $2^{-42}$  at gap  $2^{-11}$ , assuming  $|\mathbb{F}| \geq 2n$ , via quotient-core lists and the same conversion Theorem 2.6 — leaving the *error-one* grade open. The present route also never counts values, and improves each parameter of that cap: pigeonhole guarantees one heavy fiber regardless of how the values  $e_1(A)$  distribute, Theorem 2.6 converts heavy fibers into correlated-agreement error directly, and the subfield pigeonhole carries the construction to extension fields with no condition relating  $n$  and  $|\mathbb{F}|$ . The cost relative to the error-one results of [Cho26a, Cho26b] below  $2^{150}$  is quantitative: error  $\approx \frac{1}{2k}$  instead of  $1 - o(1)$ , and gap  $2^{-9}$  (uniformly  $2^{-10}$  across all four rates) instead of  $\frac{1}{64}$ . For the prize question — is  $\varepsilon_{\text{mca}} \leq 2^{-128}$  achievable above the stated gaps? — this cost is immaterial.

**Limits of the method.** Hypothesis (1) needs  $N H_2(\rho) \gtrsim 2 \log_2 q - \log_2 k$ , so the smallest certifiable gap is

$$\frac{2}{N} \approx \frac{2 H_2(\rho)}{2 \log_2 q - \log_2 k} \approx \frac{H_2(\rho)}{\log_2 q},$$

and the certified error is capped at  $\approx \frac{1}{k}$  by the shape of Theorem 2.6 (any  $\eta < 1$ ). Since  $q$  enters only as the size of the field from which the line challenge  $\gamma$  is drawn, this is also a protocol-facing floor: enlarging the challenge field cannot push a deep-point-conversion failure below gap  $\approx H_2(\rho)/\log_2 q_{\text{line}}$ , which quantitatively bounds the “larger challenge field” fallback in certificate frameworks. The construction also needs  $N \leq n$ , i.e.  $\log_2 q \lesssim H_2(\rho) n/2$ ; this covers the entire cryptographic regime  $\text{poly}(n) \leq q \leq 2^{o(n)}$ , but for  $q \geq 2^{H_2(\rho)n/2}$  the method is vacuous and the value-route caps of [Cho26a, Cho26b] are the only ones known. Within the band, the *error-one*

| Row / verdict          | Certificate                              | Verified inequality                                               |       |
|------------------------|------------------------------------------|-------------------------------------------------------------------|-------|
| KB MCA, PF             | $(c, m, w) = (16, 69748, 4211)$          | $\binom{2^{17}}{69748} > p^{4211}(q/k + 1)$                       | exact |
| KB list, PF            | $(32, 34874, 2106)$                      | $\binom{2^{16}}{34874} > p^{2106} 2^{-128} q$                     | exact |
| Circle MCA, PF         | $(32, 34873, 2104)$                      | $\binom{2^{16}}{34873} > p^{2104}(q/k + 1)$                       | exact |
| Circle list, PF        | $(32, 34874, 2106)$                      | $\binom{2^{16}}{34874} > p^{2106} 2^{-100} q$                     | exact |
| KB MCA, XH             | $(2^{12}, 258), u = X^{k+2^{13}} _D$     | $\binom{2^{56}}{129} > q/k + 1: 2^{251.6} \text{ vs } 2^{165.94}$ | exact |
| KB list, XH            | $(2^{14}, 65), \text{one head}$          | $\binom{63}{32} > 2^{-128} q: 2^{59.66} \text{ vs } 2^{57.94}$    | exact |
| Circle MCA, XH         | $(2^{12}, 258), u = T_c^{258} _D$        | $\binom{2^{56}}{129} > q/k + 1: 2^{251.6} \text{ vs } 2^{104}$    | exact |
| Circle list, XH        | $(2^{14}, 65), \text{one head}$          | $\binom{63}{32} > 2^{-100} q: 2^{59.66} \text{ vs } 2^{24}$       | exact |
| Both rows, MC          | $r = \lfloor (n - k)/2 \rfloor = 2^{19}$ | $2r \leq n - k = 2^{20}$                                          | —     |
| KB safe, HJ            | $(r, L_2) = (307121, 1001282)$           | $(1 + (r + 1)L_2)/q < 2^{-147}$                                   | exact |
| KB safe, CH IMPORT     | $r = 2^{19}$                             | $n/q < 2^{-164}$                                                  | —     |
| Circle safe, CH IMPORT | $r = 2^{19}$                             | $n/q < 2^{-102}$                                                  | —     |
| KB profile, PR         | $\kappa = 2^{54}$                        | $\lceil \binom{2^{56}}{130}/p \rceil > 1 + 2^{54} qT$             | exact |
| $i$ -free circle, RF   | $(m, w) = (2^{20} + 2^{15} + 1, 2^{16})$ | $\binom{2^{21}}{m} > p^{2^{16}} 2^{256}$                          | exact |

Table 7: Printed certificates for the deployed rows (Definition 11.13;  $T = \frac{1}{k}(1 - \frac{n}{q})$ ). Each line is one integer comparison; “exact” means it was verified in exact integer arithmetic. The two IMPORT lines are optional imported half-distance clauses — the only non-self-contained entries.

question at gap  $\frac{1}{64}$  (Open Problem 12.2) remains exactly as open as before: the present result bounds  $\delta^*$ , not the failure profile near capacity.

**Comparison with the cyclotomic sieve.** The sieve of [Cho26a] produces, for infinitely many primes, failures of error  $(\log p)^{-6}$  at gap  $\Theta(1/\log p)$ , but its prime set is given by Siegel–Walfisz and is ineffective. Corollary 5.1 trades error strength for universality: every field below  $2^{256}$ , fully effective, and elementary given Theorem 2.6. The two results triangulate the failure landscape from different directions.

**Isogeny-smooth reach and the lacunarity dichotomy.** The counting side of the method extends beyond multiplicative smoothness. The fiber lemma needs only a polynomial folding map with complete uniform fibers inside the domain (Definition 7.1 and Lemma 7.2), and the divisibility hypothesis  $a \mid k$  is removed by rounding, so the universal cap now covers mixed-radix multiplicative rows without integral  $\rho N$  (Corollary 7.6), the  $x$ -coordinate line rounds of circle FRI over  $p \equiv 3 \pmod{4}$ , and — through the torus uniformization, over any challenge field containing  $i$  — the bivariate circle codes themselves, at the deployed Mersenne-31/QM31 parameters with error  $> 2^{-23}$  at gap  $2^{-7}$  (Corollaries 7.12 and 7.13). The boundary of the method is now a clean dichotomy in the degree type  $(a, e)$  of the folding map: polynomial towers ( $e = 0$ : powers, Chebyshev) give gap  $\approx 2/N$ , while elliptic isogeny towers ( $e = a - 1$ : ECFFT) collapse the slack-two window to gap  $2/n$  (Remark 7.16). Subfield confinement and the extension-pole mechanism also carry over to circle rows (Corollary 7.14), so the deployed circle failures are again fiber-borne, genuinely extension-valued, and constructively reachable from any supplied list via extension poles.

**Pressure on the packing conjecture.** By the exact normal form of [Cho26b],  $\varepsilon_{\text{mca}}(C, \delta) = \max_t \Lambda_t^{\text{NC}}(\delta)/q$ , where  $\Lambda^{\text{NC}}$  is a noncontained residue-line packing number. Corollary 5.4 there-

fore forces  $\max_t \Lambda^{\text{NC}} \geq (q-n)/2k \approx 2^{164}$  at gap  $2^{-7}$  on the deployed sextic — superpolynomially many noncontained residue lines at  $n = 2^{21}$ . This puts no direct pressure on the final MCA conjecture of [Cho26b, Conj. “Final floor- and quotient-corrected MCA asymptotic”]: at gap  $2^{-7}$  both of its corrected-reserve hypotheses fail ( $\sigma = 2^{14}$  lies below  $Cn/\log_2 n \approx 2^{16.6}$ , and  $\sigma \log_2 q_{\text{gen}} = 2^{14} \cdot 31 \approx 2^{19}$  is far below  $\log_2 \binom{n}{k+\sigma} \approx 2^{21}$ ), so the conjectured ceiling of the form  $(n^{1+o(1)} + 2^{(\beta/H_2) Q_{H_2}(n)})/q$  is simply not asserted there — the packing mass extracted through Theorem 2.6 lives in the region the framework already marks as below-floor. The genuine structural question this paper raises for the framework of [Cho26b] is whether conversion-extracted fiber mass persists *above* the corrected reserve: whether, at reserves where the conjecture does apply, the fiber route still produces packings beyond the  $n^{1+o(1)}$  aperiodic term plus the quotient-periodic term, which would force an additional fiber-borne term into the conjectured ceiling. This is open.

**Verification caveat.** The main cap no longer depends on importing the Crites–Stewart theorem: Theorem 2.6 proves the needed deep-point list-to-CA conversion directly, including the integer-radius condition  $f \leq n-k-1$ . Theorem 2.16 remains a separate slacked fallback imported through the BCHKS/ABF route, and readers using that fallback should consult the underlying source directly. The main composition is deliberately shallow: it uses the self-contained conversion, the locator-fiber pigeonhole, and support-wise MCA monotonicity.

**What remains after the pincer and certificate closure.** The structural ledgers, extension-line accounting, scanner, and certificate grammar are closed at the level needed for the printed sandwich. The deep regime and the Johnson list regime are theorems; the near-capacity region is unsafe by the quotient and graded floors; and the genus-one and circle field-of-definition extensions are covered by Corollaries 11.2 and 11.5. What remains is the aperiodic band conjecture of Open Problem 9.11, the error-one profile question of Open Problem 12.2, and the explicit deployed-pair strengthening of Open Problem 12.1.

**Formerly open questions resolved above.** The genus-one macroscopic-cap and circle field-of-definition questions are resolved by Corollaries 11.2 and 11.5. The explicit-witness question is resolved at the first staircase step and for simple-pole certifying pairs at gap  $2^{-8}$  (Corollary 11.7 and Theorem 11.8); its full-gap strengthening is restated below.

### Remaining open problems.

**Open Problem 12.1** (explicit deployed witnesses at the full deployed gap). For the KoalaBear-sextic deployed row, exhibit explicit pairs  $(f, g)$  with MCA-bad-slope density  $> 2^{-22}$  at the gap  $2^{-7}$  certified by Corollary 5.4; on the integer-admissible subinterval, exhibit the corresponding CA-bad-slope density as well. By Corollary 6.2 such pairs exist and are not  $B$ -rational, and the natural candidates are residue-line normal forms with denominator  $E \in \mathbb{F}[X]$  not defined over  $B$ . The present paper gives explicit received words and simple-pole certifying pairs of exactly this normal form,  $E = X - \alpha$ , at the first staircase step and at gap  $2^{-8}$  (Corollary 11.7 and Theorem 11.8); the remaining task is to push the explicit pair construction to the full deployed gap, derandomize the choice of the pole  $\alpha$ , and print the deployed-size bucket tables.

**Open Problem 12.2.** Error one in the band: does  $\varepsilon_{\text{mca}}(C, 1 - \rho - \frac{1}{64}) = 1 - o(1)$  hold for all primes  $2^{150} \leq p \leq 2^{256}$ ? This is the per-fiber collision problem proper; Corollary 5.1 caps  $\delta^*$

there but says nothing above error  $\approx \frac{1}{2k}$ . Corollary 11.11 raises the certified profile to  $(1 - 2^{-54})^{\frac{1}{k}}$  at gap  $2^{-7}$ , and Remark 11.12 shows the question lies strictly beyond both mechanisms of this paper; it remains open.

**Open Problem 12.3.** Improve the threshold of Theorem 2.6: any strengthening of the hypothesis range  $\varepsilon_{\text{ca}} \leq \eta/k$  toward  $\varepsilon_{\text{ca}} \leq \eta \cdot \text{polylog}(q)/k$ , or of the list conclusion, immediately tightens the certified error of Theorem 4.1 toward  $1/\text{poly log}$ ; the gap, by contrast, is governed by the binomial in (1) and would need a denser fiber construction to shrink. Cf. Proposition 11.10: optimizing the dichotomy already attains  $(1 - o(1))/k$ , so passing  $1/k$  is exactly what a polylog strengthening would buy.

**Open Problem 12.4** (aperiodic band on genus-one and circle tangent domains). Extend the aperiodic band conjecture, in the narrowed CA-form of Corollary 10.3, to the tangent-domain Reed–Solomon models arising from genus-one isogeny towers and to the stereographic circle models over challenge fields not containing  $i$ . The macroscopic unsafe floors on these rows are supplied by Corollaries 11.2 and 11.5, and the generic safe side applies verbatim; what remains is the same safe-side band problem as in Open Problem 9.11, transported through the rational scale map.

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